R&S®CMW-B450A/B/D/H
Data Application Unit
User Manual

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User Manual

ROHDE&SCHWARZ

This user manual describes the following R&S®CMW options:

- R&S®CMW-B450A/B/D/H (data application unit H450A/B/D/H)
- R&S®CMW-KA065 (enabling test software for WLAN offloading)
- R&S®CMW-KA100 (enabling of IP-data interface for IPv4)
- R&S®CMW-KA120 (access to DAU on separate CMW)
- R&S®CMW-KA150 (extension of IP-data interface for IPv6 and IP multicast)
- R&S®CMW-KAA20 (IMS basic service)
- R&S®CMW-KAA21 (IMS advanced service)
- R&S®CMW-KAA30 (eMBMS service)
- R&S®CMW-KM050 (IP-based measurements)
- R&S®CMW-KM051 (IP traffic analysis)
- R&S®CMW-KM052 (IP connection security analysis)
- R&S®CMW-KS104 (EVS speech codec)

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The following abbreviations are used throughout this manual: R&S®CMW is abbreviated as R&S CMW.

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1 Preface

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1.1 Documentation Overview

This section provides an overview of the R&S CMW user documentation. Most documents are available in PDF format on the CMW customer web.

The CMW customer web is a section of GLORIS, the global Rohde & Schwarz information system, see https://gloris.rohde-schwarz.com. A registration is required.

1.1.1 Getting Started Manual / Quick Start Guide

Depending on your instrument model, there is either a getting started manual or a quick start guide.

The document introduces the R&S CMW and describes how to set up and start working with the product.

A PDF file is available on the CMW customer web.

1.1.2 User Manual and Help

Separate user manuals are provided for the base unit and the firmware applications:

- The base unit manual describes the base software and common features of the firmware applications. It also describes basic principles for manual operation and remote control. The manual includes the contents of the getting started manual or quick start guide.
- Additional documents, like this one, describe the firmware applications, including the graphical user interface of each application and the remote control commands.

All user manuals are available on the CMW customer web.

The contents of the user manuals are also available as help in the R&S CMW. The help offers quick, context-sensitive access to the information needed for operation and programming. If you perform a software update, the help is also updated.

1.1.3 Features & Functions

The features & functions documents list the available software options and the related features, settings and measurement results. The documents are available on the CMW customer web.

1.1.4 Basic Safety Instructions

Contains safety instructions, operating conditions and further important information. The printed document is delivered with the instrument.

1.1.5 Data Sheet and Brochure

The data sheet contains the technical specifications of the R&S CMW. It also lists the options and their order numbers and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

For download, refer to the internet page of your instrument model, for example www.rohde-schwarz.com/brochure-datasheet/CMW500.

1.1.6 Release Notes and Open Source Acknowledgment (OSA)

The release notes list new features, improvements and known issues per software version and installation package.

The open source acknowledgment provides verbatim license texts of the used open source software.

The documents are available on the CMW customer web.

1.1.7 Application Notes, Application Cards, White Papers, etc.

These documents deal with special applications or background information on particular topics.

See www.rohde-schwarz.com/appnotes.

1.2 R&S CMW Models

The R&S CMW software supports several instrument models. Depending on the model, only a subset of the software packages and software options is allowed.

To check whether this document is relevant for your instrument model and which software options are supported, refer to the preface of your model-specific base unit manual. The PDF filename of this manual starts with `CMW<Model>_UserManual`, for example `CMW500_UserManual`.

1.3 How to Read Firmware Application Chapters

Each firmware application is described in a separate chapter. These chapters can be read independently of each other. However, they are all organized as follows:

1. General description
2. Application sheets (optional)
3. GUI reference
4. Programming examples
5. Command reference

The chapters "System Overview" and "Remote Control" in the R&S CMW base unit manual provide additional important information independent of the individual firmware applications. The most important parts are referenced by the firmware application descriptions.

1.3.1 General Description

This section provides a general description of the firmware application, independent of a specific operation mode (manual or remote control). It gives a high-level introduction to the capabilities of the firmware application. Background information related to the network standard is given as far as it is directly related to administrable parameters. For measurement applications, a detailed description of measurement results and a description of configurable limits is given, including the relation to conformance requirements defined in network standard specifications.

1.3.2 Application Sheets

This optional section provides short application examples for select issues and related background information.

1.3.3 GUI Reference

The GUI reference describes the manual operation of the firmware application via the graphical user interface (GUI).

The description of a configuration dialog usually starts with a screenshot presenting the preset values of the parameters (sometimes preset values are modified to enable hidden parts of a dialog). Below the screenshot all shown parameters are described. For each single parameter, a link to the corresponding command description in the command reference is provided. Ranges for numeric parameters and reset values are given there.

For measurement results, links to the corresponding command descriptions are provided (commands to retrieve the results). The measurement results are described in detail in the general description chapter.

1.3.4 Programming Examples

The programming examples show how to control and configure the firmware application via a remote-control program and how to retrieve measurement results. The examples consist of comprehensive command sequences. You can check just a single command of a sequence to get an example for the syntax of this single command. But you can also consider an entire sequence showing the commands in the context of a command script, under consideration of dependencies and required orders of the commands.

The command sequences are written with the intention to list most commands of the firmware application. They do not show the fastest way for a given configuration task. The fastest way would use many reset values and omit the corresponding commands.

The examples are referenced by the command descriptions of the command reference.

1.3.5 Command Reference

The command reference provides information on the remote commands of the firmware application. The commands are grouped according to their function.

Each command description indicates the syntax of the command header and of the parameters. For input parameters the allowed ranges, reset values and default units are listed, for returned values the expected ranges and default units. Most commands have a command form and a query form. Exceptions are marked by "Setting only", "Query only" or "Event". Furthermore, a link to the programming examples is provided and the first software version supporting the command is indicated.

2 What's New in This Version

This documentation describes version 3.7.10 and later of the data application unit firmware application. Compared to version 3.5.52, it provides the following new features and changes:

- Remote commands for some "IP Connection Security Analysis" parts, see:
  - IP Connection Security, General Results
  - SSL/TLS Handshake Information
  - Keyword Search Settings and Results
- Handshake information enhanced, see `FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?`
- Filter also for "IP Connection Security Analysis" view, "Applications" tab, see Filter hotkey
- Command to query the status of the LAN DAU connector, see `SENSe:DATA:CONTrol:LAN:DAU:STATus?`
- Official support of R&S CMW-B450H

3 General Description

The data application unit (DAU, option R&S CMW-B450x) provides a common data testing solution for supported radio access technologies.

It allows you to test End-to-End (E2E) IP data transfer and to perform user plane (U-plane) tests for an IP connection to a mobile under test. The connection can be set up via a signaling application or a protocol test application.

The DAU is independent of the underlying radio access network. It provides a common user plane handling and ensures data continuity during handover from one radio access technology to another one.

The DAU also hosts IP services that have been optimized for high throughput and are running in an isolated controlled environment to ensure reproducible test results.

IP services

The following internal IP services are currently available:

- File transfer via file transfer protocol (FTP)
- Web browsing via hypertext transport protocol (HTTP)
- IP multimedia subsystem (IMS) server supporting voice over IMS and SMS over IMS (R&S CMW-KAA20 required)
- DNS server supporting DNS requests of type A, AAAA and SRV
- Evolved packet data gateway (ePDG) for WLAN offloading (R&S CMW-KA065 required)
- Mini service layer for eMBMS (R&S CMW-KAA30 required)

You can use the FTP and HTTP services, for example, to access the built-in DAU web server from the mobile. If desired, you can add own web pages to the server. An additional system drive integrated in the DAU allows the storage of large media files for data transfer tests.

The IMS server allows an IMS client on the mobile to register to the IMS domain. Optionally, an authentication can be performed. After successful registration, you can initiate mobile-originating and mobile-terminating voice over IMS calls. With two registered mobiles, you can perform mobile-to-mobile calls. Message transfer over IMS is also possible, via the short message service (SMS) or via the rich communication suite (RCS, option R&S CMW-KAA21 required).

The DNS server can be used to answer DNS queries for IPv4 addresses, IPv6 addresses and domains supporting a specific service. The DNS server database is configurable. Thus you can, for example, redirect the mobile to the web server of the DAU when it tries to browse a specific internet domain. DNS queries for which the local database contains no matching entry can be forwarded to an external DNS server.

The ePDG acts as gateway to an untrusted WLAN network. It provides security mechanisms required for WLAN offloading tests where an IPsec tunnel from the core network through the WLAN network to the mobile must be set up.

The mini service layer for eMBMS provides unicast and multicast services for eMBMS tests, controlled via CMWcards or via protocol test applications.

If connected to an external network, the DAU acts as IP gateway, separating the R&S CMW internal IP network from the external IP network. The mobile can use both the embedded IP services provided by the DAU and the IP services provided by the external network. For example, it can access web servers and DNS servers both in the internal network and in the external network.

DAU measurements

For DAU measurements, option R&S CMW-KM050 is required. It provides the following measurement applications for testing the properties of an IP connection to the mobile:

- Ping measurement, testing the network latency
- Iperf measurement, testing the throughput and reliability, using TCP/IP and UDP/IP
- Throughput measurement, indicating the total throughput at the DAU on IP level
- DNS request measurement, monitoring all DNS queries addressed to the internal DNS server
- IP logging application, creating log files of the IP traffic at the LAN DAU connector or between DAU and mobile
- IP analysis application, monitoring and analyzing the uplink and downlink IP traffic of the mobile (R&S CMW-KM051 required in addition to R&S CMW-KM050). For IP connection security analysis, you need also R&S CMW-KM052.
- IP replay application, replaying IP traffic from packet capture files
- Audio delay measurement, analyzing the transit time and timing error of voice call audio packets

The DAU supports the internet protocols IPv4 and IPv6. Both can be used individually or in parallel, depending on the IP connection established by the radio access network. IPv4 requires option R&S CMW-KA100, IPv6 requires also option R&S CMW-KA150.

Option R&S CMW-KA150 is also required for IP multicast traffic (IPv4 or IPv6) and thus for eMBMS tests.

Usually, the DAU is installed on the R&S CMW that establishes the IP connection to the mobile. If all expansion slots of this R&S CMW are occupied by other hardware options, a DAU installed on another R&S CMW can be used instead. For such an "external DAU" setup, option R&S CMW-KA120 is required on both instruments.

You can control the DAU manually via a graphical user interface or remotely via SCPI commands. Protocol test applications can control the DAU via the CDAU interface.

The following sections describe possible test setups and the related initial configuration of the network settings. They also provide an introduction to the DAU measurements.

3.1 Test Setup

A standalone test setup comprises the mobile, the R&S CMW and at least one RF connection between them. The mobile can access embedded DAU services, for example browse pages of the built-in web server or exchange files with the built-in FTP server.

It depends on the application used to set up the IP connection and its configuration, whether one or more RF connections are required and which RF connectors can be used. For details, refer to the documentation of the relevant signaling application or to the protocol test documentation. Set up the RF connections as described there.

The used RF path (connector and converter) is known by the used application, but not by the DAU. The DAU accesses the established IP connection via an internal Ethernet connection to the signaling units.

For scenarios involving only one downlink carrier and one uplink carrier, you use typically an RF COM connector, e.g. RF 1 COM, as shown in the following figure.

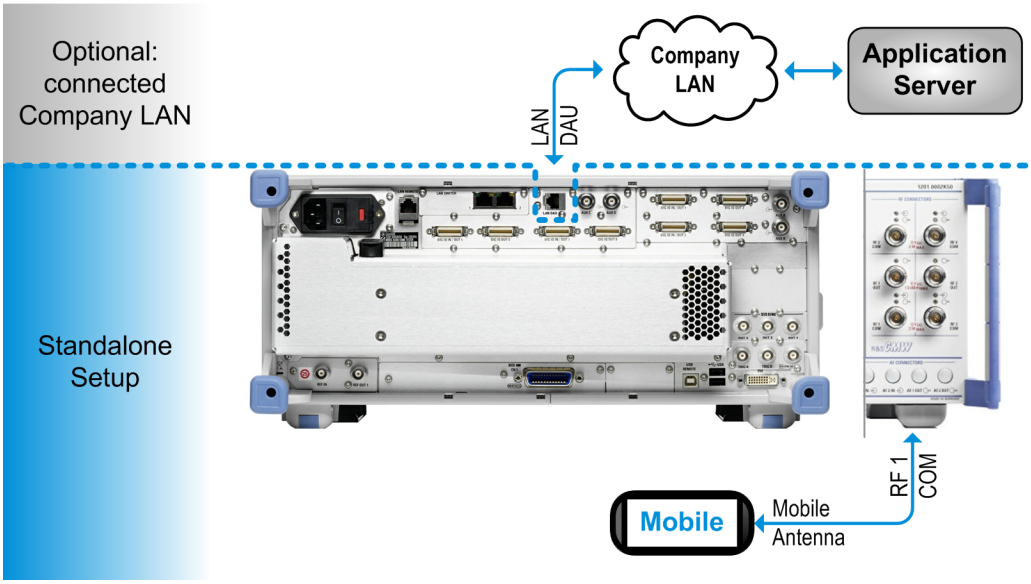


Figure 3-1: Test setup, optionally with connection to company LAN

If you want to access external services with the mobile, connect the external network to the LAN DAU connector at the rear panel of the instrument. Typically, you connect your company LAN. With this setup, the mobile can access the network connected to the LAN DAU connector and other networks reachable via this network, e.g. an application server on the Internet. It can also access embedded DAU services, as with a standalone setup.

Note that only the LAN DAU connector allows the DAU and thus the mobile to access an external network and can be used for U-plane tests. The other LAN connectors of the instrument cannot be used for this purpose. Especially do never connect your company LAN to the LAN SWITCH connector.

Connecting an external network, or not?

If you use a standalone test setup, you can use the predefined internal IP configuration. The default IP settings of the DAU are compatible to a standalone setup. No IP network configuration is required. But the mobile can only access the embedded DAU services, not an external network.

If you connect an external network, you must configure the IP network settings of the DAU compatible to the connected network. So more configuration than for a stand-alone setup is required. In return, the mobile can access the external network.

Table 3-1: Overview of the setup characteristics

Standalone test setup	Test setup with external network
No external Ethernet connection is required.	The external network must be connected to the LAN DAU connector.
The predefined DAU IP configuration is ready-to-use.	The DAU IP configuration must be adapted to the external network. Dynamic or static IP configuration is possible.
The mobile can only access IP services provided by the DAU. The setup, the services and the configuration are optimized for maximum throughput on IP level.	The mobile can access IP services provided by the DAU and also IP services running on an external application server in the company LAN or on the Internet.

Test setup with external DAU

Usually, the DAU is installed on the R&S CMW that establishes the IP connection to the mobile (internal DAU). If all expansion slots of this R&S CMW are occupied by other hardware options, a DAU installed on another R&S CMW can be used instead (external DAU).

For such an external DAU setup, you need option R&S CMW-KA120 on both R&S CMW. Both instruments must be equipped with an Ethernet switch (option R&S CMW-B660x plus -B661x).

Connect the two Ethernet switches via a LAN cable as shown in the following figure. If you have the latest Ethernet switch (R&S CMW-B661H), you can use any LAN SWITCH port (no 1 or no 2).

If you have an old Ethernet switch (R&S CMW-B661A) at the R&S CMW without DAU, connect the LAN REMOTE connector to the Ethernet switch with a short patch cable.

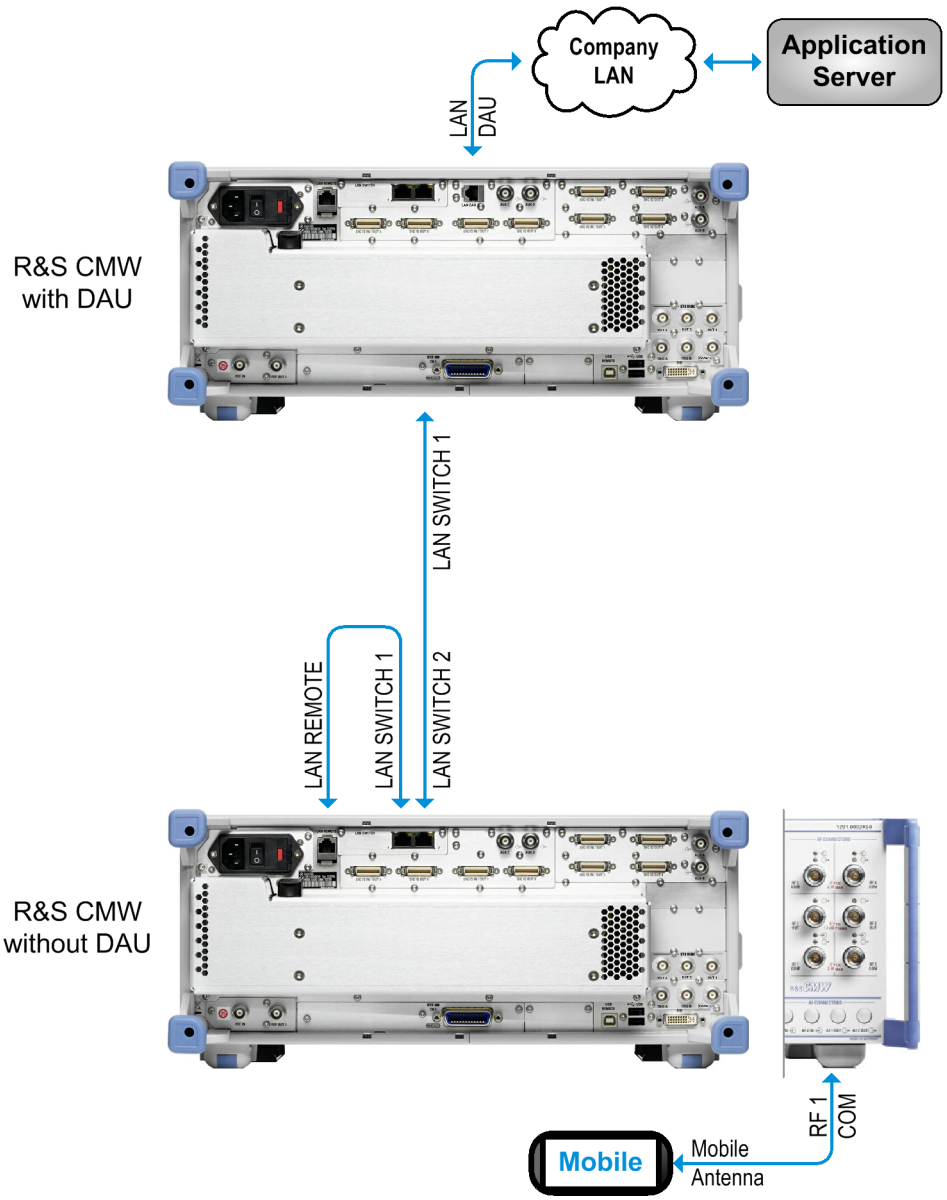


Figure 3-2: Test setup with external DAU, R&S CMW-B661A and connection to company LAN

The patch cable is only functional if the "Lan Remote" network adapter is integrated into the internal IPv4 subnet of the instrument. For configuration of a compatible "Lan Remote" IP address, refer to the description of the "Setup" dialog in the R&S CMW user manual.

In the R&S CMW without DAU, you must enable the usage of the external DAU and enter information about the subnet of the R&S CMW with DAU. These settings are provided by the application that establishes the IP connection, for example the LTE signaling application. Refer to the documentation of that application.

You can connect only one instrument without DAU to an instrument with DAU. You can use the DAU for IP connections at both instruments, even simultaneously. However,

the DAU resources are shared by all applications using the DAU. To reach high data rates, use the DAU for one application at a time.

Connecting an external USB drive

If you have the DAU version R&S CMW-B450H, you find a USB DAU connector on the rear panel. This connector provides a very fast connection to the DAU (USB 3.0 SuperSpeed).

A typical use case is to connect an external hard drive with media files. The mounted drive is accessible as folder of the samba share of the DAU, see Chapter 3.8, "Using the System Drive of the DAU", on page 42.

3.2 Adapting the DAU IP Configuration

This section introduces you to configuring the IP network settings of the DAU according to your test setup. Ensure correct settings before using the DAU.

The DAU IP configuration is specified via the "Data Application Control" dialog. You can either configure the settings manually via the "IP Config" tab of the dialog or you can use the "Initial Setup Wizard".

Opening the "Data Application Control" dialog

1. Press "Setup" to open the "Setup" dialog.
2. In section "System", press the button "Go to config".

   The "Data Application Control" dialog opens.

Configuring the "IP Config" settings via the "Initial Setup Wizard"

1. Open the "Data Application Control" dialog.
2. Press the "Wizard" key at the (soft-) front panel to open the wizard dialog.
3. On the tab "Application Wizards", run the "Initial Setup Wizard".

Configuring the "IP Config" settings manually

1. Open the "IP Config" tab of the "Data Application Control" dialog.
2. Press the "Config" hotkey to open the configuration dialog.
3. Configure all settings as desired. There is a tab for IPv4 settings and one for IPv6 settings. If you go from one tab to the other, you lose any changes. So press the "Apply" button before changing the tab.

a) For a standalone setup, select "Automatic R&S CMW500 Network (standalone)" for IPv4 and IPv6. You need no further configuration.

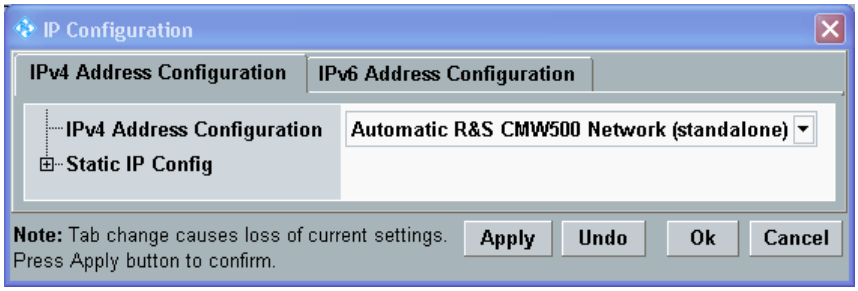


Figure 3-3: IPv4 tab, standalone setup

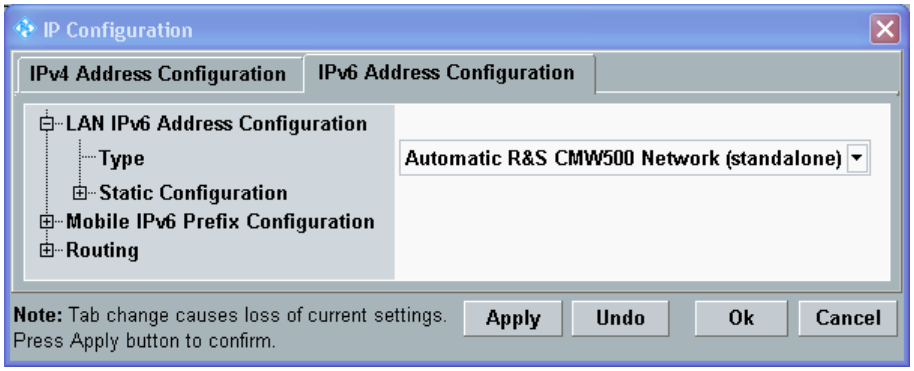


Figure 3-4: IPv6 tab, standalone setup

b) For a setup with external network, configure the settings compatible to your network, either by defining a static configuration or by using dynamic mechanisms supported by your network.
For an introduction to IP address formats and to the IP configuration methods supported by the DAU, refer to the following sections.
For a description of the "IP Config" tab and its configuration dialog, see Chapter 4.2, "IP Config Tab", on page 52.

To ensure that your changes are effective for a running protocol test, restart the test. To apply them to a signaling application already generating a signal, switch the downlink signal off and on again in the signaling application.

3.2.1 IPv4 and IPv6 Address Formats

An IPv4 address is unique within a subnet. The subnet is identified by a subnet mask. Both parameters consist of 32 bits, typically written in dot-decimal notation.

Example: address = `192.168.168.170`, subnet mask = `255.255.255.0`, all addresses from `192.168.168.0` to `192.168.168.255` belong to the subnet.

An IPv6 address consists of 128 bits, typically written in blocks of four-digit hexadecimal numbers, separated by a colon. Addresses can be abbreviated by omitting leading

zeros within a block and by replacing consecutive blocks of zeros by a double colon (only once per address).

Example - the following addresses are equivalent:

- `fcb1:cafe:0001:86c1:0000:0000:0000:0001`
- `fcb1:cafe:1:86c1:0000:0000:0000:1`
- `fcb1:cafe:1:86c1::1`

The initial bits of an address are called prefix and identify the subnet. The maximum length of the prefix is 64 bits. The remaining bits are called interface identifier and uniquely identify a link within the subnet.

The DAU must know both its own address and the prefix identifying its subnet. This information can be combined into a single string by appending the prefix length to the address. Example: The address `fcb1:cafe:1:86c1::1` with the initial 64 bits identifying the subnet can be written as `fcb1:cafe:1:86c1::1/64`.

IPv6 prefixes are written in the same notation (address + length). Prefixes with less than 64 bits indicate a whole group of 64-bit prefixes with the same initial digits.

Examples for prefixes:

- 64-bit prefix: `fcb1:cafe:0001:86c1::/64`
- Abbreviations: `ac42:45d2:0001:0000::/64` = `ac42:45d2:1:0::/64` = `ac42:45d2:1::/64`
- Prefix groups: `fcb1:cafe:0001::/48` comprises the 64-bit prefixes `fcb1:cafe:0001:0000::/64` to `fcb1:cafe:0001:ffff::/64`

3.2.2 IP Address Configuration Methods

The DAU must know at least its own IP address and it must have a pool of IPv4 addresses and/or IPv6 prefixes which it can assign to mobiles. This section describes how the DAU can get this information. It does not describe how an address is assigned to a mobile, see Chapter 3.2.3, "Assigning an IPv6 Address to a Mobile", on page 23.

In principle, there are three methods how the DAU can get IP address information:

- The DAU can use IP addresses/prefixes predefined within its software. This method is appropriate for a standalone setup. The mobile can address the DAU, but not any other host.
- You can configure static IP addresses/prefixes. With a correct configuration, the mobile can address both the DAU and also hosts in external networks, reachable via the LAN DAU connector.
- The DAU can request and receive dynamic IP addresses/prefixes from an external network via the LAN DAU connector. The mobile can address both the DAU and also hosts in external networks.

The following sections provide more detailed information about the three methods. For IPv4, the same method is used for DAU addresses and mobile addresses. For IPv6, you can also combine static and dynamic methods, i.e. use a static configuration of the DAU address and a dynamic configuration of the mobile prefixes or vice versa. The predefined configuration always applies to DAU and mobile.

The IPv4 and IPv6 configuration are independent, so you can use different methods.

Predefined automatic address configuration

This method can only be used for a standalone setup, without connection of an external network. It is available for IPv4 and IPv6 and configures all address information automatically. So you do not need to configure any IP addresses or prefixes at the DAU.

This method is active by default.

Static address configuration

This method allows you to connect an external network or host to the LAN DAU connector. Configure all addresses and prefixes so that they are compatible to the external network, typically your company network. Contact your network administrator before configuring the settings. The default values are often not compatible to an external network and they are not compatible to the Internet. But they can be used when connecting a host directly.

For IPv4, all configured static IP addresses must belong to the connected network. Enter its subnet mask, specify the DAU address and define the mobile address pool. To access internet servers, specify also the gateway address.

For IPv6, assign an IPv6 address to the DAU and define the IPv6 prefix pool for mobiles. The DAU must belong to a subnet of the external network. For communication with hosts outside of this subnet, specify an adequate default router belonging to the subnet of the DAU. Optionally, you can configure routes to other default routers or instruments equipped with a DAU, see "Routing configuration" on page 23.

Dynamic address configuration

This method allows you to connect an external network or host to the LAN DAU connector. Addresses and prefixes are requested from the external network.

For IPv4, the IP address information is configured dynamically using DHCPv4. A DHCP server must be available in the external network.

For IPv6, the DAU performs IPv6 autoconfiguration. It obtains IP address information via router advertisements (RA) and/or via DHCPv6. The external network must either contain a router supporting router advertisements or a DHCPv6 server - or both, configured compatible to each other.

Example, RA + DHCPv6: The router advertisement contains an IPv6 prefix and allows the host (the DAU) to generate its own address using this prefix. It also indicates that stateful configuration (DHCPv6) must be used to obtain optional information, like DNS server addresses.

Depending on the content of the router advertisements, it can happen that the DAU ends up with two assigned IPv6 addresses: one assigned via router advertisements and one assigned via DHCPv6. In that case, both addresses are displayed in the "IP Config" tab.

Dynamic configuration of the IPv6 prefix pool for the mobiles can be done via DHCP prefix delegation. The external network must contain a router supporting prefix delega-

tion via DHCPv6. The router delegates an IPv6 prefix to the DAU, typically with less than 64 bit. The DAU is allowed to create 64-bit prefixes from the delegated prefix and to assign the prefixes to mobiles.

Routing configuration

For IPv4, no routes need to be defined manually. The mobile, the DAU and external network nodes connected to the DAU belong to the same subnet. So they know how to reach each other. The configured gateway allows you to reach destinations outside of the subnet.

For IPv6, the mobile and the DAU belong to different subnets. Thus some administration is required to ensure that packets addressed to a specific mobile are routed to the DAU where the mobile is connected. The administration tasks to be performed depend on the use case and the configuration of the external network. Basically, you must ensure that packets with a mobile IPv6 prefix as destination are routed to the correct DAU.

At the DAU, you can configure manual routes for packets sent by the mobile. To configure routes, makes sense for destination addresses that do not belong to the subnet of the DAU and cannot be reached via the configured default router. You could, for example, route traffic for certain destinations to a different default router. Or you could route packets for another mobile connected to another R&S CMW/DAU to that DAU.

Example: The mobile prefix pool of another DAU contains 64 bit prefixes starting with `fcb1:abab:cdcd`. To create routes for all these prefixes, create a single route for the prefix `fcb1:abab:cdcd::/48` and specify the IP address of the other DAU as router address.

3.2.3 Assigning an IPv6 Address to a Mobile

According to 3GPP, the standard method for IPv6 address assignment is the IPv6 stateless address autoconfiguration. This method is used by the DAU to assign IPv6 addresses to a mobile. The optional method stateful IPv6 address assignment (DHCPv6) is not supported by the DAU.

The DAU uses stateless address autoconfiguration as specified in 3GPP TS 23.401 and 3GPP TS 23.060. The 3GPP specification of the method differs slightly from the definition in RFC 4862.

The assignment of an IPv6 address consists of the following steps:

1. The mobile requests a connection to an IP network. It sets the PDP type to IPv6 or to IPv4v6.
2. The DAU sends an IPv6 interface identifier to the mobile.

   The interface identifier is a 64-bit value, sent via a NAS signaling message. The used message depends on the RAT. Examples:
   - For GSM or WCDMA: "Activate PDP Context Accept"
   - For LTE: "Establish Default EPS Bearer Context Request"
3. The DAU sends an IPv6 prefix to the mobile.

The prefix is a 64-bit value, sent after the flow activation (bearer establishment) in an ICMPv6 router advertisement message.
Any subsequent router advertisement messages also include the prefix. The DAU sends such messages periodically and if the mobile sends a router solicitation message.

4. The mobile composes an IPv6 address.
It uses the received interface identifier to form a link-local IPv6 address as follows: `fe80::<interface identifier>`.
The mobile can also form a global IPv6 address. For the global address, the mobile must use the received prefix and can use the received interface identifier or select a different one. The address consists of the following parts:
`<prefix>::<interface identifier>`.
Example: Prefix `fcb1:abab:cdcd:efe0::/64`, interface identifier `1`
Global IPv6 address `fcb1:abab:cdcd:efe0::1/64`

3.3 Configuring and Starting DAU Services

To use an internal IP service provided by the DAU such as FTP or HTTP, you must configure and start the service via the "Data Application Control" dialog. See also "Opening the "Data Application Control" dialog" on page 19.

To configure a DAU service, select the corresponding tab of the "Data Application Control" dialog, e.g. "FTP". Then configure the settings available directly in the tab or accessible via the "Config" hotkey.

Start the DAU service by selecting the corresponding softkey, for example, "FTP Service" and press ON | OFF or RESTART | STOP to change the state of the service. Alternatively, right-click the softkey.

For a detailed description of the service tabs, see Chapter 4, "DAU Services GUI Reference", on page 50.

See also:

- Chapter 3.5, "Using the IMS Server", on page 26
- Chapter 3.6, "Using the ePDG Service for WLAN Offload", on page 40
- Chapter 3.7, "Using the eMBMS Service", on page 41

3.4 Configuring and Running DAU Measurements

As a prerequisite for any measurements, you must prepare the physical test setup and configure the DAU network settings according to your test setup (see Chapter 3.1, "Test Setup", on page 15 and Chapter 3.2, "Adapting the DAU IP Configuration", on page 19).

The configuration of the DAU network settings must be complete before switching on the downlink signal and establishing a connection to the mobile.

DAU measurements require option R&S CMW-KM050. All measurements are accessible via the tabs of the "Data Application Measurements" dialog.

For a detailed description of all parameters and results, see Chapter 5, "DAU Measurements GUI Reference", on page 111.

Performing an E2E data transfer test with a signaling application

1. Configure the signaling application as desired.
   Usually, you must enable E2E tests in the signaling application. Example: in the LTE signaling application, set parameter "Connection Type" to "Data Application".
2. Use the signaling application to establish a connection to the mobile.
3. Find out the IP address of the mobile. You need this information for configuration of most DAU measurements.
   Some signaling applications provide the assigned IP address in their main view. If you have connected a PC to the mobile, you can use the PC to retrieve the information. At a PC with Windows operating system, use the command `IPconfig`.
4. Open the "Data Application Measurements" dialog.
   For detailed steps, see "Opening the "Data Application Measurements" dialog" on page 26.
5. At the top of the GUI, select the used signaling application via the parameter "Select RAN".

   The expected maximum throughput at the physical layer is displayed for information. It results from the current signaling settings.
6. Configure and start a DAU measurement, for example "Ping" or "Iperf".
   For most measurements, you must enter the IP address of the mobile to select the IP connection that you want to analyze.
   See also "Configuring and starting a DAU measurement" on page 26.

Performing a U-plane test with a protocol test application

The following procedure assumes that the protocol test establishes a connection to the mobile at IP level.

1. Configure and start the protocol test in the usual way. There are no specific settings required for cooperation with the DAU.
2. Check the IP address of the mobile. You can, for example, check the message log files created by the protocol test.
3. Open the "Data Application Measurements" dialog.
   For detailed steps, see "Opening the "Data Application Measurements" dialog" on page 26.
4. Configure and start a DAU measurement, for example "Ping" or "IPerf".
   For most measurements, you must enter the IP address of the mobile to select the IP connection that you want to analyze.
   See also "Configuring and starting a DAU measurement" on page 26.

Protocol tests can also integrate DAU measurements into their test sequence. In that case, the protocol test controls the DAU via the so-called CDAU interface to configure the measurement settings and retrieve the measurement results.

While the CDAU interface is used, the GUI of the DAU is hidden in a similar way as during a remote control session with SCPI commands. When the GUI is displayed again, the settings valid before the CDAU session are restored automatically and the displayed results are reset.

Opening the "Data Application Measurements" dialog

1. Press MEASURE to open the "Measurement Controller" dialog.
2. Enable "Data Appl. > Measurements"
3. Press "Data Meas" on the task bar at the bottom.

   The "Data Application Measurements" dialog opens. The "Overview" tab provides a summary of the measurement states.

Configuring and starting a DAU measurement

1. Select the corresponding tab, for example, "Ping".
2. Configure the settings available directly on the tab or accessible via the "Config" hotkey.
3. Select the topmost softkey, for example "Ping". Press ON/OFF to start the measurement.

   The measurement results are provided in the same tab.

3.5 Using the IMS Server

The internal IMS server of the DAU simulates parts of an IMS network, especially a P-CSCF. It allows an IMS client on the mobile to register to the IMS domain. Option R&S CMW-KAA20 is required for usage of the IMS server.

With a registered mobile, you can test voice over IMS and messaging over IMS. You can perform mobile-originating calls and mobile-terminating calls. If you register two mobiles, you can also perform mobile-to-mobile calls.

This section provides an overview of the IMS server settings, followed by basic step-by-step procedures.

IMS server settings overview

The IMS settings are divided into the following parts:

- General settings
  The "Config" hotkey accesses the general settings. They select whether the internal IMS server or an external IMS network is used.
  The remainder of this section assumes that you use the internal IMS server.
  For a GUI description, see Chapter 4.6.5, "General IMS Settings", on page 73.

- P-CSCF settings
  The "P-CSCF" hotkey accesses settings related to the simulated P-CSCF. They define the P-CSCF address and the reaction of the P-CSCF on incoming SIP messages from a mobile.
  You can define several P-CSCF profiles with different addresses and settings. A list of all P-CSCFs is sent to the mobile. The mobile selects a P-CSCF from the list for registration.
  Two P-CSCF profiles are predefined and ready-to-use, one for IPv4 and one for IPv6.
  For a GUI description, see Chapter 4.6.6, "P-CSCF Settings", on page 75.
- Subscriber settings
  The "Subscriber" hotkey accesses settings related to the tested mobiles. They define the private user ID, the public user IDs and authentication settings.
  You can define several subscriber profiles, for example if you use SIM cards with different properties. If all your test SIM cards have the same properties, one subscriber profile is sufficient.
  The applicable profile is selected via the user ID included in SIP messages from the mobile.
  For a GUI description, see Chapter 4.6.7, "Subscriber Settings", on page 77.
- Virtual subscriber settings
  The "Virtual Subscriber" hotkey accesses settings related to simulated subscribers. A virtual subscriber can call a tested mobile and it can be called from a tested mobile.
  The settings include, for example, the public user ID via which the virtual subscriber can be reached, the reaction of the virtual subscriber to a call and codec settings.
  You can define several virtual subscriber profiles and use a specific profile for a call.
  For a GUI description, see Chapter 4.6.8.1, "Common View Elements", on page 82.

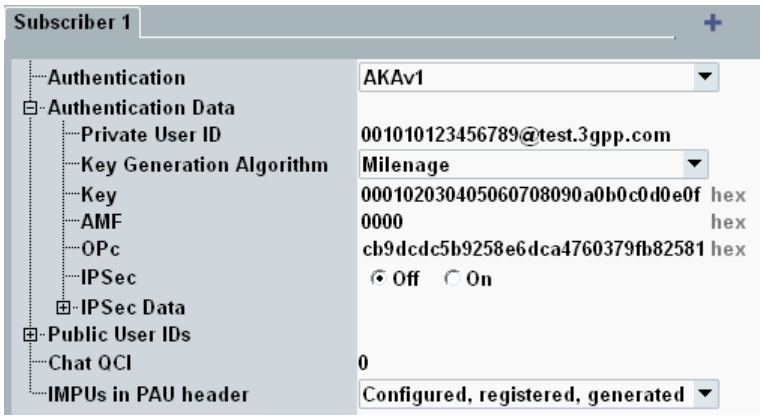
- Registering a Mobile to the IMS Server 27
- Troubleshooting a Failed Registration 29
- Setting Up a Mobile-Originating Voice over IMS Call 29
- Setting Up a Mobile-Terminating Voice over IMS Call 30
- Setting Up a Mobile-to-Mobile Voice over IMS Call 32
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3.5.1 Registering a Mobile to the IMS Server

Perform the following steps to register a mobile to the IMS server. The IMS registration is a prerequisite for voice over IMS and SMS over IMS.

Initial situation: The mobile is switched off and connected to an RF port. There is no downlink signal. The IMS server is set to default settings and switched off. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

1. Press the "Subscriber" hotkey.
   A dialog box opens. Configure a subscriber profile as follows (e.g. "Subscriber 1"):
   a) Enter the private user ID of your mobile.
   b) Configure the authentication settings compatible to your mobile.
   c) Enter all public user IDs of your mobile.
      For "Subscriber 1", this step is optional, as this profile is used for all unknown public IDs.



   d) Press OK.

2. Press ON | OFF or right-click the "IMS Service" softkey and click ON.
   The IMS service is started. Wait until the "IMS Service" softkey displays the state "ON".



3. Configure the signaling application that you want to use, for example the LTE signaling application. Usually, you must explicitly enable data tests in the signaling application. For details, refer to the documentation of the signaling application.

4. Let the mobile register/attach to the signaling application.
   Afterwards, the mobile automatically registers also to the IMS server. You can check the successful registration in the "General IMS Info" area on the IMS tab.



The displayed URI is generated automatically and registered in addition to the public user IDs configured for the subscriber profile. Each registration with a different IP address gets a unique URI, even if the same subscriber profile is used.

3.5.2 Troubleshooting a Failed Registration

If your mobile registers to the radio access network, but not to the IMS, check the following settings.

1. Is the internal IMS server used and switched on?
   a) Check the upper right corner of the "IMS" tab. It must state "intern IMS", not "extern IMS".
      To modify this setting, press the "Config" hotkey.
   b) Check the "IMS Service" softkey. It must state "ON", not "OFF".
2. Subscriber settings (hotkey "Subscriber"):
   a) Is the private user ID correct?
      If you are not sure which ID your mobile uses, you can analyze SIP register messages sent by your mobile. The private user ID is contained in the header as *username="..."*.
   b) Are the authentication settings correct?
      If the mobile does not support or require authentication, set "No Authentication".
      If the mobile requires authentication, select the AKA version and configure all "Authentication Data" settings compatible to your mobile.
3. "P-CSCF" settings (hotkey "P-CSCF").
   Search the tab containing the IP address to which the mobile sends the REGISTER message and check the settings:
   a) The "Behavior" must be "Normal".
   b) If the mobile suggests an expiration interval in its REGISTER message, check "Registration" > "Expiration Time". The value suggested by the mobile must be within the range "MIN" to "MAX".

3.5.3 Setting Up a Mobile-Originating Voice over IMS Call

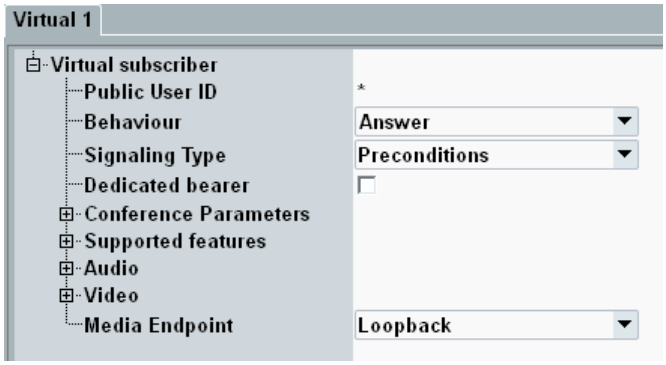
Initial situation: The mobile has registered to the IMS server. There is only one virtual subscriber profile "Virtual 1". The GUI shows the "IMS" tab of the "Data Application Control" dialog.

1. Press the "Virtual Subscriber" hotkey.

   A dialog box opens.
2. Press the leftmost button to access the basic settings.



3. Configure the "Virtual 1" tab as follows:
   a) Set the "Signaling Type" compatible to your mobile (call setup with or without SIP preconditions).
   b) For all other settings, we use the default values, especially:
      - "Public User ID" = "*"
        All calls to public user IDs not assigned to any virtual subscriber go to this subscriber.
      - "Behavior" = "Answer"
      - "Media Endpoint" = "Loopback"



4. At your mobile, initiate a voice call to an arbitrary public user ID.

   A new line is added to the "Events" area. After call setup completion, the status in that line equals "Established".



5. Perform an echo test. Speak a word into the microphone of the mobile. Listen to the echo at the loudspeaker of the mobile.

   For further tests, you can create additional virtual subscriber profiles with a different behavior or a different media endpoint. Enter a different public user ID for each profile. To call a specific virtual subscriber, call its public user ID.

3.5.4 Setting Up a Mobile-Terminating Voice over IMS Call

Initial situation: The mobile has registered to the IMS server. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

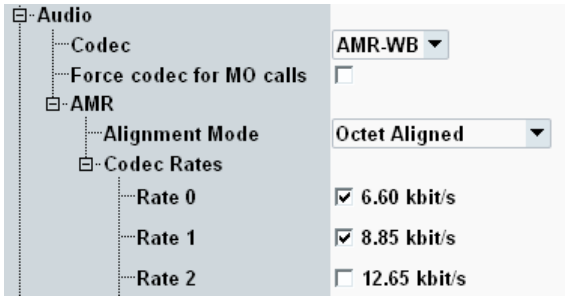
1. Press the "Virtual Subscriber" hotkey.

   A dialog box opens.

2. Press the leftmost button to access the basic settings.



3. Configure the "Virtual 1" tab as follows:
   a) Set the "Signaling Type" compatible to your mobile (call setup with or without SIP preconditions).
   b) Configure the audio codec settings compatible to your mobile.



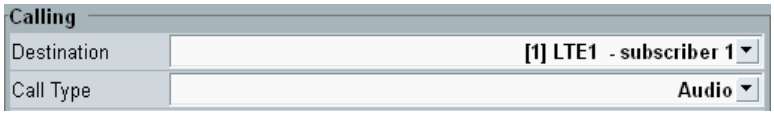
   c) For video calls, configure the video codec settings compatible to your mobile.
   d) For "Media Endpoint", we use the default value "Loopback".



4. Press the button.

   The mobile-terminating call (MTC) settings of the virtual subscriber are displayed.

5. Configure the following call settings:
   a) If there are several registered mobiles, select the "Destination" that you want to call.
   b) Select the "Call Type".

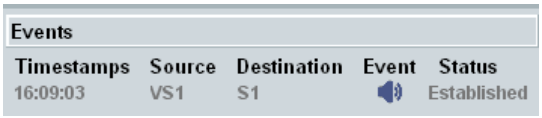


6. Press the "Call" button.

   The dialog box is closed and the mobile is alerted.
   A new line is added to the "Events" area. The status in that line equals "Ringing".

7. Accept the call at the mobile.

   The status in the "Events" area changes to "Established".



8. Perform an echo test. Speak a word into the microphone of the mobile. Listen to the echo at the loudspeaker of the mobile.

   For further tests, you can create additional virtual subscriber profiles with different call settings.

3.5.5 Setting Up a Mobile-to-Mobile Voice over IMS Call

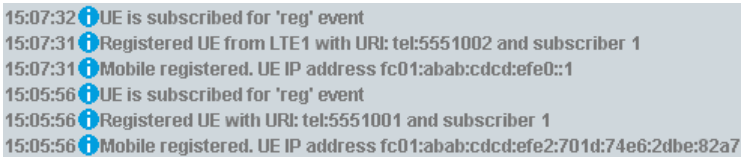
A signaling application can handle only one mobile at a time. To register two mobiles, you need two signaling application instances. To use the two instances in parallel, you must configure different hardware resources for the two instances (different signaling units, RF converters and RF connectors).

Initial situation: The mobiles are switched off and connected to different RF ports. The signaling application instances to be used are configured. The IMS server is set to default settings and switched on. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

The following procedure registers the two mobiles without authentication. The call is set up to a public user ID that is generated by the DAU. In that case, you do not need to configure subscriber profiles. The default subscriber profile with its default settings is used for both mobiles.

1. Let the mobiles register/attach to the signaling application.

   Afterwards, the mobiles automatically register also to the IMS server. You can check the successful registration in the "General IMS Info" area on the IMS tab.

   Note the "Registered UE with..." lines. Both mobiles use the default subscriber profile ("subscriber 1").
   The first registered mobile has got the automatically generated public user ID "tel: 5551001". The second mobile got the ID "tel:5551002".

2. At one mobile, initiate a call to the other mobile.
   With the public IDs assigned in this example, you can dial 5551001 at the second mobile or 5551002 at the first mobile.
   You can set up an audio call or a video call.

3. The called mobile is ringing. Accept the call.

4. Check that the call setup is complete.
   For an audio call, speak into the microphone of one mobile. Then listen to the speaker of the other mobile.
   For a video call, you can check, for example, the video quality and the video delay.

3.5.6 Setting Up a Conference Call

In the current software version, there is exactly one mobile in a conference call. All other participants are virtual subscribers. The conference can only be initiated by the mobile, not by the DAU.

Initial situation: The mobile has registered to the IMS server. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

For conference calls, option R&S CMW-KAA21 is required.

1. Press the "Virtual Subscriber" hotkey.

   A dialog box opens.

2. Press the leftmost button to access the basic settings.



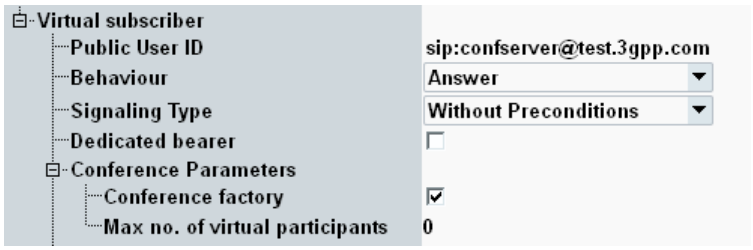
3. Press the + button repeatedly to create as many virtual subscribers as you need.

   A conference call comprises the mobile, at least two virtual subscribers and the conference server configured via a virtual subscriber profile. So you need at least three virtual subscriber profiles.

4. Configure one virtual subscriber profile as conference server as follows:

   a) Set the "Public User ID" to the conference server address used by your mobile (for example *sip:confserver@test.3gpp.com*).

   b) Enable "Conference Parameters > Conference Factory".



   c) Configure the remaining virtual subscriber settings compatible to your mobile, as for a normal call.

5. Configure the other virtual subscriber profiles as for a normal call.

6. Set up a voice over IMS call between the mobile and one virtual subscriber.

   The "Events" area lists the connection, for example, as "S1 VS1 Established".

7. At your mobile, add a second voice over IMS call to another virtual subscriber.

   The already existing call is automatically put on hold. In the "Events" area, you see both calls. For example, "S1 VS1 Hold" and "S1 VS2 Established".

8. At your mobile, initiate the conference (for example "Join" or "Merge" button).

   The calls to the virtual subscribers are released by the mobile. New calls between the conference server and all participants are established.

In the "Events" area, you see the released calls. For example as "S1 VS1 Released" and "S1 VS2 Released". The conference is listed, for example, as "S1 C1 Established", "C1 VS1 Established" and "C1 VS2 Established".

9. To add more virtual subscribers:
   a) Establish an additional call.
   b) Add the call to the conference (join/merge).
   c) Repeat these steps.
10. To remove a virtual subscriber from the conference:
    a) Press the "Release Call" softkey.
       A dialog box opens.
    b) Enable the checkbox of the connection that you want to release.
    c) Select whether the virtual subscriber leaves the conference ("Depart") or the conference server releases the connection ("Eject").
    d) Press the "Release" button.

3.5.7 Upgrading an Audio Call to a Video Call

Initial situation: You have set up an audio call. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

1. Select the audio call entry in the "Events" area of the main view. The relevant entry indicates the event and the status "Established".
2. Press the "Update Call" softkey.
   A dialog box opens.
3. Change the "Call Type" to "Video".
4. Set the "Video Codec" compatible to your mobile.
5. Press the "Update" button.
6. Check the "Events" area of the main view. If the mobile accepts the modification, the event changes to .
   You can check the established video call also at your mobile.

3.5.8 Sending a Message to the Mobile (SMS or RCS)

Initial situation: The mobile has registered to the IMS server. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

For RCS messages, option R&S CMW-KAA21 is required.

1. Press the "Virtual Subscriber" hotkey.
   A dialog box opens.
2. Press the button.

The mobile-terminating message settings of the virtual subscriber are displayed.

3. Configure all settings:
   a) Select the "Destination" for the message.
   b) Select the "Message Type".
   c) If applicable, select the "Encoding".
   d) Enter the message text.



   e) For group chats, configure the participant list.

4. Press the "Send" button.

   The dialog box is closed and the message is sent to the mobile.
   A new line is added to the "Events" area. After a successful message transfer, the status in that line equals "OK" or "Text Transferred".

5. Check the incoming message at the mobile.

6. To send additional messages for an initiated chat, use the softkey "Chat Actions".

   For further tests, you can create additional virtual subscriber profiles with different message settings.

3.5.9 Sending a Message from the Mobile (SMS or RCS)

Initial situation: The mobile has registered to the IMS server. The GUI shows the "IMS" tab of the "Data Application Control" dialog.

For RCS messages, option R&S CMW-KAA21 is required.

1. Send a message from your mobile to the DAU.
   With the default virtual subscriber settings, you can send the message to any public user ID. Virtual subscriber 1 (public user ID "*") accepts it.

2. Check the "Events" area of the main view.
   After a successful message transfer, there is a new line with the event or and the status "OK" or "Text Transferred".

3. Select this line in the "Events" area.
   The lower part of the main view displays details about the received message, for example the message text.

3.5.10 Using PCAP Streaming

For a VoIMS audio or video downlink, you can use media streaming. The streaming is synchronized with the VoIMS call. It starts and stops automatically on call establishment/termination, hold/resume. For video streaming, audio and video packets are read from the same PCAP file and streamed without drifting.

As streaming source, you need a packet capture file (PCAP file) containing the media information. Configure the streaming via the virtual subscriber profile. For details, refer to the following sections.

3.5.10.1 PCAP File Requirements and File Selection

The PCAP file must contain only RTP packets carrying the media information. The file must use the "raw IP" format, so that all packets start with the IP header. You can use the "IP Logging" application of the DAU to record such PCAP files from the U-plane IP interface.

The PCAP file used for a call must comply with the media formats negotiated between the UE and the DAU (SDP negotiation). So you need several PCAP files, one for each combination of audio/video codec and AMR alignment mode (octet-aligned, bandwidth-efficient).

The PCAP file selection is controlled by an XML meta file on the samba share of the DAU, subdirectory `ims/pcap`. The meta file contains a lookup table with <MediaConfig> elements. Each <MediaConfig> element has attributes, specifying conditions and a PCAP file to be used if the conditions are met.

Table 3-2: Attributes of element <MediaConfig>

Attribute	Type	Values	Description
AmrAligMode	Optional	AMR_OCTET_ALIGNED AMR_BANDWIDTH_EFFICIENT	AMR alignment mode of call must match.
Codec	Optional	AMR_NARROWBAND AMR_WIDEBAND	Audio codec of call must match.
PayloadTypeAudio	Optional	Integer[1)]	Audio payload type of call must match.
PayloadTypeVideo	Optional	Integer[1)]	Video payload type of call must match.
AudioPort	Mandatory	Integer	Target UDP port of audio media in PCAP file must match. For UL DAU IP logging, usually port 49200.
VideoPort	Video calls mandatory	Integer	Target UDP port of video media in PCAP file must match. For UL DAU IP logging, usually port 49250. Media configurations without this attribute are not used for video calls.

Attribute	Type	Values	Description
FileName	Mandatory	Filename with file extension	PCAP file to be used if the other attributes match. Path relative to `ims/pcap` on the samba share.

[1] Setting the attributes PayloadTypeAudio/Video is an alternative to setting the attributes AmrAligMode and Codec. The payload type number contains information about the alignment mode and the codec.

Example of an XML meta file

```
<?xml version="1.0" encoding="UTF-8"?>
<PcapEndPointConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:noNamespaceSchemaLocation="file:/etc/ims/pcap/pcapendpconfig_schema.xsd">
  <MediaConfig AmrAligMode="AMR_OCTET_ALIGNED" Codec="AMR_NARROWBAND"
    AudioPort="49200" VideoPort="49250" FileName="amr-nb_oa+h264.pcap"/>
  <MediaConfig AmrAligMode="AMR_OCTET_ALIGNED" Codec="AMR_WIDEBAND"
    AudioPort="49200" VideoPort="49250" FileName="amr-wb_oe+h264.pcap"/>
  <MediaConfig AmrAligMode="AMR_BANDWIDTH_EFFICIENT" Codec="AMR_NARROWBAND"
    AudioPort="49200" VideoPort="49250" FileName="amr-nb_be+h264.pcap"/>
  <MediaConfig AmrAligMode="AMR_BANDWIDTH_EFFICIENT" Codec="AMR_WIDEBAND"
    AudioPort="49200" VideoPort="49250" FileName="amr-wb_be+h264.pcap"/>
</PcapEndPointConfig>
```

Sample PCAP files and a sample XML meta file are available upon request.

3.5.10.2 Recording a PCAP File

To record a PCAP file with the DAU, proceed as follows.

1. Start the DAU measurement application "IP Logging".

2. Transfer the media that you want to record:
   a) Set up an audio or video call.
   b) Transfer the media.
   c) Terminate the call.

   Tip: If you want to record mid call actions like codec change, upgrade or downgrade, set up a mobile-terminating call, not a mobile-originating call.

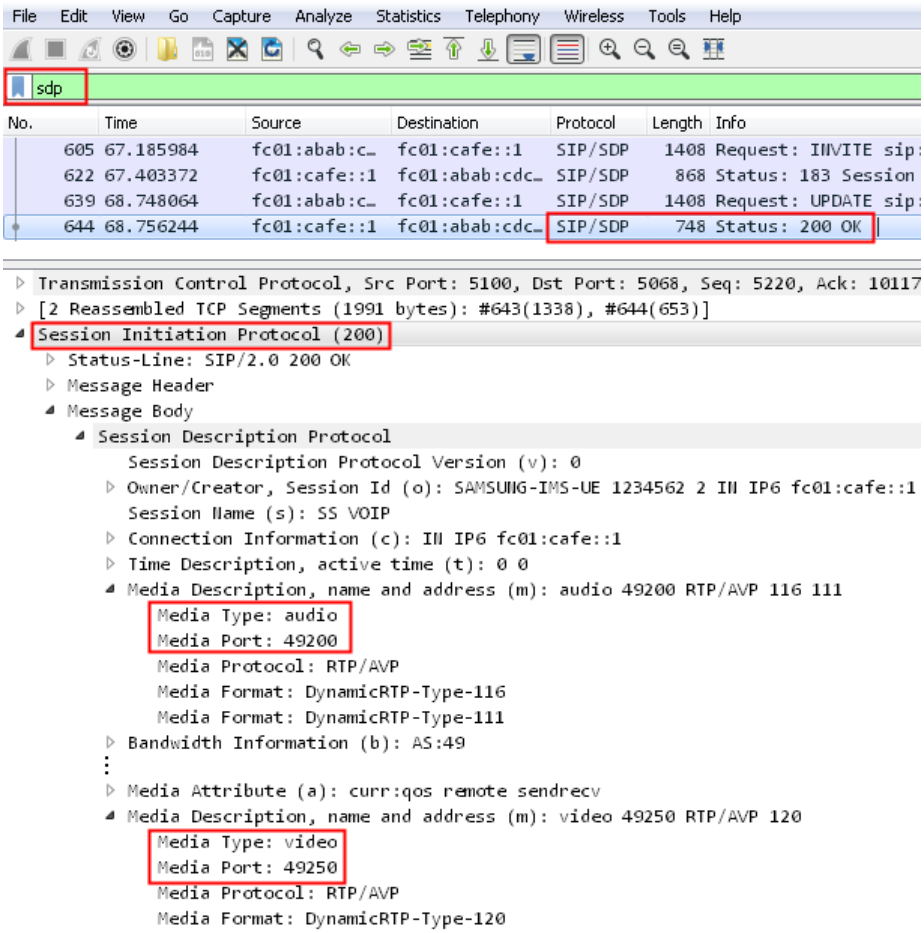
3. Stop logging.

   The recorded PCAP file is stored on the samba share, in the directory `ip_logging`.

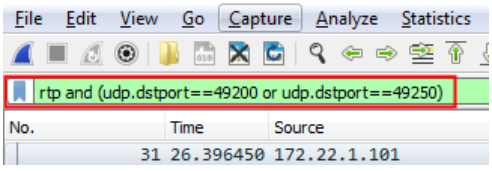
4. Open the recorded PCAP file with Wireshark.

5. Check the media ports for audio and video:

a) As filter, enter *sdp*.
If there are no SDP lines at all, see "Troubleshooting encryption settings" on page 39.
b) Select the last line of the session setup, containing "200 OK".
c) In the lower part, check the SIP message body for media description entries.
In the following example, the port for audio is 49200 and for video it is 49250.



6. Filter the file for RTP packets to the audio and video ports:
Example for filter string: *rtp and (udp.dstport==49200 or udp.dstport==49250)*



If you have set up a mobile-to-mobile call, all messages appear twice, because they are recorded when they are received from one mobile and when they are sent to the other mobile. So in that case you must enhance the filter string to remove the double messages.
For IPv4 you can check the "Time to live" of the messages. The double messages use different values. Filter one value via the string *ip.ttl==<number>*.

For IPv6 you can check the "Hop limit" of the messages. The double messages use different values. Filter one value via the string *ipv6.hlim==<number>*.

7. Save the filtered packets in a new PCAP file:
   a) Click "File" menu > "Export Specified Packets".
   b) Store the file on the samba share, in the directory `ims/pcap`.
8. Edit the XML meta file.
   Add a <MediaConfig> element referencing the PCAP file. See Chapter 3.5.10.1, "PCAP File Requirements and File Selection", on page 36.

Troubleshooting encryption settings

If you do not see any lines with the SDP protocol in Wireshark, but you see lines with the ESP protocol, you must change the Wireshark settings.

Configure the ESP settings, so that the encrypted messages are decrypted.

1. Access the ESP settings via the "Edit" menu > "Preferences" > "Protocols" > "ESP".
2. Enable the first three settings.

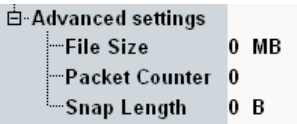


If your connection uses NULL encryption, these steps are sufficient. If your connection uses another encryption algorithm, you must provide the encryption keys to Wireshark, see Chapter 3.12, "Configuring Wireshark for IPsec", on page 48.

Troubleshooting IP logging settings

If your PCAP file is truncated, check the IP logging settings.

1. Access the IP logging settings via the "Config" hotkey of the IP logging measurement.
2. Check the advanced settings.
   To avoid truncated files, all values must equal 0.



3.5.10.3 Configuring PCAP Streaming

After preparing the PCAP files and the XML meta file on the samba share, you can use PCAP streaming.

Configure the virtual subscriber profile to be used, as follows:

1. Set "Media Endpoint" to "PCAP".
2. Use "PCAP Meta File Selection" to select the meta file that you want to use.
3. If you want to force a certain audio codec for mobile-originating calls, enable the audio setting "Force Codec for MO Calls".

   See also Chapter 4.6.8.2, "Basic Virtual Subscriber Settings", on page 83.

3.6 Using the ePDG Service for WLAN Offload

You can use the ePDG service of the DAU for offload tests from LTE to WLAN and back.

You need the WLAN signaling application to simulate a WLAN access point and the LTE signaling application to simulate an LTE cell.

At the DAU, you typically need the IMS server, the DNS server and the ePDG service. Option R&S CMW-KA065 is required for usage of the ePDG service.

Offload tests can also be performed with protocol test applications. They also use the ePDG service of the DAU and require option R&S CMW-KA065. But protocol tests do not use the graphical user interface of the DAU for ePDG configuration.

The following applies to tests with signaling applications, not to protocol tests.

General proceeding

Initially, both signaling applications are switched off. At the mobile, access to WLAN and LTE networks is switched on.

1. If your mobile performs a DNS query to derive the IP address of the ePDG, prepare the DNS service:
   a) Create an entry with the ePDG domain used by the mobile and the IP address of the internal ePDG. The ePDG IP address is configurable via the ePDG settings.
   b) Switch on the DNS service.
2. Prepare the ePDG service:
   a) Configure the ePDG settings compatible to your mobile / SIM card.
   b) Switch on the ePDG service.
3. Set up a voice over LTE/IMS call, for example a video call.
   For details, see Chapter 3.5, "Using the IMS Server", on page 26.
4. To check the established connection, you can for example use the throughput measurement of the DAU.
   You can also use the IP analysis application of the DAU and check the view "Data Pie Charts" for the used protocols ("Data per Protocol"). Most traffic must be RTP traffic.

5. To trigger a mobile-initiated offload to WLAN, switch on the WLAN signaling application.
   Note that the two signaling applications must use different TRX modules. Otherwise they cannot be active in parallel.
   The TX power of the WLAN signaling application must be high, so that the mobile can see the WLAN access point and prefers it to the LTE network.
6. Monitor the offloading. You can, for example, check the following:
   - Does the mobile list the access point within its list of available WLAN access points?
   - In the WLAN signaling application main view: Does the connection state change to "Associated"?
   - In the LTE signaling application main view: Is the list of established bearers reduced by one entry?
   - In the ePDG service main view: Is a new connection listed?
7. You can again use the throughput measurement to check the active connection. And you can use the IP analysis application and check the view "Data Pie Charts" for the used protocols ("Data per Protocol"). Most traffic must now be IPsec traffic, no longer RTP traffic.
8. To hand over the connection back to LTE, reduce the TX power of the WLAN signaling application. Alternatively, switch off the WLAN signal.
9. Monitor the process. You can, for example, check the following:
   - The mobile lists the access point with low power or not at all.
   - In the WLAN signaling application main view, the connection state changes to "Idle".
   - In the LTE signaling application main view, the list of established bearers shows a new bearer.
   - In the ePDG service main view, there is no established connection listed.

3.7 Using the eMBMS Service

The DAU option R&S CMW-KAA30 enables a mini service layer for eMBMS (evolved multimedia broadcast multicast service). The multicast option R&S CMW-KA150 is also required.

The mini service layer of the DAU provides all eMBMS services required for eMBMS tests. It supports, for example, the following features:

- Unicast services:
  - Bootstrap function
  - Associated delivery procedures, for example file repair function and reception reports
- Multicast services
  - Service announcement over the FLUTE protocol
  - File download and video streaming over the FLUTE protocol

– Replay of PCAP files, that is replay of pre-recorded eMBMS multicast content
The IP replay application of the DAU is required (included in option R&S CMW-KM050).

The eMBMS mini service layer does not provide a graphical user interface. It cannot be configured directly via the DAU software.

Instead, it is configured and controlled comfortably via CMWcards or via protocol test applications. Even the IP replay application is controlled in that way for eMBMS tests. You do not need to configure it via the DAU software.

The following figure provides a graphical overview.

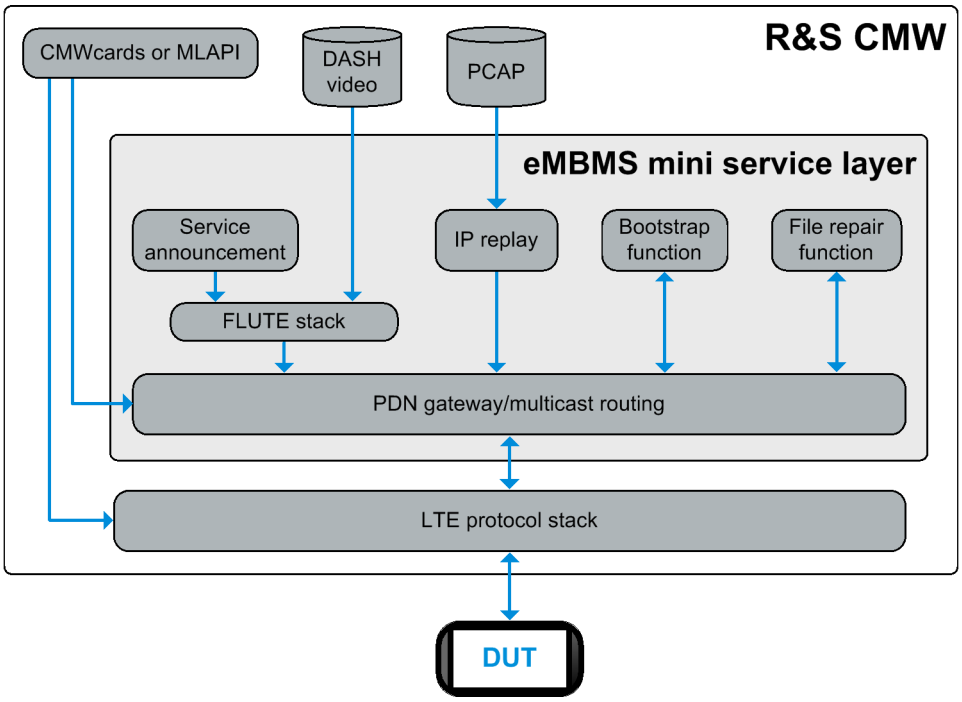


Figure 3-5: eMBMS service layer overview

3.8 Using the System Drive of the DAU

The internal system drive of the DAU can be used for file sharing. It provides a samba share for this purpose.

By default, the samba share is mapped to drive Z. You can access it via the Windows Explorer, either locally or from another host connected to the CMW internal subnet. The `multimedia` directory of the share can also be accessed via the WebDAV protocol, see "Accessing the samba share via WebDAV" on page 44.

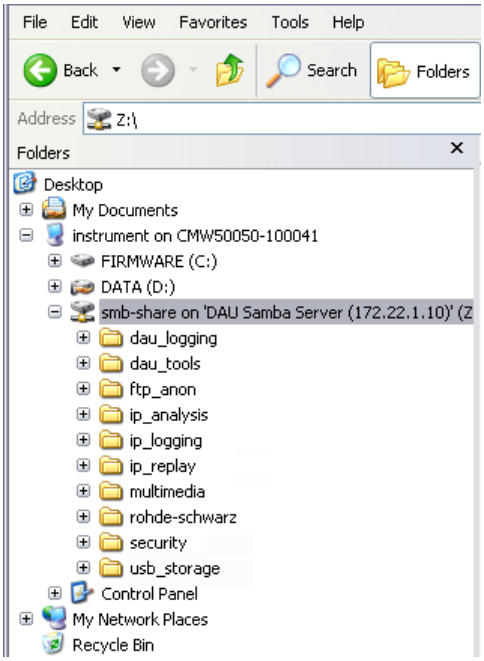


Figure 3-6: DAU samba share in Windows Explorer

To change the drive letter, use the hotkey "Network Drive Map" of the "Data Application Control" dialog.

The samba share provides the following directories:

- `dau_logging`:
  This directory contains log files for service purposes. They are intended for use by Rohde & Schwarz only.
  It also contains files with encryption keys for IPsec. You can enter these keys in a network protocol analyzer, to analyze IP log files of encrypted connections, see Chapter 3.12, "Configuring Wireshark for IPsec", on page 48.
- `dau_tools`:
  Contains tools which can be installed on the mobile for data testing. The tools are also available on the web portal of the DAU.
- `ftp_anon`:
  Anonymous FTP users have write access to this directory (upload / delete). For the other directories, they have only read access (download).
- `ims`:
  This directory is used by the IMS server.
  If you want to use the media endpoint PCAP, store metadata XML files in the directory `ims/pcap`.
  Received file transfers are stored in the directory `ims/rcs/received`.
  If you want to transfer files via MSRP or import message texts, store them in the directory `ims/rcs/samples`.
- `ip_analysis`:
  This directory contains CSV files created via the "Export DB" function of the IP analysis application, see "Export DB (hotkey)" on page 128.
- `ip_logging`:

This directory contains log files created via the "IP Logging" application, see Chapter 5.7, "IP Logging Tab", on page 123. You can delete, copy or rename these files.

- `ip_replay`:
  This directory is used by the "IP Replay" application, see Chapter 5.9, "IP Replay Tab", on page 151. If you want to replay packet capture files, store them in this directory.
- `multimedia`:
  If you add pages to the web portal of the DAU, the related files are stored in directory `multimedia/web_server/user/web_pages`.
  Video files for streaming are stored in directory `multimedia/streaming`, see also Chapter 3.10, "Streaming Videos from the Web Portal", on page 46.
- `rohde-schwarz`:
  This directory contains internal configuration files and is for use by Rohde & Schwarz only.
- `security`:
  The ePDG of the DAU reads SSL certificates from the directory `security/certificates/epdg`. If you want to use certificates, store the related PEM files in this directory.
- `usb_storage`:
  If you connect an external USB drive to the USB DAU connector on the rear panel, the contents of the drive are shown in this folder.
  The connector is only present if you have the DAU version R&S CMW-B450H.



User-specific applications are not supported

The DAU supports the embedded IP services only. Installation and execution of user-specific applications on the DAU are not supported.

If you need to use own IP services/applications, install them on a separate host, connect the host to the LAN DAU connector and configure the IP settings accordingly, see Chapter 4.2, "IP Config Tab", on page 52.

Accessing the samba share via WebDAV

The `multimedia` directory of the samba share can also be accessed via the WebDAV protocol (web-based distributed authoring and versioning). As address, enter the IP address of the DAU, followed by `/webshare`, e.g. `172.22.1.201/webshare`.

The HTTP service of the DAU must be running, see Chapter 4.5, "HTTP Tab", on page 64.

3.9 Accessing the Built-In Web Server

The DAU is shipped with a built-in web server which runs a web portal. This portal can be used to test web browsing with the mobile.

The following figure shows the main page of the web portal. The portal is the starting-point for further information, statistics and tools useful for data application testing.

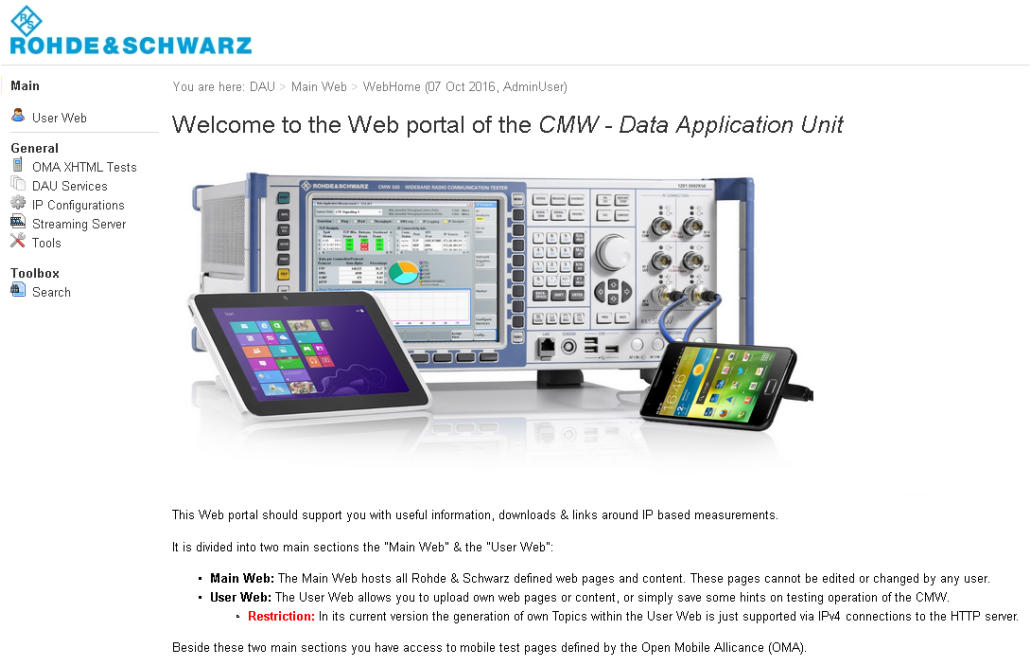


Figure 3-7: Web portal on the DAU

You can load the web pages using the "IP address" of the DAU shown in the "IP Config" tab. Type it into your web browser: `http://<IP_address>`, for example, `http://172.22.1.201`. If you want to use domain names instead of IP addresses, configure and start the DNS server, see Chapter 4.3, "DNS Tab", on page 58.

The built-in web portal on the DAU is based on a wiki solution. It is divided into two parts:

- "Main Web": This part is predefined by Rohde & Schwarz and provides the following:
  - "OMA XHTML Tests": set of XHTML tests specified by the Open Mobile Alliance
  - "DAU Services": information about DAU services, for example the DNS server
  - "IP Configurations": examples and tools for configuring IPv6 connections
  - "Streaming Server": opens the web page of the video streaming server, see Chapter 3.10, "Streaming Videos from the Web Portal", on page 46.
  - "Tools": programs which can be installed on the host behind the mobile for data testing
    For example, you can download the iperf remote configuration program which is required on the mobile for iperf measurements. See also Chapter 3.11, "Configuring Iperf on the Mobile", on page 47.
    The tools are also available on the samba share.
  - "Toolbox": search function of the wiki
- "User Web": This part is empty by default. As usual for a wiki, you can add own web pages to this part and modify them using the built-in editor.

Note: When updating the data testing support package, the "User" pages can be lost. Back up your files before updating the software. All "User" web files are stored on the system drive of the DAU in the directory `multimedia/web_server/user/web_pages/`, see Figure 3-6.

3.10 Streaming Videos from the Web Portal

You can use the web portal of the DAU to test video streaming with your mobile under test. The videos available for streaming are listed on a separate streaming server web page. To access the web page, click the menu entry "Streaming Server" in the web portal. To access the web page directly, use the web portal address plus `/streaming`, for example `http://172.22.1.201/streaming`.

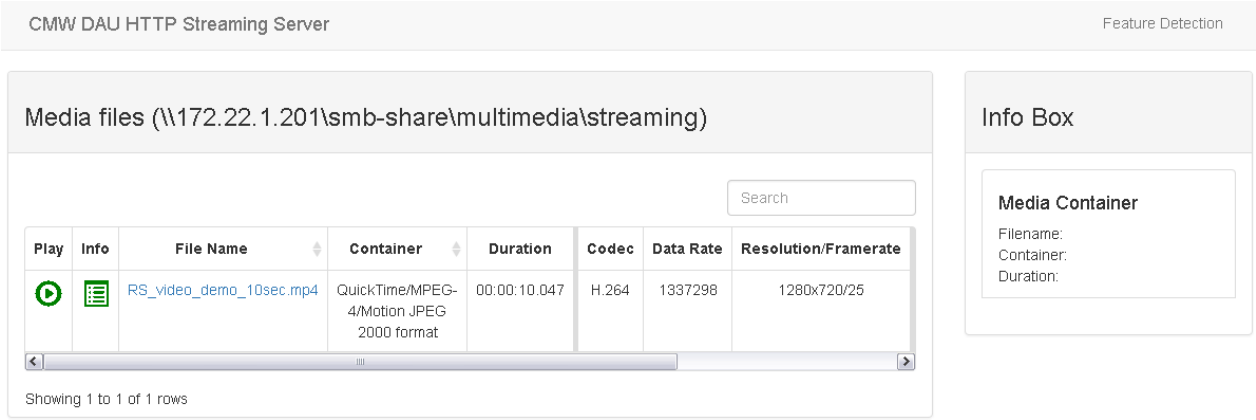


Figure 3-8: Streaming server page at http://<web portal IP address>/streaming

The page lists all available videos and their most important properties.

Possible actions:

- To start streaming, tap the play button of a video or tap the file name.
- To display detailed information about a video, tap the info button of a video. The information is shown on the right in the "Info Box" area.
- To filter the video list, enter a term into the "Search" field. All rows that do not contain the search term are hidden. The search term can be contained in any column.
- To check the capabilities of your browser or app, tap "Feature Detection". A page is displayed, listing the video and audio capabilities.

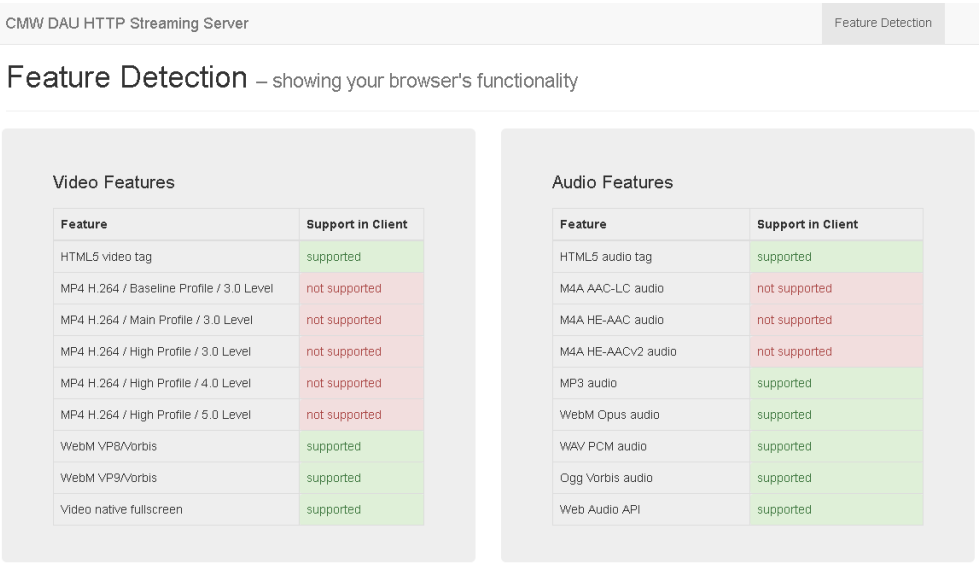


Figure 3-9: Feature detection page at http://<web portal IP address>/streaming/feature-detection

The standard installation includes video file examples. To add your own video files to the web portal, store them on the samba share of the DAU, in the directory `multimedia/streaming`. You can organize your videos in subdirectories.

The following video container formats are supported: MOV, MP4, WEBM and MPD (for MPEG-DASH).

If you store DASH videos on the samba share, keep the files of a video together in the same directory. And use a separate directory for each DASH video.

3.11 Configuring Iperf on the Mobile

For an iperf measurement, the iperf tool must be running at both ends of a connection, at the DAU side and at the mobile side.

To ensure compatibility to the iperf tool used by the DAU, use the iperf remote configuration program for Windows at the mobile side. You can download it from the tools section of the web portal or the samba share.

The following figure shows the user interface of the iperf remote configuration program. Specify the properties of the server and client instances as necessary. The settings are equal to the settings on the DAU, see Chapter 5.4, "Iperf Tab", on page 114.

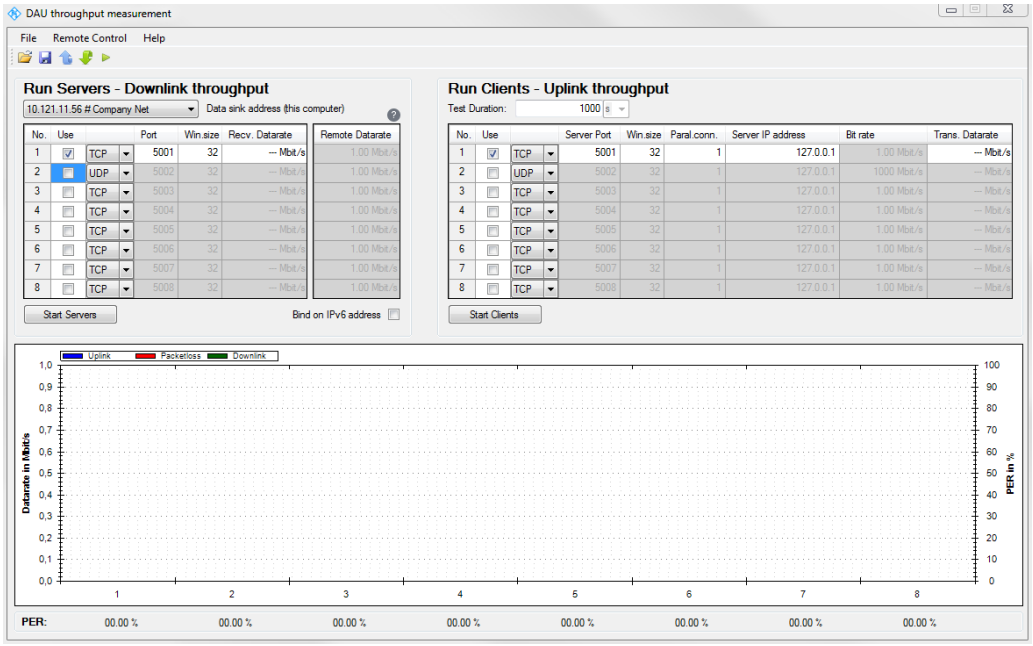


Figure 3-10: Iperf remote configuration program for the mobile

During an iperf measurement, the tool at mobile side shows the IP performance of the downlink, while the DAU shows the IP performance of the uplink.

3.12 Configuring Wireshark for IPsec

If a connection is encrypted via IPsec, log files of the connection can only be analyzed if the used encryption keys are known.

Typically, you use the "IP Logging" application of the DAU to create log files of a connection. And you use Wireshark to analyze the created PCAP files.

If the connection is encrypted, you must provide the used encryption keys to Wireshark. For this purpose, the DAU stores the used encryption keys on the samba share.

Encrypted ePDG connections

To provide encryption keys for ePDG connections to Wireshark, proceed as follows.

1. Establish the ePDG connections.
2. Access the samba share of the DAU, directory `dau_logging/logs/DAU`.
3. Open the file `epdg-esp_sa`.

   The file contains a list of encryption keys for the ESP protocol. It also indicates the path of the file `esp_sa` in the Wireshark installation.
4. Copy the encryption keys to the file `esp_sa`.
5. Open the file `epdg-ikev2_decryption_table`.

The file contains a list of encryption keys for the IKEv2 protocol. It also indicates the path of the file `ikev2_decryption_table` in the Wireshark installation.

6. Copy the encryption keys to the file `ikev2_decryption_table`.
7. Restart Wireshark.

   You can now use Wireshark to analyze log files of the ePDG connections.
   For each new ePDG connection, new keys are generated. So you must repeat the procedure for new connections.

Encrypted IMS connections

IMS connections can also be encrypted via IPsec. In that case, the same keys are used for all connections - as long as you do not modify the related authentication settings. So you need to provide the encryption keys to Wireshark only once.

As for ePDG connections, the encryption keys are stored on the samba share in the directory `dau_logging/logs/DAU`.

Copy the keys from the file `ims-esp_sa` to the file `esp_sa` of the Wireshark installation.

4 DAU Services GUI Reference

The "Data Application Control" dialog allows you to configure the DAU services and the IP network settings of the DAU.

Opening the "Data Application Control" dialog

1. Press "Setup" to open the "Setup" dialog.
2. In section "System", press the button "Go to config".

   The "Data Application Control" dialog opens.

   Alternatively, you can access the dialog from the "Data Application Measurements" dialog, using the softkey "Configure Services".

The "Data Application Control" dialog contains an overview tab, a tab for configuration of the IP settings and one tab per DAU service. Refer to the following sections for a detailed description.

- Overview Tab 50
- IP Config Tab 52
- DNS Tab 58
- FTP Tab 63
- HTTP Tab 64
- IMS Tab 65
- ePDG Tab 99
- Common Softkeys, Hotkeys and Parameters 109

4.1 Overview Tab

The tab provides an overview of the current IP settings and all DAU services, including the state of each service.

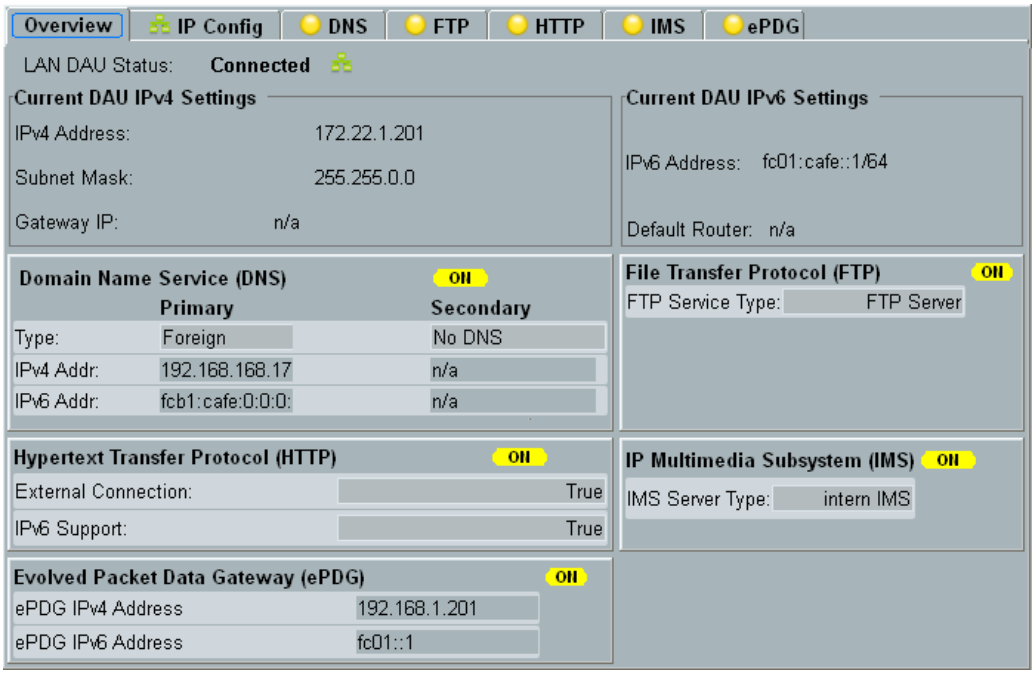


Figure 4-1: Data application control - overview tab

To open the configuration dialog, press the "Config" hotkey.

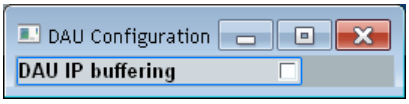


Figure 4-2: Configuration dialog

LAN DAU Status, Current DAU IPv4/IPv6 Settings 51
DNS, FTP, HTTP, IMS, ePDG 51
Service control 52
DAU IP Buffering 52

LAN DAU Status, Current DAU IPv4/IPv6 Settings
Display the most important elements of the current IP setup.

For details and configuration, see Chapter 4.2, "IP Config Tab", on page 52.

DNS, FTP, HTTP, IMS, ePDG
Shows the state of the corresponding service and the most important settings.

For details, refer to the following sections:
- Chapter 4.3, "DNS Tab", on page 58
- Chapter 4.4, "FTP Tab", on page 63
- Chapter 4.5, "HTTP Tab", on page 64
- Chapter 4.6, "IMS Tab", on page 65
- Chapter 4.7, "ePDG Tab", on page 99

Service control
To start or stop a service, select the corresponding softkey (e.g. "FTP Server") and press ON | OFF or RESTART | STOP. Alternatively, right-click the softkey.

The current service state is indicated by the softkey, in the overview and in the title of the corresponding tab.

DAU IP Buffering
This checkbox defines the behavior of layer 3, if the higher layers deliver more data than the lower layers can process (maximum throughput of lower layers reached).

- Checkbox disabled:
  Packets that cannot be handled by the lower layers are discarded.
- Checkbox enabled:
  Packets that cannot be handled by the lower layers are added to a buffer for later transmission. If the buffer is full, the higher layers are informed so that they reduce the data rate without causing packet loss.

Enable the checkbox only for TCP high-throughput scenarios where you want as low packet loss as possible and as few retransmissions as possible.

Remote command:
`CONFigure:DATA:CONTrol:ADVanced:IPBuffering`

4.2 IP Config Tab

This tab specifies the IP settings for the DAU and the mobile. The settings must be compatible to the test setup.

To apply changes of the IP settings and make them known to the signaling units, you must "restart" the generation of downlink signals by the signaling units. Switch downlink signal generation off and on again in the signaling application or restart protocol tests.

You can specify IP settings both for IPv4 and for IPv6. Which IP version is used depends on the mobile and the used signaling application or protocol test application. It is not selected by the DAU. To use IPv4, you need R&S CMW-KA100. IPv6 requires also option R&S CMW-KA150.

You can configure the DAU IP settings manually or via the "Initial Setup Wizard". For an introduction to both methods and background information concerning IPv4 and IPv6, see Chapter 3.2, "Adapting the DAU IP Configuration", on page 19. For a setup with a connected external network, contact your network administrator before configuring the IP settings.

The following sections describe the information displayed in the "IP Config" tab and the related configuration dialogs.

- Displayed Information 53
- General Configuration 54
- IPv4 Configuration 54
- IPv6 Configuration 56

4.2.1 Displayed Information

The "IP Config" tab displays the current IP configuration of the DAU.

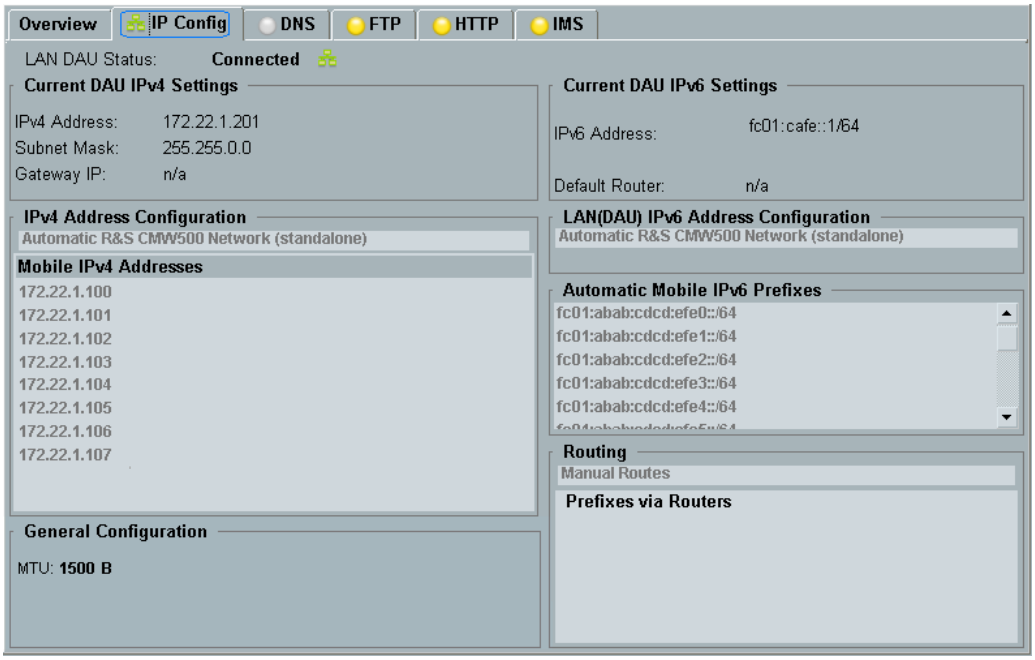


Figure 4-3: Data application control - IP Config tab

The parameter "LAN DAU Status" at the top of the tab indicates whether an active external network is connected to the LAN DAU connector or not. Note that an external network connected to the LAN DAU connector is ignored by the DAU if a standalone setup is selected in the settings, see "IPv4 Address Configuration" on page 55 and "LAN IPv6 Address Configuration" on page 57.

The current IPv4 configuration and the general IP settings are displayed in the left half of the tab, the IPv6 configuration in the right half. The values result either directly from DAU settings described in the following sections or have been assigned dynamically by a connected external network.

The current DAU IP addresses displayed at the top can be used to address the DAU, e.g. to access the internal web server or other DAU services.

If the dynamic host configuration protocol (DHCP) is selected as configuration method, an additional parameter "DHCPv4 Status" or "DHCPv6 Status" is shown. This parameter indicates via a colored icon whether the dynamic configuration via DHCP was successful and the network connection works properly (green icon) or not (yellow icon).

Related commands

The following commands allow you to query the displayed information.

For the mobile address/prefix pools, there is a separate command for each configuration method. So all pools can be queried, the displayed currently used pools and the currently inactive pools.

Remote command:

```
SENSe:DATA:CONTrol:LAN:DAU:STATus?
SENSe:DATA:CONTrol:IPVFour:CURRent:IPADdress?
SENSe:DATA:CONTrol:IPVFour:CURRent:SMASk?
SENSe:DATA:CONTrol:IPVFour:CURRent:GIP?
SENSe:DATA:CONTrol:IPVFour:STATic:ADDResses:CATalog?
SENSe:DATA:CONTrol:IPVFour:DHCP:ADDResses:CATalog?
SENSe:DATA:CONTrol:IPVFour:AUTomatic:ADDResses:CATalog?
SENSe:DATA:CONTrol:IPVSix:CURRent:IPADdress?
SENSe:DATA:CONTrol:IPVSix:CURRent:DROuter?
SENSe:DATA:CONTrol:IPVSix:STATic:PREFixes:CATalog?
SENSe:DATA:CONTrol:IPVSix:DHCP:PREFixes:CATalog?
SENSe:DATA:CONTrol:IPVSix:AUTomatic:PREFixes:CATalog?
SENSe:DATA:CONTrol:IPVSix:MANual:ROUTing:CATalog?
```

4.2.2 General Configuration

To modify the general configuration, press the "Config" hotkey. The general settings are at the top of the dialog box.



Figure 4-4: General IP configuration

Maximum Transmission Unit (MTU)
Specifies the maximum IP packet size that can be transmitted without fragmentation.

IPv6 requires an MTU of at least 1280 bytes. IPv4 is possible with any configurable value.

Remote command:

```
CONFigure:DATA:CONTrol:MTU
```

4.2.3 IPv4 Configuration

To modify the IPv4 configuration, press the "Config" hotkey and select the tab "IPv4 Address Configuration".

The available settings are described below. For additional information, see Chapter 3.2.2, "IP Address Configuration Methods", on page 21.

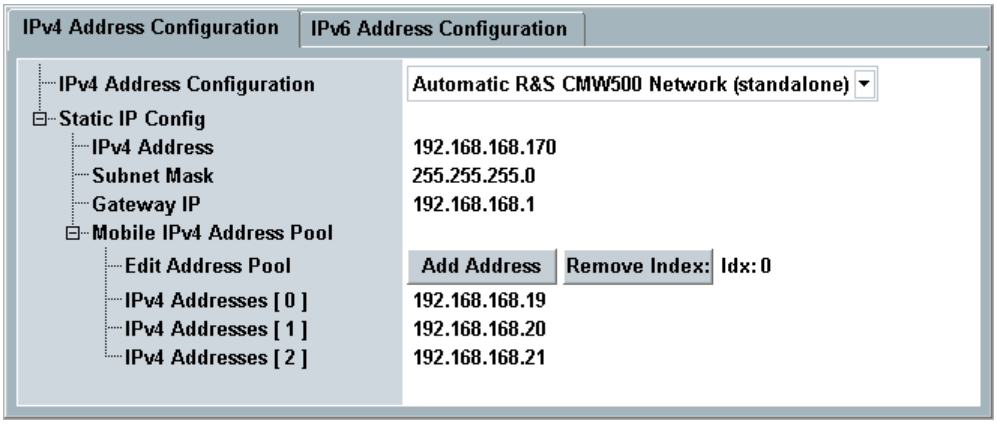


Figure 4-5: IPv4 address configuration

IPv4 Address Configuration 55
Static IP config 55
└ IPv4 Address, Subnet Mask 55
└ Gateway IP 56
└ Mobile IPv4 Address Pool 56

IPv4 Address Configuration
Selects the method to be used for IPv4 address configuration.

The following methods are available:

- **"Automatic R&S CMW500 Network (standalone)"**
  Select this value for a standalone test setup. As a consequence the predefined automatic IP configuration is used.
- **"Static IP config"**
  Select this value for a test setup with connected external network and static IP addresses.
  For configuration of the static IP configuration, see "Static IP config" on page 55.
- **"DHCPv4 from LAN (LAN DAU)"**
  Select this value for a test setup with connected external network and dynamic IP address assignment via DHCPv4.

Remote command:
`CONFigure:DATA:CONTrol:IPVFour:ADDRess:TYPE`

Static IP config
The following parameters specify the static IP configuration to be used if "Static IP config" is selected for IPv4 Address Configuration.

IPv4 Address, Subnet Mask ← Static IP config
Assign a static IPv4 address and subnet mask to the LAN DAU Ethernet adapter.

Remote command:
`CONFigure:DATA:CONTrol:IPVFour:STATic:IPADdress`
`CONFigure:DATA:CONTrol:IPVFour:STATic:SMASk`

Gateway IP ← Static IP config
If you want to test public services on the Internet, specify the IP address of the gateway connecting the external network to the Internet.

Remote command:
`CONFigure:DATA:CONTrol:IPVFour:STATic:GIP`

Mobile IPv4 Address Pool ← Static IP config
Specifies the IP address pool for mobiles. An address is assigned to the mobile upon request. A single mobile can request several addresses and thus need several entries.

Configuring entries:
- Edit existing entries directly in the dialog.
- To add an entry to the end of the list, press the "Add Address" button, then edit the new entry.
- To delete a specific entry, select the number of the entry via parameter "Idx:" and press the "Remove Index" button.

Remote command:
`CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:ADD`
`CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:DELete`
`SENSe:DATA:CONTrol:IPVFour:STATic:ADDResses:CATalog?`

4.2.4 IPv6 Configuration

To modify the IPv6 configuration, press the "Config" hotkey and select the tab "IPv6 Address Configuration".

The available settings are described below. For additional information, see Chapter 3.2.2, "IP Address Configuration Methods", on page 21.

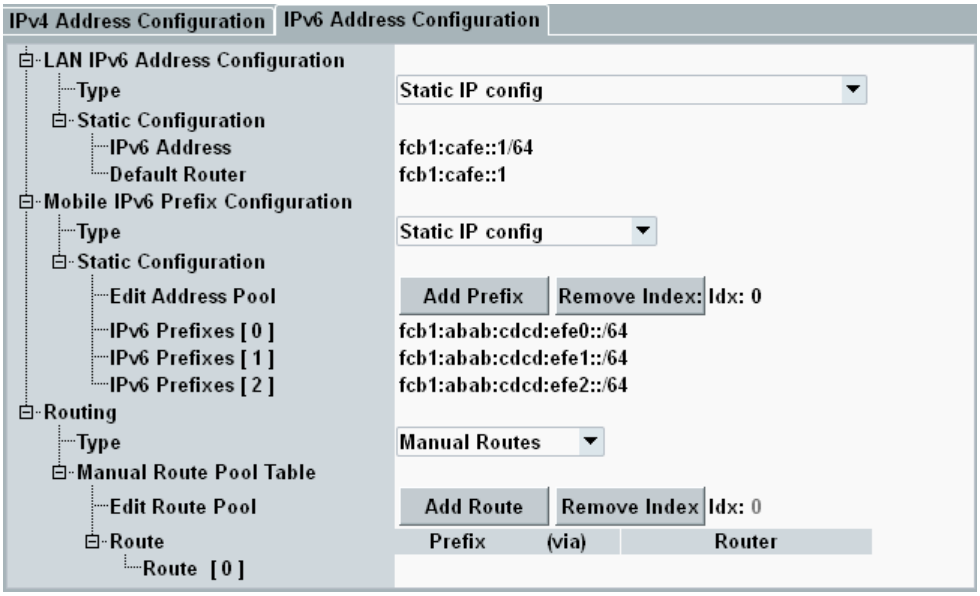


Figure 4-6: IPv6 address configuration

LAN IPv6 Address Configuration 57
Mobile IPv6 Prefix Configuration 57
Routing 58

LAN IPv6 Address Configuration
"Type" selects the method to be used for IPv6 DAU address configuration.

The following methods are available:

- **"Automatic R&S CMW500 Network (standalone)"**
  Select this value for a standalone test setup. As a consequence the predefined automatic IP configuration is used. Both the DAU address and the mobile prefixes are configured automatically. No further IP configuration is necessary for IPv6.
- **"Static IP config"**
  Select this value for a test setup with connected external network and static IP addresses.
  Specify the static IP addresses of the DAU and a default router belonging to the same subnet via parameter "Static Configuration". The DAU sends packets with unknown IPv6 prefix to this default router.
- **"Autoconfiguration (RA, DHCPv6)"**
  Select this value for a test setup with connected external network and dynamic autoconfiguration. IPv6 autoconfiguration uses router advertisements (RA) and/or DHCPv6.

Remote command:
```
CONFigure:DATA:CONTrol:IPVSix:ADDRess:TYPE
CONFigure:DATA:CONTrol:IPVSix:STATic:ADDRess
CONFigure:DATA:CONTrol:IPVSix:STATic:DROuter
```

Mobile IPv6 Prefix Configuration
This section is only relevant for test setups with connected external network. It is ignored for a standalone test setup (LAN IPv6 Address Configuration > "Type" = "Automatic R&S CMW500 Network (standalone)").

The following methods can be selected for definition of the IPv6 prefix pool for mobiles:

- **"Static IP config"**
  Select this value to use a statically defined pool of prefixes. Specify the prefix pool via parameter "Static Configuration". Only 64-bit prefixes are supported.
  The pool can be modified in the same way as the address pool for IPv4, see "Mobile IPv4 Address Pool" on page 56.
- **"DHCP Prefix Delegation"**
  Select this value to configure the prefix pool dynamically using DHCPv6 prefix delegation.

Remote command:
```
CONFigure:DATA:CONTrol:IPVSix:MOBile:PREFix:TYPE
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:ADD
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:DELete
SENSe:DATA:CONTrol:IPVSix:STATic:PREFixes:CATalog?
```

Routing
Allows you to configure manual routes for packets sent by the mobile. For destination addresses belonging to the subnet of the DAU or reachable via the default router, you do not need to configure routes.

For other destinations, specify the prefix of the destination subnet and the IPv6 address of the router to be used. The router must belong to the subnet of the DAU.

"Routing Protocols" are not supported in the current software version.

Remote command:
`CONFigure:DATA:CONTrol:IPVSix:ROUTing:TYPE`
`CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:ADD`
`CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:DELete`
`SENSe:DATA:CONTrol:IPVSix:MANual:ROUTing:CATalog?`

4.3 DNS Tab

This tab specifies the primary and secondary DNS server addresses to be sent to the mobile. It also configures and controls the local DNS server of the DAU.

Basically, the mobile can send DNS queries to a foreign DNS server located in a connected external network or to the local DNS server reachable via the DAU address. The DAU handles the queries as follows:

- DNS queries from a mobile to a foreign/external DNS server are forwarded by the DAU like any other IP packets with a foreign destination address. The DAU does not recognize them as DNS queries.
- DNS queries addressed to the DAU are handled by the local DNS server of the DAU. Optionally, the DAU can forward a query to a foreign DNS server if the local DNS server database does not contain a matching entry.
  DNS queries of type A, AAAA and SRV are supported. The DAU measurement "DNS Requests" allows you to monitor all DNS queries addressed to the DAU, see Chapter 5.6, "DNS Requests Tab", on page 122.

The DNS tab displays a summary of the DAU settings related to the DNS service.

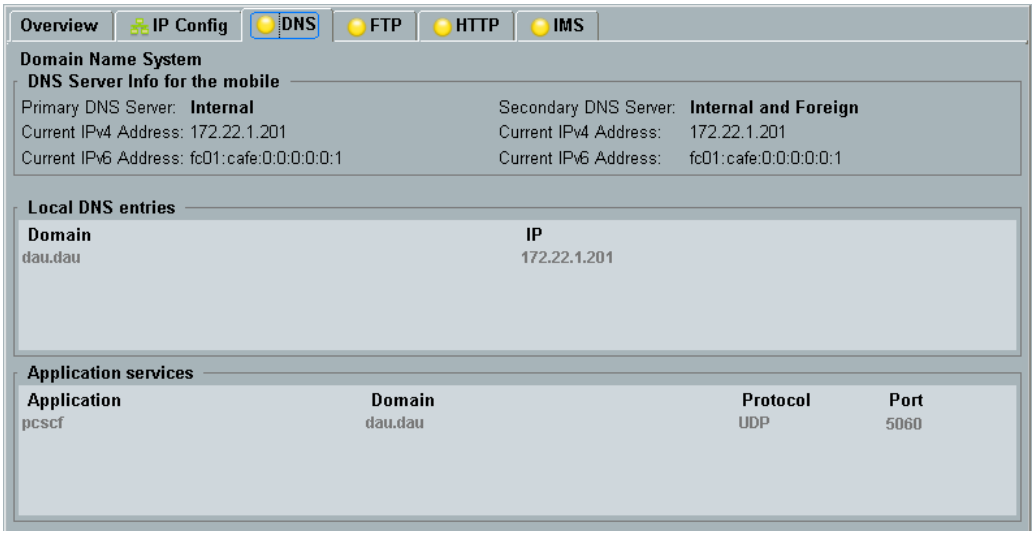


Figure 4-7: Data application control - DNS tab

To open the configuration dialog, press the "Config" hotkey.

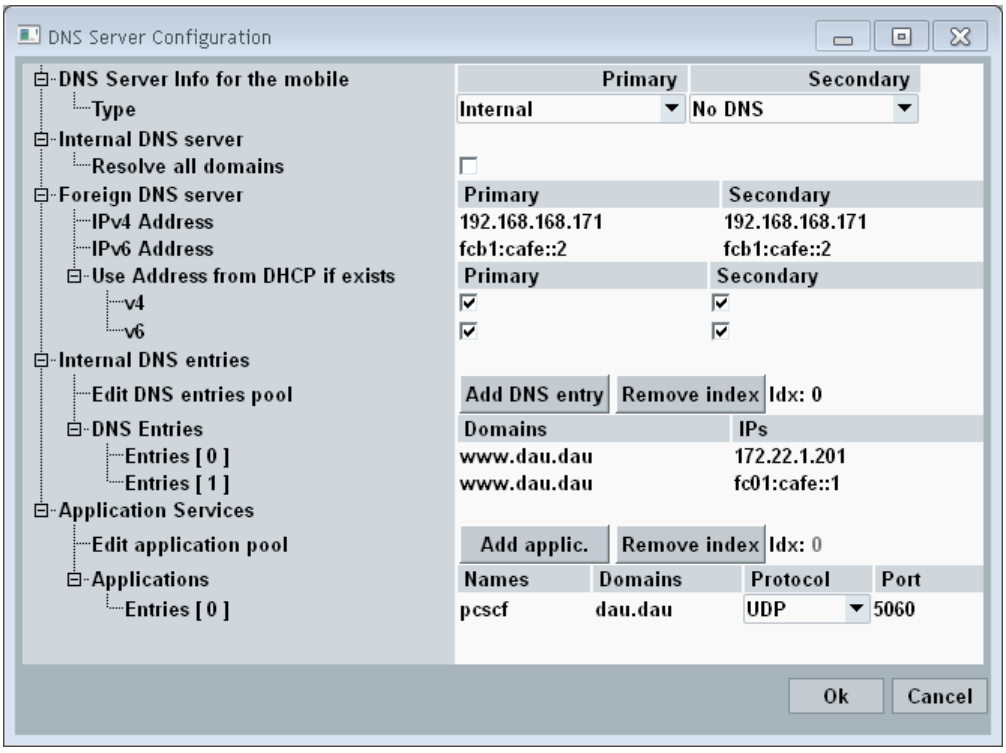


Figure 4-8: Data application control - DNS Server Configuration dialog

The settings are described below.

DNS Server Info for the mobile 60
Internal DNS server 60
Foreign DNS server 61
Internal DNS entries 61

Application Services....61
Test Primary Foreign DNS Server (softkey)....62
DNS server control....62

DNS Server Info for the mobile
The "Type" determines which addresses are sent to the mobile as primary and secondary DNS server address and how a DNS query addressed to the DAU is handled.

Note that a foreign DNS server cannot be used with a standalone setup.

Available types:

- **"No DNS"**
  No DNS server address is sent to the mobile.
- **"Internal"**
  The address of the DAU is sent to the mobile as DNS server address.
  DNS queries addressed to the DAU are handled by the local DNS server of the DAU. The foreign DNS server settings are ignored.
  If the local DNS server is not running, a DNS query fails.
- **"Internal and Foreign"**
  The address of the DAU is sent to the mobile as DNS server address.
  DNS queries addressed to the DAU are handled by the local DNS server. If the local DNS server database does not contain a matching entry, the query is forwarded to the configured foreign DNS server.
  If the local DNS server is not running, the foreign DNS server settings are ignored and a DNS query fails.
- **"Foreign"**
  The configured foreign DNS server address is sent to the mobile.
  The state and database of the local DNS server are irrelevant.

The "DNS Server Info for the mobile" at the top of the DNS tab indicates which of the listed types is active and which addresses are sent to the mobile.

A displayed address can be:

- The current IP address of the DAU, providing access to the local DNS server
- The statically configured foreign DNS server address
- A foreign DNS server address received via DHCP, overriding the configured address

The DAU sends DNS server addresses to the mobile via a signaling message, for example via an ACTIVATE PDP CONTEXT ACCEPT message.

Remote command:

```
CONFigure:DATA:CONTrol:DNS:PRIMary:STYPe
CONFigure:DATA:CONTrol:DNS:SECondary:STYPe
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:PRIMary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:SECondary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:PRIMary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:SECondary:ADDRess?
```

Internal DNS server
"Resolve All Domains" ensures that all DNS queries of type A or AAAA are answered with an IP address. If the checkbox is enabled and no database entry matching a query is found, the DAU IP address is returned.

Remote command:
CONFigure:DATA:CONTrol:DNS:RESallquery

Foreign DNS server
Specifies the IPv4 and IPv6 addresses of the foreign primary and secondary DNS servers.

If a DNS server address is received via DHCP, it can either be ignored or used instead of the statically configured address. The behavior is configurable separately for all four addresses.

Remote command:
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:PRIMary:ADDRess
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:SECondary:ADDRess
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:PRIMary:ADDRess
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:SECondary:ADDRess
CONFigure:DATA:CONTrol:DNS:FOReign:UDHCp

Internal DNS entries
Configures local DNS server database entries assigning an IPv4 or IPv6 address to a domain. The resulting DNS records are of type A for IPv4 and of type AAAA (quad-A) for IPv6.

Each entry consists of a domain and one IP address. To assign both an IPv4 and an IPv6 address to the same domain, create two entries.

A DNS query contains the domain. The answer of the DNS server contains all IPv4 addresses or all IPv6 addresses assigned to the domain.

Configuring entries:
- Edit existing entries directly in the dialog.
- To add an entry to the end of the list, press the "Add DNS entry" button, then edit the new entry.
- To delete a specific entry, select the number of the entry via parameter "Idx:" and press the "Remove index" button.

Remote command:
CONFigure:DATA:CONTrol:DNS:LOCal:ADD
CONFigure:DATA:CONTrol:DNS:LOCal:DELete
SENSe:DATA:CONTrol:DNS:LOCal:CATalog?

Application Services
Configures local DNS server database entries assigning a host to a particular service. The resulting DNS records are of type SRV (see RFC 2782).

Each entry consists of the service, the domain supporting the service, the protocol (UDP or TCP) and the port used for the service.

A type SRV DNS query contains the service and the protocol. The answer of the DNS server contains all domain/port combinations supporting the service via the requested protocol.

Configuring entries:
- Edit existing entries directly in the dialog.
- To add an entry to the end of the list, press the "Add applic." button, then edit the new entry.

- To delete a specific entry, select the number of the entry via parameter "Idx:" and press the "Remove index" button.

Remote command:
```
CONFigure:DATA:CONTrol:DNS:ASERvices:ADD
CONFigure:DATA:CONTrol:DNS:ASERvices:DELete
SENSe:DATA:CONTrol:DNS:ASERvices:CATalog?
```

Test Primary Foreign DNS Server (softkey)
The softkey opens a dialog that allows to test whether the primary foreign DNS server is reachable and whether it returns IP addresses for a specific domain.



Figure 4-9: Connection test dialog

The test uses the primary DNS server addresses resulting from section Foreign DNS server (statically defined addresses or DHCP override values).

To perform a test, enter the domain to be resolved and press the "Start Test" button.

During a test, the DAU sends a ping request to the IPv4 and IPv6 addresses of the foreign DNS server to test whether it is reachable. The result is displayed in the dialog in column "Server available".

If the server is reachable, the DAU sends four DNS queries to the foreign DNS server: type A and type AAAA queries to the IPv4 and IPv6 addresses. The results are shown in the dialog. The "A/AAAA Resource Record" columns indicate whether the server has returned an IP address (yellow dash = no result, red cross = failure, green checkmark = success).

Remote command:
```
CONFigure:DATA:CONTrol:DNS:TEST:DOMain
CONFigure:DATA:CONTrol:DNS:TEST:STARt
SENSe:DATA:CONTrol:DNS:TEST:RESults?
```

DNS server control
Start or stop the internal DNS server using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "Local DNS Server" softkey.

The softkey shows the current state.

Remote command:
```
SOURce:DATA:CONTrol:DNS:STATe
```

4.4 FTP Tab

Configures the file transfer protocol (FTP) service for file transfer between the mobile and the DAU.

The service type "FTP Server" provides access to the samba share of the DAU via FTP. With anonymous access, you can download data from the samba share and have full access rights for the subdirectory `ftp_anon` (delete, download, upload). For registered users, the access rights are configurable per user.

See also Chapter 3.8, "Using the System Drive of the DAU", on page 42.

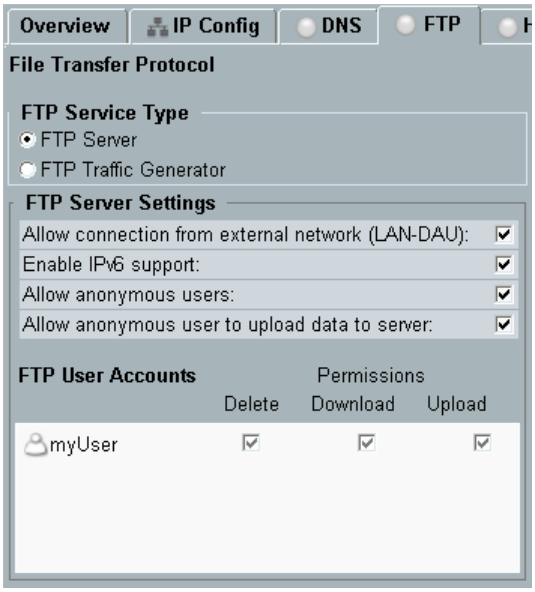


Figure 4-10: Data application control - FTP tab

FTP Service Type 63
FTP Server Settings 64
└ FTP User Accounts 64
Service control 64

FTP Service Type
Selects the FTP service type.

"FTP Server"	An FTP server runs on the R&S CMW. You can configure it via the "FTP Server Settings".
"FTP Traffic Generator"	The R&S CMW acts as a traffic generator which creates dummy data. You can download dummy data to the mobile or upload files from the mobile to the instrument. Note that uploaded files are not stored. For long-term tests, discarding the uploaded files avoids the risk of running out of data or filling up the internal disk drive of the DAU.

Remote command:
`CONFigure:DATA:CONTrol:FTP:STYPe`

FTP Server Settings
Configure the FTP server via the checkboxes.

Remote command:
```
CONFigure:DATA:CONTrol:FTP:ENConnection
CONFigure:DATA:CONTrol:FTP:IPVSix
CONFigure:DATA:CONTrol:FTP:AUSer
CONFigure:DATA:CONTrol:FTP:DUPLoad
```

FTP User Accounts ← FTP Server Settings
Use the "... user account" softkeys to configure the FTP user accounts. For each user, you can specify the user credentials and access rights for data upload, download and deletion.

Remote command:
```
CONFigure:DATA:CONTrol:FTP:USER:ADD
SENSe:DATA:CONTrol:FTP:USER:CATalog?
CONFigure:DATA:CONTrol:FTP:USER:DELete
```

Service control
Start or stop the service using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "FTP Service" softkey.

The softkey shows the current service state.

Remote command:
```
SOURce:DATA:CONTrol:FTP:STATe
```

4.5 HTTP Tab

Configures the hypertext transport protocol (HTTP) service for access to the embedded web server.

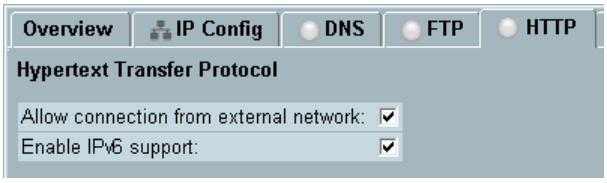


Figure 4-11: Data application control - HTTP tab

Hypertext Transfer Protocol 64
Service control 65

Hypertext Transfer Protocol
You can allow or deny HTTP connections to the web server from the external network (company LAN), and switch on/off IPv6 support.

Access to the web server from the mobile or from a host connected to the internal CMW subnet is always allowed.

Remote command:
`CONFigure:DATA:CONTrol:HTTP:ENConnection`
`CONFigure:DATA:CONTrol:HTTP:IPVSix`

Service control
Start or stop the service using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "HTTP Service" softkey.

The softkey shows the current service state.

Via remote command, you can also check for errors related to the HTTP server.

Remote command:
`SOURce:DATA:CONTrol:HTTP:STATe`

4.6 IMS Tab

This tab configures and controls the internal IP multimedia subsystem (IMS) server provided by the DAU (option R&S CMW-KAA20 required).

The IMS server allows an IMS client on the mobile to register to the IMS domain. With a registered mobile, you can test voice over IMS and messaging over IMS. For step-by-step instructions, see Chapter 3.5, "Using the IMS Server", on page 26.

If the internal IMS server is used, the IMS tab shows a log of general IMS information and an event log of calls and message transfers. The lower part shows details for the currently selected line of the event log.

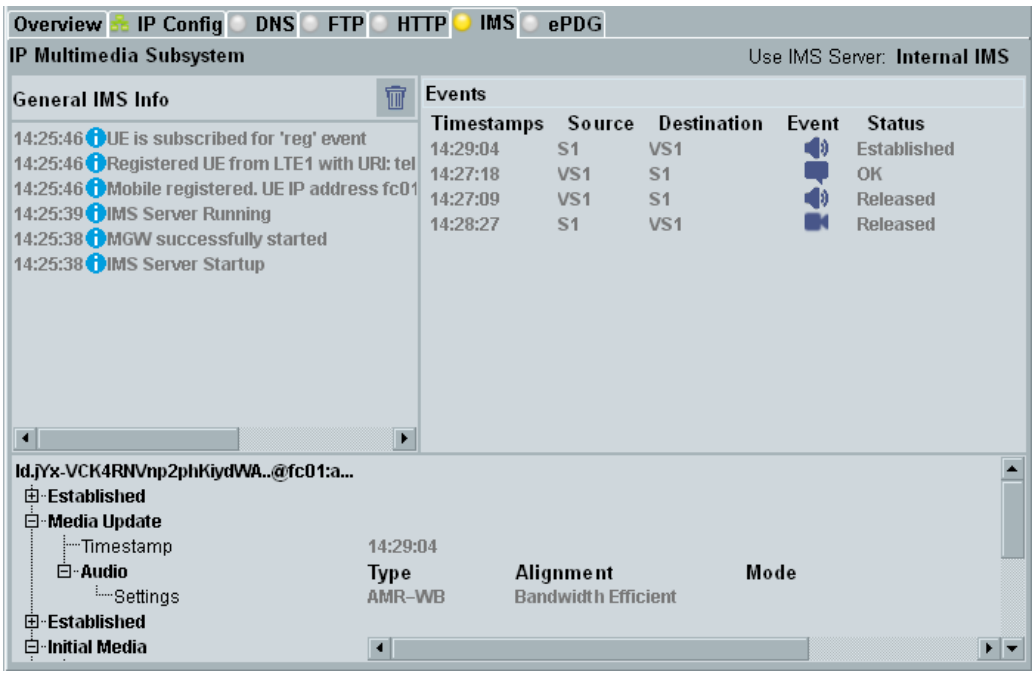


Figure 4-12: IMS tab, internal IMS server

If an external IMS network is used instead of the internal IMS server, only the address of the external P-CSCF is configured at the DAU. In that case, IMS services are transparent to the DAU. Besides the P-CSCF address, the DAU has no information about the IMS network. It does not even know whether the mobile has registered or not.

The DAU sends the P-CSCF address to the mobile via a signaling message, for example via an ACTIVATE PDP CONTEXT ACCEPT message.

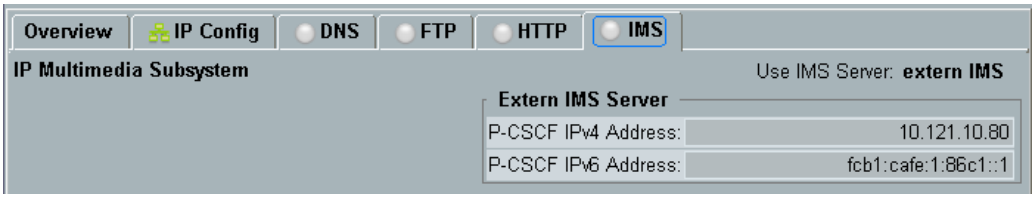


Figure 4-13: IMS tab, external IMS network

While usage of the internal IMS server requires option R&S CMW-KAA20, usage of an external IMS server is possible without this option.

The main view areas, the softkeys and the IMS server settings are described in detail in the following sections.

- IMS Server Control and Softkeys....66
- General IMS Info....70
- Event Log....70
- Event Details....72
- General IMS Settings....73
- P-CSCF Settings....75
- Subscriber Settings....77
- Virtual Subscriber Settings....81

4.6.1 IMS Server Control and Softkeys

The IMS server must be running before the mobile can register and any IMS services can be used.

Server Control....66
Establishing a call / sending a message....67
Update Call (softkey)....67
└ Outband tab....67
└ Inband tab....67
Chat Actions (softkey)....68
Release Call (softkey)....69
Deregister Mobile (softkey)....69

Server Control
To start or stop the IMS server, press the ON | OFF or RESTART | STOP key. Alternatively, right-click the "IMS Service" softkey.

The softkey shows the current service state.



Remote command:
SOURce:DATA:CONTrol:IMS2:STATe

Establishing a call / sending a message
For detailed step-by-step instructions, see Chapter 3.5, "Using the IMS Server", on page 26.

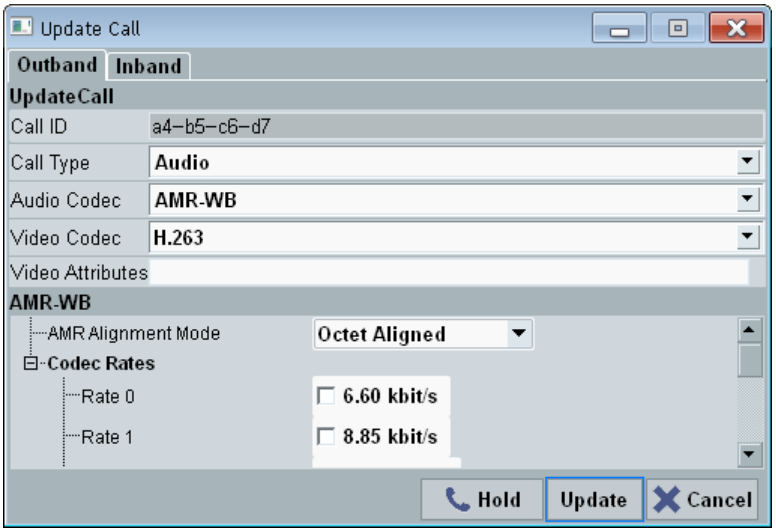
Update Call (softkey)



This softkey is active if you have selected an established call or a call on hold in the "Events" area.

The softkey opens a dialog box where you can modify call settings. The dialog box has two tabs described in the following.

Outband tab ← Update Call (softkey)
A change on the "Outband" tab causes a re-INVITE. You can, for example, reconfigure the call type, the codec type or the supported codec rates.



The tab allows you also to put an established call on hold, or to resume a call that is on hold.

Remote command:
To modify call settings, see Chapter 7.9.2, "Call Update (Outband)", on page 213.
To put a call on hold or to resume it, see CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:EVENt.

Inband tab ← Update Call (softkey)
The "Inband" tab configures codec mode requests (CMR). The tab is only relevant for calls using the audio board as media endpoint. Option R&S CMW-KAA21 is required.

For AMR, you can configure the codec rate ("Bitrate"). "No Req" means that there is no codec rate requirement.

"Repetition" selects whether the configured CMR is sent only once or continuously.



Figure 4-14: CMR settings for AMR

For EVS, select first the "Bandwidth", then the "Bitrate". The values are defined in 3GPP TS 26.445, table A.3 "Structure of CMR".

There are two special "Bandwidth" values. "No Req" means that there is no codec rate requirement (NO_REQ). "Deactivate" means that the CMR byte is removed from the header.

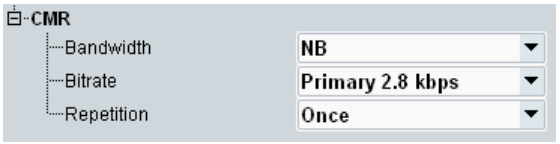


Figure 4-15: CMR settings for EVS

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRNb:CODec:RATes
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRWb:CODec:RATes
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:BW
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:CODec:RATes
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:REPetition
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:PERForm
```

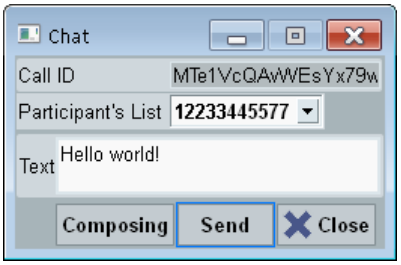
Chat Actions (softkey)



This softkey is active if you have selected an established chat in the "Events" area.

The softkey opens a dialog box that allows you to:

- Send a message to the mobile via the established chat session.
- Send "isComposing" status messages to the mobile.



The "Participant's List" is only displayed for group chats. It selects which subscriber is indicated to the mobile as originator of the message.

To send a chat message, enter the message text and press the "Send" button.

To send an "isComposing" status message, press the leftmost button.

The button shows the status "Composing" or "Idle". Pressing the button changes the status. For each status change, the application sends an "active" status message or an "idle" status message to the mobile.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:TEXT
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:PERForm
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:COMPsng:NTFCn
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:IDLE:NTFCn
```

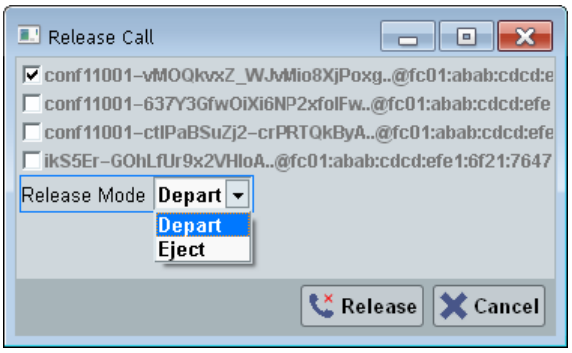
Release Call (softkey)



This softkey is active if at least one session is established (for example for a call or a chat).

The softkey opens a dialog box. To release a session or connection, enable its check-box and press the "Release" button.

For conference calls, the parameter "Release Mode" is displayed. It selects whether the connection is released by the virtual subscriber ("Depart") or by the conference server ("Eject").



Remote command:

```
SENSe:DATA:CONTrol:IMS2:RELease:LIST?
CONFigure:DATA:CONTrol:IMS2:RELease:CALL:ID
```

Deregister Mobile (softkey)



This softkey is active if at least one mobile has registered.

The softkey opens a dialog box. To deregister a mobile, enable its checkbox and press the "Deregister" button.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:MOBile<UE>:DERegister
```

4.6.2 General IMS Info

The "General IMS Info" area in the main view reports events and errors like IMS state changes, registration events and subscription events.

The button to the right clears the displayed entries.



Figure 4-16: General IMS Info area of the IMS tab

Entries
Each entry consists of a timestamp, an icon indicating the category of the event and a short text describing the event.

Meaning of the category icons: = information, warning and error

Remote command:
`SENSe:DATA:CONTrol:IMS2:GINFo?`
`CONFigure:DATA:CONTrol:IMS2:CLEan:GENeral:INFO`

4.6.3 Event Log

A new line is added to the top of the "Events" area whenever a new session is set up or a message is sent or received.

When a state changes, the corresponding line is updated.



Figure 4-17: Events area of the IMS tab

To clear the area, switch the IMS server off and on again. To view details for a call or a message, select the line and check the lower part of the main view, see Chapter 4.6.4, "Event Details", on page 72.

Event query
You can query all lines in the "Events" area via a single command.

Remote command:
`SENSe:DATA:CONTrol:IMS2:EVENts?`

Timestamps column
This column indicates the time when the status was reached.

Source, Destination columns
"Source": calling party / message sender

"Destination": called party / message recipient

"VSn": virtual subscriber number n

"Sn": subscriber number n

"Cn": conference server number n

Event column
Indicates the type of the event via one of the following icons:

- audio call
- video call
- ongoing call setup or call on hold
- sent or received message (3GPP, 3GPP2, RCS pager)
- sent or received RCS 1 to 1 chat message
- sent or received RCS group chat message
- sent or received RCS large message
- file transfer

Status column
Indicates the status of the session or message transfer.

A session is set up, for example, for VoLTE calls, RCS large message transfers, RCS chats and file transfers. Short message transfers and RCS pager mode message transfers are done without setting up a session.

"In Progress"	The session setup is in progress.
"Ringing"	The mobile is ringing during the session setup.
"Created"	An MO INVITE request or an MT call trigger was received by the IMS and the IMS started processing it.
"Initial Media"	The codec negotiation is complete (during the session setup).
"Established"	The session has been established.
"Media Update"	The codec negotiation is complete (during a call update).
"Hold"	The call is on hold.
"Resumed"	The call has been resumed after putting it on hold.
"File Transfer"	The file transfer is complete.
"Text Transferred"	An RCS message has been received or sent (RCS large, RCS chat).
"Released"	The session has been released.
"Released for SRVCC"	The call has been released due to SRVCC handover.
"Busy"	The called subscriber is busy.
"Declined"	The call has been declined by the called subscriber.

"Canceled"	The session has been canceled during the setup.
"Rejected"	The session setup failed.
"Terminated"	The session has been terminated abruptly.
"OK"	A message has been received or sent without detecting problems (3GPP, 3GPP2, RCS pager).
"Not OK"	A message has been received or sent and problems have been detected (3GPP, 3GPP2, RCS pager).

4.6.4 Event Details

The lower part of the main view shows detailed information for the currently selected entry of the "Events" area.

If the entry includes a session setup, the information covers the entire history of the session. Each status change is listed with its timestamp. Important settings are also listed.

The ID shown at the top is an automatically generated unique ID, identifying the event in the DAU.

The following figures show some examples for different types of events.

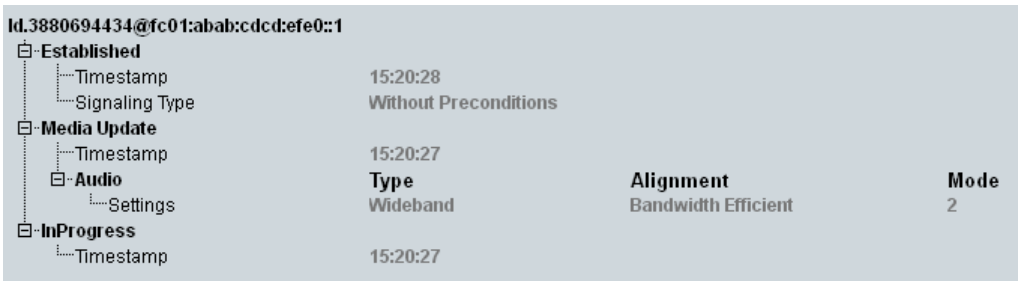


Figure 4-18: Established audio call

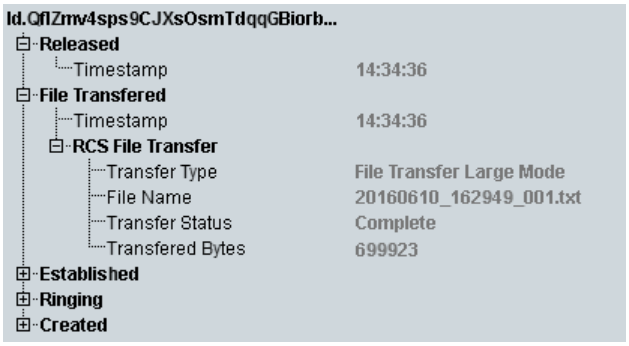


Figure 4-19: Completed file transfer



Figure 4-20: Ongoing RCS 1 to 1 chat



Figure 4-21: Sent short message

Query of event details
Use the following command.

Remote command:
`SENSe:DATA:CONTrol:IMS2:HISTory?`

4.6.5 General IMS Settings

Press the "Config" hotkey to open the "IMS Server Configuration" dialog box.

It configures general IMS server settings. Additional settings can be accessed via the other hotkeys.

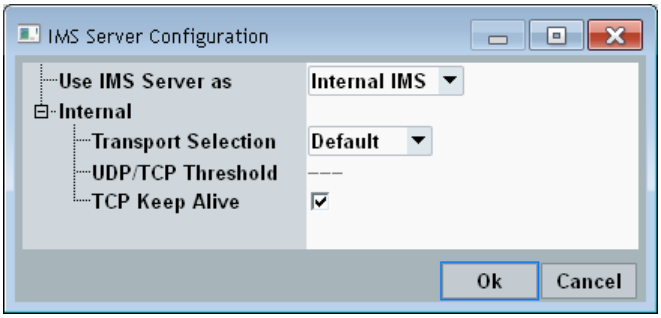


Figure 4-22: Internal IMS server settings

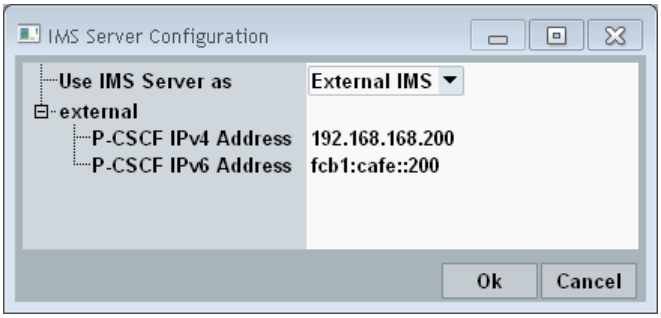


Figure 4-23: External IMS server settings

Use IMS Server as 74
Transport Selection, UDP/TCP Threshold 74
TCP Keep Alive 75
P-CSCF IPv4/IPv6 Address 75

Use IMS Server as
Selects whether the internal IMS server of the DAU or an external IMS network is used for IMS services.

For usage of an external IMS server, only the settings in this dialog box are relevant. For the internal IMS server, also all other settings of the "IMS" tab are relevant.

Usage of the internal IMS server requires option R&S CMW-KAA20.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUSage`

Transport Selection, UDP/TCP Threshold
Configures whether TCP or UDP is used by the internal IMS server.

"Default"	The threshold is fixed and equals 1300 characters per SIP message. For shorter SIP messages, UDP can be used. For longer SIP messages, TCP is used. For details, see RFC 3261, section 18.1.1.
"TCP Only"	TCP is always used, independent of the message length.
"UDP Only"	UDP is always used, independent of the message length.
"Custom"	The threshold is configurable via the setting "UDP/TCP Threshold" (number of characters per SIP message). For shorter SIP messages, UDP is used. For longer SIP messages, TCP is used.

Remote command:
CONFigure:DATA:CONTrol:IMS2:TRANsport:SELection
CONFigure:DATA:CONTrol:IMS2:THReshold:VALue

TCP Keep Alive
Selects whether the IMS server sends TCP keep-alive messages or not.

Remote command:
CONFigure:DATA:CONTrol:IMS2:TCPalive

P-CSCF IPv4/IPv6 Address
Configures the IPv4 and IPv6 address of an external P-CSCF.

Remote command:
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVFour
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVSix

4.6.6 P-CSCF Settings

Press the "P-CSCF" hotkey to open the "P-CSCF Configuration" dialog box.

It configures the P-CSCFs simulated by the internal IMS server.

You can create up to 10 P-CSCF profiles with different IP addresses and properties. Each profile is displayed on a separate tab.

Profile 1 and 2 are always available. To create profile 3 to 10, option R&S CMW-KAA21 is required.

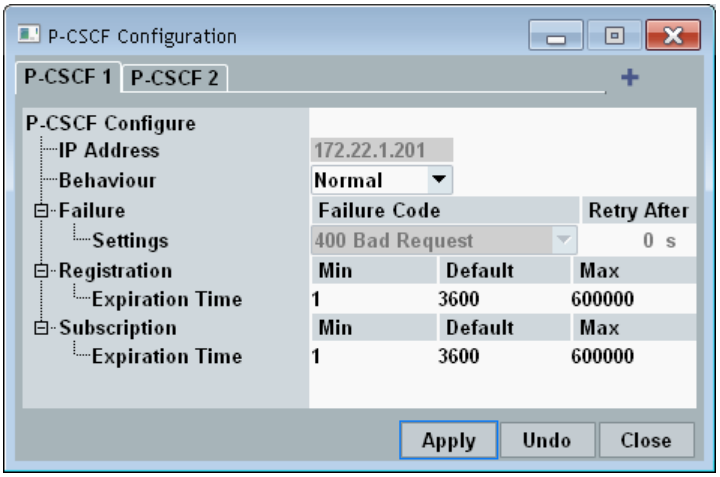


Figure 4-24: P-CSCF settings

A list of all P-CSCFs is sent to the mobile. The mobile selects a P-CSCF from the list for registration.

A possible use case is to configure P-CSCF 1 and 2 so that they reject the registration and P-CSCF 3 and 4 so that they accept the registration. Then check whether the mobile retries registration with another P-CSCF.

The settings are explained in the following.

Add or delete a tab....76
IP Address....76
Behavior....76
Failure....76
Registration....76
Subscription....77
Apply, Undo, Close (buttons)....77

Add or delete a tab

+ creates a tab / P-CSCF. The maximum number of tabs is 10.

× deletes the currently selected tab. Tab 1 and 2 cannot be deleted.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:PCSCf:ADD
CONFigure:DATA:CONTrol:IMS2:PCSCf:CREate
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:DELete
SENSe:DATA:CONTrol:IMS2:PCSCf:CATalog?
```

IP Address

Defines the IP address of the P-CSCF. You can specify an IPv4 or an IPv6 address. Assign a different address to each profile.

The addresses of profile 1 and 2 are fixed. Profile 1 uses the IPv4 address of the DAU, profile 2 the IPv6 address.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:IPADdress
```

Behavior

Defines the behavior of the P-CSCF when it receives a SIP message from the mobile.

"Normal"	The P-CSCF follows the normal call flow. For example, it performs a registration when it receives a REGISTER message.
"Failure"	The P-CSCF returns the configured failure code.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:BEHaviour
```

Failure

Selects a failure code to be returned to the mobile for "Behavior" = "Failure".

Some failure messages support the header field "Retry After". It is filled with the configured time.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:FAILurecode
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:RETRyafter
```

Registration

Defines a minimum, maximum and default value for the registration expiration time (duration of the registration).

The values are used as follows:

- If the mobile does not include an expiration interval in its REGISTER message, the configured "Default" value is used as expiration time for the registration.
- If the mobile suggests an expiration interval in its REGISTER message:
  – If the suggested value is between the configured "Min" and "Max" value, the suggested value is used as expiration time.
  – If the suggested value is outside of the configured "Min" / "Max" range, the registration is rejected.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:REGexp`

Subscription
Defines a minimum, maximum and default value for the subscription expiration time.

The logic is the same as for "Registration", with SUBSCRIBE messages instead of REGISTER messages.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:SUBexp`

Apply, Undo, Close (buttons)
"Apply" applies all changes.

"Undo" discards all not yet applied changes.

"Close" closes the dialog box. Not yet applied changes are discarded.

4.6.7 Subscriber Settings

Press the "Subscriber" hotkey to open the "Subscribers Configuration" dialog box.

It configures profiles for tested mobiles.

You can create up to five subscriber profiles with different private user IDs and properties. Each profile is displayed on a separate tab. Profile 1 is always available.

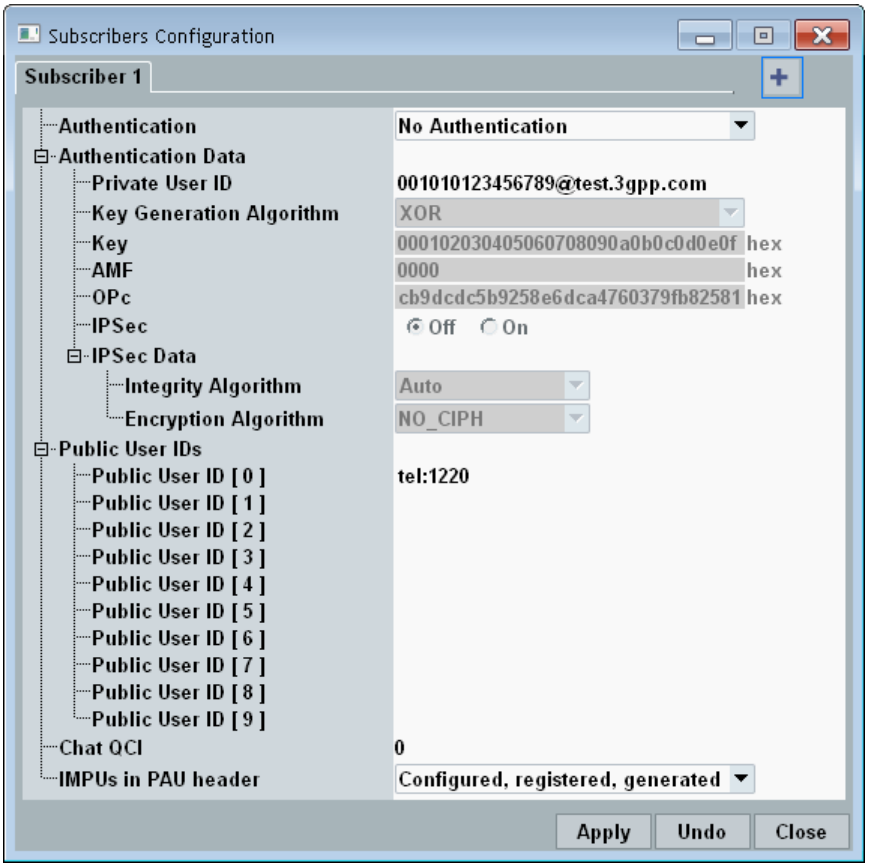


Figure 4-25: Subscriber settings

A profile is selected automatically when a SIP message from the mobile is received. The user ID included in the SIP message is compared to the user IDs in the profiles. Depending on the message type, the mobile sends a public user ID or a private user ID. A profile contains one private user ID and up to 10 public user IDs.

If you have several mobiles with the same test SIM card, you can use the same subscriber profile for all these mobiles.

The settings are explained in the following.

Add or delete a tab....79
Authentication....79
Private User ID....79
Key Generation Algorithm....79
Key....79
AMF....80
OPc....80
IPSec....80
Integrity Algorithm....80
Encryption Algorithm....80
Public User IDs....81

Chat QCI 81
IMPUs in PAU Header 81
Apply, Undo, Close (buttons) 81

Add or delete a tab

+ creates a tab / subscriber. The maximum number of tabs is five.

× deletes the currently selected tab. Tab 1 cannot be deleted.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:SUBScriber:ADD
CONFigure:DATA:CONTrol:IMS2:SUBScriber:CREate
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:DELete
SENSe:DATA:CONTrol:IMS2:SUBScriber:CATalog?
```

Authentication

Specifies whether authentication is performed during the registration procedure and selects the authentication and key agreement (AKA) version to be used (see RFC 3310 and RFC 4169).

If authentication is enabled, the settings in section "Authentication Data" must be compatible to the mobile. And the authentication and key agreement version must be supported by the mobile. Otherwise, registration fails.

Authentication often requires a test ISIM. An appropriate ISIM can be obtained from Rohde & Schwarz (R&S CMW-Z04, stock no. 1207.9901.02). The default settings are compatible to this ISIM.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:SCHeme
```

Private User ID

If you use authentication, enter the private user ID as stored on the ISIM or in the mobile. If the private user ID of the mobile is not assigned to any subscriber profile, authentication fails.

If you are not sure which ID is used by your mobile, you can analyze SIP REGISTER messages sent by your mobile. The private user ID is contained in the header as *username="..."*.

If you do not use authentication, you can enter any private user ID. But each subscriber profile must have a different private user ID.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PRIVateid
```

Key Generation Algorithm

Specifies which algorithm set is used by the ISIM: MILENAGE algorithms (see 3GPP TS 35.205) or XOR algorithms.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:ALGorithm
```

Key

The authentication key K is used for the authentication procedure (including a possible integrity check). The value is entered as 32-digit hexadecimal number.

The key K must be equal to the value stored on the test ISIM or the mobile. Otherwise authentication fails.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:KEY`

AMF
Specifies the authentication management field (AMF) to be used for the authentication procedure. The value is entered as four-digit hexadecimal number.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:AMF`

OPc
The key OP_c is used for authentication and integrity check procedures with the MILENAGE algorithm set. If you use this algorithm set, specify the OP_c used by the ISIM. The value is entered as 32-digit hexadecimal number.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:OPC`

IPSec
Specifies whether the IMS server supports IP security mechanisms.

If IPsec is supported by the IMS server but not by the mobile, the session is not rejected. It is established without IPsec.

To use Wireshark for connections with IPsec, see Chapter 3.12, "Configuring Wireshark for IPsec", on page 48.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ENABle`

Integrity Algorithm
Selects an algorithm for integrity protection via IPsec. The supported algorithms are HMAC-MD5-96 (see RFC 2403) and HMAC-SHA-1-96 (see RFC 2404).

"Auto" uses the algorithm indicated by the mobile in the REGISTER message.

Configure this setting compatible to your mobile.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:INTegrity`

Encryption Algorithm
Selects an algorithm for encryption via IPsec. The supported algorithms are DES-EDE3-CBC (see RFC 2451) and AES-CBC (see RFC 3602). NO_CIPH means no encryption/ciphering.

"Auto" uses the algorithm indicated by the mobile in the REGISTER message.

Configure this setting compatible to your mobile.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:ENCRyption`

Public User IDs
If the mobile sends a public user ID that is not assigned to any subscriber profile, the first profile is used. So for tests with this profile, you do not need to enter public user IDs.

To use another profile, enter all public user IDs stored on the ISIM or in the mobile. Do not assign the same user ID to several profiles. Each ID must start with "tel:", "sip:" or "sips:".

A four-digit number is automatically assigned to each profile.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PUBLicuserid<Index>`

Chat QCI
Specifies the QoS class identifier (QCI) used by the mobile for RCS message transfer.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:CHATqci`

IMPUs in PAU Header
Selects which public IDs are sent to the mobile in the PAU header of the 200/OK response to REGISTER messages.

IMPU stands for IP multimedia public identity. PAU stands for p-associated-URI.

Your selection determines which IDs are sent and in which order they are included in the header. You can select between combinations of the following IDs.

"Configured" The public user IDs configured in this dialog box.

"Registered" The ID sent by the mobile in the "from" header of the REGISTER message.

"Generated" The automatically generated ID starting with "tel:555", followed by a four-digit number.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IMPU:HEADer`

Apply, Undo, Close (buttons)
"Apply" applies all changes.

"Undo" discards all not yet applied changes.

"Close" closes the dialog box. Not yet applied changes are discarded.

4.6.8 Virtual Subscriber Settings

Press the "Virtual Subscriber" hotkey to open the "Virtual Subscribers Configuration" dialog box.

The dialog box configures profiles of virtual subscribers. A virtual subscriber can call the tested mobile and it can be called from the tested mobile.

You can create up to 20 virtual subscriber profiles with different properties. Each profile is displayed on a separate tab. Profile 1 is always available.

The dialog box provides several views, for the following purposes:

- Basic profile configuration, see Chapter 4.6.8.2, "Basic Virtual Subscriber Settings", on page 83
- Mobile-terminating calls, see Chapter 4.6.8.3, "Mobile-Terminating Calls", on page 93
- Mobile-terminating messages, see Chapter 4.6.8.4, "Mobile-Terminating Messages", on page 95
- File transfer, see Chapter 4.6.8.5, "File Transfer", on page 98

4.6.8.1 Common View Elements

This section describes the elements available in all virtual subscriber views, for example buttons for creating/deleting tabs and for switching between the views.

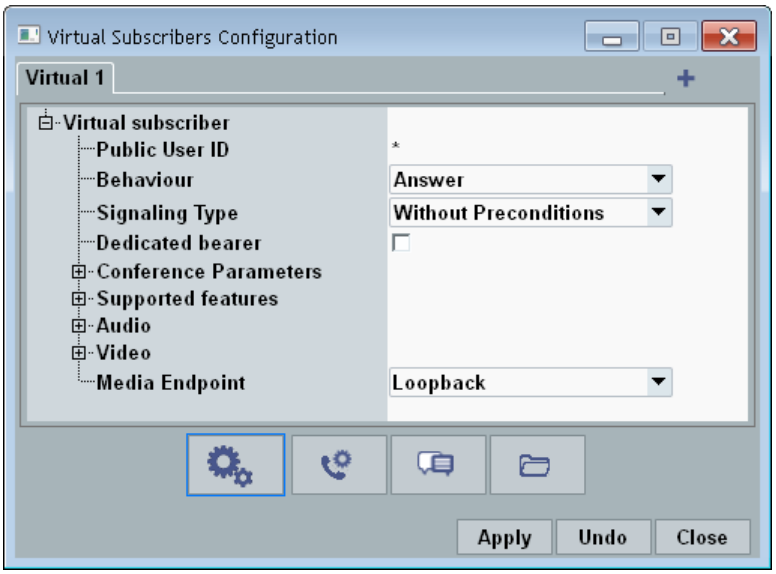


Figure 4-26: Virtual subscriber configuration dialog

Add or delete a tab....82
View access buttons....82
Apply, Undo, Close (buttons)....83

Add or delete a tab

+ creates a tab / virtual subscriber. The maximum number of tabs is 20.

× deletes the currently selected tab. Tab 1 cannot be deleted.

Remote command:

```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub:ADD
CONFigure:DATA:CONTrol:IMS2:VIRTualsub:CREate
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:DELete
SENSe:DATA:CONTrol:IMS2:VIRTualsub:CATalog?
```

View access buttons

To switch between the views of the dialog box, use the following buttons.



From left to right, they access the following views:

- Basic profile configuration, see Chapter 4.6.8.2, "Basic Virtual Subscriber Settings", on page 83
- Mobile-terminating calls, see Chapter 4.6.8.3, "Mobile-Terminating Calls", on page 93
- Mobile-terminating messages, see Chapter 4.6.8.4, "Mobile-Terminating Messages", on page 95
- File transfer, see Chapter 4.6.8.5, "File Transfer", on page 98

Apply, Undo, Close (buttons)
"Apply" applies all changes.

"Undo" discards all not yet applied changes.

"Close" closes the dialog box. Not yet applied changes are discarded.

4.6.8.2 Basic Virtual Subscriber Settings

Press the "Virtual Subscriber" hotkey to open the "Virtual Subscribers Configuration" dialog box.

The dialog box provides several views. This section describes the view shown in the following figure. The highlighted button selects this view for display.

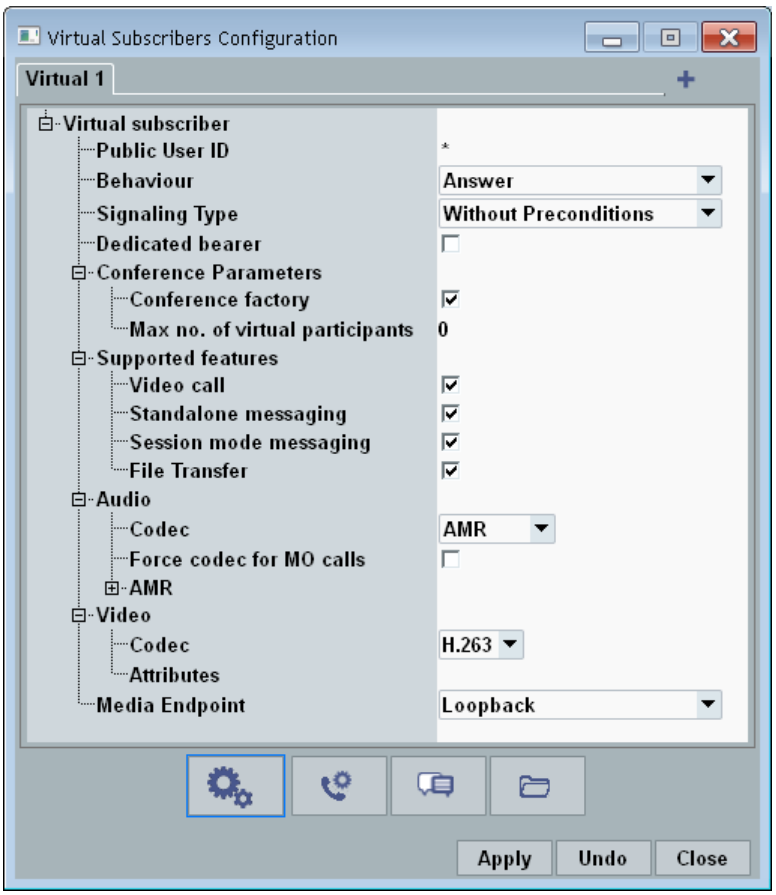


Figure 4-27: Basic virtual subscriber settings

Public User ID 85
Behavior 85
Signaling Type 85
Dedicated Bearer 86
Conference Parameters 86
└ Conference Factory 86
└ Max No. of Virtual Participants 86
Supported Features 86
Audio 87
└ Codec 87
└ Force Codec for MO Calls 87
└ AMR 87
└ AMR Alignment Mode 87
└ Codec Rates 88
└ EVS 88
└ Start Mode 88
└ HF Only 88
└ DTX 89
└ DTX Recv 89
└ AMR-WB-IO Codec Rates 89
└ Select 89

└ Bandwidth....89
└ Bit Rate....90
└ CMR....90
└ Channel-Aware Mode....90
Video....90
Media Endpoint....91
Forward....91
└ IP....91
└ RTP Port....91
└ Command Port....92
└ Endpoint AMR Alignment Mode....92
Audioboard....92
└ Enable DTX in DL....92
└ Force Mode....92
└ Audio Instance....93
└ EVS IO Mode Codec....93
PCAP....93
└ PCAP Meta File Selection....93

Public User ID
For incoming mobile-originating calls, the called virtual subscriber is selected via its public user ID.

By default, the first virtual subscriber profile has a wildcard "*" as ID. Any incoming calls, for which no profile with a matching public user ID is found, are routed to this virtual subscriber. If you want to use this mechanism, do not modify the ID.

For additional virtual subscribers, define a different public user ID for each profile.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ID`

Behavior
Defines how the virtual subscriber reacts to a call from the tested mobile.

"Answer"	Answer the call.
"NoAnswer"	Ignore the call and keep "ringing".
"Declined"	Reject the call.
"Busy Here"	The virtual subscriber is busy.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEHaviour`

Signaling Type
Specifies whether a voice call session is established with or without quality-of-service preconditions. For a call with preconditions, a certain quality of service is ensured by reserving network resources before alerting the called user. SIP preconditions are defined in RFC 3312 and RFC 4032. See also 3GPP TS 24.229.

"Preconditions"	SIP signaling as for a call with preconditions
"Without Preconditions"	SIP signaling as for a call without preconditions

"Simple"	Simplified call flow, without preconditions and without 100rel
"Require 100rel"	SIP signaling as for a call without preconditions The INVITE message contains a "Require" header field with the option tag "100rel".
"Require Pre-condition"	SIP signaling as for a call with preconditions The INVITE message contains a "Require" header field with the option tag "precondition".
"Without Pre-conditions, With 183"	SIP signaling as for a call without preconditions, with "183 Session in Progress" message

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SIGNalingtyp`

Dedicated Bearer
Enables the automatic setup of a dedicated bearer, if a VoIMS connection is established. If the checkbox is disabled, the IMS server does not set up dedicated bearers for calls to/from this virtual subscriber.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEARer`

Conference Parameters
This node configures settings for conference calls over IMS. See also Chapter 3.5.6, "Setting Up a Conference Call", on page 33.

Option R&S CMW-KAA21 is required.

Conference Factory ← Conference Parameters
If you enable the checkbox, this virtual subscriber profile configures a conference server.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:CONFerence:FACTory`

Max No. of Virtual Participants ← Conference Parameters
Limits the number of virtual subscribers allowed to participate in a conference call. The value 0 means that there is no limit.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MAX:PARTicipant`

Supported Features
Configures the service capability information of the virtual subscriber for presence-based services. The presence information is sent to the mobile.

"Standalone Messaging" includes, for example, the RCS pager mode and the RCS large mode.

"Session Mode Messaging" includes, for example, RCS chats.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:VIDeo
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:STANdalone
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:SESSionmode
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:FILetransfer
```

Audio
This node configures the codec to be used for audio calls.

Codec ← Audio
Selects the voice codec type to be used.

To use the EVS codec with the audio board, option R&S CMW-KS104 is required.

"AMR"	Narrowband adaptive multi-rate (AMR) codec
"AMR-WB"	Wideband adaptive multi-rate (AMR) codec
"EVS"	Enhanced voice services (EVS) codec

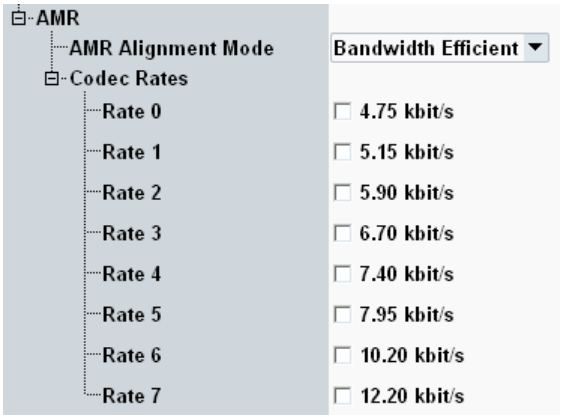
Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ADCodec:TYPE
```

Force Codec for MO Calls ← Audio
If the checkbox is enabled, the configured voice codec type is used for mobile-originating calls.

If the checkbox is disabled, the first codec offered via SDP is used.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORCemocall
```

AMR ← Audio
This node configures the wideband and narrowband AMR codec.



AMR Alignment Mode ← AMR ← Audio
Selects the payload format.

"Octet Aligned" All payload fields are aligned to octet boundaries.

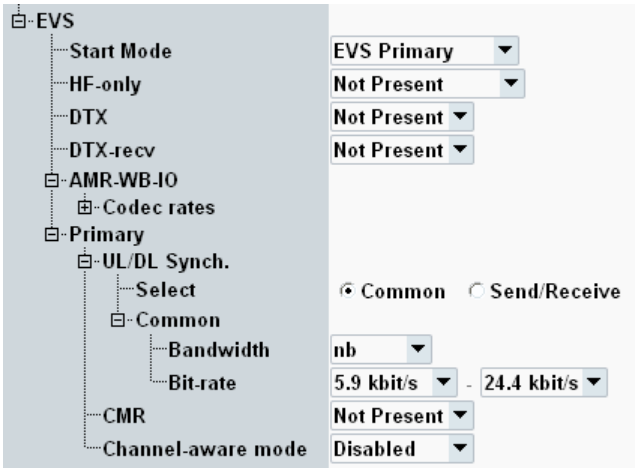
"Bandwidth Efficient" Only the full payload is octet aligned. Fewer padding bits are needed.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:ALIGnment`

Codec Rates ← AMR ← Audio
Selects the supported AMR codec rates. For the wideband AMR, there are nine codec rates (index 0 to 8). For the narrowband AMR, there are eight codec rates (index 0 to 7).

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:CODec<CodecIdx>:ENABle`

EVS ← Audio
This node configures the EVS codec.



Start Mode ← EVS ← Audio
Selects the start mode for the EVS codec. The mode can change during the session.

There is a primary mode and an interoperable (IO) mode, compatible with AMR-WB.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:STARtmode`

HF Only ← EVS ← Audio
Specifies the SDP parameter "hf-only". It indicates the allowed header format.

The SDP parameters are defined in 3GPP TS 26.445.

"Both" The "Compact" format and the "Header-Full" format are allowed (value 0).

"Header Full Only" Only the "Header-Full" format is allowed (value 1).

"Not Present" The SDP parameter is omitted (no value, same effect as value 0).

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:HFONly`

DTX ← EVS ← Audio
Specifies the SDP parameter "dtx". It enables or disables DTX for the send direction and the receive direction.

"Disable" DTX is disabled (value 0).

"Enable" DTX is enabled (value 1).

"Not Present" The SDP parameter is omitted (no value, same effect as value 1).

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTX`

DTX Recv ← EVS ← Audio
Specifies the SDP parameter "dtx-recv". It enables or disables DTX for the send direction of the receiver of the parameter.

"Disable" DTX is disabled (value 0).

"Enable" DTX is enabled (value 1).

"Not Present" The SDP parameter is omitted (no value, same effect as value 1).

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTXRecv`

AMR-WB-IO Codec Rates ← EVS ← Audio
Selects the codec rates supported for the AMR-WB IO mode.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CODec<CodecIdx>:ENABle`

Select ← EVS ← Audio
Selects a configuration mode for the bandwidth and bit-rate settings of the EVS primary mode.

"Common" The same settings apply to the downlink (sending) and the uplink (receiving). A node "Common" is displayed for configuration of the settings.

"Send/Receive" There are separate settings for the downlink and the uplink. The nodes "Send (DL)" and "Receive (UL)" are displayed for configuration of the settings.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect`

Bandwidth ← EVS ← Audio
Selects the codec bandwidths supported for the EVS primary mode:
- Narrowband only (NB)
- Wideband only (WB)
- Super wideband only (SWB)
- Fullband only (FB)
- Narrowband and wideband (NB-WB)

- Narrowband, wideband and super wideband (NB-SWB)
- Narrowband, wideband, super wideband and fullband (NB-FB)

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:BWCommon
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BW
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BW
```

Bit Rate ← EVS ← Audio
Selects the supported bit-rate range for the EVS primary mode. The offered values depend on the selected bandwidth.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:COMMon:BITRate:
RANGe
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BITRate:RANGe
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BITRate:
RANGe
```

CMR ← EVS ← Audio
Specifies the SDP parameter "cmr". It enables or disables codec mode requests.

"Disable"	Codec mode requests are disabled (value -1).
"Enable"	Codec mode requests are enabled (value 0).
"Present All"	A codec mode request must be present in each packet (value 1).
"Not Present"	The SDP parameter is omitted (no value, same effect as value 0).

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CMR
```

Channel-Aware Mode ← EVS ← Audio
Specifies the SDP parameter "ch-aw-recv".

The channel-aware mode uses partial redundant frames to improve the error resilience. The mode is only defined for a wideband or super-wideband codec with 13.2 kbit/s. For details, see 3GPP TS 26.448 and 3GPP TS 26.114.

"Disabled"	The channel-aware mode is disabled for the entire session (value -1).
"Not Used"	The channel-aware mode is not used at the start of the session (value 0).
"Not Present"	The SDP parameter is omitted (no value, same effect as value 0).
"2, 3, 5, 7"	The channel-aware mode is enabled (value 2, 3, 5, 7). The number indicates the offset of the partial copy.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CHAWmode
```

Video
Selects the video codec to be used for video calls to the tested mobile.

You can also specify codec attributes.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:CODec
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:ATTRibutes

Media Endpoint
Configures the routing of an audio signal or video signal received from the mobile. The setting applies to incoming and outgoing calls.

"Loopback" The signal from the mobile is routed back immediately. A word spoken into the microphone of the mobile is returned to the loudspeaker of the mobile.

"Forward" The RTP and RTCP packets from the mobile are forwarded to an external media endpoint. The packets returned by the media endpoint are forwarded to the mobile. This method can be used to implement an alternative loopback mechanism.

"Audioboard" The signal from the mobile is routed to the speech decoder of the audio board. The signal returned by the speech encoder of the audio board is forwarded to the mobile.
Select this value for voice over IMS audio measurements.
The audio board is provided by R&S CMW-B400B, the speech codec by R&S CMW-B405A.
To use the EVS codec with the audio board, option R&S CMW-KS104 is required.

"PCAP" The packets from the mobile are discarded. In the downlink direction, a PCAP file is played.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MEDiaendpoin

Forward
Specifies settings for Media Endpoint = "Forward".



IP ← Forward
IPv4 or IPv6 address of the media endpoint

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard

RTP Port ← Forward
The configured port is used for RTP packet forwarding.

The configured port plus one is used for RTCP packet forwarding.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard

Command Port ← Forward
Port for the command interface between the DAU and the external media endpoint.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard`

Endpoint AMR Alignment Mode ← Forward
Enter the AMR alignment mode used by the media endpoint.

You can use different modes towards the mobile and towards the media endpoint. See also "AMR Alignment Mode" on page 87.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard`

Audioboard
Specifies settings for Media Endpoint = "Audioboard".

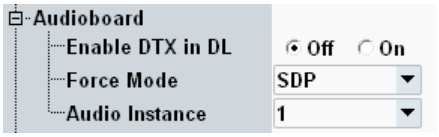


Figure 4-28: Codec AMR or AMR-WB

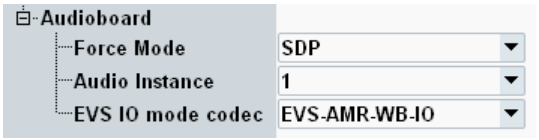


Figure 4-29: Codec EVS

Enable DTX in DL ← Audioboard
Enables voice activity detection and the generation of comfort noise parameters during silence periods. The parameters are sent to the mobile in silence indicator frames.

This setting is only relevant for AMR and AMR-WB codecs.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AUDioboard:CONFig`

Force Mode ← Audioboard
Selects the codec rate or bit rate used by the speech codec of the audio board. The offered values depend on the setting "Codec" on page 87.

Typically, audio tests are performed with a specific rate. Specify this rate here and ensure that it is enabled in the codec settings.

"SDP"	All rates are allowed. The rate requested by the mobile is used. If it sends several rates, the highest one is used.
"xxx kbit/s"	The specified AMR codec rate is used.
"Primary xxx kbps"	The specified EVS primary bit-rate is used.
"AMR-WB IO xxx kbps"	The specified EVS AMR-WB IO codec rate is used.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AUDioboard:CONFig

Audio Instance ← Audioboard
Selects the instance of the audio measurements firmware application to be used.

If your audio board is equipped with only one speech codec, use always instance 1.

Two speech codecs allow you to select the instance and implicitly the used codec hardware.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AUDioboard:CONFig

EVS IO Mode Codec ← Audioboard
Selects the speech packet type for the EVS AMR-WB IO mode. You can send AMR-WB encoded packets or EVS AMR-WB IO encoded packets to the mobile.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:IO:MODE:CONFig

PCAP
Specifies settings for Media Endpoint = "PCAP".



PCAP Meta File Selection ← PCAP
Selects an XML file with metadata. The XML file specifies which PCAP file is played, depending on session characteristics, for example the payload type.

Create metadata XML files according to your needs and store them on the samba share of the DAU, in the subdirectory `ims/pcap`.

Remote command:
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:PCAPfile:SELection

4.6.8.3 Mobile-Terminating Calls

Mobile-terminating voice over IMS calls are configured and initiated via the "Virtual Subscribers Configuration" dialog box. Press the "Virtual Subscriber" hotkey to open the dialog box.

The dialog box provides several views. This section describes the view shown in the following figure. The highlighted button selects this view for display.

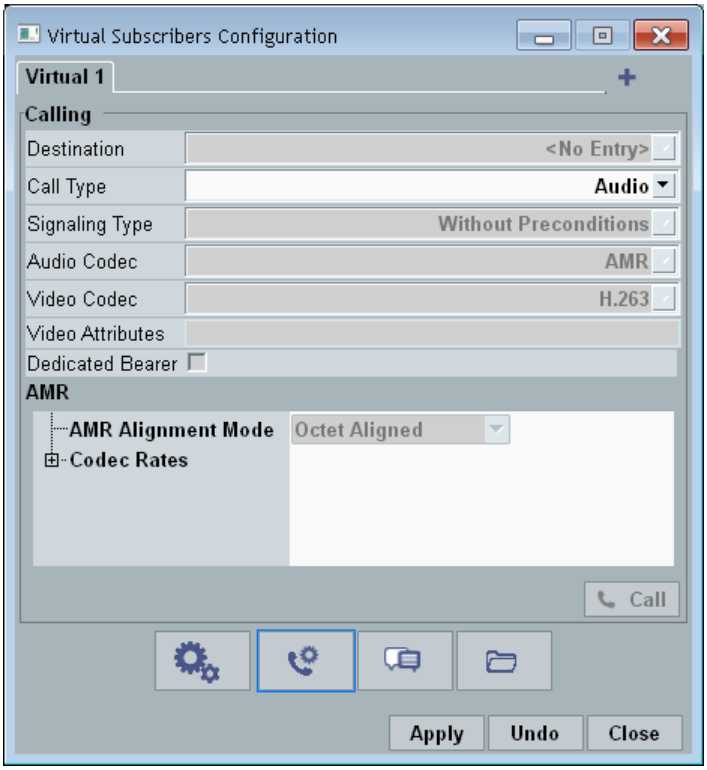


Figure 4-30: Mobile-terminating call settings

Most settings are grayed-out. They are taken from the virtual subscriber settings and displayed for information.

To initiate a call, proceed as follows:

1. Select a virtual subscriber tab at the top.

   The related call settings are displayed.

2. Select the call destination and the call type.

3. Press the "Call" button.

For more detailed instructions, see Chapter 3.5.4, "Setting Up a Mobile-Terminating Voice over IMS Call", on page 30.

The view contents are described in the following.

Destination 94
Call Type 95
Call (button) 95

Destination
Selects the destination to be called. The list is filled during registration.

Each destination indicates a contact ID in square brackets, the used signaling application, the signaling application instance and the subscriber profile.

Example: "[1] LTE1 - subscriber 1"

The contact ID is generated automatically when the UE registers. Its purpose is to distinguish between destinations with the same signaling application and subscriber profile.

Remote command:
```
SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination:LIST?
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination
```

Call Type
Selects the type of call to be set up: audio call or video call.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:TYPE
```

Call (button)
The "Call" button initiates a voice over IMS call to the mobile.

The dialog box is closed automatically. You can check the call state in the "Events" area of the main view.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:CALL
```

4.6.8.4 Mobile-Terminating Messages

Incoming mobile-originating messages sent from the mobile to a virtual subscriber are displayed in the main view, see Chapter 4.6.4, "Event Details", on page 72.

Mobile-terminating messages to be sent to the mobile are configured and initiated via the "Virtual Subscribers Configuration" dialog box. Press the "Virtual Subscriber" hotkey to open the dialog box.

The dialog box provides several views. This section describes the view shown in the following figure. The highlighted button selects this view for display.

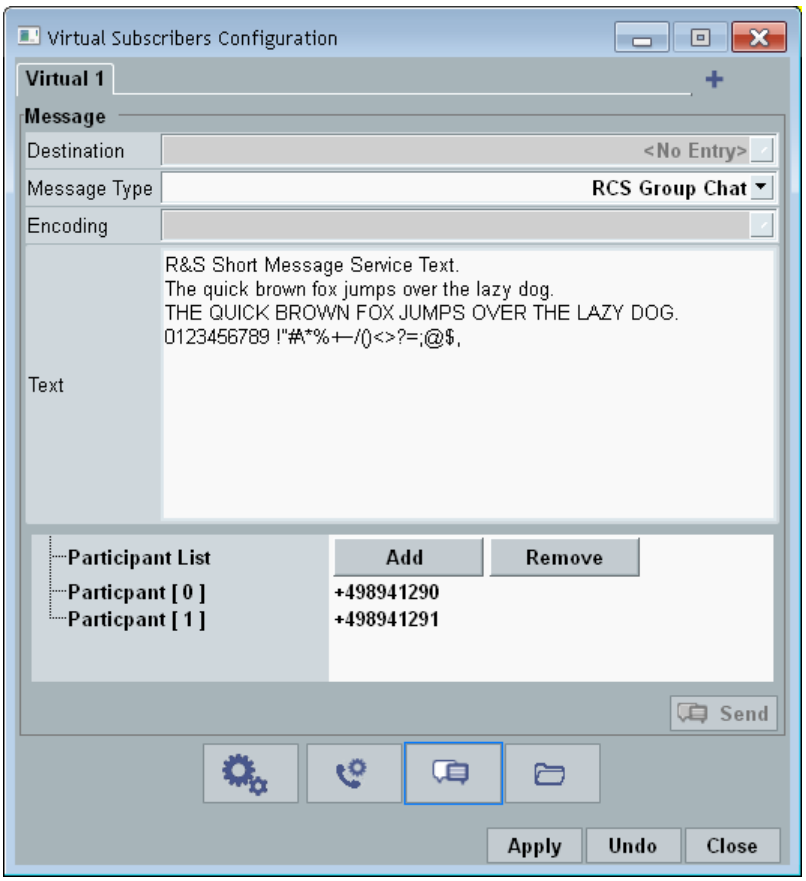


Figure 4-31: Mobile-terminating message settings

Destination 96
Message Type 96
Encoding 97
Text 97
Participant List 97
Send (button) 97
Import 97

Destination
Selects the destination to which the message is sent.

Remote command:
`SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination:LIST?`
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination`

Message Type
Selects the type of the message to be sent to the mobile.

The types belong either to the short message service (SMS, defined by 3GPP or 3GPP2) or to the rich communication suite (RCS, defined by GSMA).

For RCS, option R&S CMW-KAA21 is required.

"3GPP" SMS as defined by 3GPP

"3GGP2"	SMS as defined by 3GPP2, no delivery acknowledgment
"3GGP2 with Delivery ACK"	SMS as defined by 3GPP2, with delivery acknowledgment
"RCS Pager Mode"	RCS standalone message in pager mode
"RCS Large Mode"	RCS standalone message in large message mode
"RCS 1 to 1 Chat"	RCS 1 to 1 chat, initial message of the session
"RCS Group Chat"	RCS group chat, initial message of the session

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TYPE`

Encoding
Selects the encoding for RCS messages (pager mode and large mode).

RCS messages can be sent with Base64 encoding or as plain text without encoding.

Option R&S CMW-KAA21 is required.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:ENCoding`

Text
Specifies the text of the message to be sent to the mobile.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TEXT`

Participant List
Configures the participants of a group chat (only displayed for this message type).

The participant list is sent to the mobile during the session setup. During the session, you can send more messages to the mobile, using any of the participants as sender of the message (softkey "Chat Actions"). If the mobile sends a message, the delivery and display is automatically acknowledged for all participants.

Use the buttons to add or remove lines. Edit the entries directly in the dialog box.

Send (button)
Sends a message with the specified settings to the mobile.

The dialog box is closed automatically. You can check the message transfer in the "Events" area of the main view and at the mobile.

To send further messages in RCS chats, see "Chat Actions (softkey)" on page 68.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:SEND`

Import
This button is only available for "RCS Large Mode". It imports the message text from a file.

Create text files according to your needs and store them on the samba share of the DAU, in the directory `ims/rcs/samples`.

Remote command:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:IMPort:FILE`

4.6.8.5 File Transfer

File transfers from virtual subscribers to mobiles are configured and initiated via the "Virtual Subscribers Configuration" dialog box. Press the "Virtual Subscriber" hotkey to open the dialog box. Option R&S CMW-KAA21 is required.

The dialog box provides several views. This section describes the view shown in the following figure. The highlighted button selects this view for display.

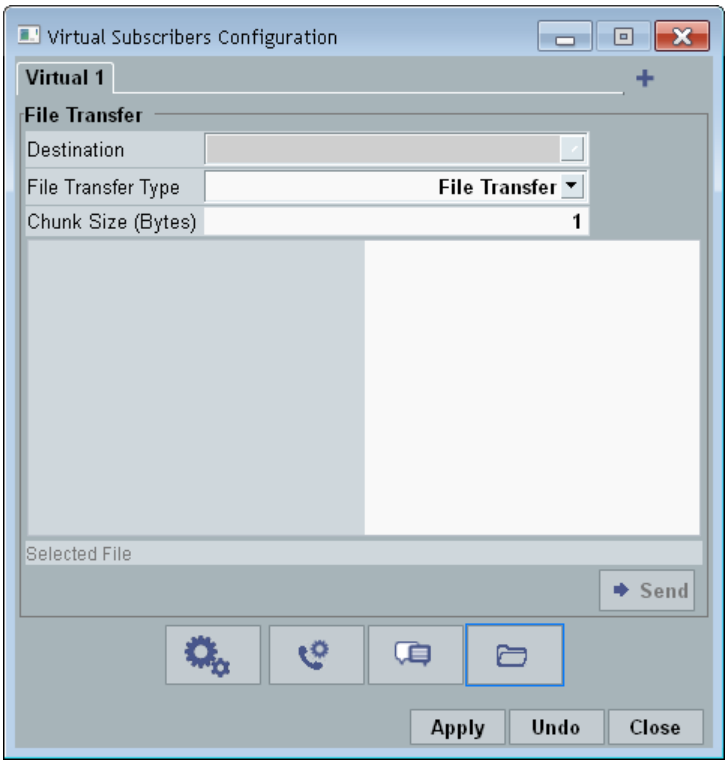


Figure 4-32: File transfer settings

A file transfer initiated via this dialog box uses the message session relay protocol (MSRP). For file transfer via FTP, see Chapter 4.4, "FTP Tab", on page 63.

Destination 98
File Transfer Type 99
Chunk Size 99
File selection 99
Send (button) 99

Destination
Selects the destination to which the file is transferred.

The list of destinations is the same as for mobile terminating calls, see "Destination" on page 94.

Remote command:
```
SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination:
LIST?
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination
```

File Transfer Type
Both transfer types use the message session relay protocol (MSRP), but with different header structures.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:TYPE
```

Chunk Size
The file is divided into chunks with the specified size. Each chunk is transferred in a separate MSRP packet.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:CHUNksize
```

File selection
Store your files on the samba share of the DAU, in the directory `ims/rcs/samples`.

The available files are listed in the lower part of the dialog box.

The currently selected file is indicated above the "Send" button.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:FILE:
SELection
```

Send (button)
Starts the file transfer.

The dialog box is closed automatically. You can check the message transfer in the "Events" area of the main view and at the mobile.

Remote command:
```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:SEND
```

4.7 ePDG Tab

This tab configures and controls the evolved packet data gateway (ePDG) of the DAU. Option R&S CMW-KA065 is required to see the tab and to use the ePDG service.

In a 3GPP core network, the ePDG is the gateway to an untrusted non-3GPP access technology, for example to a WLAN gateway. The main role of the ePDG is to provide security mechanisms. For a secure connection via the untrusted WLAN, an IPsec tunnel is established between the mobile and the ePDG.

The ePDG is, for example, used for WLAN traffic offload tests. See also Chapter 3.6, "Using the ePDG Service for WLAN Offload", on page 40.

The ePDG tab shows an event log and information about established IPsec tunnels between the ePDG and mobiles.

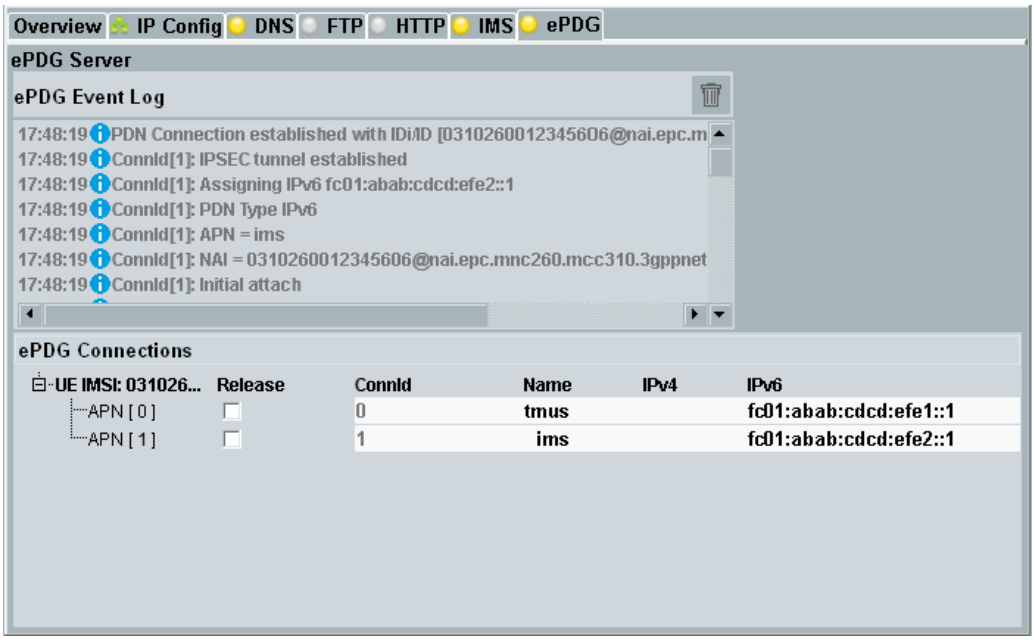


Figure 4-33: Data application control - ePDG tab

- ePDG Service Control and Softkeys 100
- ePDG Event Log 101
- ePDG Connection Information 101
- Accessing ePDG Settings 102
- ePDG IP and ID Settings 102
- IKE Settings 103
- ESP Settings 105
- P-CSCF IKEv2 Attributes 106
- Authentication Settings 106
- Dead Peer Detection Settings 108
- Certificate Settings 108

4.7.1 ePDG Service Control and Softkeys

The ePDG service must be running before you can perform WLAN offloading with signaling applications.

Service control
Start or stop the service using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "ePDG Service" softkey.

The softkey shows the current service state.

Remote command:
`SOURce:DATA:CONTrol:EPDG:STATe`

Release Connection (softkey)
This softkey releases one or more selected ePDG connections.

To select the connections to be released, use column "Release" in the connection list, see Chapter 4.7.3, "ePDG Connection Information", on page 101.

4.7.2 ePDG Event Log

The event log area reports events and errors like ePDG service state changes, IPsec tunnel establishment/release, offload events and authentication failure.

The button to the right clears the displayed entries.

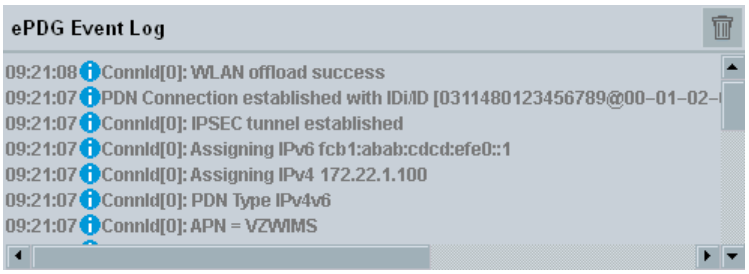


Figure 4-34: Event log on the ePDG tab

Event log entries
Each entry consists of a timestamp, an icon indicating the category of the event and a short text describing the event.

Meaning of the category icons: = information, warning and error

Remote command:
`SENSe:DATA:CONTrol:EPDG:EVENt:LOG?`
`CONFigure:DATA:CONTrol:EPDG:CLEan:GENeral:INFO`

4.7.3 ePDG Connection Information

The ePDG connections area lists all connections that are currently established via the ePDG.

For each listed connection, an IPsec tunnel has been established between the mobile and the ePDG. When a connection is released, it is removed from the list.

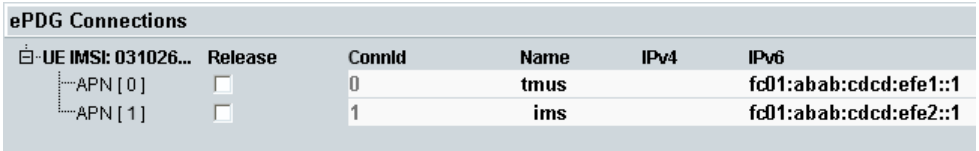


Figure 4-35: Connection information on the ePDG tab

For one IMSI, there can be up to 15 connections. If there are established connections for several mobiles (IMSIs), a separate node is displayed for each IMSI.

Connection information
There is one row per connection. The following columns are displayed:

- "UE IMSI: xxx": IMSI of the mobile
- "Release": selects the connections to be released if you press the softkey "Release Connection"
- "Connid": connection ID, used in the event log when events related to this connection occur
- "Name": access point name (APN)
- "IPv4": mobile IPv4 address
- "IPv6": mobile IPv6 address

Remote command:
`SENSe:DATA:CONTrol:EPDG:CONNections:IMSI?`
`SENSe:DATA:CONTrol:EPDG:CONNections:IMSI<Suffix>:APN?`

4.7.4 Accessing ePDG Settings

Press the "Config" hotkey to open the "ePDG Configuration" dialog box.

The dialog box provides all settings described in the following sections.

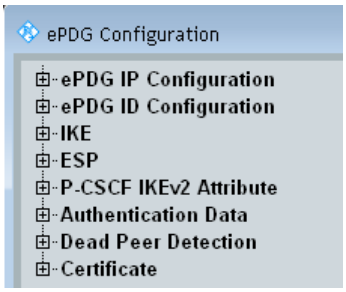


Figure 4-36: Configuration dialog box

4.7.5 ePDG IP and ID Settings

The top of the configuration tree configures the address and ID of the ePDG.

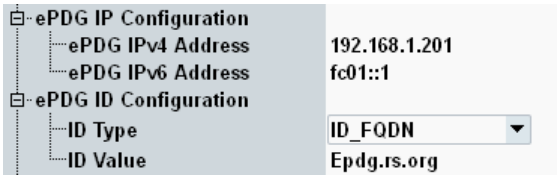


Figure 4-37: IP and ID settings

ePDG IP Configuration
Configures the IPv4 and IPv6 address of the ePDG.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVFour`
`CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVSix`

ePDG ID Configuration
Configures the type and the value of the ePDG identification.

The ePDG identifies itself according to these settings in messages to the mobile.

The ID types are specified in RFC 7296, section 3.5. The following types are supported:

- "ID_IPv4_ADDR": four-octet IPv4 address
- "ID_FQDN": fully qualified domain name string
- "ID_RFC822_ADDR": fully qualified RFC 822 email address string
- "ID_IPV6_ADDR": sixteen-octet IPv6 address
- "ID_KEY_ID": opaque octet stream

Remote command:
`CONFigure:DATA:CONTrol:EPDG:ID:TYPE`
`CONFigure:DATA:CONTrol:EPDG:ID:VALue`

4.7.6 IKE Settings

This section is related to the IPsec protocol internet key exchange version 2 (IKEv2). It is used for authentication before setup of an IPsec tunnel.



Figure 4-38: IKE settings

All enabled values are supported by the ePDG. The value to be used is negotiated with the mobile.

In each section, at least one value supported by your mobile must be enabled. Otherwise, authentication fails.

To use Wireshark for connections with IPsec, see Chapter 3.12, "Configuring Wireshark for IPsec", on page 48.

The listed values are specified in RFC 7296. That RFC references other RFCs indicated in the following in brackets.

Encryption 103
PRF 104
Integrity 104
Diffie Hellman 104

Encryption
Selects the encryption algorithms supported by the ePDG.



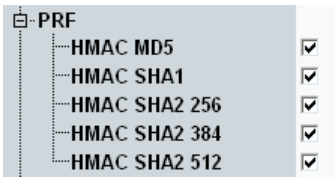
Available values:

- "ENCR_3DES" (RFC 2451)

- "ENCR_AES_CBC" (RFC 3602)

Remote command:
`CONFigure:DATA:CONTrol:EPDG:IKE:ENCRyption`

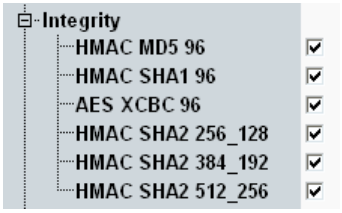
PRF
Selects the pseudorandom functions supported by the ePDG.



Available values:
- "PRF_HMAC_MD5" (RFC 2104)
- "PRF_HMAC_SHA1" (RFC 2104)
- "PRF_HMAC_SHA2_256" (RFC 4868)
- "PRF_HMAC_SHA2_384" (RFC 4868)
- "PRF_HMAC_SHA2_512" (RFC 4868)

Remote command:
`CONFigure:DATA:CONTrol:EPDG:IKE:PRF`

Integrity
Selects the integrity protection algorithms supported by the ePDG.



Available values:
- "AUTH_HMAC_MD5_96" (RFC 2403)
- "AUTH_HMAC_SHA1_96" (RFC 2404)
- "AUTH_AES_XCBC_96" (RFC 3566)
- "AUTH_HMAC_SHA2_256_128" (RFC 4868)
- "AUTH_HMAC_SHA2_384_192" (RFC 4868)
- "AUTH_HMAC_SHA2_512_256" (RFC 4868)

Remote command:
`CONFigure:DATA:CONTrol:EPDG:IKE:INTegrity`

Diffie Hellman
Selects the Diffie-Hellman groups supported by the ePDG.



Available values (specified in RFC 7296 and RFC 3526):

- "DH_GROUP_1": 768-bit MODP
- "DH_GROUP_2": 1024-bit MODP
- "DH_GROUP_5": 1536-bit MODP
- "DH_GROUP_14": 2048-bit MODP
- "DH_GROUP_15": 3072-bit MODP
- "DH_GROUP_16": 4096-bit MODP
- "DH_GROUP_17": 6144-bit MODP
- "DH_GROUP_18": 8192-bit MODP

Remote command:

`CONFigure:DATA:CONTrol:EPDG:IKE:DHGRoup`

4.7.7 ESP Settings

This section is related to the IPsec protocol encapsulating security payload (ESP). It is used for encryption of data for transfer via an IPsec tunnel.

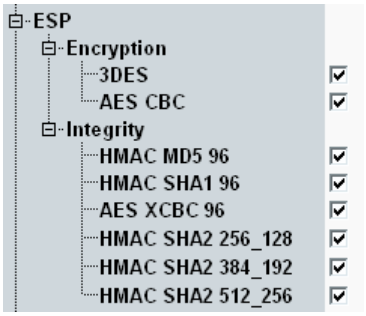


Figure 4-39: ESP settings

All enabled values are supported by the ePDG. The value to be used is negotiated with the mobile.

In each section, at least one value supported by your mobile must be enabled.

Encryption..... 105
Integrity..... 106

Encryption
Selects the encryption algorithms supported by the ePDG.

The available values are the same as for IKEv2, see "Encryption" on page 103.

Remote command:
CONFigure:DATA:CONTrol:EPDG:ESP:ENCRyption

Integrity
Selects the integrity protection algorithms supported by the ePDG.

The available values are the same as for IKEv2, see "Integrity" on page 104.

Remote command:
CONFigure:DATA:CONTrol:EPDG:ESP:INTegrity

4.7.8 P-CSCF IKEv2 Attributes

This section configures attributes related to the transport of P-CSCF addresses via IKEv2.

The default values are compliant with RFC 7651.

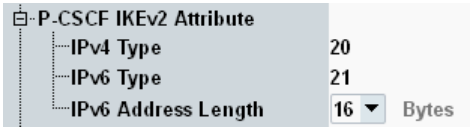


Figure 4-40: Attribute settings

IPv4 Type 106
IPv6 Type 106
IPv6 Address Length 106

IPv4 Type
Attribute type field of the P_CSCF_IP4_ADDRESS configuration attribute.

Remote command:
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVFour:TYPE

IPv6 Type
Attribute type field of the P_CSCF_IP6_ADDRESS configuration attribute.

Remote command:
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:TYPE

IPv6 Address Length
Length field of the P_CSCF_IP6_ADDRESS configuration attribute.

Remote command:
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:ADDRess:LENGth

4.7.9 Authentication Settings

The settings in this section are related to the authentication procedure. They must be compatible to the SIM of the mobile. Otherwise, authentication fails.



Figure 4-41: Authentication settings

IMSI....107
Authentication Algorithm....107
Secret Key....107
RAND....107
AMF....107
Key Type....107
Authentication OP_c....108

IMSI
15-digit international mobile subscriber identity (IMSI). It must match the IMSI on the SIM.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:AUTHentic:IMSI`

Authentication Algorithm
Key generation algorithm set used by the SIM: MILENAGE algorithms (see 3GPP TS 35.205) or XOR algorithms.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:AUTHentic:ALGorithm`

Secret Key
Authentication key K as 32-digit hexadecimal number, as stored on the SIM or the mobile.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY`

RAND
Random number to be used.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:AUTHentic:RAND`

AMF
Authentication management field as four-digit hexadecimal number.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:AUTHentic:AMF`

Key Type
Key type to be used with the MILENAGE algorithm set (fixed value).

Remote command:
CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY:TYPE

Authentication OP_c
OP_c value used by the SIM, as 32-digit hexadecimal number.

Only relevant and configurable if the MILENAGE algorithm set is used, see "Authentication Algorithm" on page 107.

Remote command:
CONFigure:DATA:CONTrol:EPDG:AUTHentic:OPC

4.7.10 Dead Peer Detection Settings

With enabled dead peer detection, the ePDG detects and releases unused IPsec tunnels.

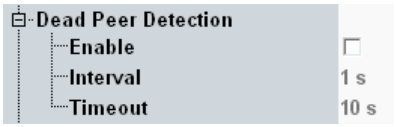


Figure 4-42: Dead peer detection settings

Enable
Enables dead peer detection.

Remote command:
CONFigure:DATA:CONTrol:EPDG:DPD:ENABle

Interval, Timeout
If the ePDG receives no packets via an established tunnel within the configured "Interval", the ePDG sends a message to the mobile.

If the ePDG receives no answer to this message within the configured "Timeout", it releases the IPsec tunnel.

Remote command:
CONFigure:DATA:CONTrol:EPDG:DPD:INTerval
CONFigure:DATA:CONTrol:EPDG:DPD:TIMeout

4.7.11 Certificate Settings

The following settings are related to SSL certificates.

If you want to use SSL without certificates, disable the checkbox.

If you want to use an SSL certificate:

- Store the server certificate file and the key file with the private key of the server on the samba share, in the directory `security/certificates/epdg`.
  Only PEM files are supported.
- Store the server certificate file on the mobile.

- Ensure that the ePDG ID used in the server certificate matches the configured ePDG ID, see "ePDG ID Configuration" on page 103.
- Enable the checkbox in the following settings.
- Select the key file and the certificate file via the following parameters. The list offers only PEM files stored in the correct directory on the samba share.



Figure 4-43: SSL certificate settings

Enable....109
Server Key, Server Certificate....109

Enable
Enables the usage of certificates for SSL.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:CERTificate:ENABle`

Server Key, Server Certificate
Select the key file and the certificate file. Only PEM files are listed.

Remote command:
`CONFigure:DATA:CONTrol:EPDG:CERTificate:KEY`
`CONFigure:DATA:CONTrol:EPDG:CERTificate:CERTificate`

4.8 Common Softkeys, Hotkeys and Parameters

Some GUI elements are displayed independent of the selected tab. They are described below.

DAU (parameter)
This information is displayed at the top of the "Data Application Control" dialog. It indicates the state of the DAU (on, off, pending).

By default, the DAU is switched on and there is no need to switch it off for regular operation. However, the DAU can be switched off and on again for troubleshooting, see "DAU (softkey)" on page 109.

Remote command:
`SOURce:DATA:CONTrol:STATe`

DAU (softkey)
To reboot the DAU for troubleshooting, select the "DAU" softkey and press ON | OFF or RESTART | STOP twice (i.e. switch the DAU off and on again). Alternatively, right-click the softkey.

The current state of the DAU is indicated by the softkey and at the top of the "Data Application Control" dialog.

Remote command:
`SOURce:DATA:CONTrol:STATe`

Select Applic (hotkey)
Selects the tab to be displayed in the "Data Application Control" dialog.

Network Drive map (hotkey)
The DAU has a system drive which can be used for data storage. This drive can be accessed from the Microsoft Windows Explorer.

The hotkey opens a dialog for changing the assigned drive letter.

5 DAU Measurements GUI Reference

Data application measurements require option R&S CMW-KM050. Activate the measurements via the "Measurement Controller" dialog, entry "Data Appl. > Measurements". You can then access the main view of the measurements dialog from the task bar.

The "Data Application Measurements" dialog contains common elements at the top, one overview tab and one tab per DAU measurement. Refer to the following sections for a detailed description. The QoS settings are also described.

- RAN Selection....111
- Overview Tab....112
- Ping Tab....113
- Iperf Tab....114
- Throughput Tab....118
- DNS Requests Tab....122
- IP Logging Tab....123
- IP Analysis/Security Tab....126
- IP Replay Tab....151
- Audio Delay Tab....155
- Quality-of-Service Settings....158
- Common Softkeys and Hotkeys....162

5.1 RAN Selection

DAU measurements can be performed for connections established via signaling applications.

Via "Select RAN" at the top of the "Data Application Measurements" dialog, you can select a signaling application. As a consequence, the expected maximum throughput for this signaling application is displayed and quick access softkeys for this signaling application are activated.

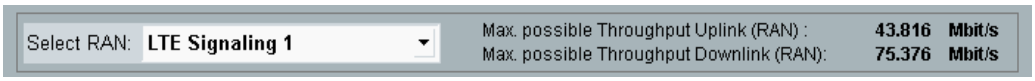


Figure 5-1: Top of dialog

The selection is mainly a convenience function. For quick access softkeys, see "Signaling Parameter (softkey)" on page 163 and "... Signaling (softkey)" on page 163.

Select RAN
Select an installed signaling application instance. Only applications supporting DAU tests are listed, e.g. LTE Signaling.

Remote command:
`CONFigure:DATA:MEAS<i>:RAN`
`CONFigure:DATA:MEAS<i>:RAN:CATaloge?`

Max. Possible Throughput
Display the expected maximum UL/DL throughput resulting from the current signaling application settings.

Remote command:
n/a, use the commands provided by the signaling application.

5.2 Overview Tab

The tab provides an overview of the DAU measurements, including the state of each measurement and selected measurement results.

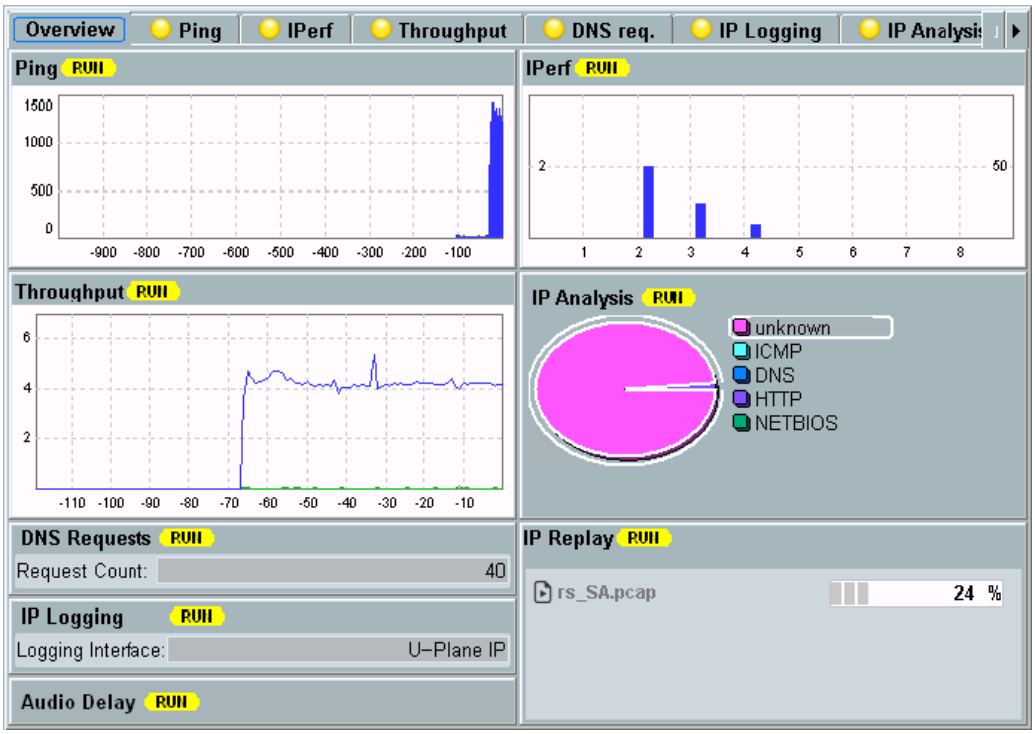


Figure 5-2: Measurements - Overview tab

Areas Ping, Throughput, ...
Each area shows the state and selected measurement results of the corresponding measurement.

For details about the measurements, refer to the following sections:

- Chapter 5.3, "Ping Tab", on page 113
- Chapter 5.4, "Iperf Tab", on page 114
- Chapter 5.5, "Throughput Tab", on page 118
- Chapter 5.6, "DNS Requests Tab", on page 122
- Chapter 5.7, "IP Logging Tab", on page 123
- Chapter 5.8, "IP Analysis/Security Tab", on page 126
- Chapter 5.9, "IP Replay Tab", on page 151
- Chapter 5.10, "Audio Delay Tab", on page 155

5.3 Ping Tab

The "Ping" measurement sends ping requests to a configurable IP address and displays the resulting ping latency in a diagram.

The "Ping" tab allows you to configure the properties of the ping command and displays the result diagram.

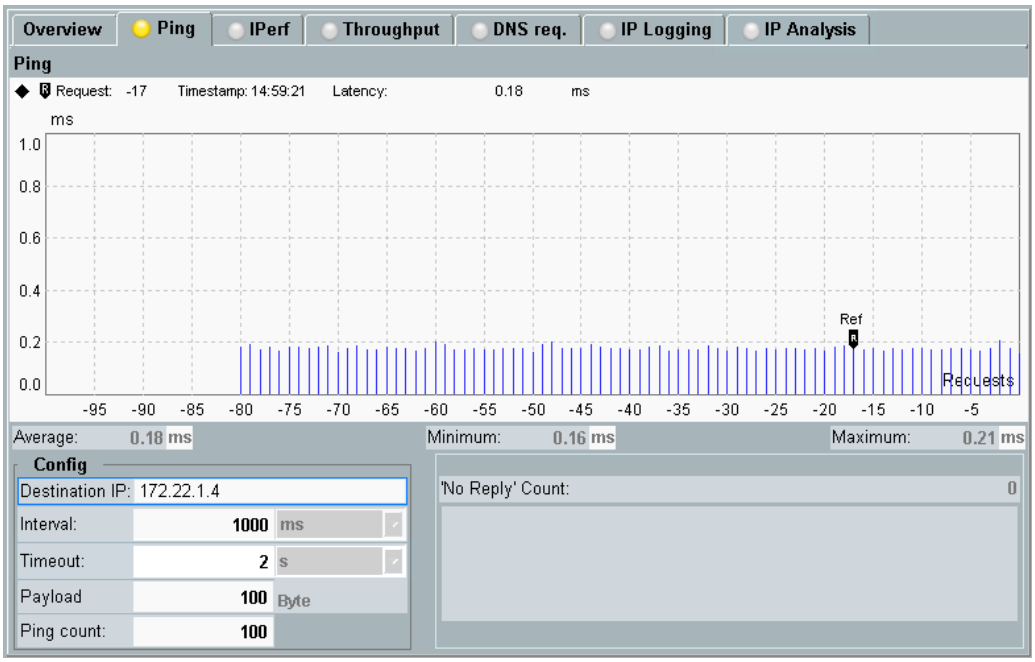


Figure 5-3: Measurements - Ping tab

Result diagram....113
Average / Minimum / Maximum....113
'No Reply' Count....114
Config....114
Measurement control....114

Result diagram
The diagram provides a graphical presentation of the measured ping latency for each ping request. The X-axis indicates the processed ping requests, with the last processed request labeled 0, the previously processed request labeled -1, and so on.

Remote command:
`FETCh:DATA:MEAS<i>:PING?`
`READ:DATA:MEAS<i>:PING?`

Average / Minimum / Maximum
The values below the diagram indicate the average, minimum and maximum of all values in the diagram.

Remote command:
`FETCh:DATA:MEAS<i>:PING:OVERall?`

'No Reply' Count
Indicates the number of ping requests with no reply. In the result diagram, such requests are displayed as gaps (ping latency = 0 ms).

The reason for the latest ping request with no reply is indicated below the parameter, e.g. timeout or no route to host.

Remote command:
```
FETCh:DATA:MEAS<i>:PING:NRCount?
```

Config
Configures the properties of the ping command.
- "Destination IP": IP address that you want to ping
- "Interval": pinging interval
- "Timeout": timeout for unanswered ping requests
- "Payload": packet size used as probe
- "Ping Count": number of echo request packets to be sent

You can change values that are not grayed out even during the measurement, for example, the "Interval".

Remote command:
```
CONFigure:DATA:MEAS<i>:PING:DIPaddress
CONFigure:DATA:MEAS<i>:PING:INTerval
CONFigure:DATA:MEAS<i>:PING:TIMeout
CONFigure:DATA:MEAS<i>:PING:PSIZe
CONFigure:DATA:MEAS<i>:PING:PCOunt
```

Measurement control
Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "Ping" softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:
```
INITiate:DATA:MEAS<i>:PING
STOP:DATA:MEAS<i>:PING
ABORt:DATA:MEAS<i>:PING
FETCh:DATA:MEAS<i>:PING:STATe?
FETCh:DATA:MEAS<i>:PING:STATe:ALL?
```

5.4 Iperf Tab

The measurement uses the open source tool iperf to evaluate the throughput and reliability of a connection in uplink and/or downlink direction.

For a measurement, the iperf tool must be running at both ends of a connection. At the end sending data, it is configured as "client" and at the end receiving data, as "server".

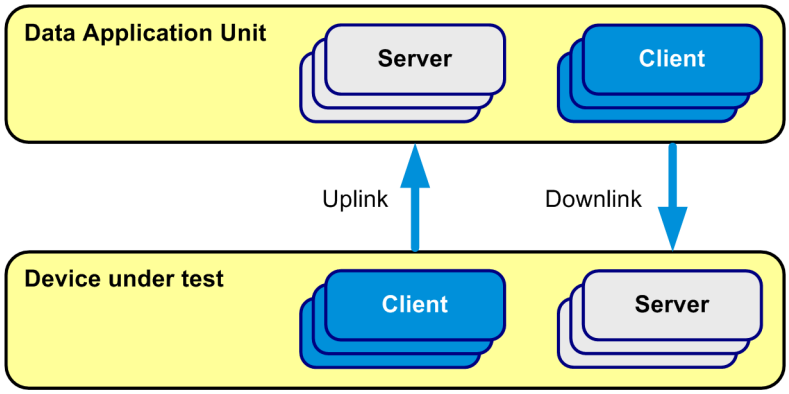


Figure 5-4: Role of servers and client instances in the context of iperf

The "Iperf" tab allows you to configure up to eight server instances and eight client instances at the DAU in parallel. For configuration of iperf at the mobile side, see Chapter 3.11, "Configuring Iperf on the Mobile", on page 47.

The main view of the "Iperf" tab shows the first three instances together with the main parameters. To display all instances, press the softkey "Expand Table".

The diagram below the instances displays the measurement results.

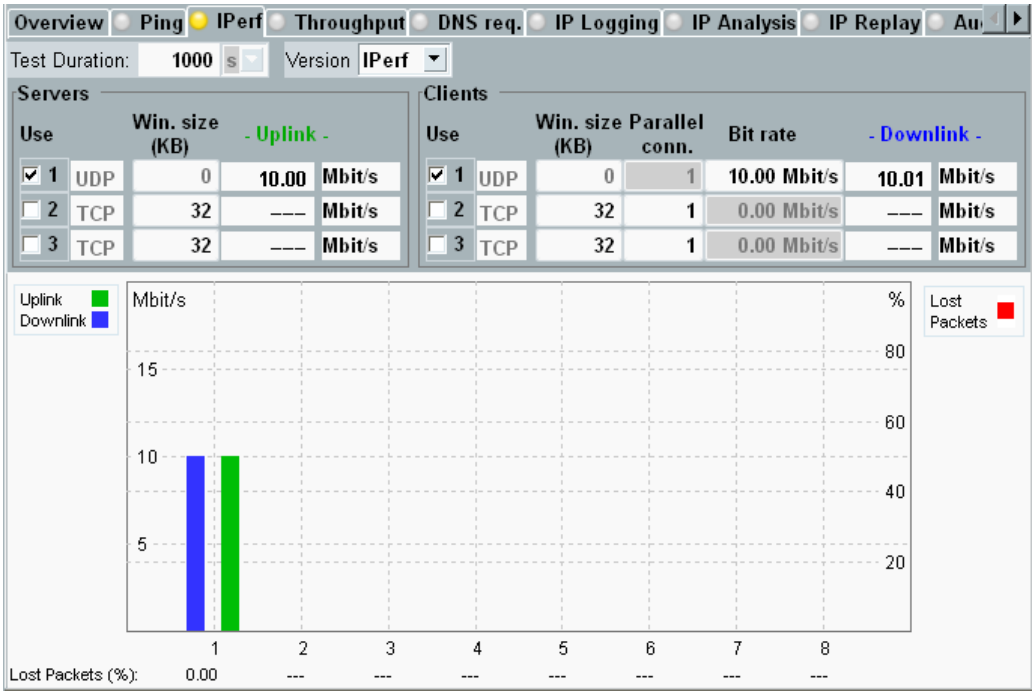


Figure 5-5: Measurements - Iperf tab

All instances can be fully configured using the "Iperf Config" dialog, which can be opened via the "Config" hotkey.

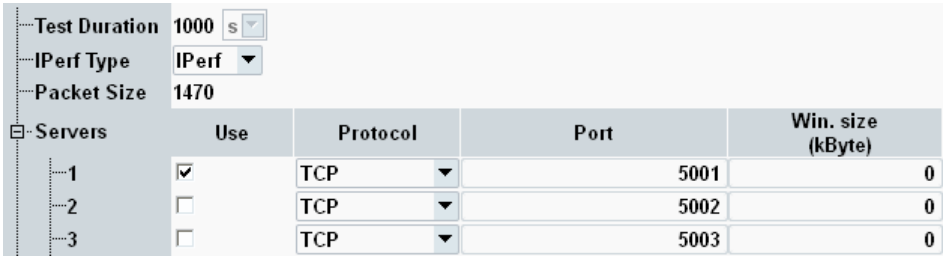


Figure 5-6: Iperf config dialog, servers

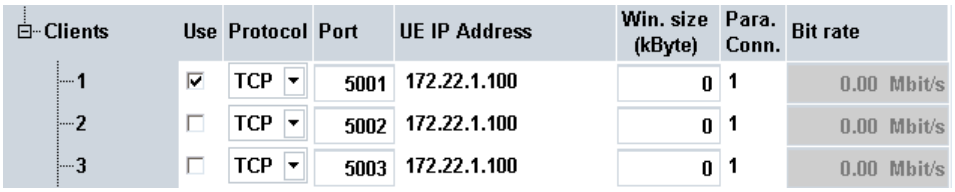


Figure 5-7: Iperf config dialog, clients

The elements of the main view and of the configuration dialog are described below.

Results 116
Test Duration 116
IPerf Type 117
Packet Size 117
Server settings 117
Client settings 118
Measurement control 118

Results

The bar graph indicates up to three result bars per instance:

- "Uplink": bit rate measured in uplink direction for active server instances
- "Downlink": bit rate transmitted in downlink direction for active client instances
- "Lost Packets": percentage of packets lost during the measurement, for active server instances with protocol type UDP

The uplink and downlink bit rates are also indicated in the instance table above the bar graph. The lost packet percentage is also indicated below the bar graph.

Remote command:

```
FETCh:DATA:MEAS<i>:IPERf:CLIent?
READ:DATA:MEAS<i>:IPERf:CLIent?
FETCh:DATA:MEAS<i>:IPERf:SERVer?
READ:DATA:MEAS<i>:IPERf:SERVer?
FETCh:DATA:MEAS<i>:IPERf:ALL?
```

Test Duration

Defines the duration of the test.

Remote command:

```
CONFigure:DATA:MEAS<i>:IPERf:TDURation
```

IPerf Type
Selects the iperf version to be used, iperf or iperf3.

Iperf3 is not compatible with older versions. Use the same iperf version on the DAU and on the mobile.

Remote command:
`CONFigure:DATA:MEAS<i>:IPERf:TYPE`

Packet Size
Defines the packet size for iperf tests (payload bytes), applicable to UDP and TDD.

The packet size is especially important if you want to use a low UDP bit rate. To reach this bit rate, configure the packet size according to the following rules:

- Convert the bit rate to byte/s and divide it by an integer number (divisor n) to derive the packet size.
- Use the smallest possible divisor n, resulting in an integer packet size.
- Ensure that the packet size is smaller than or equal to the maximum payload size, resulting from the configured MTU (MTU minus overhead), see also "Maximum Transmission Unit (MTU)" on page 54. If necessary, use a larger divisor n.

Example:

MTU = 1500 bytes, UDP bit rate = 23 kbit/s = 2875 byte/s

The smallest possible divisor n equals five. It results in a packet size of 575 bytes. This packet size is smaller than the maximum payload size.

With a packet size of 575 bytes, exactly five packets are transferred per second. The resulting bit rate is 5 * 575 byte/s * 8 bit/byte = 23 kbit/s.

Remote command:
`CONFigure:DATA:MEAS<i>:IPERf:PSIZe`

Server settings
Defines the properties of the server instances for uplink direction (DAU receives data from the mobile):

- "Use"
  Specifies whether the server instance is used.
- "UDP or TCP"
  Selects the protocol type to be used.
  This setting is configurable for iperf. For iperf3, it is dynamically set by the client at the other end.
- "Port"
  Specifies the LAN DAU port number for the connection. This value must match the mobile's client port settings.
- "Window size"
  Specifies the TCP receiving window size (for TCP only)
  This setting is configurable for iperf. For iperf3, it is dynamically set by the client at the other end.

Remote command:
`CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:ENABle`
`CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PROTocol`
`CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PORT`
`CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:WSIZe`

Client settings
Defines the properties of the client instances for downlink direction (DAU sends data to the mobile).

- "Use"
  Specifies whether the client instance is used.
- "UDP or TCP"
  Selects the protocol type to be used.
- "Port"
  Specifies the LAN DAU port number for the connection. This value must match the mobile's server port settings.
- "UE IP Address"
  Specifies the IP address of the mobile.
- "Window size"
  Specifies the size of the NACK window (in kByte), for TCP only.
- "Parallel Connections"
  Specifies the number of parallel connections for the selected port, for TCP only.
- "Bit rate"
  Specifies the maximum bit rate to be transferred, for UDP only.
  If the bit rate is not reached, check the packet size, see "Packet Size" on page 117.

Remote command:
```
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:ENABle
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PROTocol
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PORT
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:IPADdress
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:WSIZe
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PCONnection
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:BITRate
```

Measurement control
Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "Iperf" softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:
```
INITiate:DATA:MEAS<i>:IPERf
STOP:DATA:MEAS<i>:IPERf
ABORt:DATA:MEAS<i>:IPERf
FETCh:DATA:MEAS<i>:IPERf:STATe?
FETCh:DATA:MEAS<i>:IPERf:STATe:ALL?
```

5.5 Throughput Tab

The "Throughput" measurement reports the throughput at the DAU on IP level, in uplink and downlink direction. The entire packets are considered, including the IP header - in contrast to iperf measurements that consider only the payload.

Note that the measured downlink data rate on IP level and the downlink data rate on lower layers can differ. If the lower layers are not able to transfer the set IP data rates, they discard IP packets.

The "Overall Throughput" tab shows the total throughput at the DAU. Several signaling applications and mobile connections can contribute to the overall throughput.

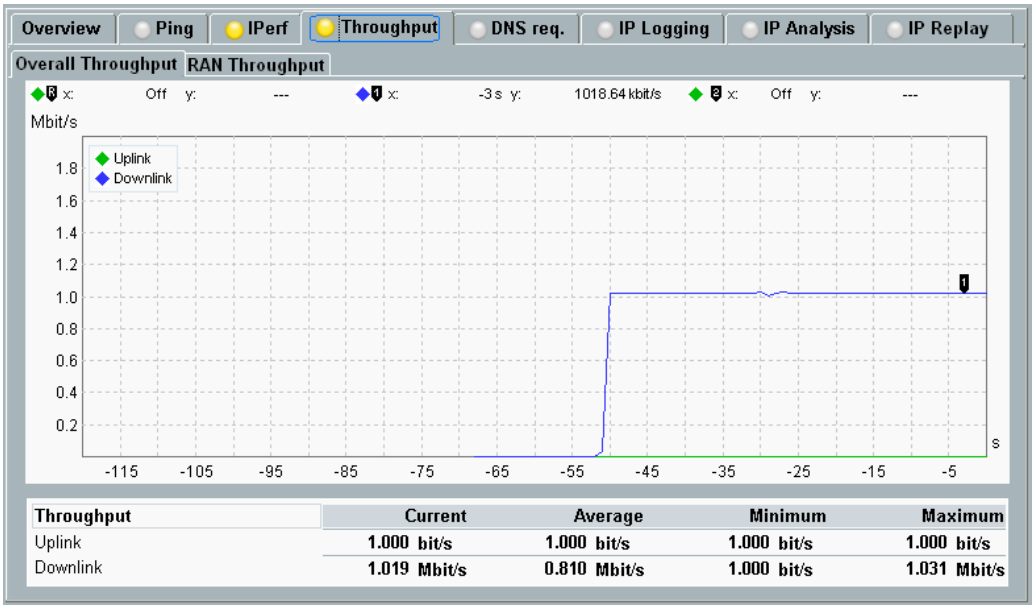


Figure 5-8: Measurements - Overall Throughput tab

The "RAN Throughput" tab shows the throughput per signaling application instance. You can measure the throughput for up to four signaling applications in parallel.

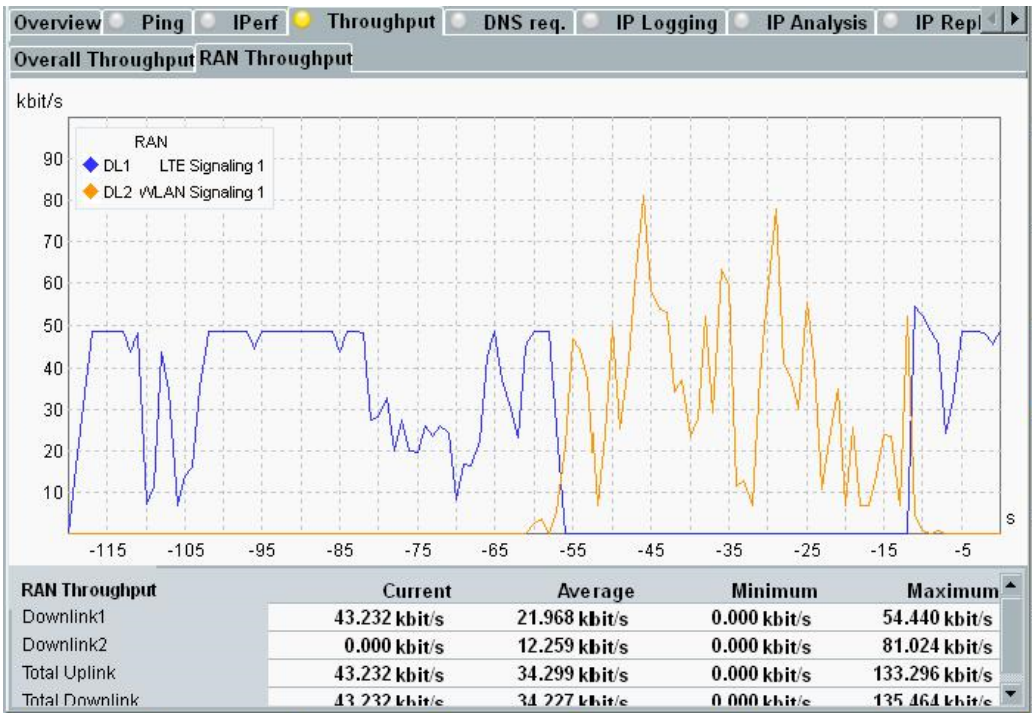


Figure 5-9: Measurements - RAN Throughput tab

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RAN Throughput results....120
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Overall Throughput results

The diagram shows the overall throughput on IP level vs. time.

"Uplink" indicates the received bit rate measured at the DAU while "Downlink" indicates the bit rate transmitted to the mobile. In downlink direction, the throughput values on RLC or MAC level can differ from the throughput on IP level.

Below the diagram, the measured/transmitted current, minimum and maximum throughput values are displayed.

Remote command:

```
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk[:CURRent]?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk[:CURRent]?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk:EXTended?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk:EXTended?
READ:DATA:MEAS<i>:THRoughput:OVERall:ULINk?
READ:DATA:MEAS<i>:THRoughput:OVERall:DLINk?
```

Plus corresponding `FETCh` commands.

RAN Throughput results

The diagram shows the throughput per signaling application instance (RAN), in uplink and downlink direction.

Remote command:
```
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:ULINk<Index>:CURRent?
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:DLINk<Index>:CURRent?
READ:DATA:MEAS<i>:THRoughput:RAN:ULINk<Index>?
READ:DATA:MEAS<i>:THRoughput:RAN:DLINk<Index>?
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:ULINk?
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:DLINk?
```
Plus corresponding `FETCh` commands.

RAN trace selection
The displayed traces are selected via the softkey "Display" > hotkey "Select Traces".

The following notes apply:
- There are four RAN slots. You can assign a RAN to each slot. For each RAN, you can display the uplink throughput and/or the downlink throughput. So there are in total up to eight traces displayed in parallel.
- If a RAN is selected at the top of the data application measurement dialog box, this RAN is automatically assigned to the first slot. See also Chapter 5.1, "RAN Selection", on page 111.
- During a running measurement, you can assign RANs to still empty slots. But you cannot remove already assigned RANs.
  To remove assigned RANs, you must stop the measurement, change the assignment and restart the measurement.

Remote command:
```
CONFigure:DATA:MEAS<i>:THRoughput:RAN:CATaloge?
CONFigure:DATA:MEAS<i>:THRoughput:RAN<Index>
CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:ULINk<Index>
CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:DLINk<Index>
```

Settings
Press the "Config" hotkey to access the measurement settings.



Figure 5-10: Measurement settings

- "Interval": displays the time interval between two throughput measurement results
- "Max. array size": specifies the maximum number of result values in the result trace

Example: Interval 1 s plus size 5 means that there are up to 5 results in the trace, at the X-axis positions "0 s" to "-4 s". Older results are discarded.

Remote command:
```
SENSe:DATA:MEAS<i>:THRoughput:INTerval?
CONFigure:DATA:MEAS<i>:THRoughput:MCOunt
CONFigure:DATA:MEAS<i>:THRoughput:RAN:MCOunt
```

Measurement control
Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "Throughput" softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:DATA:MEAS<i>:THRoughput
STOP:DATA:MEAS<i>:THRoughput
ABORt:DATA:MEAS<i>:THRoughput
FETCh:DATA:MEAS<i>:THRoughput:STATe?
FETCh:DATA:MEAS<i>:THRoughput:STATe:ALL?
```

5.6 DNS Requests Tab

The "DNS Requests" measurement monitors all DNS queries addressed to the internal DNS server of the DAU.

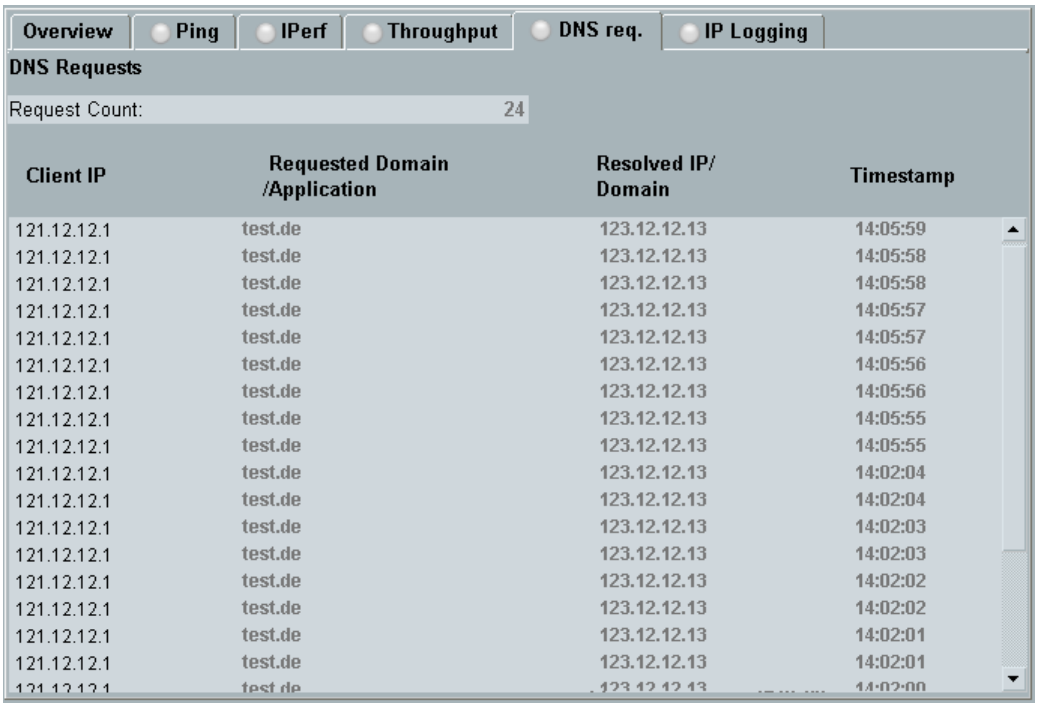


Figure 5-11: Measurements - DNS req. tab

Results 122
Settings 123
Measurement control 123

Results

The "Request Count" indicates the number of recorded DNS queries.

The table below provides a list of the recorded requests and answers, including the following information for each request:

- "Client IP"
  IP address of the client (mobile) that has sent the DNS request

- "Requested Domain / Application"
  Information to be resolved by the DNS query. A type A or type AAAA DNS query indicates a domain, while a type SRV DNS query indicates an application name.
- "Resolved IP / Domain"
  Information returned for the DNS query. A successful type A or type AAAA DNS query returns the IP address of the requested domain. A successful type SRV DNS query returns a domain supporting the requested application.
- "Timestamp"
  Time when the query was received

Remote command:
```
SENSe:DATA:MEAS<i>:DNSRequests:RCOunt?
SENSe:DATA:MEAS<i>:DNSRequests?
```

Settings
Press the "Config" hotkey to access the measurement settings.



Figure 5-12: Measurement settings

"Max. Index Count" specifies the maximum length of the result list (number of requests). The result list is stored in a ring buffer. When it is full, the first result line is deleted whenever a new result line is added to the end.

Remote command:
```
CONFigure:DATA:MEAS<i>:DNSRequests:MICount
```

Measurement control
Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "DNS Requests" softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:
```
INITiate:DATA:MEAS<i>:DNSRequests
STOP:DATA:MEAS<i>:DNSRequests
ABORt:DATA:MEAS<i>:DNSRequests
FETCh:DATA:MEAS<i>:DNSRequests:STATe?
FETCh:DATA:MEAS<i>:DNSRequests:STATe:ALL?
```

5.7 IP Logging Tab

The "IP Logging" application allows you to create signaling log files for the IP layer communication at the DAU. You can monitor either the user plane communication between the DAU and the mobile or you can monitor the communication at the LAN DAU connector.

The log files are created in a format compatible with Wireshark, a network protocol analyzer which is available free of charge. You can download Wireshark from http://www.wireshark.org. If Wireshark is installed at the instrument, you can open created log files directly from the "IP Logging" tab to analyze them in Wireshark. For encrypted connections, see also Chapter 3.12, "Configuring Wireshark for IPsec", on page 48.

The system drive operations required for IP logging impact the performance of the DAU. Thus measurement results can be distorted by IP logging. Switch off IP logging especially for throughput measurements.

The contents of the "IP Logging" tab, the related configuration dialog and the related softkeys and hotkeys are described below.

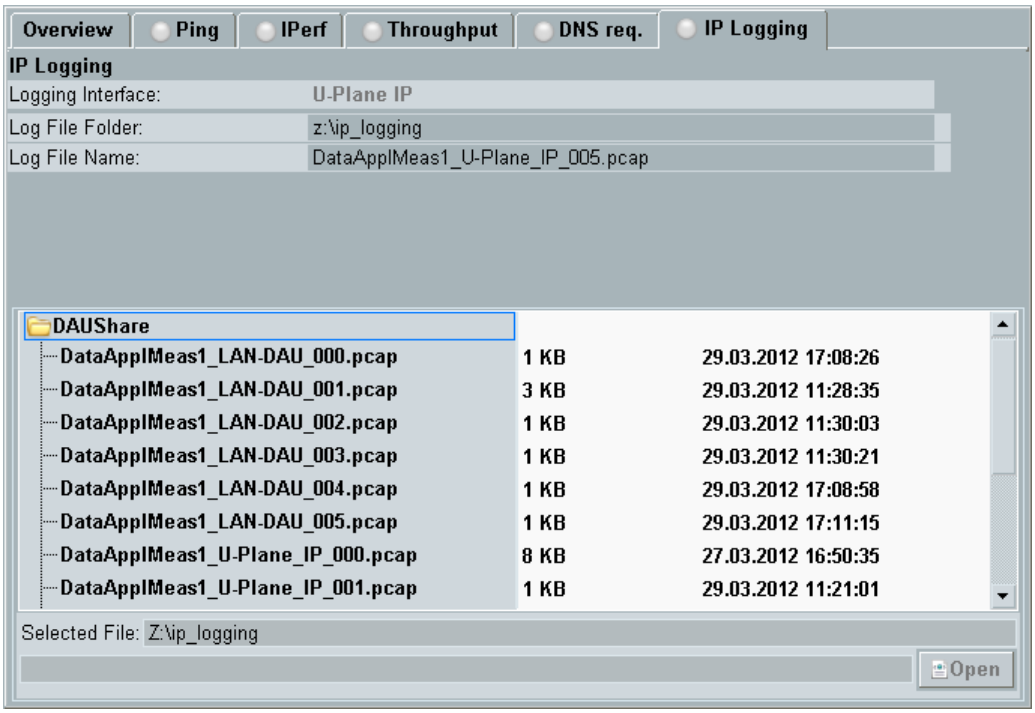


Figure 5-13: Measurements - IP Logging tab

Contents of the tab....124
Hotkeys....125
Settings....125
└ Logging Interface....125
└ Log File Name....126
└ Advanced Settings....126
Logging control....126

Contents of the tab

At the top, the "IP Logging" tab displays the configured logging interface and the folder and name of the currently used log file.

The lower part of the tab lists the log files stored on the system drive of the DAU. The files are stored in directory `ip_logging` of the samba share. Connected USB storage devices like USB sticks are also displayed.

"Selected File" at the bottom refers to the file currently selected in the list above. The button "Open" opens the selected file in Wireshark. If Wireshark is not installed, the button is grayed-out. We recommend connecting a mouse to use Wireshark. Without a mouse, you can return to the GUI of the DAU by pressing SYS at the front panel or CTRL+ALT+D at a connected keyboard.

Hotkeys
The "IP Logging" tab provides several hotkeys for file operations.



You can for example copy/paste the currently selected log file to a USB stick or delete a log file. Or you can change an automatically assigned file name.

Before pressing a hotkey, select the desired folder or file in the "IP Logging" tab.

Settings
Press the "Config" hotkey to access the IP logging settings.

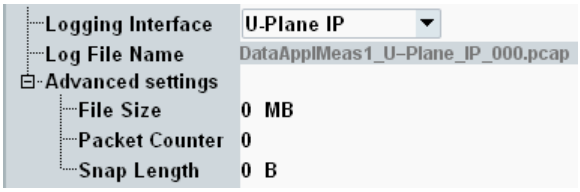


Figure 5-14: IP logging settings

Logging Interface ← Settings
Selects the interface to be monitored.

"LAN-DAU"	Communication at the LAN DAU connector
"U-Plane IP"	IP unicast traffic, for example relevant for LTE, WCDMA, GSM, WLAN and CDMA2000® 1xEV-DO (eHRPD)
"U-Plane PPP"	PPP encapsulated IP traffic, for example relevant for CDMA2000® 1xRTT / 1xEV-DO (HRPD)
"U-Plane Multi-cast"	IP multicast traffic, for example relevant for LTE With this selection, only the downlink multicast traffic is monitored. The unicast answers in the uplink are not monitored. Option R&S CMW-KA150 is required for multicast traffic.
"U-Plane IP Client"	IP traffic where the DAU acts as client and the mobile assigns an IP address to the DAU. Example: Test of a WLAN access point, where the R&S CMW simulates a WLAN station.

Remote command:
`CONFigure:DATA:MEAS<i>:IPLogging:TYPE`

Log File Name ← Settings
If IP logging is on, the name of the currently used log file is displayed. If IP logging is off, a file with the displayed name is created when logging is started.

The name is defined automatically. It contains information about the DAU measurement instance, the logging interface and an index. The index is increased automatically, so that an existing file is never overwritten.

Remote command:
`SENSe:DATA:MEAS<i>:IPLogging:FNAMe?`

Advanced Settings ← Settings
The advanced settings allow you to limit logging. "0" means no limit.

- "File Size"
  Limits the size of the log file. When this file size is reached, logging stops.
- "Packet Counter"
  Limits the number of IP packets to be logged. When this number of packets is reached, logging stops.
- "Snap Length"
  Limits the number of bytes to be logged for each IP packet. If the packet is longer, only the specified number of bytes is logged. The remaining bytes of the packet are ignored.

Remote command:
`CONFigure:DATA:MEAS<i>:IPLogging:FSIZe`
`CONFigure:DATA:MEAS<i>:IPLogging:PCOunter`
`CONFigure:DATA:MEAS<i>:IPLogging:PSLength`

Logging control
Turn logging on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "IP Logging" softkey.

The softkey shows the current state. Additional substates can be retrieved via remote control.

Remote command:
`INITiate:DATA:MEAS<i>:IPLogging`
`STOP:DATA:MEAS<i>:IPLogging`
`ABORt:DATA:MEAS<i>:IPLogging`
`FETCh:DATA:MEAS<i>:IPLogging:STATe?`
`FETCh:DATA:MEAS<i>:IPLogging:STATe:ALL?`

5.8 IP Analysis/Security Tab

The application monitors IP traffic from or to the mobile. It analyzes traffic originating at the DAU, terminating at the DAU or passing the DAU.

Typical use cases are for example:

- Analysis of reasons for too low throughput. Use the "TCP Analysis" view to check the number of retransmissions, the TCP window size and the overhead.

- Debugging of applications, for example smartphone apps, by analysis of the IP traffic caused by the applications.
- In general, getting an overview / a statistical evaluation of all IP connections

Option R&S CMW-KM051 is required for IP analysis. Without this option, the "IP Analysis/Security" tab is hidden.

The term "Security" in the tab label refers to IP connection security analysis, see Chapter 5.8.7, "View IP Connection Security", on page 141.

5.8.1 Overview

The results are displayed in an overview and in several detailed views. The overview summarizes the most important results. If you are interested in a specific part of the overview, expand this part by opening the related detailed view.

For a description of the individual results, refer to the detailed view sections.

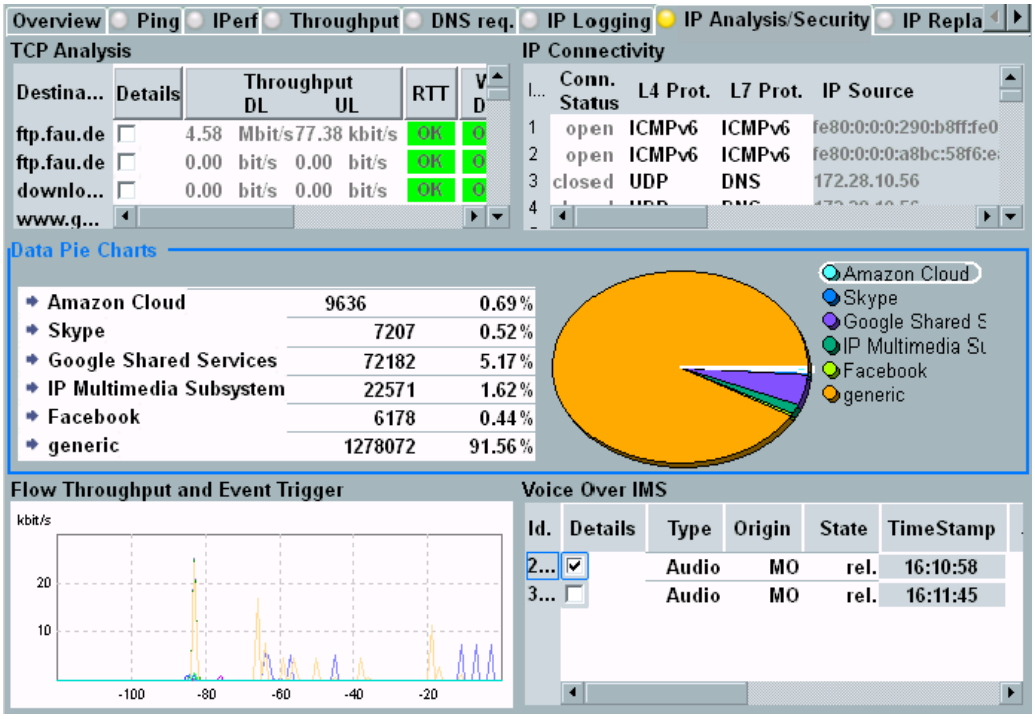


Figure 5-15: Result overview

Opening another view 127
Disabling views / Assign View hotkey 128
Export DB (hotkey) 128
Measurement control 128

Opening another view
Use the "Select View" hotkey to display a specific detailed view or to return to the overview.

Alternatively, you can select a part of the overview and press ENTER or the rotary knob to open the related detailed view.

Disabling views / Assign View hotkey
The "Assign View" hotkey enables/disables the individual detailed views. If a detailed view is disabled, the related results are not measured, the view cannot be shown and the related parts of the overview are hidden.

Configure the views before turning on the measurement. Any already measured results are lost if you change the settings (enable or disable a view).

Remote command:
```
CONFigure:DATA:MEAS<i>:IPANalysis:RESult[:ALL]
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:TCPanalysis
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:IPConnect
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:DPCP
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:FTTRigger
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:VOIMs
```

Export DB (hotkey)
Stores the IP analysis result database to a CSV file on the DAU system drive.

Remote command:
```
CONFigure:DATA:MEAS<i>:IPANalysis:EXPortdb
```

Measurement control
Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "IP Analys./Sec." softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:
```
INITiate:DATA:MEAS<i>:IPANalysis
STOP:DATA:MEAS<i>:IPANalysis
ABORt:DATA:MEAS<i>:IPANalysis
FETCh:DATA:MEAS<i>:IPANalysis:STATe?
FETCh:DATA:MEAS<i>:IPANalysis:STATe:ALL?
```

5.8.2 View TCP Analysis

The main purpose of the "TCP Analysis" view is to detect problems decreasing the performance of TCP/IP connections.

To this end, you can define thresholds and evaluate the results of the threshold checks.

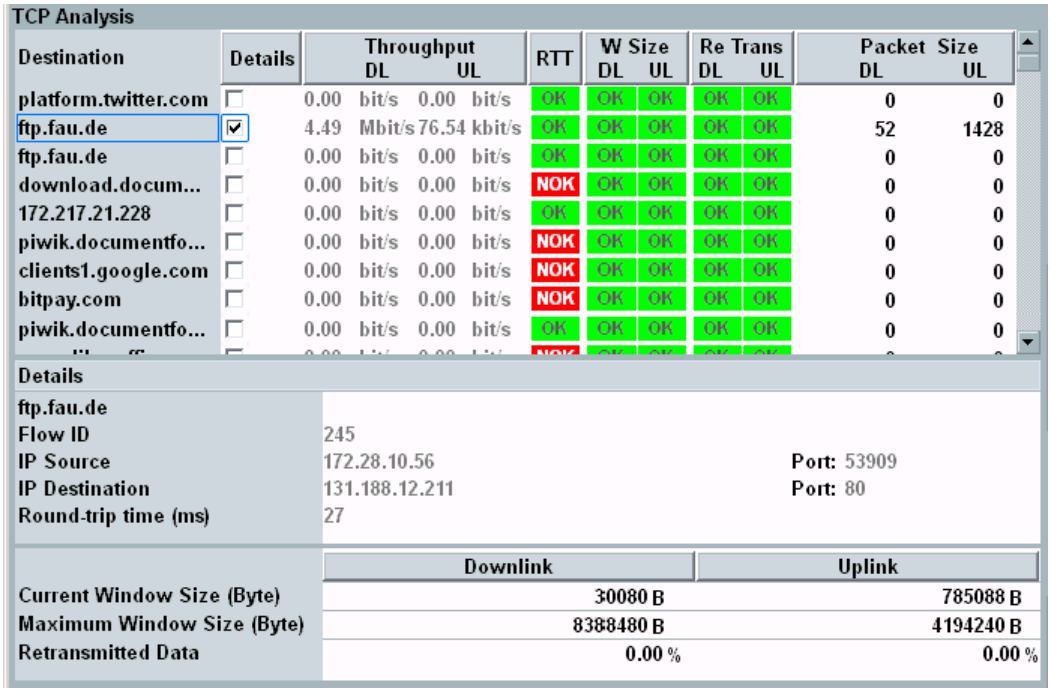


Figure 5-16: TCP Analysis view

Threshold configuration....129
Filter hotkey....130
Result overview (upper part)....130
Result details (lower part)....130

Threshold configuration

Open the configuration dialog box via the hotkey "Config".

The relevant settings are contained in the node "TCP Analysis Settings".

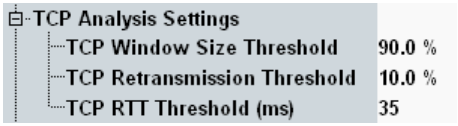


Figure 5-17: TCP analysis settings

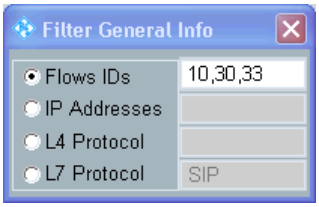
Threshold definitions:

- "Window Size":
  Current TCP window size as percentage of the negotiated maximum window size. The maximum window size defines how many bytes can be transmitted without waiting for an acknowledgment. Or in other words: how many unacknowledged bytes can be on the way simultaneously.
  If the current window size reaches the maximum window size, no more packets are sent until acknowledgments are received. So a too small maximum window size decreases the throughput.
- "Retransmission":
  Retransmissions as percentage of all transmissions. Retransmissions decrease the throughput.

- "RTT":
  Round-trip time, for TCP packet to the mobile plus acknowledgment packet from the mobile

Remote command:
```
CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TWSThreshold
CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TRTHreshold
CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:RTTThreshold
```

Filter hotkey
The hotkey "Filter" accesses a dialog with filter settings. The hotkey is available in several views and the settings apply to all these views. So you need to configure the filter only once, not per view.



Select a filter row and enter a filter text. You can enter a single value or a comma-separated list of values. Wildcards are not supported.

The colored indicator on the hotkey tells you whether the filter is active or not .

Remote command:
```
CONFigure:DATA:MEAS<i>:IPANalysis:IPConnect:FILTer
```

Result overview (upper part)
The result overview in the upper part of the tab lists all connections and indicates whether and which thresholds are exceeded.

If a threshold is exceeded, a NOK or FULL is displayed. Otherwise, an OK is displayed.

Table columns:
- "Destination": Address of the destination of the connection
- "Details": Selects the connection, for which detailed information is displayed in the lower part
- "Throughput": DL and UL throughput
- "RTT": Round-trip time threshold check result
- "W Size": Window size threshold check result for the DL and UL
- "Re Trans": Retransmissions threshold check result for the DL and UL
- "Packet Size": DL and UL layer 3 packet size in bytes

Remote command:
```
SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:FLOWid?
FETCh:DATA:MEAS<i>:IPANalysis:TCPanalysis:ALL?
```

Result details (lower part)
The "Details" area in the lower part of the tab displays additional properties for the connection selected in the upper part, column "Details".

- "Flow ID": IDs of the flows used by the connection

The same IDs are displayed in other views, for example the "IP Connectivity" view and the "Flow Throughput and Trigger" view. So you can use the flow IDs to correlate the information in different views.

- "IP Source" / "IP Destination": Address and port of the source that has initiated the connection and of the destination addressed by the source
- "Round-trip time", "Current Window Size", "Retransmitted Data": Measured values, used for the threshold checks
- "Maximum Window Size": Maximum window size negotiated for the connection

Remote command:
`SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:DETails?`

5.8.3 View IP Connectivity

This view provides an overview of all connections that have been opened since the measurement was started. Still open connections are listed as well as already closed connections.

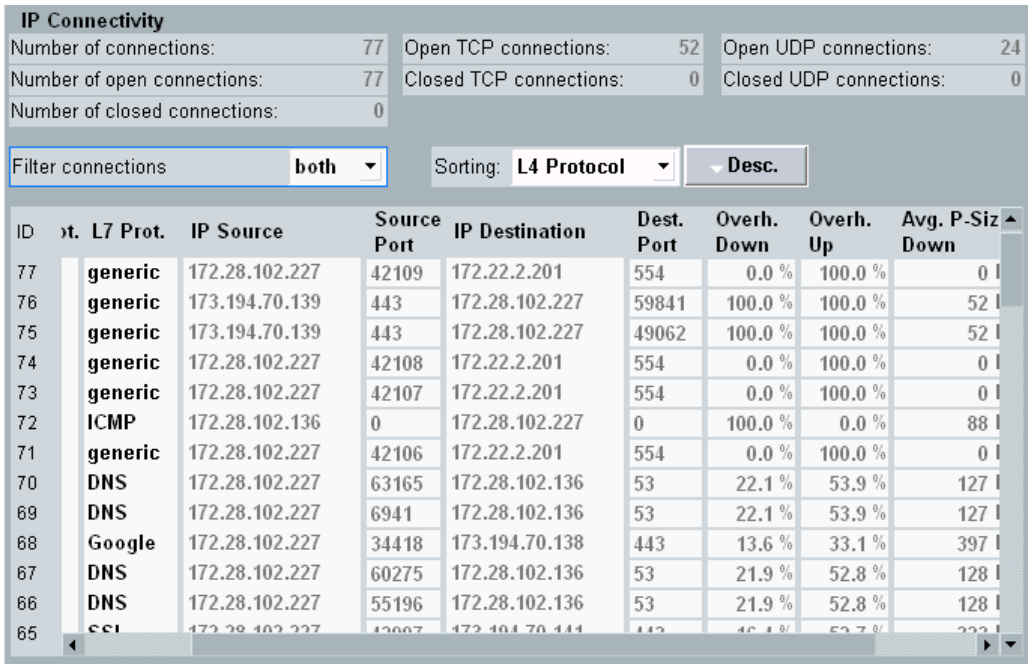


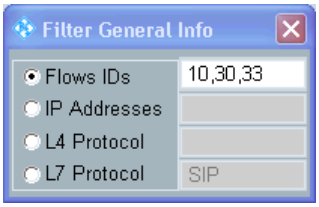
Figure 5-18: IP Connectivity view

Filtering and sorting the table
Use the settings above the table to filter or sort the table.

You can filter the table according to the connection status: "open" connections only, "closed" connections only or "both".

"Sorting" selects the table column to be used for sorting. The button toggles between ascending and descending order.

Filter hotkey
The hotkey "Filter" accesses a dialog with filter settings. The hotkey is available in several views and the settings apply to all these views. So you need to configure the filter only once, not per view.



Select a filter row and enter a filter text. You can enter a single value or a comma-separated list of values. Wildcards are not supported.

The colored indicator on the hotkey tells you whether the filter is active or not.

Remote command:
`CONFigure:DATA:MEAS<i>:IPANalysis:IPConnect:FILTer`

Results
The top of the view provides statistical information.

The table in the lower part lists the following information for each connection:
- "Conn. Status": Connection still open or already closed
- "L4 Prot.": Layer 4 protocol (transport layer)
- "L7 Prot.": Layer 7 protocol (application layer)
- "IP Source", "Source Port": IP address and port of the source that has initiated the connection
- "IP Destination", "Dest. Port": IP address and port of the destination addressed by the source
- "Overhead DL / UL": Percentage of a packet consumed by headers (non-payload part of the packet), for downlink and uplink
- "Avg. P-Size DL / UL": Average packet size for downlink and uplink

Remote command:
`SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:STATistics?`
`SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:FLOWid?`
`FETCh:DATA:MEAS<i>:IPANalysis:IPConnect:ALL?`

5.8.4 View Data Pie Charts

This view analyzes the amount of data transported since the measurement was started.

The results are displayed as a list and as a pie chart. For each list entry, the transported data is given as absolute number and as percentage of the total transported data. All values indicate the sum of downlink and uplink.

The list and the pie chart are linked. Select an entry in the list to highlight the corresponding part of the pie chart.

At the top of the view, you can select a view variant "Data per ...". The specifics of the individual categories are described in the following sections.

Data Per Connection

The view provides an overview of the amount of data transported via the individual connections since the measurement was started.

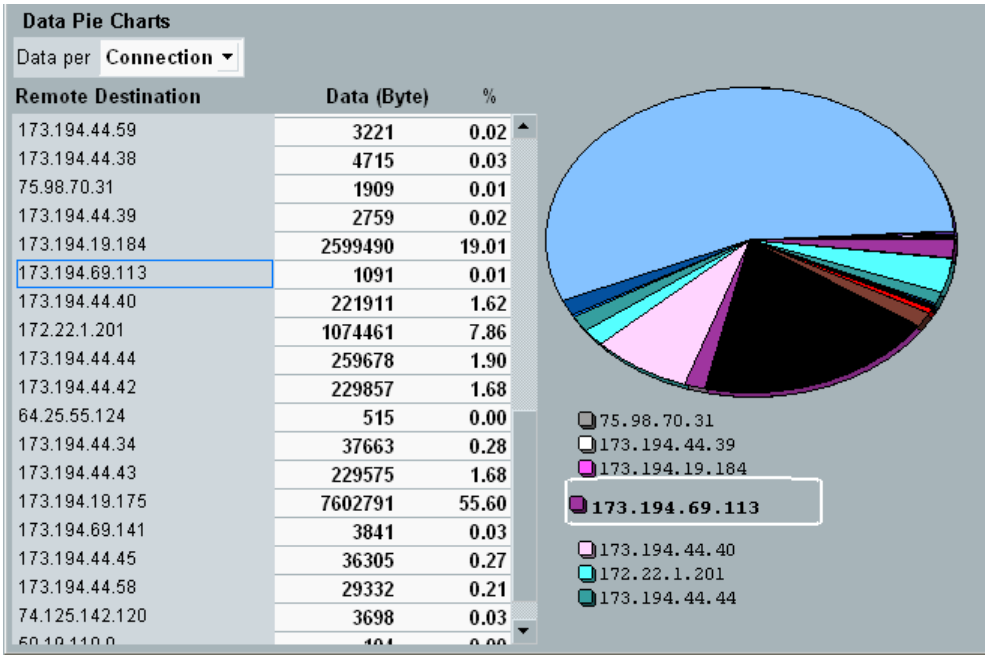


Figure 5-19: Data Pie Charts view - per connection

The term "Remote Destination" refers to the partner destination of the mobile (mobile at one end of the connection, "Remote Destination" at the other end).

Remote command:

`FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPConnection?`

Data Per Protocol

The view provides an overview of the amount of data transported via the individual protocols since the measurement was started.

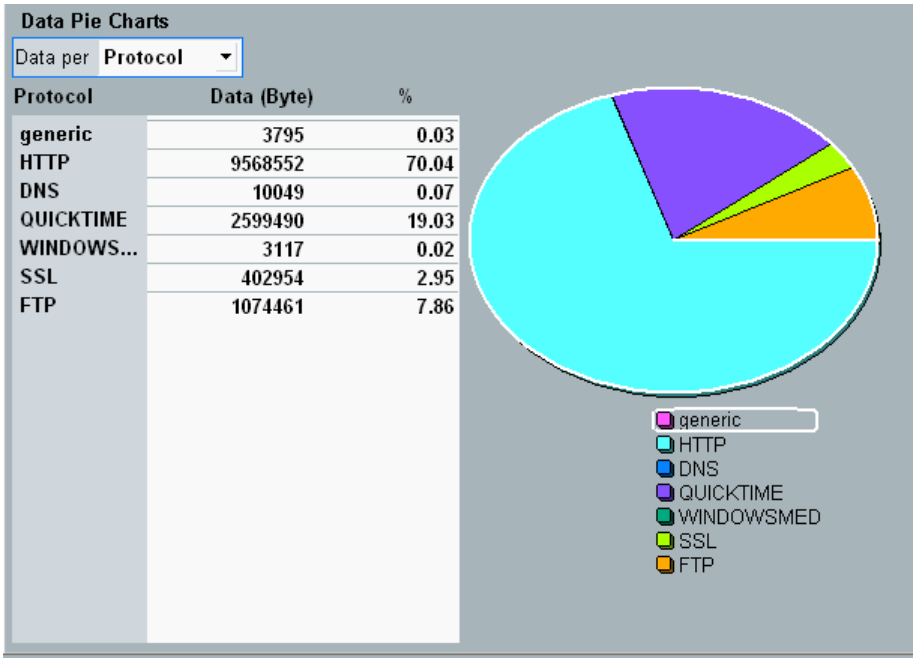


Figure 5-20: Data Pie Charts view - per protocol

Remote command:
`FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPPRotocol?`

Data Per Layer

The view provides a layer-based overview of the amount of data transported since the measurement was started.

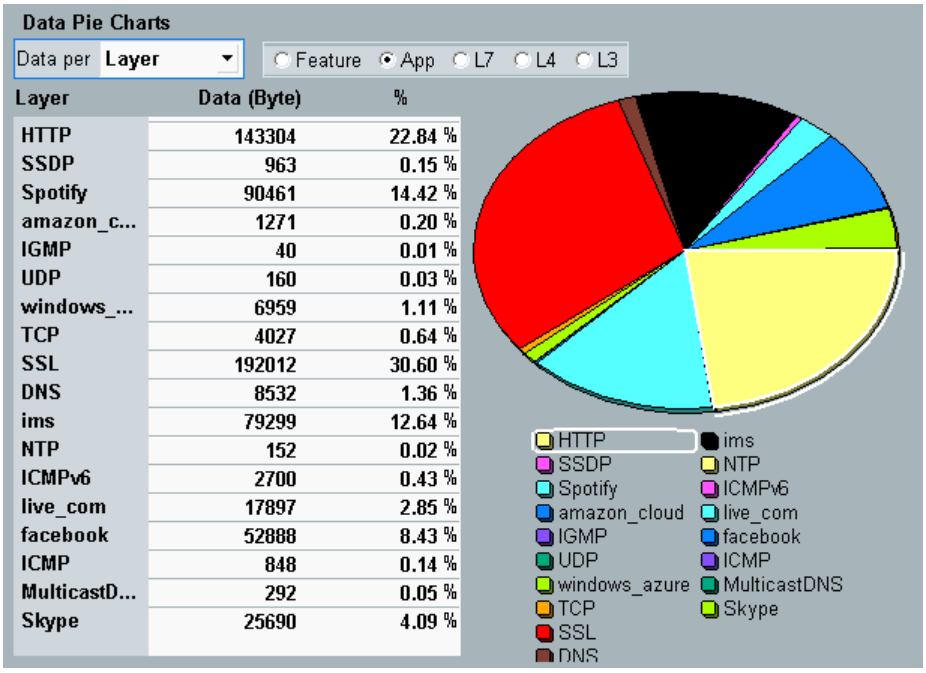


Figure 5-21: Data Pie Charts view - per layer

At the top, select the layer you are interested in.

Examples of features, applications or protocols for the individual layers:

- Feature layer: audio, video, SMS, ...
- Application layer: Facebook, IMS, YouTube, Skype, ...
- Layer 7: SIP, RTP, HTTP, ...
- Layer 4: TCP, UDP, ICMP, IGMP, ...
- Layer 3: IPv4 or IPv6

The packets are analyzed at the selected layer. If a packet cannot be assigned at this layer, it is analyzed at the next lower layer. This step is repeated until the packet can be assigned to a category (feature, application or protocol).

If you select the application layer for example, the results list mainly applications. For packets that cannot be assigned to an application, the L7 protocol is listed. And for packets that cannot be assigned to a L7 protocol, the layer 4 protocol is listed.

Remote command:

```
CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer:LAYer
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer?
```

Data Per Application

The view provides an overview of the amount of data transported via the individual applications since the measurement was started.

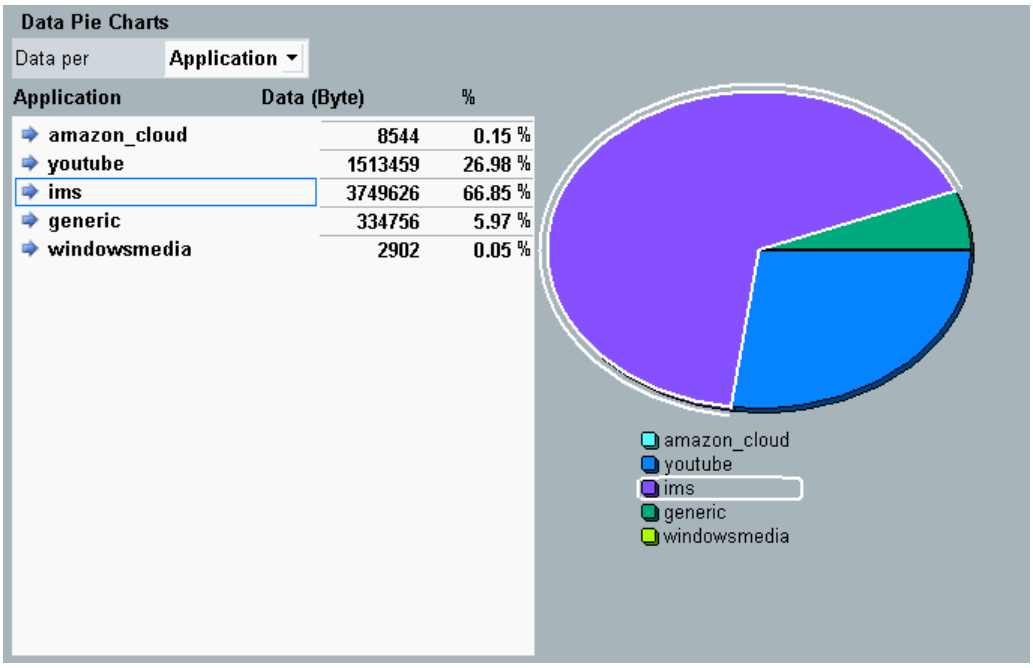


Figure 5-22: Data Pie Charts view - per application

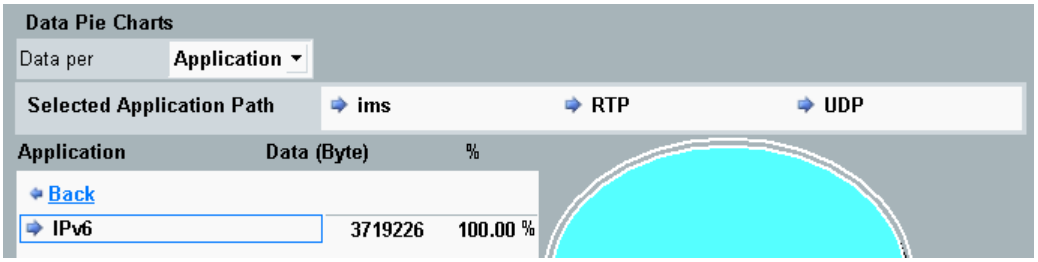
You can select an application and display information for a lower layer.

To navigate to the next lower layer:

- Double-click an entry in the "Application" column, or
- Select an entry in the "Application" column and press the hotkey "Detailed View"

In the following example, this action has been performed three times, to navigate from the application layer (IMS) through layer 7 (RTP) and layer 4 (UDP) to layer 3 (IPv6).

This path is displayed as "Selected Application Path".



To navigate back to the next higher layer:

- Click "Back" or press the hotkey "Back"

Remote command:

```
CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic:APP
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic?
```

5.8.5 View Flow Throughput and Trigger

The upper diagram provides a throughput history per connection. You can select up to 10 connections to be displayed in parallel.

The lower diagram displays a history of the open and close events (number of connections opened or closed per second).

The history covers the previous hour. Configure the X-axis to select the time period you are interested in. The two diagrams are synchronized and show the same time period.

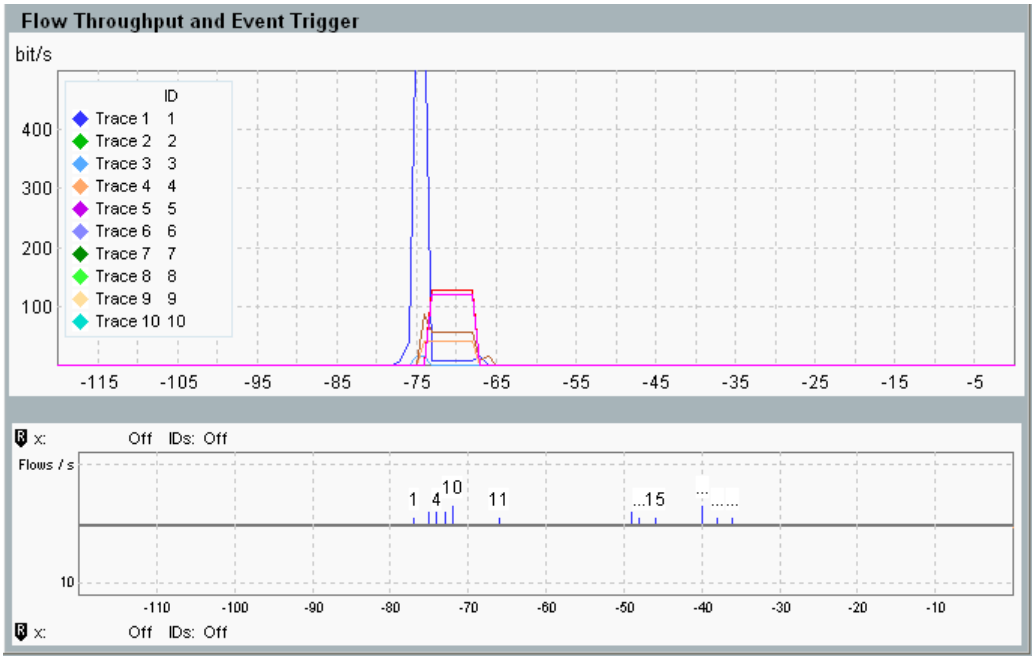


Figure 5-23: Flow Throughput and Event Trigger view

Selecting the connections displayed in the upper diagram

1. Press the softkey "Display" then the hotkey "Traces <-> Flows".
2. Enable/disable the traces as desired.
3. Enter the IDs of the connections that you want to display.
   The "IP Connectivity" view provides a good overview of all connections and their IDs.

Configuring diagram axes

► Press the softkey "Display" then the hotkey "X Scale" or "Y Scale".
"X Scale" configures the X-axis of both diagrams, "Y Scale" the Y-axis of the upper diagram.

Using markers to display IDs in the lower diagram

1. Press the softkey "Marker" then the hotkey "Ref Marker".
2. Configure the "Start Events" marker for "open" events and the "End Events" marker for "close" events.

   As a result, the IDs of all connections opened at the "Start Events" marker position are listed above the diagram. The IDs of all connections closed at the "End Events" marker position are listed below the diagram.

Results

The upper diagram displays the total throughput (uplink plus downlink) of selected connections vs. time. The X-axis indicates the time in seconds, with 0 corresponding to "now".

The lower diagram indicates the number of connections opened per second as positive bars and the number of connections closed per second as negative bars. The labels above/below the bars indicate the ID of one opened/closed connection. To improve the display of labels, zoom into the diagram (scale the X-axis). To display all IDs of a selected bar, see Using markers to display IDs in the lower diagram.

Remote command:

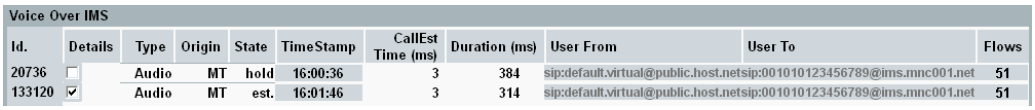
```
CONFigure:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACe<TraceIndex>:TFLowid
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACes<TraceIndex>:FTHRoughput?
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:STARt?
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:END?
```

5.8.6 View Voice over IMS

This view provides information about all voice over IMS audio or video calls that have been established via the internal IMS server since the measurement was started.

In the upper part, the view contains an overview table that lists all voice over IMS calls. The tabs in the lower part provide additional information for one or more calls, selected in the table column "Details".

Table (upper part)
The table contains one row per voice over IMS call.


Voice Over IMS

Id.	Details	Type	Origin	State	TimeStamp	CallEst Time (ms)	Duration (ms)	User From	User To	Flows
20736	☐	Audio	MT	hold	16:00:36	3	384	sip:default.virtual@public.host.net	sip:001010123456789@ims.mnc001.net	51
133120	☑	Audio	MT	est.	16:01:46	3	314	sip:default.virtual@public.host.net	sip:001010123456789@ims.mnc001.net	51

Figure 5-24: Voice Over IMS view - upper part

Table columns:
- "Id.": Call ID
- "Details": Selects the calls, for which detailed information is displayed in the lower part
- "Type": Call type audio, video or emergency
- "Origin": Mobile-terminating (MT) call initiated by the DAU or mobile-originating (MO) call initiated by the mobile
- "State": One of the following call states:
  – Ringing - MT call setup started, mobile is ringing
  – Established - call established
  – Released - established call has been released
  – Hold - call on hold
- "TimeStamp": Time when the call setup was initiated (INVITE message sent/ received)
- "CallEst Time (ms)": Duration of the call setup procedure (from INVITE message to first RTP packet)
- "Duration": Duration of the call (from INVITE message to BYE/Cancel)
- "User From": User ID or phone number of the calling party
  For an MO call, the calling party is the mobile. For an MT call, the calling party is the DAU, simulating a calling mobile.
- "User To": User ID or phone number of the called party
  For an MT call, the called party is the mobile. For an MO call, the called party is the DAU, simulating a called mobile.
- "Flows": IDs of the flows used by the call
  The same IDs are displayed in other views, for example the "IP Connectivity" view and the "Flow Throughput and Trigger" view. So you can use the flow IDs to correlate the information in different views.

Remote command:
`FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?`

Flows Info tab (lower part)
This tab contains a table for audio flows and a table for video flows. The listed flows are related to the calls selected in the upper part, column "Details". There is one line per flow and direction.

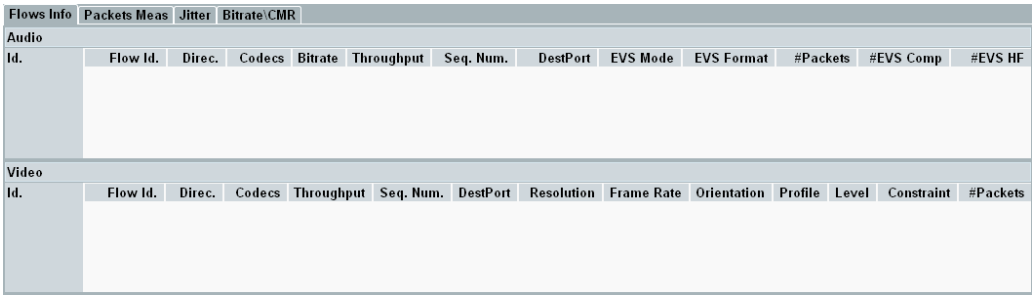


Figure 5-25: Voice Over IMS view - Flows Info tab

To hide or show the individual columns, press the hotkey "Configure Column".



Table columns, for audio and video:

- "Id.": Call ID, same value as in the upper part
- "Flow ID": Same values as in the upper part, column "Flows".
- "Direc.": Flow direction, uplink or downlink
  The two directions are always evaluated separately, even for a bidirectional flow.
- "Codecs": Name of used codec
- "Throughput": Current audio or video data throughput at the RTP level
- "Seq. Num.": Sequence number of the currently processed packet
- "DestPort": Port used at the flow destination
- "#Packets": Number of already processed packets

Table columns, audio only:

- "Bitrate": Audio codec rate
- "EVS Mode": Primary mode or AMR-WB-IO mode
- "EVS Format": Header-full format or compact format
- "#EVS Comp": Number of EVS packets with compact format
- "#EVS HF": Number of EVS packets with header-full format

Table columns, video only:

- "Resolution": Video resolution
- "Frame Rate": Video frame rate
- "Orientation": Counter-clockwise video rotation
- "Profile": H.264 profile, "profile_idc" (baseline, extended, main, high 10, ...)
- "Level": H.264 level (1, 1b, 1.1, ...)
- "Constraint": H.264 constraint set (set n)

Remote command:

`SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:FLOWs?`

Packets Meas tab (lower part)

This tab displays the results of packet measurements for the calls selected in the upper part, column "Details". You can display the bar graphs of up to four calls in parallel. Each X-axis label corresponds to one call.

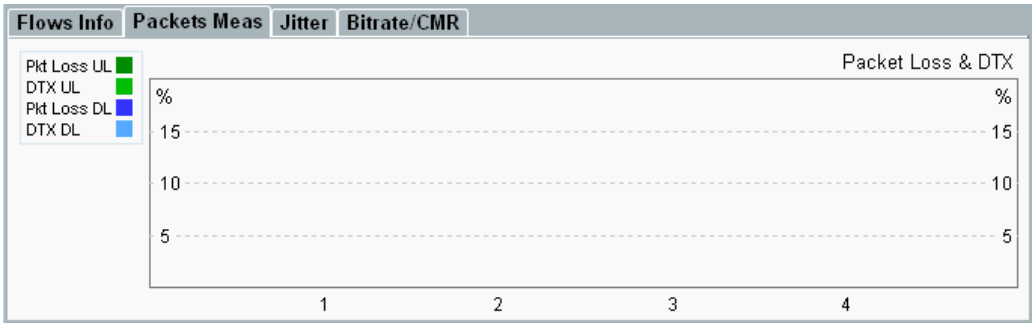


Figure 5-26: Voice Over IMS view - Packets Meas tab

The following measurement results are displayed per direction:

- "Pkt Loss": Number of lost packets as percentage of the total number of packets
- "DTX": Discontinuous transmission rate, that is the number of silent packets as percentage of the total number of packets

Remote command:

`SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:PERDtx?`

Jitter tab (lower part)

This tab displays the jitter measured for the calls selected in the upper part, column "Details". You can display the bar graphs of up to four calls in parallel. Each X-axis label corresponds to one call.

The jitter is measured separately for the uplink and the downlink direction. The maximum, average and minimum jitters for the entire call duration are displayed.

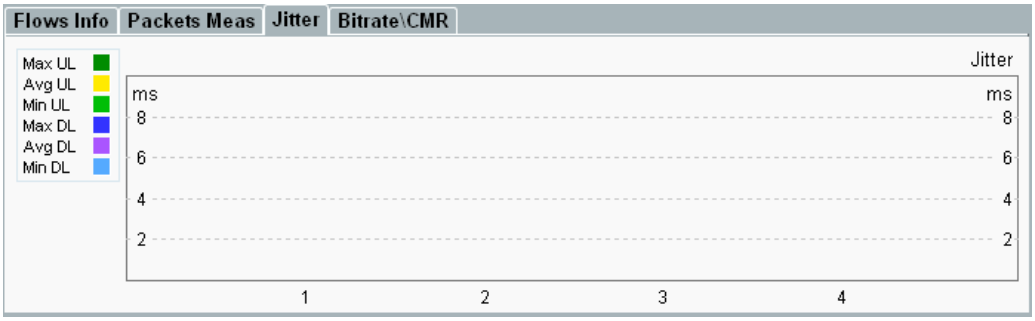


Figure 5-27: Voice Over IMS view - Jitter tab

Remote command:

`SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:JITTer?`

Bitrate\CMR

This tab displays measured bitrates and information from codec mode requests (CMR), for the calls selected in the upper part, column "Details".

You can use the view, for example, to check whether the measured bitrate matches the bitrate requested via CMR.

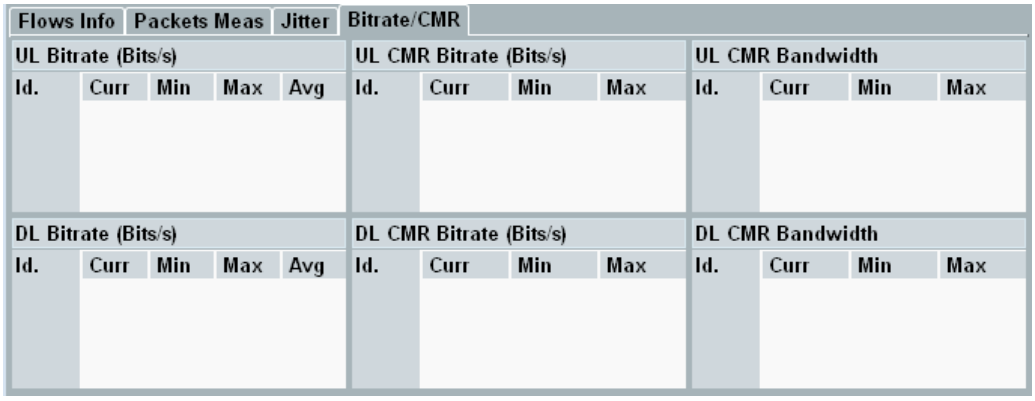


Figure 5-28: Voice Over IMS view - Bitrate\CMR tab

Areas:

- "UL Bitrate": Measured uplink bitrate
- "DL Bitrate": Measured downlink bitrate
- "UL CMR Bitrate": Downlink bitrate requested via uplink CMR
- "DL CMR Bitrate": Uplink bitrate requested via downlink CMR
- "UL CMR Bandwidth": Downlink bandwidth requested via uplink CMR
- "DL CMR Bandwidth": Uplink bandwidth requested via downlink CMR

If you want to compare bitrates with requests, compare UL bitrates with DL CMR bitrates and DL bitrates with UL CMR bitrates.

Table columns:

- "Id.": Call ID, same value as in the upper part
- "Curr.": Current result. No result is displayed, for example, if there is no transmission or if the header contains no codec mode request.
- "Min", "Max", "Avg": Minimum, maximum and average of the measured current results for the entire call duration

Remote command:

`SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:BITRate:CMR?`

5.8.7 View IP Connection Security

The IP connection security analysis evaluates the traffic between a mobile and the Internet under IP security aspects. It provides an overview of security-relevant parameters for each connection.

The provided information includes for example:

- Whether a connection is encrypted at all
- Whether critical information like a password is transmitted unencrypted
- Which SSL/TLS protocol version is used
- Details about the SSL/TLS handshake, including the used key lengths, the parameters offered by the client and the values selected by the server
- Certificates sent by the server
- Country, city and domain of the server end point
- Open IP ports at the mobile

The information allows you to detect security problems of a mobile or security risks of specific apps.

To use the IP connection security analysis, you need option R&S CMW-KM052. Without this option, the "IP Connection Security" view is hidden. Option R&S CMW-KM051 is a prerequisite for option R&S CMW-KM052.

A restart of the measurement clears all results.

In the context of this application, the term "destination" refers to the connection partner of the mobile, typically a server on the Internet. The term is used independent of which side initiated a connection and in which direction the traffic is transmitted.

5.8.7.1 Applications Tab

The "Applications" tab provides an overview of all observed IP connections. It groups the connections for easier analysis, according to applications and destinations. For SSL/TLS connections, you can select a single connection and display handshake and certificate information.

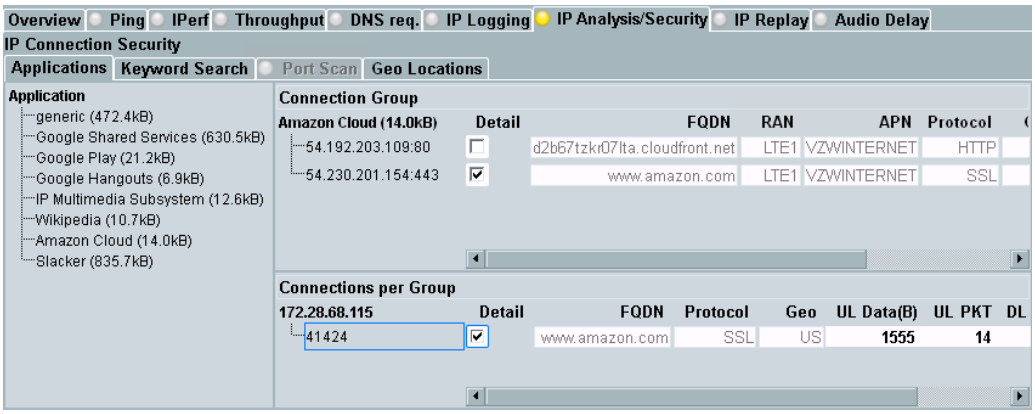


Figure 5-29: Applications tab with three areas

Navigate within the tab as follows:

1. Select an application on the left.

   The area "Connection Group" on the right lists the connection groups for the selected application.

2. Select a connection group (checkbox in column "Detail").

   The area "Connections per Group" below lists all connections of the selected group.

3. Select a connection (checkbox in column "Detail").

   One or two hotkeys are displayed. The checkbox is only available if there is handshake or certificate information for the connection.

4. Press the hotkey "SSL/TLS Handshake" or "SSL/TLS Certificates".

   The handshake or certificates information is displayed for the selected connection.

The areas of the "Applications" tab are described in the following.

Application area 143
Connection Group area 143
Connections per Group area 144
Hotkeys 144
Filter hotkey 145

Application area

The area on the left lists all applications resulting from the IP packet analysis on the application layer.

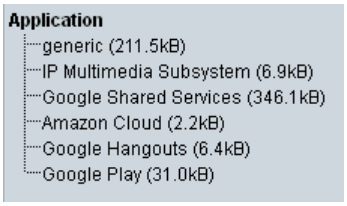


Figure 5-30: Application area

The information in the brackets indicates the total layer 3 data exchanged in UL plus DL direction per application, since the measurement is running.

Remote command:

`SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:APPLications?`

Connection Group area

The area lists all connection groups for the application selected on the left.

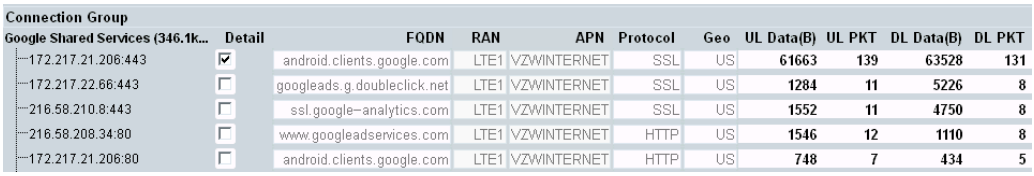


Figure 5-31: Connection Group area

There is one line per connection group. The columns list the following information.

First column	Destination address and destination port, for example 216.58.210.8:443. The application is listed above the first line.
"Detail"	Selects one connection group for the display of details in the lower part.
"FQDN"	Fully qualified domain name of the destination
"RAN"	Used radio access network, typically a signaling application instance like "LTE1"
"APN"	Access point name
"Protocol"	Used protocol, for example SSL or HTTP SSL means here SSL or TLS. If the connections of the group use different protocols, an asterisk * is displayed.
"Geo"	Country of the destination, as two-letter country code (ISO 3166-1)

"UL/DL Data" Layer 3 UL or DL data exchanged via the group, since the measurement is running (in bytes).

"UL/DL PKT" Number of packets exchanged via the group, since the measurement is running.

Remote command:
`FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication?`

Connections per Group area

The area lists all connections for the group selected in the upper part.

All connections of a group use the same IP address at the mobile side, but different port numbers.

Connections per Group

172.28.10.117	Detail	FQDN	Protocol	Geo	UL Data(B)	UL PKT	DL Data(B)	DL PKT
42972	☑	android.clients.google.com	SSL	US	10238	45	14612	39
42973	☐	android.clients.google.com	SSL	US	2116	17	6630	16
42976	☐	android.clients.google.com	SSL	US	18764	24	6594	16
42987	☐	android.clients.google.com	SSL	US	18595	20	2108	19

Figure 5-32: Connections per Group area

There is one line per connection. The columns have the following meaning.

First column Port used at the mobile side. The IP address of the mobile is listed above the first line.

"Detail" Selects one connection for the display of details, see "Hotkeys" on page 144.
The checkbox is only available if there is handshake or certificate information for the connection.

Remaining columns Same information as in the "Connection Group" area, per connection instead of per connection group

Remote command:
`FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication?`

Hotkeys

The following hotkeys are specific for the "Applications" tab.

SSL/TLS Handshake	SSL/TLS Certificates

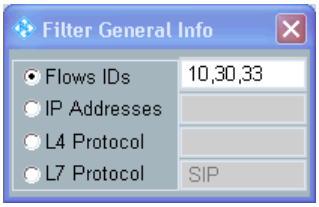
The hotkeys are only present, if you select a connection in the "Connections per Group" area, column "Detail". If there is no handshake information or no certificate information, the corresponding hotkey is hidden.

The hotkeys open views with additional information about the connection, see:

- Chapter 5.8.7.2, "SSL/TLS Handshake View", on page 145
- Chapter 5.8.7.3, "SSL/TLS Certificate Details View", on page 148

While one of these views is open, the hotkey "Back to connection info" is shown, for return to the "Applications" tab.

Filter hotkey
The hotkey "Filter" accesses a dialog with filter settings. The hotkey is available in several views and the settings apply to all these views. So you need to configure the filter only once, not per view.



Select a filter row and enter a filter text. You can enter a single value or a comma-separated list of values. Wildcards are not supported.

The colored indicator on the hotkey tells you whether the filter is active or not.

Remote command:
`CONFigure:DATA:MEAS<i>:IPANalysis:IPConnect:FILTer`

5.8.7.2 SSL/TLS Handshake View

The view is displayed via the hotkey "SSL/TLS Handshake", see "Hotkeys" on page 144.

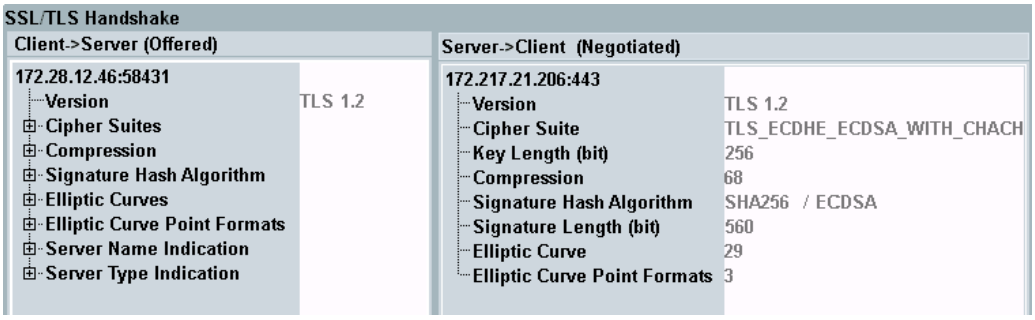


Figure 5-33: SSL/TLS Handshake view

The view shows information about the SSL/TLS handshake between the client (the mobile) and the server (the destination). The handshake is done at the beginning of a secure connection.

The left part lists the values offered by the client. The right part lists the values selected by the server. The client identification and the server identification are displayed at the top of the trees.

The following handshake parameters are listed.

Version 146
Cipher Suites 146
Key Length 146
Compression 147
Signature Hash Algorithm 147

Signature Length....147
Elliptic Curves, Elliptic Curve Point Formats, Elliptic Curve Point Type....147
Server Name Indication, Server Type Indication....147

Version
The protocol version is the most important indicator for the security of a connection. Ideally, an application uses the highest version supported by the mobile, not a lower version. Any protocol version below TLS 1.1 can be considered as insecure.

Table 5-1: Versions with release year, security decreasing from left to right

TLS 1.3	TLS 1.2	TLS 1.1	SSL 3.1 / TLS 1.0	SSL 3.0	SSL 2.0
Draft	2008	2006	1999	1996	1995

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:
HANDshake:OFFered:VERSion?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:
HANDshake:NEGotiated?
```

Cipher Suites
A cipher suite specifies the set of used security methods / algorithms. The suites are listed in IANA notation.

Example for a cipher suite name with all possible components:

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

This string is explained in the following table.

Value	Information type	Purpose
TLS	Protocol	
ECDHE	Key exchange method	Keeping secrets (confidentiality)
RSA	Authentication method	Verifying identities (authenticity)
AES	Encryption algorithm	Keeping secrets (confidentiality)
128 (bit)	Encryption key length	
GCM	Cipher mode	Transport safety (integrity)
SHA256	MAC or PRF algorithm	

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:
HANDshake:OFFered:CIPSuite?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:
HANDshake:NEGotiated?
```

Key Length
Length of the public key of the server in bits. Long keys are more secure than short keys.

You can use the following internet resource to evaluate the key length: https://www.keylength.com/en/compare/.

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?
```

Compression
Number of the compression method. 0 means no compression.

A connection with compression is more vulnerable to attacks than a connection without compression.

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:COMPression?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?
```

Signature Hash Algorithm
Pairs of hash algorithm (for example SHA256) and signature algorithm (for example ECDSA) for digital signatures, see RFC 5246, section 7.4.1.4.1.

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SHALgorithm?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?
```

Signature Length
Length of the server signature in bits.

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?
```

Elliptic Curves, Elliptic Curve Point Formats, Elliptic Curve Point Type
Additional parameters for cipher suites using elliptic curve cryptography (ECC), see RFC 4492, section 5.1.

Remote command:
```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECURve?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECPFormat?
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated?
```

Server Name Indication, Server Type Indication
The server name indication (SNI) specifies the name of the server that the client wants to contact. The identification can, for example, be a hostname or an IP address.

Remote command:
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SRINdication?

5.8.7.3 SSL/TLS Certificate Details View

The view is displayed via the hotkey "SSL/TLS Certificates", see "Hotkeys" on page 144.

The view shows information about the certificates provided by the destination.

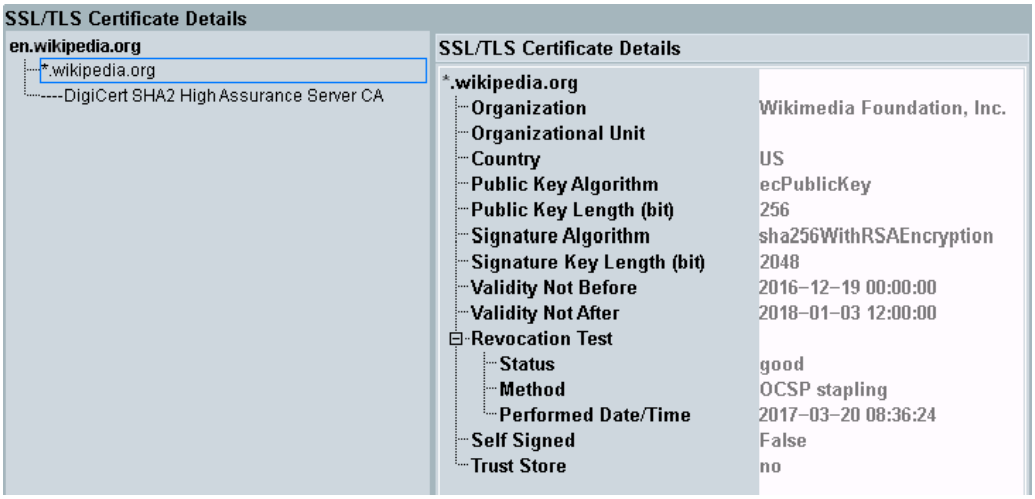


Figure 5-34: SSL/TLS Certificate Details view

The left part lists a tree of certificates. The root of the tree indicates the destination. The other entries show the certificate chain, from the end-user certificate of the domain via intermediate certificates down to the trusted root certificate.

Select an entry in the tree on the left to show the certificate details on the right.

5.8.7.4 Keyword Search Tab

Critical data like passwords, PINs or IMSIs must be kept secret. Only encrypted transmission is allowed for such data. Privacy must be ensured.

To check whether unencrypted transmissions contain critical data, you can use the "Keyword Search" tab.

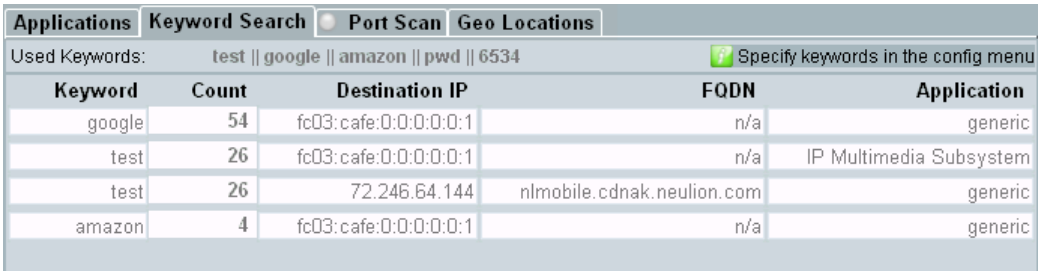


Figure 5-35: Keyword Search tab

Use the tab as follows:

1. While the measurement is off, press the hotkey "Config".

   The configuration dialog box opens.

2. Configure the list of critical keywords that must not appear in unencrypted transmissions.
   For details, see "Keyword configuration settings" on page 149.

   Your current list of keywords is listed at the top of the "Keyword Search" tab, as "Used Keywords".

3. Start the measurement.

4. Generate the traffic that you want to search for keywords.

   If a keyword is found in an unencrypted transmission, the keyword is listed in the result table together with connection information.
   For further analysis, you can create log files with the IP logging application of the DAU and analyze them with Wireshark. See also Chapter 5.7, "IP Logging Tab", on page 123.

The settings and the result table are described in the following.

Keyword configuration settings 149
Keyword search results 150

Keyword configuration settings
Open the configuration dialog box via the hotkey "Config", while the measurement is off.

The relevant settings are contained in the node "Keyword Search".



Figure 5-36: Keyword search settings

"Add Keyword" adds an entry to the keyword list. Enter the keyword and press the "+" button or the ENTER key.

"Remove Keyword" removes an entry from the list. Select the entry and press the "-" button.

Remote command:

```
CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:ADD
CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:
DELete
SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:LIST?
```

Keyword search results

The result table contains one line for each keyword found in unencrypted transmissions. Both traffic directions are searched (from the Internet to the mobile and vice versa). The results do not distinguish between the two directions.

Keyword	Count	Destination IP	FQDN	Application
google	54	fc03:cafe:0:0:0:0:0:1	n/a	generic
test	26	fc03:cafe:0:0:0:0:0:1	n/a	IP Multimedia Subsystem
test	26	72.246.64.144	nlmobile.cdnak.neulion.com	generic
amazon	4	fc03:cafe:0:0:0:0:0:1	n/a	generic

Figure 5-37: Result table

The table contains the following columns:

- "Keyword": found keyword
- "Count": how often the keyword was found
- "Destination IP": IP address of the destination
- "FQDN": FQDN of the destination
- "Application": application using the connection

Remote command:

```
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch?
```

5.8.7.5 Geo Locations Tab

This tab allows you to analyze the locations of the servers addressed by the mobile. The locations are displayed as dots on a world map. Move your mouse over a location, to see connection information.

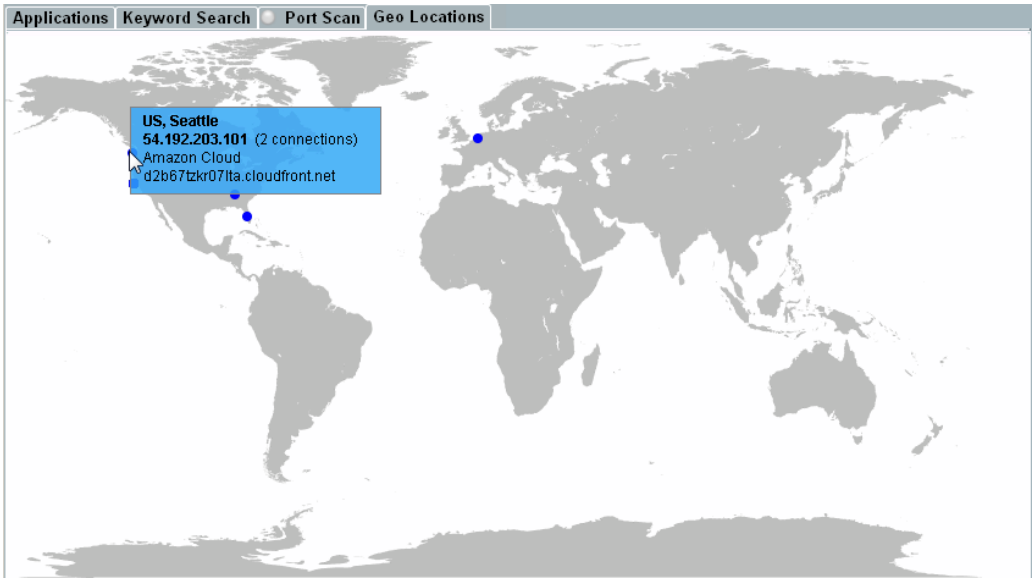


Figure 5-38: Geo Locations tab

The map considers all destinations for which the country and the city can be determined. Only destinations on the Internet are shown, not local destinations within the DAU.

Location details
The map shows a single dot for each city with destinations. A dot can be related to several destinations and to several connections per destination.

If you move your mouse over a dot, a popup lists information for all destinations in the city:

- Country and city
- IP address of the destination
- Number of connections to the destination
- Application
- FQDN of the destination

Remote command:
`FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication?`

5.9 IP Replay Tab

The "IP Replay" application allows you to replay previously captured IP traffic. This feature is, for example, useful for multicast tests. You record the multicast (for example a video stream) into a file and then replay it to the mobile under test.

The IP traffic must be captured into a PCAP file (packet capture file). You can use Wireshark for this purpose. The "IP Logging" application of the DAU also creates PCAP files.

You can replay a single PCAP file or several PCAP files in parallel. The traffic can be sent to the mobile or to the LAN DAU connector.

If you want to replay traffic to a mobile, establish a connection via a signaling application or a protocol test application, before proceeding.

To replay captured files, proceed as follows:

1. Store the files to the samba share of the DAU, into the folder `ip_replay`.
2. Press the "Edit Playlist" hotkey. Configure the playlist.

   All files in the playlist are played simultaneously.
3. Start the "IP Replay" application.
4. If the playlist contains more than one file, sequentially select each file in the upper part of the tab. This triggers file analysis and display of the resulting information. Wait until information is available for all files.
5. Press the "Play" hotkey to start the playlist.

The contents of the "IP Replay" tab and the related dialogs, softkeys and hotkeys are described below.

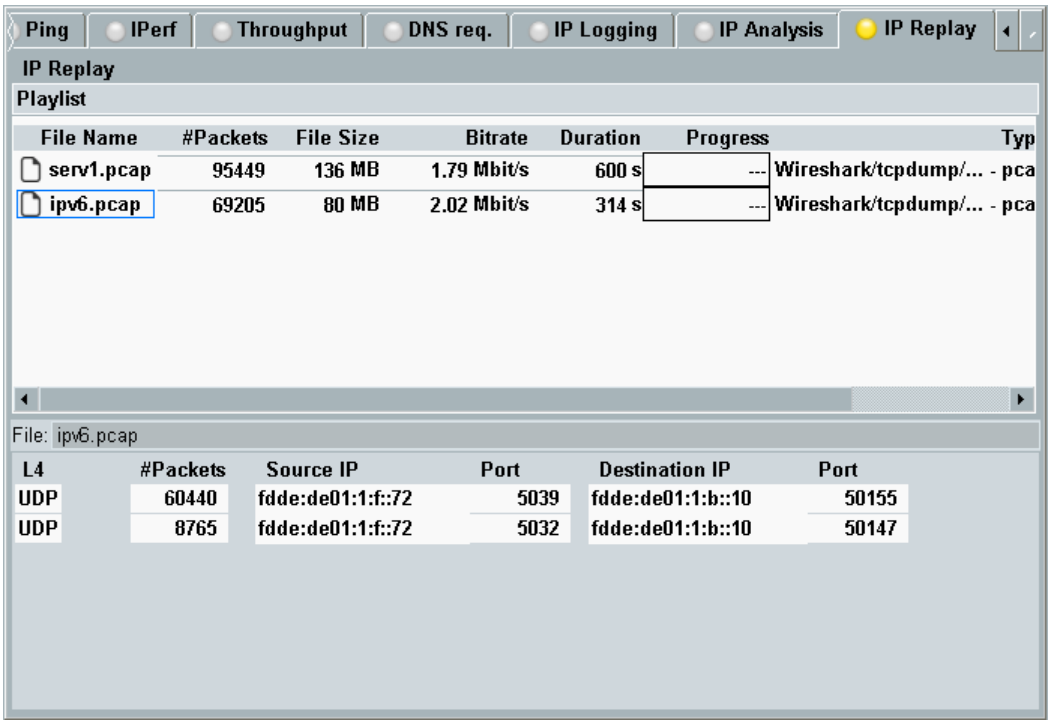


Figure 5-39: Measurements - IP Replay tab

Playlist (upper part of tab)..... 153
Connection information (lower part of tab)..... 153
Play/Stop (hotkeys)..... 153
Clear Playlist (hotkey)..... 153
Edit Playlist (hotkey)..... 154
Application control..... 154

Playlist (upper part of tab)
The upper part of the tab lists information about all files in the playlist. To derive this information, the application must analyze the files.

The first file is analyzed automatically when the "IP Replay" application is started. The other files are only analyzed if you select them in the list.

Analyzing a large file takes some time. Analyzing many large files in parallel can affect the performance of the DAU. The results of measurements running in parallel can be distorted.

The information listed for each file includes:
- File name, number of IP packets in the file, file size, bit rate and duration of the file
- "Progress" displays a progress bar for each file, while the files are replayed.
- "Type" indicates the file type and information about the application used for capturing the file.
  Only PCAP files can be replayed. If a different type is displayed, the file is no PCAP file, despite its PCAP file-name extension.
- "Encapsulation" indicates whether the file contains raw IP traffic or IP traffic plus Ethernet headers.

Remote command:
`SENSe:DATA:MEAS<i>:IPReplay:INFofile?`
`SENSe:DATA:MEAS<i>:IPReplay:PROGress?`

Connection information (lower part of tab)
The lower part of the tab lists all IP connections contained in the PCAP file that is selected in the upper part.

The information listed for each IP connection includes:
- The layer 4 protocol and the number of IP packets
- The IP address and port number of the sender (source) and of the recipient (destination) of the IP packets

Remote command:
`SENSe:DATA:MEAS<i>:IPReplay:TRAFficfile?`

Play/Stop (hotkeys)
The "Play" hotkey starts replaying the playlist. The "Stop" hotkey aborts replaying. It is not possible to pause and continue a replay.



The "Play" hotkey is only active if the "IP Replay" application is running and the playlist is not empty.

Remote command:
`CONFigure:DATA:MEAS<i>:IPReplay:PLAYall`
`CONFigure:DATA:MEAS<i>:IPReplay:STOPall`

Clear Playlist (hotkey)
The "Clear Playlist" hotkey is only active if the "IP Replay" application is switched off.



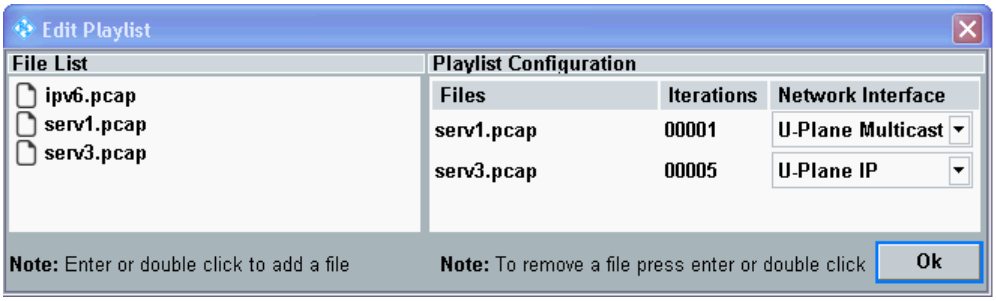
The hotkey removes all files from the playlist.

Remote command:
`CONFigure:DATA:MEAS<i>:IPReplay:REMovelist`

Edit Playlist (hotkey)
The "Edit Playlist" hotkey is only active if the "IP Replay" application is switched off.



It opens a dialog for configuration of the playlist.



The left part lists all PCAP files stored in the directory `ip_replay` of the samba share.

The right part displays the playlist contents.

Double-click a file on the left to add it to the playlist. Double-click a file in the playlist to remove it from the list.

For each file in the list, you can configure how often the file is replayed and to which network interface:

- "U-Plane IP": IP unicast traffic to the mobile
- "U-Plane Multicast": IP multicast traffic to the mobile
- "LAN-DAU": IP traffic to the LAN DAU connector

Remote command:
`FETCh:DATA:MEAS<i>:IPReplay:FILelist?`
`CONFigure:DATA:MEAS<i>:IPReplay:CREatelist`
`CONFigure:DATA:MEAS<i>:IPReplay:ITERation`
`CONFigure:DATA:MEAS<i>:IPReplay:INTerface`

Application control
Turn the "IP Replay" application on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "IP Replay" softkey.

The softkey shows the current state. Additional substates can be retrieved via remote control.

Remote command:
`INITiate:DATA:MEAS<i>:IPReplay`
`STOP:DATA:MEAS<i>:IPReplay`
`ABORt:DATA:MEAS<i>:IPReplay`

```
FETCh:DATA:MEAS<i>:IPReplay:STATe?
FETCh:DATA:MEAS<i>:IPReplay:STATe:ALL?
```

5.10 Audio Delay Tab

The "audio delay" measurement analyzes an established voice over IMS call. It determines how long the audio information needs to pass the R&S CMW. And it determines the timing error of audio packets received in the uplink direction.

Set up only a single call to get meaningful measurement results.

There are two measurement modes, depending on the media endpoint used by the call. The used media endpoint is configured in the settings of the virtual subscriber involved in the call, see "Media Endpoint" on page 91.

For calls with the media endpoint "Audioboard", "Uplink" and "Downlink" results are displayed.



Figure 5-40: Audio Delay tab, media endpoint Audioboard

For calls with the media endpoint "Loopback", only "Loopback" results are displayed.

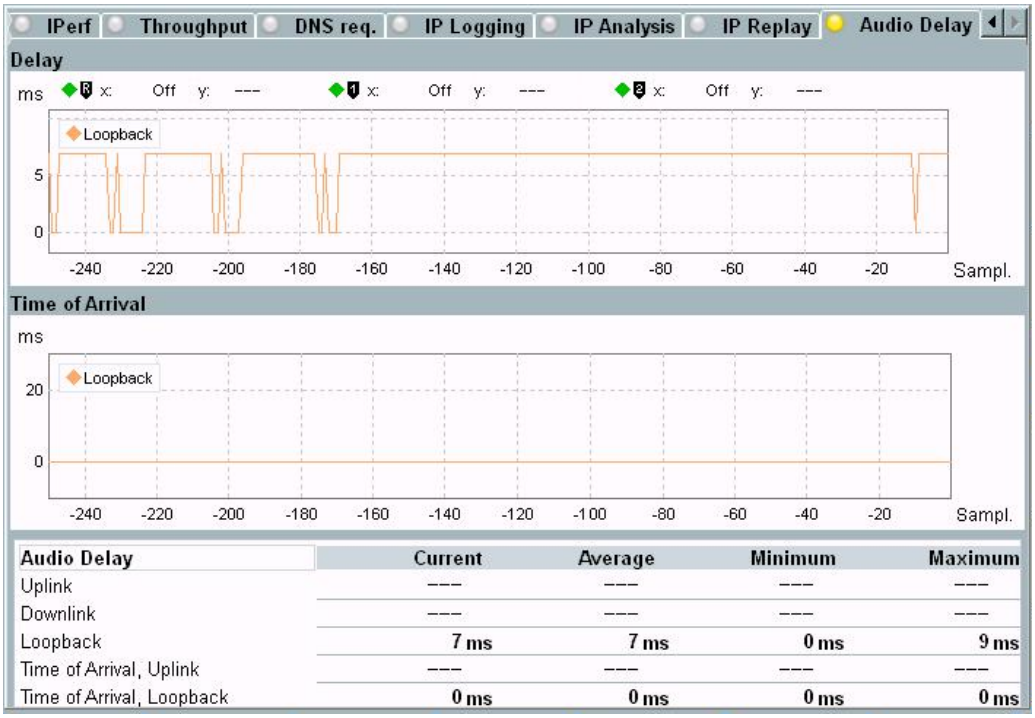


Figure 5-41: Audio Delay tab, media endpoint Loopback

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Time of Arrival result diagram 157
Result table 157
Settings 157
Measurement control 158

Delay result diagram

The upper diagram shows the measured audio delays vs. the received samples. Each sample corresponds to 20 ms.

The audio delay is the system simulator delay of the R&S CMW. It is measured according to the following definition specified in 3GPP TS 26.132 Rel-12:

- "Uplink":
  System simulator delay T_{TES} in the uplink direction (mobile sending). The delay starts with the last bit of a speech frame at the system simulator antenna (R&S CMW RF input connector). It ends with the first electrical event at the electrical access point of the test equipment (R&S CMW audio output connector).
- "Downlink":
  System simulator delay T_{TER} in the downlink direction (mobile receiving). The delay starts with the first electrical event at the electrical access point of the test equipment (R&S CMW audio input connector). It ends with the first bit of the corresponding speech frame at the system simulator antenna (R&S CMW RF output connector).
- "Loopback":
  System simulator delay in echo mode T_{SS}. The delay starts with the last bit of a received speech frame at the system simulator antenna (R&S CMW RF input con-

nector). It ends with the first bit of the looped back speech frame at the system simulator antenna (R&S CMW RF output connector).

For "Uplink" and "Loopback", ensure that the mobile has enough audio input to send an audio packet every 20 ms. Silence periods can result in skipped audio packets and peaks in the trace. A peak every 8 samples, for example, means that the mobile has sent an audio packet every 160 ms.

Remote command:

```
READ:DATA:MEAS<i>:ADELay:TRACe:ULINk[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:DLINk[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:LOOPback[:CURRent]?
```

Plus corresponding `FETCh` commands.

Time of Arrival result diagram

The lower diagram shows the timing error of audio packets received in the uplink direction. Ideally, an audio packet is received every 20 ms and a flat line at 0 ms is shown.

Remote command:

```
READ:DATA:MEAS<i>:ADELay:TRACe:TAULink[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:TALoopback[:CURRent]?
```

Plus corresponding `FETCh` commands.

Result table

The table below the diagrams provides a statistical evaluation for all measurement results.

The "Current" value refers to the sample number 0 in the diagrams.

The "Average", "Minimum" and "Maximum" values are calculated from all results since the beginning of the measurement.

Remote command:

```
READ:DATA:MEAS<i>:ADELay:ULINk?
READ:DATA:MEAS<i>:ADELay:DLINk?
READ:DATA:MEAS<i>:ADELay:LOOPback?
READ:DATA:MEAS<i>:ADELay:TAULink?
READ:DATA:MEAS<i>:ADELay:TALoopback?
```

Plus corresponding `FETCh` commands.

Settings

Press the "Config" hotkey to access the measurement settings.

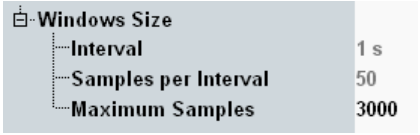


Figure 5-42: Measurement settings

- "Interval": displays the fixed time interval between two display updates
- "Samples per Interval": displays the fixed number of measurement samples per interval

- "Maximum Samples": configures the maximum size of the X-axis of the diagrams. Measured data is kept within that range. Older data is discarded.

Remote command:

```
CONFigure:DATA:MEAS<i>:ADELay:SPINterval?
CONFigure:DATA:MEAS<i>:ADELay:SAMPles?
CONFigure:DATA:MEAS<i>:ADELay:MSAMples
```

Measurement control

Turn the measurement on or off using the ON | OFF or RESTART | STOP keys. Alternatively, right-click the "Audio Delay" softkey.

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:DATA:MEAS<i>:ADELay
STOP:DATA:MEAS<i>:ADELay
ABORt:DATA:MEAS<i>:ADELay
FETCh:DATA:MEAS<i>:ADELay:STATe?
FETCh:DATA:MEAS<i>:ADELay:STATe:ALL?
```

5.11 Quality-of-Service Settings

The quality-of-service (QoS) settings apply to IP traffic from the DAU to a specific destination IP address. The destination can, for example, be a mobile or a host at the LAN DAU connector.

You can prioritize the traffic and you can apply network impairments. Real networks are not perfect. IP packets can, for example, be lost, delayed, reordered, duplicated or corrupted. Such network impairments can be simulated by the DAU.

The QoS settings can be configured and controlled from all measurement tabs. Press the "Quality of Service" softkey to open the configuration dialog box.

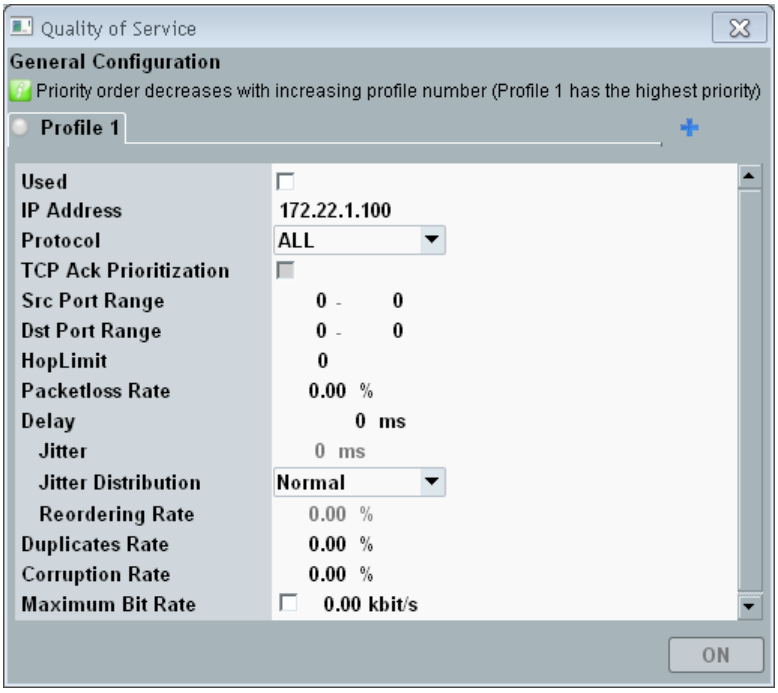


Figure 5-43: QoS configuration dialog

You can create several profiles (tabs) with different settings. Each profile contains filter settings and settings to be applied to IP traffic that matches the filter settings.

A profile is only used, if it is enabled ("Used" checkbox) and if the QoS feature is activated ("ON" button). The state of each profile is shown via a colored icon on the tab header:

- Gray: profile disabled
- Green: profile enabled, QoS feature deactivated
- Yellow: profile enabled, QoS feature activated

If the QoS feature is activated, the outgoing IP traffic is prioritized according to the enabled profiles:

- Highest priority: Traffic that matches the enabled profile with the lowest number
- Second-highest priority: Traffic that matches the enabled profile with the second-lowest number
  All enabled profiles build a priority chain, from the profile with the lowest number to the profile with the highest number.
- Lowest priority: Traffic that does not match any enabled profile

Example: There are seven profiles. Only the profiles 2, 4 and 7 are used. Traffic that matches the filter settings of profile 2 gets the highest priority. Then comes traffic for profile 4, then traffic for profile 7, finally any other traffic.

The QoS settings are described in the following. For a more detailed description of the network impairment settings, see Chapter 9, "Annex: Network Impairments", on page 359.

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└ Reordering Rate....162
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└ Corruption Rate....162
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QoS activation....162

Add or delete a tab

creates a tab / profile.

deletes the currently selected tab. Tab 1 cannot be deleted.

Remote command:

```
CONFigure:DATA:MEAS<i>:QOS:FILTer:ADD
SOURce:DATA:MEAS<i>:QOS:FILTer:CATalog?
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:REMove
```

Used

A profile is only applied if this parameter is enabled.

Remote command:

```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:ENABle
```

Filter criteria

The following settings define a filter. The filter actions are applied to traffic matching the filter criteria.

IP Address ← Filter criteria

IPv4 or IPv6 address of the traffic destination

As IPv6 address, you can enter an IPv6 address or an IPv6 prefix. Examples:

- IPv6 address: `fc01:abab:cdcd:efe0::2`
- IPv6 prefix: `fc01:abab:cdcd:efe0::/64`

Remote command:

```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:IPADdress
```

Protocol ← Filter criteria

TCP traffic only, UDP traffic only or no filtering.

Remote command:

```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PROTocol
```

Src Port Range / Dst Port Range ← Filter criteria
Specifies a port range used at the DAU ("Src") or at the traffic destination ("Dst"). The port range 0 to 0 disables the filter criterion.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:SRCPrange
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PRANge
```

Filter Actions
The following settings are applied to traffic matching the filter criteria.

TCP ACK Prioritization ← Filter Actions
This setting is configurable if "Protocol" equals "TCP".

With enabled checkbox, TCP acknowledgments are prioritized over other packets.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:TCPackprio:ENABle
```

Hop Limit ← Filter Actions
Number of allowed hops before the packet is discarded.

The QoS feature writes the configured value to the hop limit field (IPv6) or to the time to live field (IPv4) of the packet. If zero is configured, the QoS feature does not modify the packet.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:HOPLmt
```

Packetloss Rate ← Filter Actions
On average, the specified percentage of outgoing IP packets is discarded.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PLRate
```

Delay ← Filter Actions
Delays all outgoing IP packets by the specified time.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DELay
```

Jitter, Jitter Distribution ← Filter Actions
The jitter specifies a packet delay variation. The maximum accepted jitter value equals the specified delay time.

The distribution of the total delays is defined via the "Jitter Distribution". You can configure a uniform distribution, a normal (Gaussian) distribution, a pareto or a pareto normal distribution.

For details, see Chapter 9.1, "Delay and Jitter", on page 359.

Remote command:
```
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JITTer
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JDIStribut
```

Reordering Rate ← Filter Actions
On average, the indicated percentage of outgoing packets is reordered. Reordering is only possible, if a sufficient delay time is specified.

Remote command:
`CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:RRATe`

Duplicates Rate ← Filter Actions
On average, the indicated percentage of outgoing packets is duplicated.

Remote command:
`CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DRATe`

Corruption Rate ← Filter Actions
On average, the indicated percentage of outgoing packets is corrupted, by introducing a single bit error anywhere in the packet (header or payload).

Remote command:
`CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:CRATe`

Maximum Bit Rate ← Filter Actions
Maximum bit rate of the connection. The checkbox enables/disables the limitation.

Remote command:
`CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:BITRate`

QoS activation
To use the QoS feature, you must configure and enable a profile (one tab) and activate the QoS feature.

You can activate/deactivate the QoS feature via the button "ON/OFF" at the bottom of the configuration dialog box.

You can also right-click the "Quality of Service" softkey. The softkey shows the current state.

Remote command:
`SOURce:DATA:MEAS<i>:QOS:STATe`

5.12 Common Softkeys and Hotkeys

Some softkeys and hotkeys are displayed independent of the selected tab or are available for several measurements. These keys are described below. For measurement-specific hotkeys, refer to the description of the measurement tab.

Configure Services (softkey)
Opens the "Data Application Control" dialog, see Chapter 4, "DAU Services GUI Reference", on page 50.

Use this softkey for quick access, for example to configure the IP configuration or embedded DAU services.

Quality of Service (softkey)
Opens the "Quality of Service" dialog, see Chapter 5.11, "Quality-of-Service Settings", on page 158.

Display (softkey)
Some measurements provide a "Display" softkey for configuration of result diagrams. Press the softkey to display the related hotkeys. Which of the following hotkeys are available, depends on the measurement.

Hotkey	Description
"Select Traces ..."	Select the trace types to be displayed in the diagram.
"Scale X / Scale Y"	Configure the X-axis and the Y-axis.

Marker (softkey)
Some measurements provide a "Marker" softkey for configuration of markers in result diagrams. Press the softkey to display the related hotkeys. Which of the following hotkeys are available, depends on the measurement.

Hotkey	Description
"Ref. Marker"	Enable or disable the reference marker and select the marker position. If several traces can be displayed, a trace can also be selected.
"Marker 1/2"	Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker). If several traces can be displayed, a trace can also be selected.

Signaling Parameter (softkey)
Provides quick access to essential settings of a coupled signaling application, see "Select RAN" on page 111.

While the softkey is selected, the "Config" hotkey opens the configuration dialog of the signaling application, not the configuration dialog of the DAU measurement.

... Signaling (softkey)
This softkey controls the coupled signaling application, see "Select RAN" on page 111.

The softkey label indicates the currently coupled application, for example "LTE 1 Signaling".

Select the softkey and press ON | OFF to turn the downlink signal transmission on or off. Alternatively, right-click the softkey.

Press the softkey two times (select it and press it again) to switch to the signaling application.

While the softkey is selected, the "Config" hotkey opens the configuration dialog of the signaling application, not the configuration dialog of the DAU measurement.

Select Applic (hotkey)
Selects the tab to be displayed.

Remote command:
`CONFigure:DATA:MEAS<i>:SELect:APP`

6 Programming

The following sections provide programming examples for the data application unit.

The examples contain SCPI commands supported by the R&S CMW and the following symbolic scripting commands:

- `// <comment>`:
  A <comment> ignored by the used programming tool
- `WHILE <expression>`:
  Waits until the <expression> is not / no longer true. The <expression> usually evaluates a value returned by a query.
- `WAITKEY <message>`:
  Displays a dialog box with a <message> and waits until you close the box.
- `PAUSE <time>`:
  Waits a certain number of seconds.

See also: "Remote Control" in the R&S CMW base unit manual

- Data Application Control 164
- Data Application Measurements 175

6.1 Data Application Control

This section provides programming examples for the SCPI commands with the syntax `...:DATA:CONTrol:....`

6.1.1 Rebooting the DAU

During normal operation, it is not required to reboot the DAU. But it can make sense for troubleshooting. In that case, proceed as follows.

```
// ******************************************************************************
// Switch the DAU off and wait until the procedure is complete.
// Then switch the DAU on again and wait until the reboot is complete.
// ******************************************************************************
SOURce:DATA:CONTrol:STATe OFF; *OPC?
SOURce:DATA:CONTrol:STATe ON; *OPC?
```

6.1.2 Configuring General IP Settings

```
// ******************************************************************************
// Configure the maximum transmission unit.
// ******************************************************************************
CONFigure:DATA:CONTrol:MTU 1700
```

6.1.3 Configuring IPv4 Settings

```
// ***************************************************************************
// Enable static IPv4 address configuration and configure the subnet mask,
// the address of the DAU and the address of the gateway.
// ***************************************************************************
CONFigure:DATA:CONTrol:IPVFour:ADDRess:TYPE STATic
CONFigure:DATA:CONTrol:IPVFour:STATic:SMASk '255.255.255.0'
CONFigure:DATA:CONTrol:IPVFour:STATic:IPADdress '192.168.168.180'
CONFigure:DATA:CONTrol:IPVFour:STATic:GIP '192.168.168.1'

// ***************************************************************************
// Verify which addresses are really currently used.
// ***************************************************************************
SENSe:DATA:CONTrol:IPVFour:CURRent:IPADdress?
SENSe:DATA:CONTrol:IPVFour:CURRent:SMASk?
SENSe:DATA:CONTrol:IPVFour:CURRent:GIP?

// ***************************************************************************
// Query the mobile address pool, delete the first entry and add a new entry.
// ***************************************************************************
SENSe:DATA:CONTrol:IPVFour:STATic:ADDResses:CATalog?
CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:DELete 0
CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:ADD '192.168.168.22'

// ***************************************************************************
// If you configure the mobile address pool via DHCP or use the automatic
// standalone configuration, you can query the pool via the following commands.
// ***************************************************************************
SENSe:DATA:CONTrol:IPVFour:DHCP:ADDResses:CATalog?
SENSe:DATA:CONTrol:IPVFour:AUTomatic:ADDResses:CATalog?
```

6.1.4 Configuring IPv6 Settings

```
// ***************************************************************************
// Enable static IPv6 address configuration and configure
// the address of the DAU and the address of the default router.
// ***************************************************************************
CONFigure:DATA:CONTrol:IPVSix:ADDRess:TYPE STATic
CONFigure:DATA:CONTrol:IPVSix:STATic:ADDRess 'fcb1:c014::1/64'
CONFigure:DATA:CONTrol:IPVSix:STATic:DROuter 'fcb1:c014::1'

// ***************************************************************************
// Verify which addresses are really currently used.
// ***************************************************************************
SENSe:DATA:CONTrol:IPVSix:CURRent:IPADdress?
SENSe:DATA:CONTrol:IPVSix:CURRent:DROuter?

// ***************************************************************************
```

```
// Enable static configuration of the mobile address pool.
// Query the mobile address pool, delete the first entry and add a new entry.
// ********************************************************************************
CONFigure:DATA:CONTrol:IPVSix:MOBile:PREFix:TYPE STATic
SENSe:DATA:CONTrol:IPVSix:STATic:PREFixes:CATalog?
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:DELete 0
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:ADD 'fcb1:abab:cdcd:efe3::/64'

// ********************************************************************************
// If you configure the mobile address pool via DHCP or use the automatic
// standalone configuration, you can query the pool via the following commands.
// ********************************************************************************
SENSe:DATA:CONTrol:IPVSix:DHCP:PREFixes:CATalog?
SENSe:DATA:CONTrol:IPVSix:AUTomatic:PREFixes:CATalog?

// ********************************************************************************
// Enable manual configuration of routes and query the existing routes.
// Add two new routes, then delete the first one.
// ********************************************************************************
CONFigure:DATA:CONTrol:IPVSix:ROUTing:TYPE MAN
SENSe:DATA:CONTrol:IPVSix:MANual:ROUTing:CATalog?
CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:ADD 'fcb1:014f::/64','fcb1:abcd::1'
CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:ADD 'fcb1:afce::/64','fcb1:abcd::1'
CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:DELete 'fcb1:014f::/64'
```

6.1.5 Configuring DNS Server Settings

```
// ********************************************************************************
// Configure server types: primary internal and foreign, secondary foreign
// ********************************************************************************
CONFigure:DATA:CONTrol:DNS:PRIMary:STYPe IAForeign
CONFigure:DATA:CONTrol:DNS:SECondary:STYPe FOReign

// ********************************************************************************
// Query used primary and secondary DNS server addresses for IPv4 and IPv6.
// ********************************************************************************
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:PRIMary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:SECondary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:PRIMary:ADDRess?
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:SECondary:ADDRess?

// ********************************************************************************
// Configure primary and secondary addresses for foreign DNS servers.
// Specify whether an address received via DHCP overrides these addresses.
// ********************************************************************************
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:PRIMary:ADDRess '192.168.168.181'
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:SECondary:ADDRess '192.168.168.182'
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:PRIMary:ADDRess 'fcb1:c014::2'
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:SECondary:ADDRess 'fcb1:c014::3'
```

```
CONFigure:DATA:CONTrol:DNS:FOReign:UDHCp ON, OFF, ON, OFF

// ****************************************************************************
// Query the entries of the internal DNS server for type A/AAAA DNS queries.
// Add three new entries and delete the last one.
// Ensure that all queries are answered successfully.
// ****************************************************************************
SENSe:DATA:CONTrol:DNS:LOCal:CATalog?
CONFigure:DATA:CONTrol:DNS:LOCal:ADD 'www.DAU-Home.com', '192.168.168.180'
CONFigure:DATA:CONTrol:DNS:LOCal:ADD 'www.DAU-Home.com', 'fcb1:c014::1'
CONFigure:DATA:CONTrol:DNS:LOCal:ADD 'nonsense', '1.1.1.1'
CONFigure:DATA:CONTrol:DNS:LOCal:DELete 'nonsense'
CONFigure:DATA:CONTrol:DNS:RESallquery ON

// ****************************************************************************
// Query the entries of the internal DNS server for type SRV DNS queries.
// Add two new entries and delete the last one.
// ****************************************************************************
SENSe:DATA:CONTrol:DNS:ASERvices:CATalog?
CONFigure:DATA:CONTrol:DNS:ASERvices:ADD 'pcscf', 'www.DAU-Home.com', UDP, 5060
CONFigure:DATA:CONTrol:DNS:ASERvices:ADD 'typo', 'www.x.y', UDP, 5070
CONFigure:DATA:CONTrol:DNS:ASERvices:DELete 'typo'

// ****************************************************************************
// Perform a test of the primary foreign DNS server:
// Specify the domain to be resolved, start the test and retrieve the results.
// ****************************************************************************
CONFigure:DATA:CONTrol:DNS:TEST:DOMain 'www.example.com'
CONFigure:DATA:CONTrol:DNS:TEST:STARt
SENSe:DATA:CONTrol:DNS:TEST:RESults?

// ****************************************************************************
// Start the internal DNS server.
// ****************************************************************************
SOURce:DATA:CONTrol:DNS:STATe ON
```

6.1.6 Configuring the FTP Service

```
// ****************************************************************************
// Select the FTP service type, allow access from external network,
// enable IPv6 support, allow access for anonymous users - but not upload.
// ****************************************************************************
CONFigure:DATA:CONTrol:FTP:STYPe SERV
CONFigure:DATA:CONTrol:FTP:ENConnection ON
CONFigure:DATA:CONTrol:FTP:IPVSix ON
CONFigure:DATA:CONTrol:FTP:AUSer ON
CONFigure:DATA:CONTrol:FTP:DUPLoad OFF

// ****************************************************************************
```

```
// Query information about existing FTP users.
// Create two new FTP users and delete the second one.
// ********************************************************************************
SENSe:DATA:CONTrol:FTP:USER:CATalog?
CONFigure:DATA:CONTrol:FTP:USER:ADD 'user1', 'newpwdxyz', ON, ON, ON
CONFigure:DATA:CONTrol:FTP:USER:ADD 'usernametypo', 'pwd', ON, ON, ON
CONFigure:DATA:CONTrol:FTP:USER:DELete 'usernametypo'

// ********************************************************************************
// Start the FTP service.
// ********************************************************************************
SOURce:DATA:CONTrol:FTP:STATe ON
```

6.1.7 Configuring the HTTP Service

```
// ********************************************************************************
// Allow access to internal Web server from external network and
// enable IPv6 support.
// ********************************************************************************
CONFigure:DATA:CONTrol:HTTP:ENConnection ON
CONFigure:DATA:CONTrol:HTTP:IPVSix ON

// ********************************************************************************
// Start the HTTP service.
// ********************************************************************************
SOURce:DATA:CONTrol:HTTP:STATe ON
```

6.1.8 Configuring IMS Services

Using an external IMS server

```
// ********************************************************************************
// Specify that an external IMS server shall be used and configure the
// IPv4 and IPv6 addresses of the external P-CSCF.
// ********************************************************************************
CONFigure:DATA:CONTrol:IMS2:SUSage EXTernal
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVFour '10.121.10.85'
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVSix 'fcb1:cafe:1:c1::1'
```

The remaining commands in this section are only relevant if the internal IMS server is used.

Using the internal IMS server

```
// ********************************************************************************
// Specify that the internal IMS server shall be used,
// configure the UDP/TCP threshold and enable keep-alive messages.
// ********************************************************************************
```

```
CONFigure:DATA:CONTrol:IMS2:SUSage INTernal
CONFigure:DATA:CONTrol:IMS2:TRANsport:SELection CUSTom
CONFigure:DATA:CONTrol:IMS2:THReshold:VALue 1000
CONFigure:DATA:CONTrol:IMS2:TCPalive ON
```

Configuring P-CSCF profiles

```
// ******************************************************************************
// Configure the first IPv4 P-CSCF so that it rejects requests
// with failure code 500.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:PCSCf1:BEHaviour FAILure
CONFigure:DATA:CONTrol:IMS2:PCSCf1:FAILurecode 500
CONFigure:DATA:CONTrol:IMS2:PCSCf1:RETRyafter 0

// ******************************************************************************
// Create a new IPv4 P-CSCF that accepts requests.
// As there are two default profiles, the new profile is number 3.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:PCSCf:ADD
CONFigure:DATA:CONTrol:IMS2:PCSCf3:IPADdress '172.22.1.205'
CONFigure:DATA:CONTrol:IMS2:PCSCf3:BEHaviour NORMal
CONFigure:DATA:CONTrol:IMS2:PCSCf3:REGexp 1, 1800, 60000
CONFigure:DATA:CONTrol:IMS2:PCSCf3:SUBexp 1, 1800, 60000

// ******************************************************************************
// Update the P-CSCF profile database and query a list of all profile names.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:PCSCf:CREate
SENSe:DATA:CONTrol:IMS2:PCSCf:CATalog?
```

Configuring subscriber profiles

```
// ******************************************************************************
// Configure the first subscriber profile compatible to the mobile.
// ******************************************************************************
CONF:DATA:CONTrol:IMS2:SUBScriber1:PRIVateid '001010123456789@test.3gpp.com'
CONF:DATA:CONTrol:IMS2:SUBScriber1:AUTHenticati:SCHeme AKA1
CONF:DATA:CONTrol:IMS2:SUBScriber1:AUTHenticati:ALGorithm MIL
CONF:DATA:CONTrol:IMS2:SUBScriber1:AUTHenticati:KEY '0x000102030405060708090a0b0c0d0e0f'
CONF:DATA:CONTrol:IMS2:SUBScriber1:AUTHenticati:AMF '0x0000'
CONF:DATA:CONTrol:IMS2:SUBScriber1:AUTHenticati:OPC '0xcb9dcdc5b9258e6dca4760379fb82581'
CONF:DATA:CONTrol:IMS2:SUBScriber1:IPSec:ENABle ON
CONF:DATA:CONTrol:IMS2:SUBScriber1:IPSec:ALGorithm:INTegrity HMSH
CONF:DATA:CONTrol:IMS2:SUBScriber1:IPSec:ALGorithm:ENCRyption DES
CONF:DATA:CONT:IMS2:SUBS:PUBL2 'sip:001010123456789@ims.mnc001.mcc001.3gppnetwork.org'
CONF:DATA:CONT:IMS2:SUBS:PUBLicuserid3 'sip:+11234567890@test.3gpp.com'
CONF:DATA:CONT:IMS2:SUBS:PUBLicuserid4 'sip:user@test.3gpp.com'
CONF:DATA:CONTrol:IMS2:SUBScriber1:CHATqci 11
CONF:DATA:CONTrol:IMS2:SUBScriber1:IMPU:HEADer RECoge
```

```
// ****************************************************************************
// Update the subscriber profile database and query a list of all profile names.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:SUBScriber:CREate
SENSe:DATA:CONTrol:IMS2:SUBScriber:CATalog?
```

Configuring virtual subscriber profiles

```
// ****************************************************************************
// Configure the first virtual subscriber profile compatible to the mobile.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:BEHaviour ANSWer
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:SIGNalingtyp NOPRecondit
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:BEARer ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:SUPPorted:FEATures:SESSionmode ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:SUPPorted:FEATures:STANdalone ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:SUPPorted:FEATures:VIDeo ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:SUPPorted:FEATures:FILetransfer ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:FORCemocall ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:ADCodec:TYPE WIDeband
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:AMR:ALIGnment OCTetaligned
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:AMR:CODec3:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:AMR:CODec5:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:AMR:CODec7:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:STARtmode EPRimary
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:HFONly BOTH
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:DTX ENABle
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:DTXRecv ENABle
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:CODec3:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:CODec5:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:CODec7:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:SYNCh:SELect COMMon
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:BWCommon NBWB
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:COMMon:BITRate:RANGe R59, R1280
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:CMR ENABle
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:EVS:CHAWmode NP
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:VIDeo:CODec H264
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:VIDeo:ATTRibutes ''
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MEDiaendpoin LOOP

// ****************************************************************************
// Create a new virtual subscriber using the audio board as media endpoint.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub:ADD
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:ID 'sip:audio@test.3gpp.com'
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:SIGNalingtyp NOPRecondit
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:ADCodec:TYPE WIDeband
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:AMR:ALIGnment OCTetaligned
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:AMR:CODec3:ENABle ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:MEDiaendpoin AUD
```

```
CONFigure:DATA:CONTrol:IMS2:VIRTualsub2:AUDioboard:CONF INST1, ON, TWO, TWO, SDP

// ****************************************************************************
// Create a virtual subscriber to be used as conference server.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub3:ID 'sip:confserver@test.3gpp.com'
CONFigure:DATA:CONTrol:IMS2:VIRTualsub3:CONFerence:FACTory ON
CONFigure:DATA:CONTrol:IMS2:VIRTualsub3:MAX:PARTicipant 5

// ****************************************************************************
// Update the virtual subscriber profile database and query a list of all
// profile names.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub:CREate
SENSe:DATA:CONTrol:IMS2:VIRTualsub:CATalog?
```

Starting the internal IMS server and registering a mobile

The following assumes that a cell signal is available. You can, for example, use the LTE signaling application to provide an LTE cell signal. The mobile registers first to the LTE cell, then to the IMS.

```
// ****************************************************************************
// Start the internal IMS server.
// ****************************************************************************
SOURce:DATA:CONTrol:IMS2:STATe ON

// ****************************************************************************
// Switch the mobile on and wait until it has registered to the IMS server.
// You can for example query the information in the "Info" area until it
// reports successful registration. Or insert a pause into your script.
// ****************************************************************************
SENSe:DATA:CONTrol:IMS2:GINFo?
```

Performing a mobile-terminating voice over IMS call

```
// ****************************************************************************
// Configure additional call settings for virtual subscriber 1.
// ****************************************************************************
SENSe:DATA:CONTrol:IMS2:VIRTualsub1:MTCall:DESTination:LIST?
CONF:DATA:CONTrol:IMS2:VIRTualsub1:MTCall:DESTination '[1] LTE1 - subscriber 1'
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTCall:TYPE AUDio

// ****************************************************************************
// Initiate the call.
// ****************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTCall:CALL
```

Updating an established call

```
// ****************************************************************************
// "Outband" update: Select a call, modify the settings and trigger the update.
```

```
// ***************************************************************************
CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID '4Mjh7g2CFmd0pMG9Gi93j-8lyly7fzPA'
CONFigure:DATA:CONTrol:IMS2:UPDate:ADCodec:TYPE WIDeband
CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:ALIGnment OCT
CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:CODec8:ENABle ON
CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:TYPE VIDeo
CONFigure:DATA:CONTrol:IMS2:UPDate:VIDeo:CODec H264
CONFigure:DATA:CONTrol:IMS2:UPDate:PERForm

// ***************************************************************************
// "Inband" update: Select a call, configure the CMR and trigger the update.
// ***************************************************************************
CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID '4Mjh7g2CFmd0pMG9Gi93j-8lyly7fzPA'
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRWb:CODec:RATes R1985
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:REPetition ONCE
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:PERForm
```

Updating a chat

```
// ***************************************************************************
// Define a chat message and send it to the mobile.
// ***************************************************************************
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:TEXT "Hello world!"
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:PERForm

// ***************************************************************************
// Toggle the "isComposing" status:
// Send "active" notification. Send "idle" notification.
// ***************************************************************************
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:COMPsng:NTFCn
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:IDLE:NTFCn
```

Releasing a call

```
// ***************************************************************************
// Query a list of all established calls and release one call.
// ***************************************************************************
SENSe:DATA:CONTrol:IMS2:RELease:LIST?
CONFigure:DATA:CONTrol:IMS2:RELease:CALL:ID '4Mjh7g2CFmd0pMG9Gi93j-8lyly7fzPA'
```

Sending a short message over IMS to the mobile

```
// ***************************************************************************
// Configure the short message.
// ***************************************************************************
SENSe:DATA:CONTrol:IMS2:VIRTualsub1:MTSMs:DESTination:LIST?
CONF:DATA:CONTrol:IMS2:VIRTualsub1:MTSMs:DESTination '[1] LTE1 - subscriber 1'
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTSMs:TYPE GPP
CONF:DATA:CONT:IMS2:VIRTualsub1:MTSMs:TEXT 'Testing SMS 012!.#\*%+-/()<>=;@$,'

// ***************************************************************************
```

```
// Send the message.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTSMs:SEND

// ******************************************************************************
// Query the event log contents.
// Query details for one event log entry.
// ******************************************************************************
SENSe:DATA:CONTrol:IMS2:EVENts?
SENSe:DATA:CONTrol:IMS2:HISTory? 'Id.3880694434@fc01:abab:cdcd:efe0::1'
```

Transferring a file to the mobile

```
// ******************************************************************************
// Configure the file transfer.
// ******************************************************************************
SENSe:DATA:CONTrol:IMS2:VIRTualsub1:MTFiletfr:DESTination:LIST?
CONF:DATA:CONT:IMS2:VIRTualsub1:MTFiletfr:DESTination '[1] LTE1 - subscriber 1'
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTFiletfr:TYPE FILetransfer
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTFiletfr:CHUNksize 1000
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTFiletfr:FILE:SELection "myFile.zip"

// ******************************************************************************
// Transfer the file.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:VIRTualsub1:MTFiletfr:SEND
```

Deregistering a mobile

```
// ******************************************************************************
// Deregister the first mobile.
// ******************************************************************************
CONFigure:DATA:CONTrol:IMS2:MOBile1:DERegister
```

6.1.9 Configuring the ePDG Service

```
// ******************************************************************************
// Configure ePDG addresses.
// ******************************************************************************
CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVFour '192.168.1.202'
CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVSix 'fc01::2'

// ******************************************************************************
// Configure the ePDG ID.
// ******************************************************************************
CONFigure:DATA:CONTrol:EPDG:ID:TYPE FQDN
CONFigure:DATA:CONTrol:EPDG:ID:VALue 'Epdg.rs.org'

// ******************************************************************************
// Configure IKEv2 settings.
```

```
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:IKE:ENCRyption ON, ON
CONFigure:DATA:CONTrol:EPDG:IKE:PRF ON, ON, ON, ON, ON
CONFigure:DATA:CONTrol:EPDG:IKE:INTegrity ON, ON, ON, ON, ON, ON
CONFigure:DATA:CONTrol:EPDG:IKE:DHGRoup ON, ON, ON, ON, ON, ON, ON, ON

// ****************************************************************************
// Configure ESP settings.
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:ESP:ENCRyption ON, ON
CONFigure:DATA:CONTrol:EPDG:ESP:INTegrity ON, ON, ON, ON, ON, ON

// ****************************************************************************
// Configure IKEv2 attribute settings.
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVFour:TYPE 20
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:TYPE 21
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:ADDRess:LENGth L16

// ****************************************************************************
// Configure authentication settings.
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:AUTHentic:IMSI '001010123456063'
CONFigure:DATA:CONTrol:EPDG:AUTHentic:ALGorithm MILenage
CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY #H102030405060708090A0B0C0D0E0F
CONFigure:DATA:CONTrol:EPDG:AUTHentic:RAND #H12ACF4321
CONFigure:DATA:CONTrol:EPDG:AUTHentic:AMF #H8000
CONFigure:DATA:CONTrol:EPDG:AUTHentic:OPC #H1233CB71954A9F4E48A5994E37A02BAF

// ****************************************************************************
// Configure dead peer detection.
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:DPD:ENABle ON
CONFigure:DATA:CONTrol:EPDG:DPD:INTerval 5
CONFigure:DATA:CONTrol:EPDG:DPD:TIMeout 30

// ****************************************************************************
// Configure the certificate usage for SSL.
// ****************************************************************************
CONFigure:DATA:CONTrol:EPDG:CERTificate:ENABle ON
CONFigure:DATA:CONTrol:EPDG:CERTificate:KEY "myKeyFile.pem"
CONFigure:DATA:CONTrol:EPDG:CERTificate:CERTificate "myCertificateFile.pem"

// ****************************************************************************
// Start the ePDG service.
// ****************************************************************************
SOURce:DATA:CONTrol:EPDG:STATe ON
```

6.2 Data Application Measurements

This section provides programming examples for the SCPI commands with the syntax `...:DATA:MEAS:....`

Most measurements require a signaling connection to the mobile, for example an LTE connection, set up via the LTE signaling application. To set up the signaling connection, use the commands provided by the application.

6.2.1 Configuring Global Settings

```
// ******************************************************************************
// Query a list of available signaling applications and select one of them.
// ******************************************************************************
CONFigure:DATA:MEAS:RAN:CATaloge?
CONFigure:DATA:MEAS:RAN 'LTE Sig1'
```

6.2.2 Configuring QoS Profiles

```
// ******************************************************************************
// Enable the QoS feature. Enable QoS profile number 1.
// ******************************************************************************
SOURce:DATA:MEAS:QOS:STATe ON
CONFigure:DATA:MEAS:QOS:FILTer1:ENABle ON

// ******************************************************************************
// Configure filter criteria of QoS profile number 1.
// ******************************************************************************
CONFigure:DATA:MEAS:QOS:FILTer1:IPADdress '172.22.1.110'
CONFigure:DATA:MEAS:QOS:FILTer1:PROTocol TCP
CONFigure:DATA:MEAS:QOS:FILTer1:SRCPrange 0, 0
CONFigure:DATA:MEAS:QOS:FILTer1:PRANge 0, 0

// ******************************************************************************
// Configure actions for QoS profile number 1.
// ******************************************************************************
CONFigure:DATA:MEAS:QOS:FILTer1:TCPackprio:ENABle ON
CONFigure:DATA:MEAS:QOS:FILTer1:HOPLmt 10

// ******************************************************************************
// Configure network impairments for QoS profile number 1.
// ******************************************************************************
CONFigure:DATA:MEAS:QOS:FILTer1:PLRate 10
CONFigure:DATA:MEAS:QOS:FILTer1:DELay 1
CONFigure:DATA:MEAS:QOS:FILTer1:JITTer 0.5
CONFigure:DATA:MEAS:QOS:FILTer1:JDIStribut UNIForm
CONFigure:DATA:MEAS:QOS:FILTer1:RRATe 10
CONFigure:DATA:MEAS:QOS:FILTer1:DRATe 0
```

```
CONFigure:DATA:MEAS:QOS:FILTer1:CRATe 1
CONFigure:DATA:MEAS:QOS:FILTer1:BITRate 100000

// ******************************************************************************
// Create Q0S profile number 2, enable it and assign a target address
// ******************************************************************************
CONFigure:DATA:MEAS:QOS:FILTer:ADD
CONFigure:DATA:MEAS:QOS:FILTer2:ENABle ON
CONFigure:DATA:MEAS:QOS:FILTer2:IPADdress '172.22.4.220'
```

6.2.3 Performing a Ping Measurement

```
// ******************************************************************************
// Configure the measurement:
// Specify destination IP address, interval between two requests, timeout,
// payload size and ping count.
// ******************************************************************************
CONFigure:DATA:MEAS:PING:DIPaddress '172.22.1.110'
CONFigure:DATA:MEAS:PING:INTerval 0.5
CONFigure:DATA:MEAS:PING:TIMeout 3
CONFigure:DATA:MEAS:PING:PSIZe 50
CONFigure:DATA:MEAS:PING:PCOunt 10

// ******************************************************************************
// Start the measurement.
// ******************************************************************************
INIT:DATA:MEAS:PING

// ******************************************************************************
// Wait until the measurement is complete ("RDY").
// ******************************************************************************
WHILE FETCh:DATA:MEAS:PING:STATe? <> 'RDY'

// ******************************************************************************
// Query all results.
// ******************************************************************************
FETCh:DATA:MEAS:PING?
FETCh:DATA:MEAS:PING:OVERall?
FETCh:DATA:MEAS:PING:NRCount?
```

6.2.4 Performing an Iperf Measurement

The following SCPI commands are related to the iperf tool at the DAU side of the connection. In parallel, you must configure the iperf tool installed at the mobile side and start the measurement also there. The mobile side also provides measurement results.

```
// ******************************************************************************
// Specify the test duration, the iperf version and the packet size.
```

```
// ********************************************************************************
CONFigure:DATA:MEAS:IPERf:TDURation 60
CONFigure:DATA:MEAS:IPERf:TYPE IPERf
CONFigure:DATA:MEAS:IPERf:PSIZe 1500

// ********************************************************************************
// Enable server instance 1 and select protocol type, port number and
// window size for this server instance.
// ********************************************************************************
CONFigure:DATA:MEAS:IPERf:SERVer1:ENABle ON
CONFigure:DATA:MEAS:IPERf:SERVer1:PROTocol TCP
CONFigure:DATA:MEAS:IPERf:SERVer1:PORT 5001
CONFigure:DATA:MEAS:IPERf:SERVer1:WSIZe 64

// ********************************************************************************
// Enable client instance 1 and configure the client instance:
// Select protocol type, port number, mobile IP address, window size,
// parallel connections and maximum UDP bit rate.
// ********************************************************************************
CONFigure:DATA:MEAS:IPERf:CLIent1:ENABle ON
CONFigure:DATA:MEAS:IPERf:CLIent1:PROTocol TCP
CONFigure:DATA:MEAS:IPERf:CLIent1:PORT 5003
CONFigure:DATA:MEAS:IPERf:CLIent1:IPADdress '172.22.1.110'
CONFigure:DATA:MEAS:IPERf:CLIent1:WSIZe 64
CONFigure:DATA:MEAS:IPERf:CLIent1:PCONnection 2
CONFigure:DATA:MEAS:IPERf:CLIent1:BITRate 2E+6

// ********************************************************************************
// Start the measurement at the DAU side (also start the mobile side now).
// ********************************************************************************
INIT:DATA:MEAS:IPERf

// ********************************************************************************
// Implement a loop and insert one of the following commands into the loop
// to query/monitor the results. Iperf delivers new results once per second.
// The first two commands query only a subset of results. The third command
// queries all results.
// ********************************************************************************
FETCh:DATA:MEAS:IPERf:SERVer?
FETCh:DATA:MEAS:IPERf:CLIent?
FETCh:DATA:MEAS:IPERf:ALL?
```

6.2.5 Performing a Throughput Measurement

```
// ********************************************************************************
// Specify the test duration for overall and RAN throughput.
// ********************************************************************************
CONFigure:DATA:MEAS:THRoughput:MCOunt 10
CONFigure:DATA:MEAS:THRoughput:RAN:MCOunt 10
```

```
// ********************************************************************************
// Configure the RAN slot number 2.
// ********************************************************************************
CONFigure:DATA:MEAS:THRoughput:RAN:CATaloge?
CONFigure:DATA:MEAS:THRoughput:RAN2 "LTE Signaling 2"
CONFigure:DATA:MEAS:THRoughput:RAN:TRACe:ULINk2 ON
CONFigure:DATA:MEAS:THRoughput:RAN:TRACe:DLINk2 ON

// ********************************************************************************
// Start the measurement.
// ********************************************************************************
INIT:DATA:MEAS:THRoughput

// ********************************************************************************
// Wait 12 seconds, so that a complete trace can be measured.
// ********************************************************************************
PAUSE 12

// ********************************************************************************
// Query the overall throughput results.
// ********************************************************************************
FETCh:DATA:MEAS:THRoughput:OVERall:ULINk?
FETCh:DATA:MEAS:THRoughput:OVERall:DLINk?
FETCh:DATA:MEAS:THRoughput:TRACe:OVERall:ULINk?
FETCh:DATA:MEAS:THRoughput:TRACe:OVERall:DLINk?

// ********************************************************************************
// Alternatively, query the trace results with x-values.
// ********************************************************************************
FETCh:DATA:MEAS:THRoughput:TRACe:OVERall:ULINk:EXTended?
FETCh:DATA:MEAS:THRoughput:TRACe:OVERall:DLINk:EXTended?

// ********************************************************************************
// Query the results for RAN slot number 2 and the sum of all active RAN slots.
// ********************************************************************************
FETCh:DATA:MEAS:THRoughput:RAN:ULINk2?
FETCh:DATA:MEAS:THRoughput:RAN:DLINk2?
FETCh:DATA:MEAS:THRoughput:RAN:TOTal:SUM:ULINk?
FETCh:DATA:MEAS:THRoughput:RAN:TOTal:SUM:DLINk?
FETCh:DATA:MEAS:THRoughput:TRACe:RAN:ULINk2:CURRent?
FETCh:DATA:MEAS:THRoughput:TRACe:RAN:DLINk2:CURRent?
```

6.2.6 Monitoring DNS Requests

```
// ********************************************************************************
// Specify the maximum number of DNS requests in the result list.
// ********************************************************************************
CONFigure:DATA:MEAS:DNSRequests:MICount 900
```

```
// ********************************************************************************
// Start monitoring.
// ********************************************************************************
INIT:DATA:MEAS:DNSRequests

// ********************************************************************************
// Wait until at least 10 requests have been monitored.
// ********************************************************************************
WHILE SENSe:DATA:MEAS:DNSRequests:RCOunt? < 10

// ********************************************************************************
// Query information about the already monitored DNS requests.
// You can repeat this command any time to query updated results.
// ********************************************************************************
SENSe:DATA:MEAS:DNSRequests?
```

6.2.7 Performing IP Logging

```
// ********************************************************************************
// Specify the interface to be monitored, the maximum log file size, the maximum
// number of IP packets and the maximum number of bytes per packet to be logged.
// ********************************************************************************
CONFigure:DATA:MEAS:IPLogging:TYPE UPIP
CONFigure:DATA:MEAS:IPLogging:FSIZe 10E+6
CONFigure:DATA:MEAS:IPLogging:PCOunter 1E+3
CONFigure:DATA:MEAS:IPLogging:PSLength 30E+3

// ********************************************************************************
// Start logging and query the name of the log file. Wait until logging has
// really started (e.g. before initiating actions to be logged).
// ********************************************************************************
INIT:DATA:MEAS:IPLogging
SENSe:DATA:MEAS:IPLogging:FNAMe?
WHILE FETCh:DATA:MEAS:IPLogging:STATe? <> 'RUN'
```

6.2.8 Performing IP Replay

```
// ********************************************************************************
// Query a list of all available files.
// Add two of the files to the playlist.
// ********************************************************************************
FETCh:DATA:MEAS:IPReplay:FILelist?
CONFigure:DATA:MEAS:IPReplay:CREatelist 'myfile.pcap',3,MULTicast
CONFigure:DATA:MEAS:IPReplay:CREatelist 'file42.pcap',1,MULTicast

// ********************************************************************************
// Start the IP replay application and wait until it is running.
```

```
// ****************************************************************************
INIT:DATA:MEAS:IPReplay
WHILE FETCh:DATA:MEAS:IPReplay:STATe? <> 'RUN'

// ****************************************************************************
// Trigger analysis of the files. Use a loop to query the results
// repeatedly until the analysis is finished and the results are available.
// ****************************************************************************
WHILE SENSe:DATA:MEAS:IPReplay:INFofile? 'myfile.pcap' = INV,INV,INV,INV,INV,INV
WHILE SENSe:DATA:MEAS:IPReplay:INFofile? 'file42.pcap' = INV,INV,INV,INV,INV,INV

// ****************************************************************************
// Query information about the IP connections in the files.
// ****************************************************************************
SENSe:DATA:MEAS:IPReplay:TRAFficfile? 'myfile.pcap'
SENSe:DATA:MEAS:IPReplay:TRAFficfile? 'file42.pcap'

// ****************************************************************************
// Start playlist execution.
// Continue until the user closes a dialog box. Query the playlist progress.
// ****************************************************************************
CONFigure:DATA:MEAS:IPReplay:PLAYall
WAITKEY >Check the progress?<
SENSe:DATA:MEAS:IPReplay:PROGress?

// ****************************************************************************
// Stop the playlist. Modify the number of iterations for one file and the
// interface for the other file. Play the playlist again.
// ****************************************************************************
CONFigure:DATA:MEAS:IPReplay:STOPall
CONFigure:DATA:MEAS:IPReplay:ITERation 'file42.pcap',4
CONFigure:DATA:MEAS:IPReplay:INTerface 'myfile.pcap',IP
CONFigure:DATA:MEAS:IPReplay:PLAYall

// ****************************************************************************
// Switch off the IP replay application and clear the list.
// ****************************************************************************
ABORt:DATA:MEAS:IPReplay
CONFigure:DATA:MEAS:IPReplay:REMovelist
```

6.2.9 Performing an Audio Delay Measurement

```
// ****************************************************************************
// Specify the size of the measurement record.
// Query the measurement interval duration and the number of samples per interval.
// ****************************************************************************
CONFigure:DATA:MEAS:ADELay:MSAMples 1500
CONFigure:DATA:MEAS:ADELay:SAMPles?
CONFigure:DATA:MEAS:ADELay:SPINterval?
```

```
// ********************************************************************************
// Start the measurement.
// ********************************************************************************
INIT:DATA:MEAS:ADELay

// ********************************************************************************
// Wait 30 seconds, so that a complete trace can be measured.
// ********************************************************************************
PAUSE 30

// ********************************************************************************
// Query the results relevant for the media endpoint type "Audioboard".
// ********************************************************************************
FETCh:DATA:MEAS:ADELay:TRACe:ULINk:CURRent?
FETCh:DATA:MEAS:ADELay:TRACe:DLINk:CURRent?
FETCh:DATA:MEAS:ADELay:TRACe:TAULink:CURRent?
FETCh:DATA:MEAS:ADELay:ULINk?
FETCh:DATA:MEAS:ADELay:DLINk?
FETCh:DATA:MEAS:ADELay:TAULink?

// ********************************************************************************
// Query the results relevant for the media endpoint type "Loopback".
// ********************************************************************************
FETCh:DATA:MEAS:ADELay:TRACe:LOOPback:CURRent?
FETCh:DATA:MEAS:ADELay:TRACe:TALoopback:CURRent?
FETCh:DATA:MEAS:ADELay:LOOPback?
FETCh:DATA:MEAS:ADELay:TALoopback?
```

6.2.10 Performing IP Analysis

```
// ********************************************************************************
// Enable analysis of all results
// ********************************************************************************
CONF:DATA:MEAS:IPANalysis:RESult:ALL ON,ON,ON,ON,ON

// ********************************************************************************
// Configure the OK/NOK thresholds for the "TCP Analysis" results.
// ********************************************************************************
CONFigure:DATA:MEAS:IPANalysis:TCPanalysis:TWSThreshold 80
CONFigure:DATA:MEAS:IPANalysis:TCPanalysis:TRTHreshold 9
CONFigure:DATA:MEAS:IPANalysis:TCPanalysis:TOTHreshold 40
CONFigure:DATA:MEAS:IPANalysis:TCPanalysis:RTTThreshold 40

// ********************************************************************************
// Start the IP Analysis application.
// ********************************************************************************
INIT:DATA:MEAS:IPANalysis
```

```
// ********************************************************************************
// Query the "TCP Analysis" results:
//     threshold check results for connection number 3
//     threshold check results for all connections
//     details for connection number 3
// ********************************************************************************
SENSe:DATA:MEAS:IPANalysis:TCPanalysis:FLOWid? 3
FETCh:DATA:MEAS:IPANalysis:TCPanalysis:ALL?
SENSe:DATA:MEAS:IPANalysis:TCPanalysis:DETails? 3

// ********************************************************************************
// Configure filter: only connections with L7 protocol = DNS
// Query the "IP Connectivity" results:
//     overall statistical connection information
//     connection-specific results for all connections
//     connection-specific results for connection number 3
// ********************************************************************************
CONFigure:DATA:MEAS:IPANalysis:IPConnect:FILTer L7PR, "DNS"
SENSe:DATA:MEAS:IPANalysis:IPConnect:STATistics?
SENSe:DATA:MEAS:IPANalysis:IPConnect:FLOWid? 3
FETCh:DATA:MEAS:IPANalysis:IPConnect:ALL?

// ********************************************************************************
// Query the "Data Pie Charts" results:
//     data per connection
//     data per protocol
//     data per layer
//
// Query the data per application results of the application layer.
// Navigate to the next lower layer for the IMS application.
// Query the data per application results of layer 7 (for the IMS application).
// ********************************************************************************
FETCh:DATA:MEAS:IPANalysis:DPCP:DPConnection?
FETCh:DATA:MEAS:IPANalysis:DPCP:DPPRotocol?
FETCh:DATA:MEAS:IPANalysis:DPCP:DPLayer? APP
FETCh:DATA:MEAS:IPANalysis:DPCP:DPAPplic?
CONFigure:DATA:MEAS:IPANalysis:DPCP:DPAPplic:APP 'IMS'
FETCh:DATA:MEAS:IPANalysis:DPCP:DPAPplic?

// ********************************************************************************
// Assign connection number 3 to the throughput trace number 3.
// Query the "Flow Throughput and Event Trigger" results:
//     throughput trace number 3
//     "open" events trace
//     "close" events trace
// ********************************************************************************
CONFigure:DATA:MEAS:IPANalysis:FTTRigger:TRACe3:TFLowid 3
FETCh:DATA:MEAS:IPANalysis:FTTRigger:TRACes3:FTHRoughput?
FETCh:DATA:MEAS:IPANalysis:FTTRigger:TRIGger:STARt?
FETCh:DATA:MEAS:IPANalysis:FTTRigger:TRIGger:END?
```

```
// ****************************************************************************
// Query the "Voice Over IMS" results:
//     information at call level
//     information at flow level
//     measured DTX and PER
//     measured jitter
//     measured bitrates / codec mode requests
// ****************************************************************************
FETCh:DATA:MEAS:IPANalysis:VOIMs:ALL?
SENSe:DATA:MEAS:IPANalysis:VOIMs:FLOWs? 34, 36, DL, VIDeo
SENSe:DATA:MEAS:IPANalysis:VOIMs:PERDtx? 132
SENSe:DATA:MEAS:IPANalysis:VOIMs:JITTer? 132
SENSe:DATA:MEAS:IPANalysis:VOIMs:BITRate:CMR? 34, 36, DL

// ****************************************************************************
// Store the IP analysis result database to a file
// ****************************************************************************
CONFigure:DATA:MEAS:IPANalysis:EXPortdb
```

6.2.11 Performing IP Security Tests

```
// ****************************************************************************
// Add an entry to the keyword list. Query the keyword list.
// ****************************************************************************
CONFigure:DATA:MEAS:IPANalysis:IPCSecurity:KYWord:SEARch:ADD "pwd"
SENSe:DATA:MEAS:IPANalysis:IPCSecurity:KYWord:SEARch:LIST?

// ****************************************************************************
// Start the IP Analysis application.
// ****************************************************************************
INIT:DATA:MEAS:IPANalysis

// ****************************************************************************
// Query the results on the applications tab.
// ****************************************************************************
SENSe:DATA:MEAS:IPANalysis:IPCSecurity:APPLications?
FETCh:DATA:MEAS:IPANalysis:IPCSecurity:CAPPlication? "Wikipedia"

// ****************************************************************************
// Query handshake information for a connection.
// ****************************************************************************
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:VERSion? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:CIPSuite? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:COMPression? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:SHALgorithm? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:ECURve? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:ECPFormat? 3
FETCh:DATA:MEAS:IPAN:IPCSecurity:CAPPlication:HANDshake:OFFered:SRINdication? 3
```

```
// ****************************************************************************
// Query the keyword search results.
// ****************************************************************************
FETCh:DATA:MEAS:IPANalysis:IPCSecurity:KYWord:SEARch?
```

7 DAU Services Command Reference

The following sections provide detailed reference information on the remote control commands related to the DAU services (`...:DATA:CONTrol:...` commands).

The commands configure the IP settings, configure and control the DAU services and control the DAU unit.

Reset

The data application control settings are considered as system settings and are not affected by the command `*RST`.

But you can reset data application control settings via other commands, for example via `SYSTem:RESet "Data Appl. Control"`. The "*RST:" information in the command descriptions indicates the result of such commands.

Refer to the base software manual for a description of the reset commands.

- Conventions and General Information 185
- General Settings 186
- General IP Settings 187
- IPv4 Settings 187
- IPv6 Settings 191
- DNS Server Settings 196
- FTP Service 204
- HTTP Service 208
- IMS Service 209
- ePDG Service 262

7.1 Conventions and General Information

The following sections describe the most important conventions and general information concerning the command reference.

7.1.1 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**
  If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise the command usage is stated as follows:
  - "Setting only": Command can only be used to set parameters.
  - "Query only": Command can only be used to query parameters.
  - "Event": Command initiates an event.
- **Parameter usage**
  The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query
- "Setting parameters" are only sent with a setting command
- "Query parameters" are only sent with a query command (to refine the query)
- "Return values" are only returned as the result of a query

- **Firmware/Software:**
  Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

7.2 General Settings

The following command controls the DAU, retrieve its current state and configure settings of the overview tab.

SOURce:DATA:CONTrol:STATe 186
SENSe:DATA:CONTrol:LAN:DAU:STATus? 186
CONFigure:DATA:CONTrol:ADVanced:IPBuffering 187

SOURce:DATA:CONTrol:STATe <Control>

Switches the DAU on or off. These actions are irrelevant for normal operation of the DAU. For troubleshooting, a reboot of the DAU can be initiated by switching if off and on again.

Setting parameters:

<Control> ON | OFF

Switch DAU **ON** or **OFF**

Return values:

<DAUState> OFF | PENDing | ON

OFF: DAU switched off
PEND: DAU has been switched on and is booting
ON: DAU switched on and ready for operation

*RST: ON

Example: See Rebooting the DAU

Firmware/Software: V3.0.10

Manual operation: See "DAU (parameter)" on page 109

SENSe:DATA:CONTrol:LAN:DAU:STATus?

Queries the state of the LAN DAU connector.

Return values:

<Status> NOTConn | CONN

NOTConn: no external network connected
CONN: active external network connected

Usage: Query only

Firmware/Software: V3.7.10

Manual operation: See "Related commands" on page 53

CONFigure:DATA:CONTrol:ADVanced:IPBuffering <BufferingCnfg>

Enables or disables IP buffering.

Parameters:

<BufferingCnfg> OFF | ON
*RST: OFF

Firmware/Software: V3.5.51

Manual operation: See "DAU IP Buffering" on page 52

7.3 General IP Settings

The following commands configure IP settings applicable to IPv4 and to IPv6.

CONFigure:DATA:CONTrol:MTU <MaxTransUnit>

Specifies the MTU, that is the maximum IP packet size that can be transmitted without fragmentation.

Parameters:

<MaxTransUnit> Range: 552 bytes to 4096 bytes
*RST: 1500 bytes
Default unit: bytes

Example: See Configuring General IP Settings

Firmware/Software: V3.2.20

Manual operation: See "Maximum Transmission Unit (MTU)" on page 54

7.4 IPv4 Settings

The following commands configure the IPv4 settings and query the resulting current IP configuration.

CONFigure:DATA:CONTrol:IPVFour:ADDRess:TYPE 188
CONFigure:DATA:CONTrol:IPVFour:STATic:IPADdress 188
CONFigure:DATA:CONTrol:IPVFour:STATic:SMASk 188
CONFigure:DATA:CONTrol:IPVFour:STATic:GIP 189
SENSe:DATA:CONTrol:IPVFour:CURRent:IPADdress? 189
SENSe:DATA:CONTrol:IPVFour:CURRent:SMASk? 189
SENSe:DATA:CONTrol:IPVFour:CURRent:GIP? 189
CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:ADD 190

CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:DELete 190
SENSe:DATA:CONTrol:IPVFour:STATic:ADDResses:CATalog? 190
SENSe:DATA:CONTrol:IPVFour:DHCP:ADDResses:CATalog? 191
SENSe:DATA:CONTrol:IPVFour:AUTomatic:ADDResses:CATalog? 191

CONFigure:DATA:CONTrol:IPVFour:ADDRess:TYPE <AddressType>

Selects the type of the IPv4 configuration.

Parameters:

<AddressType> AUTomatic | STATic | DHCPv4

AUTomatic: predefined internal IP configuration
STATic: user-defined static IP configuration defined via the commands
`CONFigure:DATA:CONTrol:IPVFour:STATic:...`
DHCPv4: the IPv4 address is obtained from a DHCP server in the company LAN

*RST: AUT

Example: See Configuring IPv4 Settings

Firmware/Software: V1.0.15.20

Manual operation: See "IPv4 Address Configuration" on page 55

CONFigure:DATA:CONTrol:IPVFour:STATic:IPADdress <IPAddress>

Sets the IP address of the DAU to be used for static IPv4 configuration.

Parameters:

<IPAddress> IP address as string

Range: '0.0.0.0' to '255.255.255.255'
*RST: '192.168.168.170'

Example: See Configuring IPv4 Settings

Firmware/Software: V1.0.15.20

Manual operation: See "IPv4 Address, Subnet Mask" on page 55

CONFigure:DATA:CONTrol:IPVFour:STATic:SMASk <SubnetMask>

Defines the subnet mask for static IPv4 configuration.

Parameters:

<SubnetMask> Subnet mask as string

Range: '0.0.0.0' to '255.255.255.255'
*RST: '255.255.255.0'

Example: See Configuring IPv4 Settings

Firmware/Software: V1.0.15.20

Manual operation: See "IPv4 Address, Subnet Mask" on page 55

CONFigure:DATA:CONTrol:IPVFour:STATic:GIP <GatewayIP>

Defines the address of an external gateway to be used for static IPv4 configuration.

Parameters:

<GatewayIP> IP address as string
Range: '0.0.0.0' to '255.255.255.255'
*RST: '192.168.168.1'

Example: See Configuring IPv4 Settings

Firmware/Software: V1.0.15.20

Manual operation: See "Gateway IP" on page 56

SENSe:DATA:CONTrol:IPVFour:CURRent:IPADdress?

Queries the current IPv4 address of the DAU.

Return values:

<IPAddress> IP address as string
Range: '0.0.0.0' to '255.255.255.255'

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVFour:CURRent:SMASk?

Queries the subnet mask of the current IPv4 data testing configuration.

Return values:

<SubnetMask> Subnet mask as string
Range: '0.0.0.0' to '255.255.255.255'

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVFour:CURRent:GIP?

Queries the current IPv4 address of the gateway server.

Return values:

<GatewayIP> IP address as string
Range: '0.0.0.0' to '255.255.255.255'

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:ADD <IPAddress>

Adds an IP address to the mobile IPv4 address pool for static IPv4 configuration.

Setting parameters:

<IPAddress> IP address as string

Range: '0.0.0.0' to '255.255.255.255'

Example: See Configuring IPv4 Settings

Usage: Event

Firmware/Software: V1.0.15.20

Manual operation: See "Mobile IPv4 Address Pool" on page 56

CONFigure:DATA:CONTrol:IPVFour:STATic:ADDResses:DELete <IPAddress>

Deletes an address from the mobile IPv4 address pool for static IPv4 configuration.

Setting parameters:

<IPAddress> IP address to be deleted, either identified via its index number or as string

Range: 0 to <total number of entries> - 1 | '0.0.0.0' to '255.255.255.255'

Example: See Configuring IPv4 Settings

Usage: Event

Firmware/Software: V1.0.15.20

Manual operation: See "Mobile IPv4 Address Pool" on page 56

SENSe:DATA:CONTrol:IPVFour:STATic:ADDResses:CATalog?

Queries the current mobile IPv4 address pool configured statically.

Return values:

<IPAddress> Comma-separated list of strings. Each string represents a static mobile IPv4 address.

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVFour:DHCP:ADDResses:CATalog?

Queries the current mobile IPv4 address pool configured via DHCPv4.

Return values:

<IPAddress> Comma-separated list of strings. Each string represents a static mobile IPv4 address.

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVFour:AUTomatic:ADDResses:CATalog?

Queries the current mobile IPv4 address pool configured automatically (standalone setup).

Return values:

<IPAddress> Comma-separated list of strings. Each string represents a static mobile IPv4 address.

Example: See Configuring IPv4 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

7.5 IPv6 Settings

The following commands configure the IPv6 settings.

CONFigure:DATA:CONTrol:IPVSix:ADDRess:TYPE....192
CONFigure:DATA:CONTrol:IPVSix:STATic:ADDRess....192
CONFigure:DATA:CONTrol:IPVSix:STATic:DROuter....192
SENSe:DATA:CONTrol:IPVSix:CURRent:IPADdress?....192
SENSe:DATA:CONTrol:IPVSix:CURRent:DROuter?....193
CONFigure:DATA:CONTrol:IPVSix:MOBile:PREFix:TYPE....193
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:ADD....193
CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:DELete....194
SENSe:DATA:CONTrol:IPVSix:STATic:PREFixes:CATalog?....194
SENSe:DATA:CONTrol:IPVSix:DHCP:PREFixes:CATalog?....194
SENSe:DATA:CONTrol:IPVSix:AUTomatic:PREFixes:CATalog?....194
CONFigure:DATA:CONTrol:IPVSix:ROUTing:TYPE....195
CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:ADD....195
CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:DELete....195
SENSe:DATA:CONTrol:IPVSix:MANual:ROUTing:CATalog?....196

CONFigure:DATA:CONTrol:IPVSix:ADDRess:TYPE <AddressType>

Selects the method to be used for IPv6 DAU address configuration.

Parameters:

<AddressType> AUTO | STATic | ACONf

AUTO: predefined automatic configuration (standalone setup)
STATic: static IP configuration
ACONf: dynamic autoconfiguration

*RST: AUTO

Example: See Configuring IPv6 Settings

Firmware/Software: V1.0.15.20
V3.0.10: all parameter values changed

Manual operation: See "LAN IPv6 Address Configuration" on page 57

CONFigure:DATA:CONTrol:IPVSix:STATic:ADDRess <IPAddress>

Sets the IP address of the DAU to be used for static IPv6 configuration.

Parameters:

<IPAddress> IPv6 address as string

*RST: 'fcb1:cafe::1/64'

Example: See Configuring IPv6 Settings

Firmware/Software: V3.0.10

Manual operation: See "LAN IPv6 Address Configuration" on page 57

CONFigure:DATA:CONTrol:IPVSix:STATic:DROuter <Router>

Sets the IP address of the external default router to be used for static IPv6 configuration.

Parameters:

<Router> IPv6 address as string

*RST: 'fcb1:cafe::1'

Example: See Configuring IPv6 Settings

Firmware/Software: V3.0.10

Manual operation: See "LAN IPv6 Address Configuration" on page 57

SENSe:DATA:CONTrol:IPVSix:CURRent:IPADdress?

Queries the current IPv6 address of the DAU.

Return values:

<IPAddress> IPv6 address as string, e.g. `'fcb1:abab:cdcd:efe0::1/64'`

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVSix:CURRent:DROuter?

Queries the IPv6 address currently used to address the default router.

Return values:
<DefRouter> IPv6 address as string, e.g. `'fcb1:abab:cdcd:efe0::1'`

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

CONFigure:DATA:CONTrol:IPVSix:MOBile:PREFix:TYPE <PrefixType>

Selects the method to be used for mobile IPv6 prefix configuration.

This setting is only relevant for test setups with connected external network. It is ignored for a standalone test setup (`AUTO` set via `CONFigure:DATA:CONTrol:IPVSix:ADDRess:TYPE` on page 192).

Parameters:
<PrefixType> STATic | DHCP

STATic: static IP configuration
DHCP: DHCP prefix delegation

*RST: STAT

Example: See Configuring IPv6 Settings

Firmware/Software: V3.0.10

Manual operation: See "Mobile IPv6 Prefix Configuration" on page 57

CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:ADD <Prefix>

Adds a new 64-bit prefix to the mobile IPv6 prefix pool for static IPv6 configuration.

Setting parameters:
<Prefix> String, e.g. `'fcb1:abab:cdcd:efe0::/64'`

Example: See Configuring IPv6 Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Mobile IPv6 Prefix Configuration" on page 57

CONFigure:DATA:CONTrol:IPVSix:STATic:PREFixes:DELete <Prefix>

Deletes an address from the mobile IPv6 prefix pool for static IPv6 configuration.

Setting parameters:

<Prefix> Entry to be deleted, either identified via its index number or its prefix string

Range: 0 to <total number of entries> - 1 | '<prefix>'

Example: See Configuring IPv6 Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Mobile IPv6 Prefix Configuration" on page 57

SENSe:DATA:CONTrol:IPVSix:STATic:PREFixes:CATalog?

Queries the current mobile IPv6 prefix pool configured statically.

Return values:

<Prefixes> Comma-separated list of strings, each string representing a pool entry

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVSix:DHCP:PREFixes:CATalog?

Queries the current mobile IPv6 prefix pool configured via DHCP prefix delegation.

Return values:

<Prefixes> Comma-separated list of strings, each string representing a pool entry

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

SENSe:DATA:CONTrol:IPVSix:AUTomatic:PREFixes:CATalog?

Queries the current mobile IPv6 prefix pool configured automatically (standalone setup).

Return values:
<Prefixes> Comma-separated list of strings, each string representing a pool entry

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

CONFigure:DATA:CONTrol:IPVSix:ROUTing:TYPE <RoutingType>

Selects the mechanism to be used for IPv6 route configuration.

The routes are only relevant for mobile-originating packets with destination addresses that do not belong to the subnet of the DAU and that are not reachable via the default router.

Parameters:
<RoutingType> MANual
In the current software version, the value is fixed.
MANual: manually configured routes
*RST: MANual

Example: See Configuring IPv6 Settings

Firmware/Software: V3.0.10

Manual operation: See "Routing" on page 58

CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:ADD <Prefix>, <Router>

Adds a route to the pool of manual routes for IPv6. If the destination address of a packet matches the <Prefix>, it is routed to the <Router>.

Setting parameters:
<Prefix> String, e.g. `'fcb1:abab:cdcd:efe0::/64'`, 64-bit prefixes and shorter prefixes allowed

<Router> Router address as string, e.g. 'fcb1:abcd:17c5:efe0::1'

Example: See Configuring IPv6 Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Routing" on page 58

CONFigure:DATA:CONTrol:IPVSix:MANual:ROUTing:DELete <Prefix>

Deletes an entry from the pool of manual routes for IPv6.

Setting parameters:

<Prefix> Entry to be deleted, either identified via its index number or its prefix string

Range: 0 to <total number of entries> - 1 | '<prefix>'

Example: See Configuring IPv6 Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Routing" on page 58

SENSe:DATA:CONTrol:IPVSix:MANual:ROUTing:CATalog?

Queries the pool of manual routes for IPv6.

The two values listed below are returned for each route: {<Prefixes>, <Routers>}$_{entry\ 0}$, {<Prefixes>, <Routers>}$_{entry\ 1}$, ...

Return values:

<Prefixes> String specifying an IPv6 prefix

<Routers> IPv6 address of assigned router as string

Example: See Configuring IPv6 Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Related commands" on page 53

7.6 DNS Server Settings

The following commands configure the DNS server settings and start/stop the local DNS server.

CONFigure:DATA:CONTrol:DNS:PRIMary:STYPe ... 197
CONFigure:DATA:CONTrol:DNS:SECondary:STYPe ... 197
CONFigure:DATA:CONTrol:DNS:RESallquery ... 197
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:PRIMary:ADDRess? ... 197
SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:SECondary:ADDRess? ... 198
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:PRIMary:ADDRess? ... 198
SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:SECondary:ADDRess? ... 198
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:PRIMary:ADDRess ... 198
CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:SECondary:ADDRess ... 199
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:PRIMary:ADDRess ... 199
CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:SECondary:ADDRess ... 199
CONFigure:DATA:CONTrol:DNS:FOReign:UDHCp ... 200
CONFigure:DATA:CONTrol:DNS:LOCal:ADD ... 200
CONFigure:DATA:CONTrol:DNS:LOCal:DELete ... 201
SENSe:DATA:CONTrol:DNS:LOCal:CATalog? ... 201

CONFigure:DATA:CONTrol:DNS:ASERvices:ADD 201
CONFigure:DATA:CONTrol:DNS:ASERvices:DELete 202
SENSe:DATA:CONTrol:DNS:ASERvices:CATalog? 202
CONFigure:DATA:CONTrol:DNS:TEST:DOMain 202
CONFigure:DATA:CONTrol:DNS:TEST:STARt 203
SENSe:DATA:CONTrol:DNS:TEST:RESults? 203
SOURce:DATA:CONTrol:DNS:STATe 203

CONFigure:DATA:CONTrol:DNS:PRIMary:STYPe <SType>
CONFigure:DATA:CONTrol:DNS:SECondary:STYPe <SType>

Select the primary and secondary DNS server type.

Parameters:

<SType> NONE | INTernal | IAForeign | FOReign

NONE: no DNS server address sent to the mobile
INTernal: use local DNS server
IAForeign: use local DNS server, if no entry found then foreign DNS server
FOReign: use foreign DNS server

*RST: primary INT, secondary NONE

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "DNS Server Info for the mobile" on page 60

CONFigure:DATA:CONTrol:DNS:RESallquery <DNSRESOLVEALL>

Configures the response of the internal DNS server, if no matching database entry is found for a DNS query of type A or AAAA.

Parameters:

<DNSRESOLVEALL> OFF | ON

OFF: Return no IP address
ON: Return the DAU IP address

*RST: OFF

Example: See Configuring DNS Server Settings

Firmware/Software: V3.5.30

Manual operation: See "Internal DNS server" on page 60

SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:PRIMary:ADDRess?

Queries the IPv4 address sent to the mobile as primary DNS server address.

Return values:

<CDNSPrimIP4> IPv4 address as string

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "DNS Server Info for the mobile" on page 60

SENSe:DATA:CONTrol:DNS:CURRent:IPVFour:SECondary:ADDRess?

Queries the IPv4 address sent to the mobile as secondary DNS server address.

Return values:
<CDNSSecIP4> IPv4 address as string

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "DNS Server Info for the mobile" on page 60

SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:PRIMary:ADDRess?

Queries the IPv6 address sent to the mobile as primary DNS server address.

Return values:
<CDNSPrimIP6> IPv6 address as string

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "DNS Server Info for the mobile" on page 60

SENSe:DATA:CONTrol:DNS:CURRent:IPVSix:SECondary:ADDRess?

Queries the IPv6 address sent to the mobile as secondary DNS server address.

Return values:
<CDNSSecIP6> IPv6 address as string

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "DNS Server Info for the mobile" on page 60

CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:PRIMary:ADDRess
<FDNSPrimIP4>

Specifies the IPv4 address of the foreign primary DNS server.

Parameters:
<FDNSPrimIP4> IPv4 address as string
Range: '0.0.0.0' to '255.255.255.255'
*RST: '192.168.168.171'

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "Foreign DNS server" on page 61

CONFigure:DATA:CONTrol:DNS:FOReign:IPVFour:SECondary:ADDRess
<FDNSSecIP4>

Specifies the IPv4 address of the foreign secondary DNS server.

Parameters:
<FDNSSecIP4> IPv4 address as string
Range: '0.0.0.0' to '255.255.255.255'
*RST: '192.168.168.171'

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "Foreign DNS server" on page 61

CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:PRIMary:ADDRess
<FDNSPrimIP6>

Specifies the IPv6 address of the foreign primary DNS server.

Parameters:
<FDNSPrimIP6> IPv6 address as string
*RST: 'fcb1:cafe::2'

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "Foreign DNS server" on page 61

CONFigure:DATA:CONTrol:DNS:FOReign:IPVSix:SECondary:ADDRess
<FDNSSecIP6>

Specifies the IPv6 address of the foreign secondary DNS server.

Parameters:
<FDNSSecIP6> IPv6 address as string
*RST: 'fcb1:cafe::2'

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "Foreign DNS server" on page 61

CONFigure:DATA:CONTrol:DNS:FOReign:UDHCp <PrimIP4>, <PrimIP6>, <SecIP4>, <SecIP6>

Specifies whether an IP address received via DHCPv4 / DHCPv6 is used (if available) instead of the IPv4 / IPv6 address configured statically for the foreign DNS server.

Parameters:

<PrimIP4> OFF | ON
DHCPv4 address, primary DNS server
*RST: ON

<PrimIP6> OFF | ON
DHCPv6 address, primary DNS server
*RST: ON

<SecIP4> OFF | ON
DHCPv4 address, secondary DNS server
*RST: ON

<SecIP6> OFF | ON
DHCPv6 address, secondary DNS server
*RST: ON

Example: See Configuring DNS Server Settings

Firmware/Software: V3.2.20

Manual operation: See "Foreign DNS server" on page 61

CONFigure:DATA:CONTrol:DNS:LOCal:ADD <URL>, <IP>

Adds an entry to the database of the local DNS server for type A or type AAAA DNS queries.

Each entry consists of two strings, one specifying a domain and the other indicating the assigned IP address.

Setting parameters:

<URL> String specifying the URL of a domain, e.g. `'www.example.com'`

<IP> Assigned IPv4 address or IPv6 address as string, e.g. `'192.168.168.170'` or `'fcb1:abab:1::1'`

Example: See Configuring DNS Server Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Internal DNS entries" on page 61

CONFigure:DATA:CONTrol:DNS:LOCal:DELete <URLorIP>

Deletes an entry from the database of the local DNS server for type A or type AAAA DNS queries.

Each entry consists of two strings, one specifying a domain and the other indicating the assigned IP address. Enter one of these strings to select the entry to be deleted.

Setting parameters:

<URLorIP> String selecting the entry to be deleted

Example: See Configuring DNS Server Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Internal DNS entries" on page 61

SENSe:DATA:CONTrol:DNS:LOCal:CATalog?

Queries the entries of the local DNS server database for type A or type AAAA DNS queries.

The two values listed below are returned for each database entry: {<Url>, <IP>}$_{entry\ 0}$, {<Url>, <IP>}$_{entry\ 1}$, ...

Return values:

<Url> String specifying the URL of a domain

<IP> Assigned IPv4 address or IPv6 address as string

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Internal DNS entries" on page 61

CONFigure:DATA:CONTrol:DNS:ASERvices:ADD <Name>, <Url>, <Protocol>, <Port>

Adds an entry to the database of the local DNS server for type SRV DNS queries.

Setting parameters:

<Name> String specifying the service name, e.g. `'pcscf'`

<Url> String specifying the URL of the domain, e.g. `'www.example.com'`

<Protocol> UDP | TCP

<Port> Range: 0 to 65654

Example: See Configuring DNS Server Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Application Services" on page 61

CONFigure:DATA:CONTrol:DNS:ASERvices:DELete <Name>

Deletes an entry from the database of the local DNS server for type SRV DNS queries.

Setting parameters:

<Name> String specifying the service name of the entry to be deleted

Example: See Configuring DNS Server Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Application Services" on page 61

SENSe:DATA:CONTrol:DNS:ASERvices:CATalog?

Queries the entries of the local DNS server database for type SRV DNS queries.

The four values listed below are returned for each database entry: {<Name>, <Url>, <Protocol>, <Port>}$_{entry\ 0}$, {...}$_{entry\ 1}$, ...

Return values:

<Name> String specifying the service name

<Url> String specifying the URL of the domain

<Protocol> UDP | TCP

<Port> Range: 0 to 65654

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Application Services" on page 61

CONFigure:DATA:CONTrol:DNS:TEST:DOMain <Domain>

Specifies the domain to be resolved during a test of the foreign DNS server.

Parameters:

<Domain> String specifying the URL of the domain

*RST: 'www.rohde-schwarz.com'

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "Test Primary Foreign DNS Server (softkey)" on page 62

CONFigure:DATA:CONTrol:DNS:TEST:STARt

Starts a test of the foreign DNS server.

Example: See Configuring DNS Server Settings

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Test Primary Foreign DNS Server (softkey)" on page 62

SENSe:DATA:CONTrol:DNS:TEST:RESults?

Queries the results of a foreign DNS server test.

NONE indicates that no result is available (e.g. no test started yet).

"Successful query" means that the domain could be resolved and the DNS server has returned an IP address.

Return values:

<V4SAvailable> NONE | SUCCeded | FAILed
Indicates whether the server was reachable via its IPv4 address

<V4AResRec> NONE | SUCCeded | FAILed
Indicates whether a query type A, sent to the IPv4 address was successful

<V44AResRec> NONE | SUCCeded | FAILed
Indicates whether a query type AAAA, sent to the IPv4 address was successful

<V6SAvailable> NONE | SUCCeded | FAILed
Indicates whether the server was reachable via its IPv6 address

<V6AResRec> NONE | SUCCeded | FAILed
Indicates whether a query type A, sent to the IPv6 address was successful

<V64AResRec> NONE | SUCCeded | FAILed
Indicates whether a query type AAAA, sent to the IPv6 address was successful

Example: See Configuring DNS Server Settings

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Test Primary Foreign DNS Server (softkey)" on page 62

SOURce:DATA:CONTrol:DNS:STATe <Control>

Starts or stops the local DNS server.

Setting parameters:

<Control> ON | OFF

Switch the service **ON** or **OFF**

Return values:

<State> OFF | ON | PENDing

OFF: service switched off
ON: service switched on
PEND: service activation or deactivation ongoing

*RST: OFF

Example: See Configuring DNS Server Settings

Firmware/Software: V3.0.10

Manual operation: See "DNS server control" on page 62

7.7 FTP Service

The following commands configure and start/stop the FTP service.

CONFigure:DATA:CONTrol:FTP:STYPe 204
CONFigure:DATA:CONTrol:FTP:ENConnection 205
CONFigure:DATA:CONTrol:FTP:IPVSix 205
CONFigure:DATA:CONTrol:FTP:AUSer 205
CONFigure:DATA:CONTrol:FTP:DUPLoad 205
CONFigure:DATA:CONTrol:FTP:USER:ADD 206
SENSe:DATA:CONTrol:FTP:USER:CATalog? 206
CONFigure:DATA:CONTrol:FTP:USER:DELete 207
SOURce:DATA:CONTrol:FTP:STATe 207

CONFigure:DATA:CONTrol:FTP:STYPe <ServiceType>

Selects the FTP service type.

The other `CONFigure:DATA:CONTrol:FTP:...` commands configure the FTP server. They are not relevant, if the service type `TGENerator` is selected.

Parameters:

<ServiceType> SERVer | TGENerator

SERVer: an FTP server runs on the R&S CMW
TGENerator: the R&S CMW acts as a traffic generator

*RST: SERV

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "FTP Service Type" on page 63

CONFigure:DATA:CONTrol:FTP:ENConnection <ExtNetConn>

Specifies whether access to the FTP server is allowed from an external network (via LAN DAU).

Parameters:

<ExtNetConn> OFF | ON

OFF: not allowed
ON: allowed

*RST: ON

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "FTP Server Settings" on page 64

CONFigure:DATA:CONTrol:FTP:IPVSix <IPv6Enable>

Specifies whether the FTP server supports IPv6.

Parameters:

<IPv6Enable> OFF | ON

OFF: IPv4 support only
ON: support of IPv4 and IPv6

*RST: ON

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20
V3.0.10: *RST value changed

Manual operation: See "FTP Server Settings" on page 64

CONFigure:DATA:CONTrol:FTP:AUSer <Anonymous>

Specifies whether access to the FTP server is allowed for anonymous users.

Parameters:

<Anonymous> OFF | ON

OFF: not allowed
ON: allowed

*RST: ON

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "FTP Server Settings" on page 64

CONFigure:DATA:CONTrol:FTP:DUPLoad <DataUpload>

Specifies whether data upload to the FTP server is allowed for anonymous users.

Parameters:

<DataUpload> OFF | ON

OFF: not allowed
ON: allowed

*RST: ON

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "FTP Server Settings" on page 64

CONFigure:DATA:CONTrol:FTP:USER:ADD <User>, <Password>, <DeleteAllowed>, <DownloadAllowed>, <UploadAllowed>

Creates an FTP user account.

Setting parameters:

<User> User name as string

<Password> Password as string

<DeleteAllowed> OFF | ON

<DownloadAllowed> OFF | ON

<UploadAllowed> OFF | ON

Example: See Configuring the FTP Service

Usage: Event

Firmware/Software: V1.0.15.20

Manual operation: See "FTP User Accounts" on page 64

SENSe:DATA:CONTrol:FTP:USER:CATalog?

Queries the existing FTP user accounts and their permissions.

The four values listed below are returned for each FTP user: {<User>, <DeleteAllowed>, <DownloadAllowed>, <UploadAllowed>}$_{user\ 1}$, {...}$_{user\ 2}$, ..., {...}$_{user\ n}$.

Return values:

<User> FTP user name as string

<DeleteAllowed> OFF | ON

OFF: delete forbidden
ON: delete allowed

<DownloadAllowed> OFF | ON

OFF: download forbidden
ON: download allowed

<UploadAllowed> OFF | ON

OFF: upload forbidden
ON: upload allowed

Example: See Configuring the FTP Service

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "FTP User Accounts" on page 64

CONFigure:DATA:CONTrol:FTP:USER:DELete <User>[, <Password>]

Deletes an FTP user account.

Setting parameters:

<User> FTP user name as string

<Password> Password of the FTP user as string - supported for backward compatibility of the command, can be omitted

Example: See Configuring the FTP Service

Usage: Event

Firmware/Software: V1.0.15.20
V3.0.10: <Password> now optional

Manual operation: See "FTP User Accounts" on page 64

SOURce:DATA:CONTrol:FTP:STATe <Control>

Starts or stops the FTP service.

Setting parameters:

<Control> ON | OFF

Switch the service **ON** or **OFF**

Return values:

<State> OFF | ON | PENDing

OFF: service switched off
ON: service switched on
PEND: service activation or deactivation ongoing

*RST: OFF

Example: See Configuring the FTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "Service control" on page 64

7.8 HTTP Service

The following commands configure and start/stop the HTTP service.

CONFigure:DATA:CONTrol:HTTP:ENConnection....208
CONFigure:DATA:CONTrol:HTTP:IPVSix....208
SOURce:DATA:CONTrol:HTTP:STATe....208

CONFigure:DATA:CONTrol:HTTP:ENConnection <ExtNetConn>

Specifies whether access to the internal web server is allowed from an external network (via LAN DAU).

Parameters:

<ExtNetConn> OFF | ON

OFF: not allowed
ON: allowed

*RST: ON

Example: See Configuring the HTTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "Hypertext Transfer Protocol" on page 64

CONFigure:DATA:CONTrol:HTTP:IPVSix <IPv6Enable>

Specifies whether the internal web server supports IPv6.

Parameters:

<IPv6Enable> OFF | ON

OFF: IPv4 support only
ON: support of IPv4 and IPv6

*RST: ON

Example: See Configuring the HTTP Service

Firmware/Software: V1.0.15.20
V3.0.10: *RST value changed

Manual operation: See "Hypertext Transfer Protocol" on page 64

SOURce:DATA:CONTrol:HTTP:STATe <Control>

Starts or stops the HTTP service.

Setting parameters:

<Control> ON | OFF

Switch the service **ON** or **OFF**

Return values:

<State> OFF | ON | PENDing

OFF: service switched off
ON: service switched on
PEND: service activation or deactivation ongoing

*RST: OFF

Example: See Configuring the HTTP Service

Firmware/Software: V1.0.15.20

Manual operation: See "Service control" on page 65

7.9 IMS Service

The commands described in this section configure and control the IMS service. It provides, for example, voice over IMS calls and SMS over IMS.

Do not combine the described `...:IMS2:...` commands with older `...:IMS:...` commands from software versions before 3.5.10. Scripts using only the old commands are still supported.

- Service Control and Information....209
- Call Update (Outband)....213
- Call Update (Inband)....219
- Chat Actions....223
- Call Actions and Deregistration....224
- General Settings....225
- P-CSCF Settings....227
- Subscriber Settings....231
- Virtual Subscriber Settings....237

7.9.1 Service Control and Information

The following commands control the IMS service and query status information.

SOURce:DATA:CONTrol:IMS2:STATe....209
SENSe:DATA:CONTrol:IMS2:GINFo?....210
CONFigure:DATA:CONTrol:IMS2:CLEan:GENeral:INFO....210
SENSe:DATA:CONTrol:IMS2:EVENts?....211
SENSe:DATA:CONTrol:IMS2:HISTory?....211
SENSe:DATA:CONTrol:IMS2:MOBile<UE>:STATus?....213
SENSe:DATA:CONTrol:IMS2:MOBile<UE>:CIPaddress?....213

SOURce:DATA:CONTrol:IMS2:STATe <Control>

Starts or stops the IMS service and the IMS server.

Setting parameters:

<Control> ON | OFF

Switch the service **ON** or **OFF**

Return values:

<State> OFF | ON | PENDing

OFF: service switched off
ON: service switched on
PEND: service activation or deactivation ongoing

*RST: OFF

Example: See Starting the internal IMS server and registering a mobile

Firmware/Software: V3.5.10

Manual operation: See "Server Control" on page 66

SENSe:DATA:CONTrol:IMS2:GINFo?

Queries all entries of the "General IMS Info" area.

For each entry, three parameters are returned, from oldest to latest entry: {<Timestamp>, <InfoType>, <GenericInfo>}$_{entry\ 1}$, {<Timestamp>, <InfoType>, <GenericInfo>}$_{entry\ 2}$, ...

Return values:

<Timestamp> Timestamp of the entry as string in the format `"hh:mm:ss"`

<InfoType> NONE | INFO | WARNing | ERRor

Category of the entry
NONE means that no category is assigned. If no entry at all is available, the answer is `"",NONE,""`.

<GenericInfo> Text string describing the event

Example: See Starting the internal IMS server and registering a mobile

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Entries" on page 70

CONFigure:DATA:CONTrol:IMS2:CLEan:GENeral:INFO

Clears the "General IMS Info" area.

Usage: Event

Firmware/Software: V3.5.31

Manual operation: See "Entries" on page 70

SENSe:DATA:CONTrol:IMS2:EVENts?

Queries all entries of the event log.

For each entry, six parameters are returned: {<ID>, <Timestamps>, <Source>, <Destination>, <Type>, <State>}$_{entry\ 1}$, {<ID>, ..., <State>}$_{entry\ 2}$, ...

Return values:

<ID> String identifying the event log entry
Use this ID to query event details via `SENSe:DATA:CONTrol:IMS2:HISTory?`

<Timestamps> Timestamp as string in the format `"hh:mm:ss"`

<Source> Originating party as string

<Destination> Terminating party as string

<Type> AUDio | VIDeo | SMS | INValid | CALL | RCSLmsg | FILetransfer | FILetralarge

AUDio: audio call
VIDeo: video call
SMS: sent or received short message
CALL: call setup, released call or call on hold
RCSLmsg: sent or received RCS large message
FILetransfer: file transfer
FILetralarge: file transfer large mode

<State> OK | NOK | PROGgres | RINGing | ESTablished | HOLD | RESumed | RELeased | MEDiaupdate | BUSY | DECLined | INITialmedia | FILetransfer | SRVCcrelease | TERMinated | CANCeled | REJected | CREated

Status of the session or message transfer

Example: See Sending a short message over IMS to the mobile

Usage: Query only

Firmware/Software: V3.5.10, some values have been added in later releases

Manual operation: See "Event query" on page 70

SENSe:DATA:CONTrol:IMS2:HISTory? <ID>

Queries details for a selected event log entry.

The returned sequence depends on the type of the entry. Examples:

- Four values are returned for a message entry of the type 3GPP, 3GPP2 or RCS pager mode:
  `<SMSTimestamps>, <SMSType>, <SMSEncoding>, <SMSText>`
- Eight values are returned for each recorded state of a call entry:
  {`<HistoryState>, <HistoryTimestamps>, <SignalingType>, <AudioCodecType>, <AMRAlignMode>, <AMRMode>, <VideoCodec>, <VideoAttributes>`}$_{state\ 1}$, {...}$_{state\ 2}$, ..., {...}$_{state\ n}$

If a parameter is not relevant for a state, `INV` is returned for this parameter.

Query parameters:

<ID> String selecting the event log entry
To query IDs, see `SENSe:DATA:CONTrol:IMS2:EVENts?` on page 211.

Return values:

<SMSTimestamps> Timestamp of the message transfer as string in the format `"hh:mm:ss"`

<SMSType> TGPP | TGPP2 | OGPP | OGPP2 | OPAGer | TPAGer
TGPP: mobile-terminating message, 3GPP
TGPP2: mobile-terminating message, 3GPP2
OGPP: mobile-originating message, 3GPP
OGPP2: mobile-originating message, 3GPP2
OPAGer: mobile-originating message, RCS pager mode
TPAGer: mobile-terminating message, RCS pager mode

<SMSEncoding> GSM7 | GSM8 | UCS | ASCI | IAF5 | NENC | BASE64
Encoding of the message

<SMSText> Message text as string

<HistoryState> OK | NOK | PROGgres | RINGing | ESTablished | HOLD | RESumed | RELeased | MEDiaupdate | BUSY | DECLined | RCSTxt | INITialmedia | FILetransfer | SRVCcrelease | TERMinated | CANCeled | REJected
Status of the call

<HistoryTimestamps> Timestamp of the call as string in the format `"hh:mm:ss"`

<SignalingType> PRECondit | NOPRecondit | SIMPle | REQU100 | REQuprecondi | WOTPrec183
Signaling type of the call

<AudioCodecType> NARRowband | WIDeband | EVS
Audio codec type of the call

<AMRAlignMode> OCTetaligned | BANDwidtheff
AMR voice codec alignment mode of the call

<AMRMode> Codec mode as string

<VideoCodec> H263 | H264
Video codec of the video call

<VideoAttributes> Video codec attributes of the video call

Example: See Sending a short message over IMS to the mobile

Usage: Query only

Firmware/Software: V3.5.10, some values have been added in later releases

Manual operation: See "Query of event details" on page 73

SENSe:DATA:CONTrol:IMS2:MOBile<UE>:STATus?

Queries the state of a subscriber.

Suffix:
<UE> 1..5
Profile number of the subscriber to be queried

Return values:
<Status> UNRegistered | REGistered | EXPired
Mobile unregistered, registered or registration expired

Usage: Query only

Firmware/Software: V3.5.10

SENSe:DATA:CONTrol:IMS2:MOBile<UE>:CIPaddress?

Queries the IP addresses of a subscriber.

Suffix:
<UE> 1..5
Profile number of the subscriber to be queried

Return values:
<IPAddress> IP addresses as string

Usage: Query only

Firmware/Software: V3.5.51

7.9.2 Call Update (Outband)

The commands in this section modify the settings of an established call via a re-INVITE.

CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID....214
CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:TYPE....214
CONFigure:DATA:CONTrol:IMS2:UPDate:ADCodec:TYPE....214
CONFigure:DATA:CONTrol:IMS2:UPDate:VIDeo:CODec....214
CONFigure:DATA:CONTrol:IMS2:UPDate:VIDeo:ATTRibutes....215
CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:ALIGnment....215
CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:CODec<CodecIdx>:ENABle....215
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:STARtmode....215
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:HFONly....215
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:DTX....216
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:DTXRecv....216
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CODec<CodecIdx>:ENABle....216
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SYNCh:SELect....216
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:BWCommon....217
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:COMMon:BITRate:RANGe....217
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SEND:BW....217
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SEND:BITRate:RANGe....218

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:RECeive:BW....218
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:RECeive:BITRate:RANGe....218
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CMR....219
CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CHAWmode....219
CONFigure:DATA:CONTrol:IMS2:UPDate:PERForm....219

CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID <ID>

Selects the call to be updated. To query a list of IDs, see `CONFigure:DATA:CONTrol:IMS2:RELease:CALL:ID` on page 225.

All other `UPDate` commands affect the call selected via this command.

Parameters:

<ID> Call ID as string, selecting the call to be updated

Example: See Updating an established call

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:TYPE <CallType>

Selects the new call type for a call update.

Parameters:

<CallType> AUDio | VIDeo

Example: See Updating an established call

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:ADCodec:TYPE <AudioCodec>

Selects the new audio codec type for a call update.

Parameters:

<AudioCodec> NARRowband | WIDeband | EVS
AMR NB, AMR WB, EVS

Example: See Updating an established call

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:VIDeo:CODec <VideoCodec>

Selects the video codec for a call update.

Parameters:

<VideoCodec> H263 | H264
H.263 or H.264 codec

Example: See Updating an established call

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:VIDeo:ATTRibutes <VideoAttributes>

Configures video codec attributes for a call update.

Parameters:
<VideoAttributes> Codec attributes as string

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:ALIGnment <AMRAlignment>

Selects the new AMR voice codec alignment mode for a call update.

Parameters:
<AMRAlignment> OCTetaligned | BANDwidtheff

OCTetaligned: octet-aligned
BANDwidtheff: bandwidth-efficient

Example: See Updating an established call

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:AMR:CODec<CodecIdx>:ENABle <CodecRate>

Enables or disables an AMR codec rate for a call update.

Suffix:
<CodecIdx> 1..9
Selects the codec rate via its index.

Parameters:
<CodecRate> OFF | ON

OFF: codec rate not supported
ON: codec rate supported

Example: See Updating an established call

Firmware/Software: V3.5.10

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:STARtmode <STARTMODE>

Selects the start mode for the EVS codec, for a call update.

Parameters:
<STARTMODE> EPRimary | EAMRwbio
EVS primary or EVS AMR-WB IO

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:HFONly <HF>

Specifies the SDP parameter "hf-only" for a call update.

Parameters:

<HF> BOTH | HEADfull | NP
Both, header-full only, not present

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:DTX <DTX>

Specifies the SDP parameter "dtx" for a call update.

Parameters:

<DTX> DISable | ENABle | NP
Disable, enable, not present

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:DTXRecv <DTXRECV>

Specifies the SDP parameter "dtx-recv" for a call update.

Parameters:

<DTXRECV> DISable | ENABle | NP
Disable, enable, not present

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CODec<CodecIdx>:ENABle
<CodecRate>

Enables or disables a codec rate for the AMR-WB IO mode of the EVS codec, for a call update.

Suffix:

<CodecIdx> 1..9
Selects the codec rate via its index.

Parameters:

<CodecRate> OFF | ON
OFF: codec rate not supported
ON: codec rate supported

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SYNCh:SELect <BwRanges>

Selects a configuration mode for the bandwidth and bit-rate settings of the EVS primary mode, for a call update.

Parameters:

<BwRanges> COMMon | SENDrx
Common configuration or send/receive configured separately

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:BWCommon <BWCommon>

Selects the codec bandwidths supported in the EVS primary mode, for a call update and common configuration.

Parameters:

<BWCommon> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:COMMon:BITRate:RANGe <BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode, for a call update and common configuration.

Parameters:

<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SEND:BW <SENDBW>

Selects the codec bandwidths supported in the EVS primary mode in the downlink (send) direction, for a call update and separate configuration.

Parameters:

<SENDBW> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:SEND:BITRate:RANGe
<BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode in the downlink (send) direction, for a call update and separate configuration.

Parameters:

<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:RECeive:BW <RXBW>

Selects the codec bandwidths supported in the EVS primary mode in the uplink (receive) direction, for a call update and separate configuration.

Parameters:

<RXBW> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:RECeive:BITRate:RANGe
<BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode in the uplink (receive) direction, for a call update and separate configuration.

Parameters:

<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CMR <CMR>

Specifies the SDP parameter "cmr" for the EVS codec, for a call update.

Parameters:

<CMR> DISable | ENABle | PRESent | NP
Disable, enable, present all, not present

Firmware/Software: V3.5.30

CONFigure:DATA:CONTrol:IMS2:UPDate:EVS:CHAWmode <CHAWMODE>

Specifies the SDP parameter "ch-aw-recv" for the EVS codec, for a call update.

Parameters:

<CHAWMODE> DIS | NUSed | TWO | THRee | FIVE | SEVen | NP
Disabled, not used, 2, 3, 5, 7, not present
*RST: DIS

Firmware/Software: V3.5.30
V3.5.31: added TWO | THRee | FIVE | SEVen

CONFigure:DATA:CONTrol:IMS2:UPDate:PERForm

Initiates an "outband" call update.

The settings configured via the `UPDate` commands are applied to the call selected via `CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID` on page 214.

Example: See Updating an established call

Usage: Event

Firmware/Software: V3.5.10

7.9.3 Call Update (Inband)

The commands in this section modify the settings of an established call via codec mode requests (CMR).

For call selection, see `CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID` on page 214.

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:BW....220
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:CODec:RATes....220
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRNb:CODec:RATes....221
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRWb:CODec:RATes....222
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:REPetition....222
CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:PERForm....222

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:BW <BW>

Configures the EVS codec bandwidth to be requested via CMR.

Parameters:

<BW> NB | IO | WB | SWB | FB | WBCA | SWBCa | NOReq | DEAC

IO: AMR-WB IO mode
NB: primary, narrowband
WB: primary, wideband
SWB: primary, super wideband
FB: primary, fullband
WBCA: primary, WB, channel-aware mode
SWBCa: primary, SWB, channel-aware mode
NOReq: NO_REQ, no codec rate requirement
DEAC: CMR byte removed from header

*RST: NB

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:EVS:CODec:RATes <EVSBitrate>

Configures an EVS codec rate or bit rate to be requested via CMR. For BW=DEAC, you cannot set a rate.

Parameters:

<EVSBitrate> NOReq | AW66 | AW885 | AW1265 | AW1425 | AW1585 | AW1825 | AW1985 | AW2305 | AWB2385 | PR59 | PR72 | PR80 | PR96 | P132 | P164 | P244 | P320 | P480 | P640 | P960 | P1280 | SLO2 | SLO3 | SLO5 | SLO7 | SHO2 | SHO3 | SHO5 | SHO7 | WLO2 | WLO3 | WLO5 | WLO7 | WHO2 | WHO3 | WHO5 | WHO7

NOReq
No codec rate requirement (NO_REQ)
Only for BW=NOReq

AW66 to AWB2385
AMR-WB IO mode, 6.6 kbit/s to 23.85 kbit/s
Only for BW=IO

PR59 to P1280
Primary mode, 5.9 kbit/s to 128.0 kbit/s
For BW=NB: PR59 to P244
For BW=WB: PR59 to P1280
For BW=SWB: PR96 to P1280
For BW=FB: P164 to P1280

SLO2 to SLO7
SWB with channel-aware mode, CA-L-O2 to CA-L-O7
Only for BW=SWBCa

SHO2 to SHO7
SWB with channel-aware mode, CA-H-O2 to CA-H-O7
Only for BW=SWBCa

WLO2 to WLO7
WB with channel-aware mode, CA-L-O2 to CA-L-O7
Only for BW=WBCA

WHO2 to WHO7
WB with channel-aware mode, CA-H-O2 to CA-H-O7
Only for BW=WBCA

*RST: PR59

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRNb:CODec:RATes
<AMRNBBitrate>

Configures an AMR narrowband codec rate to be requested via CMR.

Parameters:

<AMRNBBitrate> R475 | R515 | R590 | R670 | R740 | R795 | R1020 | R1220 | NOReq

R475 to R1220: 4.75 kbit/s to 12.20 kbit/s
NOReq: no codec rate requirement (NO_REQ)

*RST: NOR

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:AMRWb:CODec:RATes <AMRWBBitrate>

Configures an AMR wideband codec rate to be requested via CMR.

Parameters:

<AMRWBBitrate> R660 | R885 | R1265 | R1425 | R1585 | R1825 | R1985 | R2305 | RA2385 | NOReq

R660 to RA2385: 6.60 kbit/s to 23.85 kbit/s
NOReq: no codec rate requirement (NO_REQ)

*RST: R660

Example: See Updating an established call

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:REPetition <Repetition>

Selects whether a CMR is sent only once or continuously.

Parameters:

<Repetition> ENDLess | ONCE

*RST: ENDL

Example: See Updating an established call

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

CONFigure:DATA:CONTrol:IMS2:UPDate:INBand:PERForm

Initiates an "inband" call update.

The settings configured via the `UPDate` commands are applied to the call selected via `CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID` on page 214.

Example: See Updating an established call

Usage: Event

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Inband tab" on page 67

7.9.4 Chat Actions

The commands in this section allow you to perform the following actions for an established chat:

- Send a message to the mobile
- Send "isComposing" status messages ("idle" / "active")

CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:TEXT 223
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:PERForm 223
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:COMPsng:NTFCn 224
CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:IDLE:NTFCn 224

CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:TEXT <Text>

Defines a message text to be sent to the mobile via an established chat session.

Initiate the message transfer via `CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:PERForm`.

Parameters:

<Text> Message as string

Example: See Updating a chat

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Chat Actions (softkey)" on page 68

CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:CHAT:PERForm

Initiates a chat message transfer to the mobile.

Example: See Updating a chat

Usage: Event

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Chat Actions (softkey)" on page 68

CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:COMPsng:NTFCn

Send an "active" notification to the mobile as "isComposing" status message.

Example: See Updating a chat

Usage: Event

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Chat Actions (softkey)" on page 68

CONFigure:DATA:CONTrol:IMS2:UPDate:RCS:IDLE:NTFCn

Send an "idle" notification to the mobile as "isComposing" status message.

Example: See Updating a chat

Usage: Event

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Chat Actions (softkey)" on page 68

7.9.5 Call Actions and Deregistration

The commands in this section allow you to perform the following actions:

- Put a call on hold; resume a call
- Release a call / session
- Deregister a mobile

CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:EVENt 224
SENSe:DATA:CONTrol:IMS2:RELease:LIST? 225
CONFigure:DATA:CONTrol:IMS2:RELease:CALL:ID 225
CONFigure:DATA:CONTrol:IMS2:MOBile<UE>:DERegister 225

CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:EVENt <UpdateCallEvent>

Puts a call on hold or resumes a call that has been put on hold.

To select the call, use `CONFigure:DATA:CONTrol:IMS2:UPDate:CALL:ID` on page 214.

Parameters:
<UpdateCallEvent> HOLD | RESume

Firmware/Software: V3.5.30

Manual operation: See "Outband tab" on page 67

SENSe:DATA:CONTrol:IMS2:RELease:LIST?

Queries the IDs of all established calls.

Return values:
<Ids> Comma-separated list of ID strings, one string per established call

Example: See Releasing a call

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Release Call (softkey)" on page 69

CONFigure:DATA:CONTrol:IMS2:RELease:CALL:ID <ID>

Queries a list of call IDs or releases a call selected via its ID.

Setting parameters:
<ID> ID as string, selecting the call to be released

Return values:
<IDs> Comma-separated list of ID strings, one string per established call

Example: See Releasing a call

Firmware/Software: V3.5.10

Manual operation: See "Release Call (softkey)" on page 69

CONFigure:DATA:CONTrol:IMS2:MOBile<UE>:DERegister

Deregisters a registered subscriber from the IMS server.

Suffix:
<UE> 1..5
Profile number of the subscriber to be deregistered

Example: See Deregistering a mobile

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Deregister Mobile (softkey)" on page 69

7.9.6 General Settings

The following commands configure basic IMS service settings.

CONFigure:DATA:CONTrol:IMS2:SUSage 226
CONFigure:DATA:CONTrol:IMS2:TRANsport:SELection 226
CONFigure:DATA:CONTrol:IMS2:THReshold:VALue 226

CONFigure:DATA:CONTrol:IMS2:TCPalive 227
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVFour 227
CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVSix 227

CONFigure:DATA:CONTrol:IMS2:SUSage <ServerUsage>

Selects whether the internal IMS server of the DAU or an external IMS network is used for IMS services.

Parameters:

<ServerUsage> INTernal | EXTernal

*RST: INT (if allowed)

Example: See Using the internal IMS server

Firmware/Software: V3.5.10

Options: R&S CMW-KAA20 required for internal IMS server

Manual operation: See "Use IMS Server as" on page 74

CONFigure:DATA:CONTrol:IMS2:TRANsport:SELection <TransportSelection>

Configures whether TCP or UDP is used by the internal IMS server.

Parameters:

<TransportSelection> DEFault | TCP | UDP | CUSTom

DEFault: Fixed threshold as defined in RFC 3261
TCP: Only TCP is used
UDP: Only UDP is used
CUSTom: UDP for short messages, TCP for long messages, threshold configurable via `CONFigure:DATA:CONTrol:IMS2:THReshold:VALue`

*RST: DEF

Example: See Using the internal IMS server

Firmware/Software: V3.5.21

Manual operation: See "Transport Selection, UDP/TCP Threshold" on page 74

CONFigure:DATA:CONTrol:IMS2:THReshold:VALue <Threshold Value>

Configures a threshold for the usage of UDP (below threshold) and TCP (above threshold).

The setting is only relevant for `CONFigure:DATA:CONTrol:IMS2:TRANsport:SELection CUSTom`.

Parameters:

<Threshold Value> Number of characters per SIP message

Range: 1 to 65535
*RST: 1

Example: See Using the internal IMS server

Firmware/Software: V3.5.21

Manual operation: See "Transport Selection, UDP/TCP Threshold" on page 74

CONFigure:DATA:CONTrol:IMS2:TCPalive <TCPKEEP>

Selects whether the IMS server sends TCP keep-alive messages or not.

Parameters:

<TCPKEEP> OFF | ON

*RST: ON

Example: See Using the internal IMS server

Firmware/Software: V3.5.30

Manual operation: See "TCP Keep Alive" on page 75

CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVFour <IPv4Addr>

Specifies the IPv4 address of an external P-CSCF.

Parameters:

<IPv4Addr> IPv4 address string

*RST: '192.168.168.200'

Example: See Using an external IMS server

Firmware/Software: V3.5.10

Manual operation: See "P-CSCF IPv4/IPv6 Address" on page 75

CONFigure:DATA:CONTrol:IMS2:EXTern:PCSCf:ADDRess:IPVSix <IPv6Addr>

Specifies the IPv6 address of an external P-CSCF.

Parameters:

<IPv6Addr> IPv6 address string

*RST: 'fcb1:cafe::200'

Example: See Using an external IMS server

Firmware/Software: V3.5.10

Manual operation: See "P-CSCF IPv4/IPv6 Address" on page 75

7.9.7 P-CSCF Settings

The following commands configure the simulated P-CSCFs.

CONFigure:DATA:CONTrol:IMS2:PCSCf:ADD....228
CONFigure:DATA:CONTrol:IMS2:PCSCf:CREate....228
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:DELete....228
SENSe:DATA:CONTrol:IMS2:PCSCf:CATalog?....229
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:IPADdress....229
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:BEHaviour....229
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:FAILurecode....230
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:RETRyafter....230
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:REGexp....230
CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:SUBexp....231

CONFigure:DATA:CONTrol:IMS2:PCSCf:ADD

Creates a P-CSCF profile.

See also `CONFigure:DATA:CONTrol:IMS2:PCSCf:CREate` on page 228

Example: See Configuring P-CSCF profiles

Usage: Event

Firmware/Software: V3.5.10

Options: R&S CMW-KAA21

Manual operation: See "Add or delete a tab" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf:CREate

Updates the internal list of P-CSCF profiles.

If your command script adds P-CSCF profiles, you must insert this command before you can use the new profiles.

It is sufficient to insert the command once, after adding the last P-CSCF profile / before using the profiles.

Example: See Configuring P-CSCF profiles

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:DELete

Deletes the P-CSCF profile number <p>.

Suffix:
<p> 3..10

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 76

SENSe:DATA:CONTrol:IMS2:PCSCf:CATalog?

Queries a list of all P-CSCF profile names.

Return values:

<Catalog> String listing all profile names, separated by commas
String example: `"P-CSCF 1,P-CSCF 2"`

Example: See Configuring P-CSCF profiles

Usage: Query only

Firmware/Software: V3.5.11

Manual operation: See "Add or delete a tab" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:IPADdress <IPAddress>

Defines the IP address of the P-CSCF number <p>.

Suffix:

<p> 1..10

Parameters:

<IPAddress> IPv4 or IPv6 address string
Example: '172.22.1.201' or 'fc01:cafe::1/64'

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "IP Address" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:BEHaviour <Behaviour>

Defines the behavior of the P-CSCF number <p> when it receives a SIP message from the mobile.

Suffix:

<p> 1..10

Parameters:

<Behaviour> NORMal | FAILure

NORMal: normal behavior
FAILure: return failure code

*RST: NORM

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "Behavior" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:FAILurecode <FailureCode>

Defines a failure code for the P-CSCF number <p>, behavior = `FAIL`.

Suffix:
<p> 1..10

Parameters:
<FailureCode> BADRequest | FORBidden | NOTFound | INTerror | UNAVailable | BUSYeveryw

BADRequest: "400 Bad Request"
FORBidden: "403 Forbidden"
NOTFound: "404 Not Found"
INTerror: "500 Server Internal Error"
UNAVailable: "503 Service Unavailable"
BUSYeveryw: "600 Busy Everywhere"

*RST: BADR

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "Failure" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:RETRyafter <RetryAfter>

Defines the contents of the "Retry After" header field for the P-CSCF number <p>, behavior = `FAIL`.

Suffix:
<p> 1..10

Parameters:
<RetryAfter> *RST: 0 s
Default unit: s

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "Failure" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:REGexp <RegExpMin>, <RegExpDefault>, <RegExpMax>

Defines registration expiration times for the P-CSCF number <p>.

Suffix:
<p> 1..10

Parameters:
<RegExpMin> Minimum acceptable expiration time

*RST: 1 s
Default unit: s

<RegExpDefault> Default value, used if the mobile does not suggest an expiration time

*RST: 3600 s
Default unit: s

<RegExpMax> Maximum acceptable expiration time

*RST: 600E+3 s
Default unit: s

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "Registration" on page 76

CONFigure:DATA:CONTrol:IMS2:PCSCf<p>:SUBexp <SubsExpMin>, <SubsExpDefault>, <SubsExpMax>

Defines subscription expiration times for the P-CSCF number <p>.

Suffix:
<p> 1..10

Parameters:
<SubsExpMin> Minimum acceptable expiration time

*RST: 1 s
Default unit: s

<SubsExpDefault> Default value, used if the mobile does not suggest an expiration time

*RST: 3600 s
Default unit: s

<SubsExpMax> Maximum acceptable expiration time

*RST: 600E+3 s
Default unit: s

Example: See Configuring P-CSCF profiles

Firmware/Software: V3.5.10

Manual operation: See "Subscription" on page 77

7.9.8 Subscriber Settings

The following commands configure subscriber profiles for the tested mobiles.

CONFigure:DATA:CONTrol:IMS2:SUBScriber:ADD 232
CONFigure:DATA:CONTrol:IMS2:SUBScriber:CREate 232
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:DELete 232
SENSe:DATA:CONTrol:IMS2:SUBScriber:CATalog? 233
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:SCHeme 233
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PRIVateid 233

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:ALGorithm....234
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:KEY....234
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:AMF....234
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:OPC....235
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ENABle....235
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:INTegrity....235
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:ENCRyption....236
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PUBLicuserid<Index>....236
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:CHATqci....236
CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IMPU:HEADer....237

CONFigure:DATA:CONTrol:IMS2:SUBScriber:ADD

Creates a subscriber profile.

See also `CONFigure:DATA:CONTrol:IMS2:SUBScriber:CREate` on page 232.

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber:CREate

Updates the internal list of subscriber profiles.

If your command script adds subscriber profiles, you must insert this command before you can use the new profiles.

It is sufficient to insert the command once, after adding the last subscriber profile / before using the profiles.

Example: See Configuring subscriber profiles

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:DELete

Deletes the subscriber profile number <s>.

Suffix:
<s> 2..5

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 79

SENSe:DATA:CONTrol:IMS2:SUBScriber:CATalog?

Queries a list of all subscriber profile names.

Return values:
<Catalog> String listing all profile names, separated by commas
String example: `"Subscriber 1,Subscriber 2"`

Example: See Configuring subscriber profiles

Usage: Query only

Firmware/Software: V3.5.11

Manual operation: See "Add or delete a tab" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:SCHeme <AuthScheme>

Specifies whether authentication is performed for the subscriber profile number <s> and selects the authentication and key agreement version to be used.

Suffix:
<s> 1..5

Parameters:
<AuthScheme> AKA1 | AKA2 | NOAuthentic
AKA1: authentication with AKA version 1
AKA2: authentication with AKA version 2
NOAuthentic: no authentication
*RST: NOA

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Authentication" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PRIVateid <PrivateId>

Specifies the private user ID of the subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<PrivateId> Private user ID as string
*RST: '001010123456789@test.3gpp.com'

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Private User ID" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:ALGorithm <AuthKeyGenAlg>

Specifies which algorithm set is used for the subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<AuthKeyGenAlg> XOR | MILenage
*RST: XOR

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Key Generation Algorithm" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:KEY <AuthKey>

Defines the authentication key K for the subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<AuthKey> Key as string (32-digit hexadecimal number)
*RST: '000102030405060708090a0b0c0d0e0f'

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Key" on page 79

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:AMF <AuthAmf>

Specifies the authentication management field (AMF) for the subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<AuthAmf> AMF as string (four-digit hexadecimal number)
*RST: '0000'

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "AMF" on page 80

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:AUTHenticati:OPC <AuthOPc>

Specifies the key OP_c for the subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<AuthOPc> Key as string (32-digit hexadecimal number)
*RST: 'cb9dcdc5b9258e6dca4760379fb82581'

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "OPc" on page 80

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ENABle <IPSec>

Enables or disables support of the IP security mechanisms by the IMS server for subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<IPSec> OFF | ON
*RST: OFF

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "IPSec" on page 80

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:INTegrity <IntegrityAlg>

Selects an integrity protection algorithm for subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<IntegrityAlg> HMMD | HMSH | AUTO
HMMD: HMAC-MD5-96
HMSH: HMAC-SHA-1-96
AUTO: as indicated in REGISTER message
*RST: AUTO

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10, V3.5.50 added AUTO

Manual operation: See "Integrity Algorithm" on page 80

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IPSec:ALGorithm:ENCRyption <EncryptionAlg>

Selects an encryption algorithm for subscriber profile number <s>.

Suffix:
<s> 1..5

Parameters:
<EncryptionAlg> DES | AES | NOC | AUTO

DES: DES-EDE3-CBC
AES: AES-CBC
NOC: NULL, no encryption
AUTO: as indicated in REGISTER message

*RST: NOC

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10, V3.5.50 added AUTO

Manual operation: See "Encryption Algorithm" on page 80

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:PUBLicuserid<Index> <PublicUserIds>

Defines public user ID number <Index> for the subscriber profile number <s>.

Suffix:
<s> 1..5

<Index> 1..10

Parameters:
<PublicUserIds> Public user ID as string

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Public User IDs" on page 81

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:CHATqci <ChatQCI>

Specifies the QCI used by the mobile for RCS message transfer.

Suffix:
<s> 1..5

Parameters:
<ChatQCI> Range: 0 to 255
*RST: 0

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Chat QCI" on page 81

CONFigure:DATA:CONTrol:IMS2:SUBScriber<s>:IMPU:HEADer <PAUHeader>

Selects which public IDs are sent to the mobile in the PAU header of the 200/OK response to REGISTER messages:

- Configured: public user IDs configured in the subscriber settings
- Registered: ID sent by the mobile in the "from" header of the REGISTER message
- Generated: automatically generated ID starting with "tel:555"

Suffix:
<s> 1..5

Parameters:
<PAUHeader> CONRege | RECoge | CORE | RECN | COGE | REGE | CONF | REGD

CONRege: configured, registered, generated
RECoge: registered, configured, generated
CORE: configured, registered
RECN: registered, configured
COGE: configured, generated
REGE: registered, generated
CONF: configured
REGD: registered

Example: See Configuring subscriber profiles

Firmware/Software: V3.5.51

Manual operation: See "IMPUs in PAU Header" on page 81

7.9.9 Virtual Subscriber Settings

The following commands configure virtual subscriber profiles.

Most commands include the mnemonic `VIRTualsub<v>`. They apply to the virtual subscriber profile number <v>.

- Profile Administration 237
- Basic Profile Settings 239
- Mobile-Terminating Calls 255
- Mobile-Terminating Messages 256
- File Transfer 259

7.9.9.1 Profile Administration

The following commands create or delete virtual subscriber profiles.

CONFigure:DATA:CONTrol:IMS2:VIRTualsub:ADD....238
CONFigure:DATA:CONTrol:IMS2:VIRTualsub:CREate....238
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:DELete....238
SENSe:DATA:CONTrol:IMS2:VIRTualsub:CATalog?....238

CONFigure:DATA:CONTrol:IMS2:VIRTualsub:ADD

Creates a new virtual subscriber profile.

See also `CONFigure:DATA:CONTrol:IMS2:VIRTualsub:CREate` on page 238

Example: See Configuring virtual subscriber profiles

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 82

CONFigure:DATA:CONTrol:IMS2:VIRTualsub:CREate

Updates the internal list of virtual subscriber profiles.

If your command script adds virtual subscriber profiles, you must insert this command before you can use the new profiles.

It is sufficient to insert the command once, after adding the last virtual subscriber profile / before using the profiles.

Example: See Configuring virtual subscriber profiles

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 82

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:DELete

Deletes the virtual subscriber profile number <v>.

Suffix:
<v> 2..20

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Add or delete a tab" on page 82

SENSe:DATA:CONTrol:IMS2:VIRTualsub:CATalog?

Queries a list of all virtual subscriber profile names.

Return values:

<Catalog> String listing all profile names, separated by commas
String example: `"Virtual 1,Virtual 2"`

Example: See Configuring virtual subscriber profiles

Usage: Query only

Firmware/Software: V3.5.11

Manual operation: See "Add or delete a tab" on page 82

7.9.9.2 Basic Profile Settings

The following commands configure basic virtual subscriber profile settings.

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ID....240
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEHaviour....240
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SIGNalingtyp....240
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEARer....241
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:CONFerence:FACTory....241
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MAX:PARTicipant....242
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:VIDeo....242
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:STANdalone....243
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:SESSionmode....243
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:FILetransfer....243
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ADCodec:TYPE....244
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORCemocall....244
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:ALIGnment....244
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:CODec<CodecIdx>:ENABle....245
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:STARtmode....245
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:HFONly....245
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTX....246
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTXRecv....246
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CODec<CodecIdx>:ENABle....246
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect....247
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:BWCommon....248
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:COMMon:BITRate:RANGe....248
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BW....249
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BITRate:RANGe....249
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BW....250
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BITRate:RANGe....250
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CMR....251
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CHAWmode....251
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:CODec....252
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:ATTRibutes....252
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MEDiaendpoin....252
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard....253
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AUDioboard:CONFig....253
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:IO:MODE:CONFig....254
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:PCAPfile:SELection....255

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ID <ID>

Specifies the public user ID of the virtual subscriber number <v>.

Suffix:
<v> 1..20

Parameters:
<ID> Public user ID as string

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Public User ID" on page 85

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEHaviour <Behaviour>

Defines the reaction of virtual subscriber number <v> to incoming calls.

Suffix:
<v> 1..20

Parameters:
<Behaviour> ANSWer | NOANswer | DECLined | BUSY

ANSWer: answer the call
NOANswer: keep "ringing"
DECLined: reject call
BUSY: subscriber busy

*RST: ANSW

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Behavior" on page 85

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SIGNalingtyp <SigType>

Specifies whether a voice call session is established with or without quality-of-service preconditions.

Suffix:
<v> 1..20

Parameters:

<SigType> PRECondit | NOPRecondit | SIMPle | REQU100 | REQuprecondi | WOTPrec183

PRECondit
With preconditions

NOPRecondit
Without preconditions

SIMPle
Simplified call flow, without preconditions

REQU100
Require 100rel, without preconditions

REQuprecondi
Require precondition, with preconditions

WOTPrec183
With 183, without preconditions

*RST: NOPR

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10
V3.5.21: REQU100, REQuprecondi, WOTPrec183

Manual operation: See "Signaling Type" on page 85

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:BEARer <Bearer>

Enables the automatic setup of a dedicated bearer.

Suffix:

<v> 1..20

Parameters:

<Bearer> OFF | ON

*RST: OFF

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.21

Manual operation: See "Dedicated Bearer" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:CONFerence:FACTory <Conference Factory>

Selects whether the virtual subscriber profile configures a conference server.

Suffix:

<v> 1..20

Parameters:

<Conference Factory>
OFF | ON
OFF: ordinary virtual subscriber
ON: conference server
*RST: OFF

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.51

Options: R&S CMW-KAA21

Manual operation: See "Conference Factory" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MAX:PARTicipant
<NoParticapant>

Configures the maximum number of virtual subscribers allowed to participate in a conference call.

Suffix:

<v> 1..20

Parameters:

<NoParticapant>
The value 0 means that there is no limit.
Range: 0 to 10
*RST: 0

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.51

Options: R&S CMW-KAA21

Manual operation: See "Max No. of Virtual Participants" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:VIDeo
<Video>

Specifies whether the virtual subscriber number <v> supports video calls.

Suffix:

<v> 1..20

Parameters:

<Video>
OFF | ON
*RST: ON

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.50

Manual operation: See "Supported Features" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:STANdalone <StandAlone>

Specifies whether the virtual subscriber number <v> supports standalone messaging.

Suffix:
<v> 1..20

Parameters:
<StandAlone> OFF | ON
*RST: ON

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.50

Manual operation: See "Supported Features" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:SESSionmode <SessionMode>

Specifies whether the virtual subscriber number <v> supports session mode messaging.

Suffix:
<v> 1..20

Parameters:
<SessionMode> OFF | ON
*RST: ON

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.50

Manual operation: See "Supported Features" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:SUPPorted:FEATures:FILetransfer <FileTransfer>

Specifies whether the virtual subscriber number <v> supports file transfer via MSRP.

Suffix:
<v> 1..20

Parameters:
<FileTransfer> OFF | ON
*RST: ON

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.50

Manual operation: See "Supported Features" on page 86

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ADCodec:TYPE <Type>

Selects the audio codec type for voice over IMS calls.

Suffix:
<v> 1..20

Parameters:
<Type> NARRowband | WIDeband | EVS
AMR NB, AMR WB, EVS
*RST: NARR

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Codec" on page 87

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORCemocall <ForceCodecMo>

Forces a specific codec type for mobile-originating calls.

Suffix:
<v> 1..20

Parameters:
<ForceCodecMo> OFF | ON
OFF
The first codec type offered via SDP is used.
ON
The voice codec type configured for the virtual subscriber is used.
*RST: OFF

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Force Codec for MO Calls" on page 87

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:ALIGnment <AlignMode>

Selects the AMR voice codec alignment mode for voice over IMS calls.

Suffix:
<v> 1..20

Parameters:
<AlignMode> OCTetaligned | BANDwidtheff
OCTetaligned: octet-aligned
BANDwidtheff: bandwidth-efficient
*RST: OCT

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "AMR Alignment Mode" on page 87

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AMR:CODec<CodecIdx>:ENABle <CodecRate>

Enables or disables a codec rate for the currently active AMR type, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:ADCodec:TYPE`.

Suffix:
<v> 1..20

<CodecIdx> 1..9
Selects the codec rate via its index.

Parameters:
<CodecRate> OFF | ON

OFF: codec rate not supported
ON: codec rate supported

*RST: OFF

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Codec Rates" on page 88

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:STARtmode <STARTMODE>

Selects the start mode for the EVS codec.

Suffix:
<v> 1..20

Parameters:
<STARTMODE> EPRimary | EAMRwbio

EVS primary or EVS AMR-WB IO

*RST: EPR

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Start Mode" on page 88

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:HFONly <HF>

Specifies the SDP parameter "hf-only" for the EVS codec.

Suffix:
<v> 1..20

Parameters:

<HF> BOTH | HEADfull | NP
Both, header-full only, not present
*RST: NP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "HF Only" on page 88

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTX <DTX>

Specifies the SDP parameter "dtx" for the EVS codec.

Suffix:

<v> 1..20

Parameters:

<DTX> DISable | ENABle | NP
Disable, enable, not present
*RST: NP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "DTX" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:DTXRecv <DTXRECV>

Specifies the SDP parameter "dtx-recv" for the EVS codec.

Suffix:

<v> 1..20

Parameters:

<DTXRECV> DISable | ENABle | NP
Disable, enable, not present
*RST: NP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "DTX Recv" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CODec<CodecIdx>:ENABle <CodecRate>

Enables or disables a codec rate for the AMR-WB IO mode of the EVS codec.

Suffix:

<v> 1..20

<CodecIdx> 1..9
Selects the codec rate via its index.

Parameters:
<CodecRate> OFF | ON
OFF: codec rate not supported
ON: codec rate supported
*RST: OFF

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "AMR-WB-IO Codec Rates" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect
<BWRANGE>

Selects a configuration mode for the bandwidth and bit-rate settings of the EVS primary mode.

The uplink (receive) and the downlink (send) can be configured together (`COMMon`) or separately (`SENDrx`).

Suffix:
<v> 1..20

Parameters:
<BWRANGE> COMMon | SENDrx

COMMon
The following commands apply:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:BWCommon`
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:COMMon:BITRate:RANGe`

SENDrx
The following commands apply:
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BW`
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BITRate:RANGe`
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BW`
`CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BITRate:RANGe`

*RST: COMM

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Select" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:BWCommon
<BWCommon>

Selects the codec bandwidths supported in the EVS primary mode.

The setting applies only if the uplink (receive) and the downlink (send) are configured together, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:
<BWCommon> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

*RST: NB

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Bandwidth" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:COMMon:BITRate:RANGe
<BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode. The value must be compatible to the selected bandwidth.

The setting applies only if the uplink (receive) and the downlink (send) are configured together, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:
<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R59

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R244

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "Bit Rate" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BW <SENDBW>

Selects the codec bandwidths supported in the EVS primary mode in the downlink (send) direction.

The setting applies only if the uplink and the downlink are configured separately, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:
<SENDBW> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

*RST: NB

Firmware/Software: V3.5.30

Manual operation: See "Bandwidth" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SEND:BITRate:RANGe <BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode in the downlink (send) direction. The value must be compatible to the selected bandwidth.

The setting applies only if the uplink and the downlink are configured separately, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:
<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R59

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R244

Firmware/Software: V3.5.30

Manual operation: See "Bit Rate" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BW <RXBW>

Selects the codec bandwidths supported in the EVS primary mode in the uplink (receive) direction.

The setting applies only if the uplink and the downlink are configured separately, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:
<RXBW> NB | WB | SWB | FB | NBWB | NBSWb | NBFB

NB: narrowband only
WB: wideband only
SWB: super wideband only
FB: fullband only
NBWB: narrowband and wideband
NBSWb: narrowband, wideband and super wideband
NBFB: narrowband, wideband, super wideband and fullband

*RST: NB

Firmware/Software: V3.5.30

Manual operation: See "Bandwidth" on page 89

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:RECeive:BITRate:RANGe <BitrateLower>, <BitrateHigher>

Selects the bit-rate range supported in the EVS primary mode in the uplink (receive) direction. The value must be compatible to the selected bandwidth.

The setting applies only if the uplink and the downlink are configured separately, see `CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:SYNCh:SELect` on page 247.

Suffix:
<v> 1..20

Parameters:

<BitrateLower> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Lower end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R59

<BitrateHigher> R59 | R72 | R80 | R96 | R132 | R164 | R244 | R320 | R480 | R640 | R960 | R1280

Upper end of the range, 5.9 kbit/s to 128 kbit/s

*RST: R244

Firmware/Software: V3.5.30

Manual operation: See "Bit Rate" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CMR <CMR>

Specifies the SDP parameter "cmr" for the EVS codec.

Suffix:

<v> 1..20

Parameters:

<CMR> DISable | ENABle | PRESent | NP

Disable, enable, present all, not present

*RST: NP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30

Manual operation: See "CMR" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:CHAWmode <CHAWMODE>

Specifies the SDP parameter "ch-aw-recv" for the EVS codec.

Suffix:

<v> 1..20

Parameters:

<CHAWMODE> DIS | NUSed | TWO | THRee | FIVE | SEVen | NP

Disabled, not used, 2, 3, 5, 7, not present

*RST: DIS

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.30
V3.5.31: added TWO | THRee | FIVE | SEVen

Manual operation: See "Channel-Aware Mode" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:CODec <Codec>

Selects the codec for video calls.

Suffix:
<v> 1..20

Parameters:
<Codec> H263 | H264
H.263 or H.264 codec
*RST: H263

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Video" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:VIDeo:ATTRibutes <Attributes>

Configures codec attributes for video calls.

Suffix:
<v> 1..20

Parameters:
<Attributes> Codec attributes as string

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10

Manual operation: See "Video" on page 90

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MEDiaendpoin <MediaEndpoint>

Configures the media endpoint.

Suffix:
<v> 1..20

Parameters:
<MediaEndpoint> LOOPback | FORWard | AUDioboard | PCAP
LOOPback: Loop back to the mobile
FORWard: Route to an external media endpoint
AUDioboard: Route to the speech codec of the audio board
PCAP: Play a PCAP file
*RST: LOOP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.10
V3.5.30: added PCAP

Manual operation: See "Media Endpoint" on page 91

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:FORWard <IPAddress>, <Port>, <CmdPort>, <AMRAlign>

Configures an external media endpoint.

Suffix:
<v> 1..20

Parameters:
<IPAddress> IPv4 or IPv6 address of the media endpoint as string
*RST: '172.22.2.2'

<Port> Port for RTP packet forwarding
*RST: 1000

<CmdPort> Port for the command interface to the media endpoint
*RST: 1000

<AMRAlign> OCTetaligned | BANDwidtheff
AMR alignment mode used by the media endpoint
OCTetaligned: octet-aligned
BANDwidtheff: bandwidth-efficient
*RST: BAND

Firmware/Software: V3.5.10

Manual operation: See "IP" on page 91

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:AUDioboard:CONFig <Instance>, <DTXEnable>, <ForceModeNB>, <ForceModeWB>, <ForceModeEVS>

Configures the audio board.

A query returns only the values that are relevant for the active codec:

- NB AMR: `<Instance>, <DTXEnable>, <ForceModeNB>`
- WB AMR: `<Instance>, <DTXEnable>, <ForceModeWB>`
- EVS: `<Instance>, <ForceModeEVS>`

Suffix:
<v> 1..20

Parameters:
<Instance> INST1 | INST2
Audio software instance 1 or 2
*RST: INST1

<DTXEnable> OFF | ON
Enable comfort noise in the downlink for AMR codecs
*RST: OFF

<ForceModeNB> ZERO | ONE | TWO | THRE | FOUR | FIVE | SIX | SEVN | FREE

Index of the codec rate to be used if the AMR narrowband codec is active
FREE means that no specific codec rate is forced

*RST: FREE

<ForceModeWB> ZERO | ONE | TWO | THRE | FOUR | FIVE | SIX | SEVN | EIGH | FREE

Index of the codec rate to be used if the AMR wideband codec is active
FREE means that no specific codec rate is forced

*RST: FREE

<ForceModeEVS> SDP | P28 | P72 | P80 | P96 | P132 | P164 | P244 | P320 | P480 | P640 | P960 | P1280 | A660 | A885 | A1265 | A1425 | A1585 | A1825 | A1985 | A2305 | A2385

Start mode and rate to be used if the EVS codec is active
SDP: no specific codec rate forced
P28 to P1280: EVS primary mode, 2.8 kbit/s to 128 kbit/s
A660 to A2385: AMR-WB IO mode, 6.6 kbit/s to 23.85 kbit/s

*RST: SDP

Example: See Configuring virtual subscriber profiles

Firmware/Software: V3.5.31

Manual operation: See "Enable DTX in DL" on page 92

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:EVS:IO:MODE:CONFig
<EVSIOModeCnfg>

Selects the speech packet type to be sent from the audio board to the mobile for the EVS AMR-WB IO mode.

Suffix:
<v> 1..20

Parameters:
<EVSIOModeCnfg> EVSamrwb | AMRWb

AMRWb: AMR-WB encoded packets
EVSamrwb: EVS AMR-WB IO encoded packets

*RST: EVS

Firmware/Software: V3.5.51

Manual operation: See "EVS IO Mode Codec" on page 93

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:PCAPfile:SELection
<PCAPFILE>

Selects an XML file with metadata for the media endpoint type "PCAP". The file must be on the samba share of the DAU, in the subdirectory `ims/pcap`.

Suffix:
<v> 1..20

Parameters:
<PCAPFILE> File name as string

Firmware/Software: V3.5.31

Manual operation: See "PCAP Meta File Selection" on page 93

7.9.9.3 Mobile-Terminating Calls

The following commands configure and control mobile-terminating voice over IMS calls from virtual subscriber profiles to the mobile.

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination:LIST?.....255
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination.....255
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:TYPE..... 256
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:CALL.....256

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination:LIST?

Queries a list of all possible destination strings for mobile-terminating calls.

Suffix:
<v> 1..20

Return values:
<Destinations> Comma-separated list of strings, one per destination

Example: See Performing a mobile-terminating voice over IMS call

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Destination" on page 94

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination
<Destination>

Specifies the destination to be called by virtual subscriber number <v>.

To query a list of all possible destination strings, see `SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:DESTination:LIST?` on page 255.

Suffix:
<v> 1..20

Parameters:
<Destination> Destination string

Example: See Performing a mobile-terminating voice over IMS call

Firmware/Software: V3.5.10

Manual operation: See "Destination" on page 94

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:TYPE <Type>

Selects the type of call to be initiated: audio call or video call.

Suffix:
<v> 1..20

Parameters:
<Type> AUDio | VIDeo
*RST: AUD

Example: See Performing a mobile-terminating voice over IMS call

Firmware/Software: V3.5.10

Manual operation: See "Call Type" on page 95

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTCall:CALL

Initiates the setup of a voice over IMS call, from virtual subscriber number <v> to the mobile.

Suffix:
<v> 1..20

Example: See Performing a mobile-terminating voice over IMS call

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Call (button)" on page 95

7.9.9.4 Mobile-Terminating Messages

The following commands configure and control mobile-terminating message transfers from virtual subscriber profiles to the mobile.

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination:LIST? 257
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination 257
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TYPE 257
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:ENCoding 258
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TEXT 258
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:IMPort:FILE 259
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:SEND 259

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination:LIST?

Queries a list of all possible destination strings for mobile-terminating messages.

Suffix:
<v> 1..20

Return values:
<Destinations> Comma-separated list of strings, one per destination

Example: See Sending a short message over IMS to the mobile

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Destination" on page 96

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination <Destination>

Specifies the destination to which messages are sent by virtual subscriber number <v>.

To query a list of all possible destination strings, see `SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:DESTination:LIST?` on page 257.

Suffix:
<v> 1..20

Parameters:
<Destination> Destination string

Example: See Sending a short message over IMS to the mobile

Firmware/Software: V3.5.10

Manual operation: See "Destination" on page 96

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TYPE <Type>

Selects the type of messages to be sent by virtual subscriber number <v>.

Suffix:
<v> 1..20

Parameters:

<Type> GPP | GPP2 | ACK | PAGer | LARGe | RCSChat | RCSGrpchat

GPP: 3GPP
GPP2: 3GPP2 without delivery ACK
ACK: 3GPP2 with delivery ACK
PAGer: RCS pager mode
LARGe: RCS large mode
RCSChat: RCS 1 to 1 chat
RCSGrpchat: RCS group chat

*RST: GPP

Example: See Sending a short message over IMS to the mobile

Firmware/Software: V3.5.10, V3.5.30 added PAGer, LARGe
V3.5.30 added RCSChat, V3.5.51 added RCSGrpchat

Options: R&S CMW-KAA21 for PAGer, LARGe, RCSChat, RCSGrpchat

Manual operation: See "Message Type" on page 96

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:ENCoding
<ENCODING>

Selects the encoding for RCS messages.

Suffix:

<v> 1..20

Parameters:

<ENCODING> NENCoding | BASE64

No encoding or Base64 encoding

*RST: NENC

Firmware/Software: V3.5.30

Options: R&S CMW-KAA21

Manual operation: See "Encoding" on page 97

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:TEXT <Text>

Defines the text for messages to be sent by virtual subscriber number <v>.

Suffix:

<v> 1..20

Parameters:

<Text> Message text as string

RST: "R&S Short Message Service Text. The quick brown fox jumps over the lazy dog. THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG. 0123456789 !"#%+-/()<>?=;@$,"

Example: See Sending a short message over IMS to the mobile

Firmware/Software: V3.5.10

Manual operation: See "Text" on page 97

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:IMPort:FILE <Selected File>

Imports the message text for "RCS Large Mode" transfer from a file, for virtual subscriber number <v>. The file must be on the samba share of the DAU, in the subdirectory `ims/rcs/samples`.

Suffix:
<v> 1..20

Parameters:
<Selected File> File name as string

Firmware/Software: V3.5.50

Manual operation: See "Import" on page 97

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTSMs:SEND

Initiates a message transfer from virtual subscriber number <v> to the mobile.

Suffix:
<v> 1..20

Example: See Sending a short message over IMS to the mobile

Usage: Event

Firmware/Software: V3.5.10

Manual operation: See "Send (button)" on page 97

7.9.9.5 File Transfer

The following commands configure and control file transfers from virtual subscriber profiles to the mobile.

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination:LIST? 259
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination 260
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:TYPE 260
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:CHUNksize 261
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:FILE:SELection 261
CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:SEND 261

SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination:LIST?

Queries a list of all possible destination strings for file transfers by virtual subscriber <v>.

Suffix:
<v> 1..20

Return values:
<Destinations> Comma-separated list of strings, one per destination

Example: See Transferring a file to the mobile

Usage: Query only

Firmware/Software: V3.5.40

Options: R&S CMW-KAA21

Manual operation: See "Destination" on page 98

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination <Destination>

Specifies the destination for file transfers by virtual subscriber number <v>.

To query a list of all possible destination strings, see `SENSe:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:DESTination:LIST?` on page 259.

Suffix:
<v> 1..20

Parameters:
<Destination> Destination string

Example: See Transferring a file to the mobile

Firmware/Software: V3.5.40

Options: R&S CMW-KAA21

Manual operation: See "Destination" on page 98

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:TYPE <File Transfer Type>

Selects the transfer type for file transfers by virtual subscriber number <v>.

Suffix:
<v> 1..20

Parameters:
<File Transfer Type> FILetransfer | LARGe
File transfer or file transfer large mode
*RST: FIL

Example: See Transferring a file to the mobile

Firmware/Software: V3.5.40

Options: R&S CMW-KAA21

Manual operation: See "File Transfer Type" on page 99

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:CHUNksize
<ChunkSize>

Specifies the size of the chunks into which the file is divided for transfer.

Suffix:
<v> 1..20

Parameters:
<ChunkSize> Range: 1 byte to 2.147483647E+9 bytes
*RST: 1 byte
Default unit: byte

Example: See Transferring a file to the mobile

Firmware/Software: V3.5.50

Options: R&S CMW-KAA21

Manual operation: See "Chunk Size" on page 99

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:FILE:SELection
<Selected File>

Selects a file for file transfers by virtual subscriber number <v>. The file must be on the samba share of the DAU, in the subdirectory `ims/rcs/samples`.

Suffix:
<v> 1..20

Parameters:
<Selected File> File name as string

Example: See Transferring a file to the mobile

Firmware/Software: V3.5.40

Options: R&S CMW-KAA21

Manual operation: See "File selection" on page 99

CONFigure:DATA:CONTrol:IMS2:VIRTualsub<v>:MTFiletfr:SEND

Initiates a file transfer from virtual subscriber number <v> to the mobile.

Suffix:
<v> 1..20

Example: See Transferring a file to the mobile

Usage: Event

Firmware/Software: V3.5.40

Options: R&S CMW-KAA21

7.10 ePDG Service

The commands described in this section configure and control the ePDG service.

- Service Control and Information....262
- ePDG IP and ID Settings....264
- IKE Settings....265
- ESP Settings....268
- P-CSCF IKEv2 Attributes....269
- Authentication Settings....270
- Dead Peer Detection Settings....273
- Certificate Settings....274

7.10.1 Service Control and Information

The following commands control the ePDG service and query status information.

SOURce:DATA:CONTrol:EPDG:STATe....262
SENSe:DATA:CONTrol:EPDG:EVENt:LOG?....262
CONFigure:DATA:CONTrol:EPDG:CLEan:GENeral:INFO....263
SENSe:DATA:CONTrol:EPDG:CONNections:IMSI?....263
SENSe:DATA:CONTrol:EPDG:CONNections:IMSI<Suffix>:APN?....263

SOURce:DATA:CONTrol:EPDG:STATe <Control>

Starts or stops the ePDG service.

Setting parameters:

<Control> ON | OFF

Switch the service **ON** or **OFF**

Return values:

<State> OFF | ON | PENDing

OFF: service switched off
ON: service switched on
PEND: service activation or deactivation ongoing

*RST: OFF

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.10

Manual operation: See "Service control" on page 100

SENSe:DATA:CONTrol:EPDG:EVENt:LOG?

Queries all entries of the event log.

For each entry, three parameters are returned, from oldest to latest entry: {<Timestamps>, <Type>, <Info>}$_{entry\ 1}$, {<Timestamps>, <Type>, <Info>}$_{entry\ 2}$, ...

Return values:

<Timestamps> Timestamp of the entry as string in the format `"hh:mm:ss"`

<Type> NONE | INFO | WARNing | ERRor
Category of the entry
NONE means that no category is assigned. If no entry at all is available, the answer is `"",NONE,""`.

<Info> Text string describing the event

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Event log entries" on page 101

CONFigure:DATA:CONTrol:EPDG:CLEan:GENeral:INFO

Clears the "ePDG Event Log" area.

Usage: Event

Firmware/Software: V3.5.51

Manual operation: See "Event log entries" on page 101

SENSe:DATA:CONTrol:EPDG:CONNections:IMSI?

Queries all IMSIs for which ePDG connections are established.

Return values:

<IMSI> Comma-separated list of 10 values
Each IMSI is returned as a string value. The remaining values are filled with INV.

Usage: Query only

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Connection information" on page 102

SENSe:DATA:CONTrol:EPDG:CONNections:IMSI<Suffix>:APN?

Queries the connection list for a selected IMSI.

The list contains 15 connections. If there are fewer connections, the remaining entries are filled with INV. Three parameters are returned for each of the 15 connections:

$\{<Name>, <IPv4>, <IPv6>\}_{conn\ 1}, \{...\}_{conn\ 2}, ..., \{...\}_{conn\ 15}$

Suffix:

<Suffix> 1..10
Selects the IMSI via its index

Return values:

<Name> Access point name (APN) as string

<IPv4> IPv4 address as string

<IPv6> IPv6 address as string

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Connection information" on page 102

7.10.2 ePDG IP and ID Settings

The following commands configure the address and ID of the ePDG.

CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVFour....264
CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVSix....264
CONFigure:DATA:CONTrol:EPDG:ID:TYPE....265
CONFigure:DATA:CONTrol:EPDG:ID:VALue....265

CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVFour <IPv4Address>

Specifies the IPv4 address of the ePDG.

Parameters:

<IPv4Address> IPv4 address string

*RST: '192.168.1.201'

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "ePDG IP Configuration" on page 102

CONFigure:DATA:CONTrol:EPDG:ADDRess:IPVSix <IPv6Address>

Specifies the IPv6 address of the ePDG.

Parameters:

<IPv6Address> IPv6 address string

*RST: 'fc01::1'

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "ePDG IP Configuration" on page 102

CONFigure:DATA:CONTrol:EPDG:ID:TYPE <IDType>

Configures the type of the ePDG identification.

Parameters:

<IDType> IPVF | FQDN | RFC | IPVS | KEY

IPVF: ID_IPv4_ADDR
FQDN: ID_FQDN
RFC: ID_RFC822_ADDR
IPVS: ID_IPV6_ADDR
KEY: ID_KEY_ID

*RST: FQDN

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "ePDG ID Configuration" on page 103

CONFigure:DATA:CONTrol:EPDG:ID:VALue <IDValue>

Configures the value of the ePDG identification.

Parameters:

<IDValue> Identification as string

*RST: 'Epdg.rs.org'

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "ePDG ID Configuration" on page 103

7.10.3 IKE Settings

The following commands configure settings for the IKEv2 protocol.

CONFigure:DATA:CONTrol:EPDG:IKE:ENCRyption....265
CONFigure:DATA:CONTrol:EPDG:IKE:PRF....266
CONFigure:DATA:CONTrol:EPDG:IKE:INTegrity....267
CONFigure:DATA:CONTrol:EPDG:IKE:DHGRoup....267

CONFigure:DATA:CONTrol:EPDG:IKE:ENCRyption <AESCBC>, <ENCR_3DES>

Selects the supported encryption algorithms for the IKEv2 protocol.

Parameters:

<AESCBC> OFF | ON
ENCR_AES_CBC
*RST: ON

<ENCR_3DES> OFF | ON
ENCR_3DES
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20
V3.5.30: added <ENCR_3DES>

Options: R&S CMW-KA065

Manual operation: See "Encryption" on page 103

CONFigure:DATA:CONTrol:EPDG:IKE:PRF <PRFMD5>, <PRFSHA1>, <SHA2_256>, <SHA2_384>, <SHA2_512>

Selects the supported pseudorandom functions for the IKEv2 protocol.

Parameters:

<PRFMD5> OFF | ON
PRF_HMAC_MD5
*RST: ON

<PRFSHA1> OFF | ON
PRF_HMAC_SHA1
*RST: ON

<SHA2_256> OFF | ON
PRF_HMAC_SHA2_256
*RST: ON

<SHA2_384> OFF | ON
PRF_HMAC_SHA2_384
*RST: ON

<SHA2_512> OFF | ON
PRF_HMAC_SHA2_512
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20
V3.5.30: added <SHA2_...>

Options: R&S CMW-KA065

Manual operation: See "PRF" on page 104

CONFigure:DATA:CONTrol:EPDG:IKE:INTegrity <HMAC_MD5_96>, <HMAC_SHAI_96>, <AES_XCB_96>, <SHA2_256_128>, <SHA2_384_192>, <SHA2_512_256>

Selects the supported integrity protection algorithms for the IKEv2 protocol.

Parameters:

<HMAC_MD5_96> OFF | ON
AUTH_HMAC_MD5_96
*RST: ON

<HMAC_SHAI_96> OFF | ON
AUTH_HMAC_SHA1_96
*RST: ON

<AES_XCB_96> OFF | ON
AUTH_AES_XCBC_96
*RST: ON

<SHA2_256_128> OFF | ON
AUTH_HMAC_SHA2_256_128
*RST: ON

<SHA2_384_192> OFF | ON
AUTH_HMAC_SHA2_384_192
*RST: ON

<SHA2_512_256> OFF | ON
AUTH_HMAC_SHA2_512_256
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20
V3.5.30: added <SHA2_...>

Options: R&S CMW-KA065

Manual operation: See "Integrity" on page 104

CONFigure:DATA:CONTrol:EPDG:IKE:DHGRoup <DHGroup1>, <DHGroup2>, <DHGroup5>, <DHGroup14>[, <DHGroup15>, <DHGroup16>, <DHGroup17>, <DHGroup18>]

Selects the supported Diffie-Hellman groups for the IKEv2 protocol.

Parameters:

<DHGroup1> OFF | ON
*RST: ON

<DHGroup2> OFF | ON
*RST: ON

<DHGroup5> OFF | ON
*RST: ON

<DHGroup14> OFF | ON
*RST: ON

<DHGroup15> OFF | ON
*RST: ON

<DHGroup16> OFF | ON
*RST: ON

<DHGroup17> OFF | ON
*RST: ON

<DHGroup18> OFF | ON
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20, V3.5.31: groups 15 to 18

Options: R&S CMW-KA065

Manual operation: See "Diffie Hellman" on page 104

7.10.4 ESP Settings

The following commands configure settings for the ESP protocol.

CONFigure:DATA:CONTrol:EPDG:ESP:ENCRyption 268
CONFigure:DATA:CONTrol:EPDG:ESP:INTegrity 269

CONFigure:DATA:CONTrol:EPDG:ESP:ENCRyption <AESCBC>, <ENCR_3DES>

Selects the supported encryption algorithms for the ESP protocol.

Parameters:

<AESCBC> OFF | ON
ENCR_AES_CBC
*RST: ON

<ENCR_3DES> OFF | ON
ENCR_3DES
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20
V3.5.30: added <ENCR_3DES>

Options: R&S CMW-KA065

Manual operation: See "Encryption" on page 105

CONFigure:DATA:CONTrol:EPDG:ESP:INTegrity <MD596>, <SHAI96>, <XCBC96>, <SHA256>, <SHA384>, <SHA512>

Selects the supported integrity protection algorithms for the ESP protocol.

Parameters:

<MD596> OFF | ON
AUTH_HMAC_MD5_96
*RST: ON

<SHAI96> OFF | ON
AUTH_HMAC_SHA1_96
*RST: ON

<XCBC96> OFF | ON
AUTH_AES_XCBC_96
*RST: ON

<SHA256> OFF | ON
AUTH_HMAC_SHA2_256_128
*RST: ON

<SHA384> OFF | ON
AUTH_HMAC_SHA2_384_192
*RST: ON

<SHA512> OFF | ON
AUTH_HMAC_SHA2_512_256
*RST: ON

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20
V3.5.30: added <SHA2_...>

Options: R&S CMW-KA065

Manual operation: See "Integrity" on page 106

7.10.5 P-CSCF IKEv2 Attributes

The following commands configure settings related to the transport of P-CSCF addresses via IKEv2.

CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVFour:TYPE....269
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:TYPE....270
CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:ADDRess:LENGth....270

CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVFour:TYPE <PCSCFIPV4TYP>

Sets the attribute type field of the P_CSCF_IP4_ADDRESS configuration attribute.

Parameters:

<PCSCFIPV4TYP> Range: 0 to 65535
*RST: 20

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.21

Manual operation: See "IPv4 Type" on page 106

CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:TYPE <PCSCFIPV6TYP>

Sets the attribute type field of the P_CSCF_IP6_ADDRESS configuration attribute.

Parameters:

<PCSCFIPV6TYP> Range: 0 to 65535
*RST: 21

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.21

Manual operation: See "IPv6 Type" on page 106

CONFigure:DATA:CONTrol:EPDG:PCSCf:IPVSix:ADDRess:LENGth <IPVSixADDLGH>

Sets the length field of the P_CSCF_IP6_ADDRESS configuration attribute.

Parameters:

<IPVSixADDLGH> L16 | L17
*RST: L16

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.21

Manual operation: See "IPv6 Address Length" on page 106

7.10.6 Authentication Settings

The following commands configure settings for the authentication procedure.

CONFigure:DATA:CONTrol:EPDG:AUTHentic:IMSI 270
CONFigure:DATA:CONTrol:EPDG:AUTHentic:ALGorithm 271
CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY 271
CONFigure:DATA:CONTrol:EPDG:AUTHentic:RAND 271
CONFigure:DATA:CONTrol:EPDG:AUTHentic:AMF 272
CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY:TYPE 272
CONFigure:DATA:CONTrol:EPDG:AUTHentic:OPC 272

CONFigure:DATA:CONTrol:EPDG:AUTHentic:IMSI <AuthIMSI>

Specifies the IMSI of the SIM card.

Parameters:
<AuthIMSI> String value, containing 15 digits
*RST: '001010123456789'

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "IMSI" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:ALGorithm <AuthAlg>

Specifies the key generation algorithm set used by the SIM.

Parameters:
<AuthAlg> XOR | MILenage
*RST: XOR

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Authentication Algorithm" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY <AuthKey>

Defines the secret key K as 32-digit hexadecimal number. Leading zeros can be omitted.

Parameters:
<AuthKey> Range: #H0 to #HFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
*RST: #H000102030405060708090A0B0C0D0E0F

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Secret Key" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:RAND <AuthRAND>

Defines the random number RAND as 32-digit hexadecimal number. Leading zeros can be omitted.

Parameters:
<AuthRAND> Range: #H0 to #HFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
*RST: #H55555555555555555555555555555555

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "RAND" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:AMF <AuthAMF>

Specifies the authentication management field (AMF) as four-digit hexadecimal number. Leading zeros can be omitted.

Parameters:

<AuthAMF>	Range:	#H0 to #HFFFF
	*RST:	#H0000

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "AMF" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:KEY:TYPE <AuthKeyType>

Selects the key type to be used with the MILENAGE algorithm set.

Currently, only OP_c is supported.

Parameters:

<AuthKeyType>	OPC

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Key Type" on page 107

CONFigure:DATA:CONTrol:EPDG:AUTHentic:OPC <AuthOPC>

Specifies the key OP_c as 32-digit hexadecimal number. Leading zeros can be omitted.

Parameters:

<AuthOPC>	Range:	#H00000000000000000000000000000000 to #HFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
	*RST:	#HCD63CB71954A9F4E48A5994E37A02BAF

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Authentication OP_c" on page 108

7.10.7 Dead Peer Detection Settings

The following commands configure settings for dead peer detection.

CONFigure:DATA:CONTrol:EPDG:DPD:ENABle....273
CONFigure:DATA:CONTrol:EPDG:DPD:INTerval....273
CONFigure:DATA:CONTrol:EPDG:DPD:TIMeout....273

CONFigure:DATA:CONTrol:EPDG:DPD:ENABle <DPDENABLE>

Enables dead peer detection.

Parameters:
<DPDENABLE> OFF | ON
*RST: OFF

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Enable" on page 108

CONFigure:DATA:CONTrol:EPDG:DPD:INTerval <DPDInterval>

Configures the inactivity time interval for dead peer detection.

Parameters:
<DPDInterval> Range: 1 s to 100 s
*RST: 1 s
Default unit: s

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Interval, Timeout" on page 108

CONFigure:DATA:CONTrol:EPDG:DPD:TIMeout <DPDTimeout>

Configures the no answer timeout for dead peer detection.

Parameters:
<DPDTimeout> Range: 1 s to 10E+3 s
*RST: 10 s
Default unit: s

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Interval, Timeout" on page 108

7.10.8 Certificate Settings

The following commands configure settings related to SSL certificates.

CONFigure:DATA:CONTrol:EPDG:CERTificate:ENABle 274
CONFigure:DATA:CONTrol:EPDG:CERTificate:KEY 274
CONFigure:DATA:CONTrol:EPDG:CERTificate:CERTificate 274

CONFigure:DATA:CONTrol:EPDG:CERTificate:ENABle <CertificateEnable>

Enables the usage of certificates for SSL.

Parameters:
<CertificateEnable> OFF | ON
*RST: OFF

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Enable" on page 109

CONFigure:DATA:CONTrol:EPDG:CERTificate:KEY <CertificateKeyFile>

Selects the server key file to be used for SSL.

Parameters:
<CertificateKeyFile> Filename as string, without path

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Server Key, Server Certificate" on page 109

CONFigure:DATA:CONTrol:EPDG:CERTificate:CERTificate <CertificateFile>

Selects the server certificate file to be used for SSL.

Parameters:
<CertificateFile> Filename as string, without path

Example: See Configuring the ePDG Service

Firmware/Software: V3.5.20

Options: R&S CMW-KA065

Manual operation: See "Server Key, Server Certificate" on page 109

8 DAU Measurements Command Reference

The following sections provide detailed reference information on the remote control commands related to the DAU measurements (`...:DATA:MEAS<i>:...` commands).

- Conventions and General Information 275
- General 280
- Ping Measurement 281
- Iperf Measurement 285
- Throughput Measurement 294
- DNS Requests Measurement 303
- IP Logging 306
- IP Analysis and Security 310
- IP Replay 340
- Audio Delay Measurement 346
- Quality-of-Service Settings 351

8.1 Conventions and General Information

The following sections describe the most important conventions and general information concerning the command reference.

8.1.1 MEAS<i>

MEAS<i> is used as abbreviation of "MEASurement<instance>". For better readability only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same firmware application. It can be omitted if the instrument supports only one instance, or to address the first instance.

See also: "Firmware Applications" in the R&S CMW base unit manual, chapter "Remote Control"

8.1.2 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**
  If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise the command usage is stated as follows:
  - "Setting only": Command can only be used to set parameters.
  - "Query only": Command can only be used to query parameters.
  - "Event": Command initiates an event.

- **Parameter usage**
  The parameter usage is indicated by the keyword preceding the parameters:
  - "Parameters" are sent with a setting or query command and are returned as the result of a query
  - "Setting parameters" are only sent with a setting command
  - "Query parameters" are only sent with a query command (to refine the query)
  - "Return values" are only returned as the result of a query
- **Firmware/Software:**
  Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

8.1.3 Reliability Indicator

The first value in the output arrays of `FETCh...?, READ...?` and `CALCulate...?` queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

The reliability indicator has one of the following values:

- **0 (OK)**:
  Measurement values available, no error detected.
- **1 (measurement timeout)**:
  The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.
- **2 (capture buffer overflow)**:
  The measurement configuration results in a capture length that exceeds the available memory.
- **3 (overdriven) / 4 (underdriven)**:
  The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 (trigger timeout)**:
  The measurement could not be started or continued because no trigger event was detected.
- **7 (acquisition error)**:
  The R&S CMW could not properly decode the RF input signal.
- **8 (sync error)**:
  The R&S CMW could not synchronize to the RF input signal.
- **9 (uncalibrated)**:
  Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the specified data sheet limits.
- **15 (reference frequency error)**:
  The instrument has been configured to use an external reference signal. But the reference oscillator could not be phase-locked to the external signal (for example signal level too low, frequency out of range or reference signal not available at all).

- **16 (RF not available)**:
  The measurement could not be started because the configured RF input path was not active. This problem can occur if a measurement is started in combined signal path mode and the master application has not yet activated the input path. The LEDs above the RF connectors indicate whether the input and output paths are active.
- **17 (RF level not settled) / 18 (RF frequency not settled)**:
  The measurement could not be started because the R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 (call not established)**:
  For measurements: The measurement could not be started because no signaling connection to the DUT was established.
  For DAU IMS service: Establishing a voice over IMS call failed.
- **20 (call type not usable)**:
  For measurements: The measurement could not be started because the established signaling connection had wrong properties.
  For DAU IMS service: The voice over IMS settings could not be applied.
- **21 (call lost)**:
  For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.
  For DAU IMS service: The voice over IMS call was lost.
- **23 (missing option)**:
  The ARB file cannot be played by the GPRF generator due to a missing option.
- **24 (invalid RF setting)**:
  The desired RF TX level or RF RX reference level could not be applied.
- **25 (level overrange)**:
  The RF TX level is in overrange. The signal quality can be degraded.
- **26 (resource conflict)**:
  The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.
  Stop the application that has allocated the conflicting resources and try again.
- **27 (no sensor connected)**:
  The GPRF external power sensor measurement could not be started due to missing power sensor.
- **28 (unexpected parameter change)**:
  One or more measurement configuration parameters were changed while the measurement completed. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.
- **30 (file not found)**:
  The specified file could not be found.
- **31 (no DTM reply)**:
  The EUT did not reply to the direct test mode (DTM) command.
- **32 (ACL disconnected)**:
  The ACL connection has been disconnected or lost.

- **40 (ARB file CRC error)**:
  The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.
- **42 (ARB header tag invalid)**:
  The ARB file selected in the GPRF generator contains an invalid header tag.
- **43 (ARB segment overflow)**:
  The number of segments in the multi-segment ARB file is higher than the allowed maximum.
- **44 (ARB file not found)**:
  The selected ARB file could not be found.
- **45 (ARB memory overflow)**:
  The ARB file length is greater than the available memory.
- **46 (ARB sample rate out of range)**:
  The clock rate of the ARB file is either too high or too low.
- **47 (ARB cycles out of range)**:
  The repetition mode equals "Single Shot" and the playback length is greater than 40 s. Reduce the playback length or set the repetition mode to "Continuous".
  *<Length> = (<Cycles> * <Samples> + <Additional Samples>) / <Clock Rate>*
- **50 (startup error)**:
  The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 (no reply)**:
  The DAU has received no response, for example for a ping request.
- **52 (connection error)**:
  The DAU could not establish a connection to internal components. Restart the instrument.
- **53 (configuration error)**:
  The current DAU configuration is incomplete or wrong and could not be applied. Check especially the IP address configuration.
- **54 (filesystem error)**:
  The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **70 (wrong standard)**
  The standard of the measured signal does not match the configured standard.
- **71 (wrong bandwidth)**
  The bandwidth of the measured signal does not match the configured bandwidth.
- **72 (wrong burst type)**
  The burst type of the measured signal does not match the configured burst type.
- **73 (MIMO signal detected)**
  The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 (more streams than antennas)**
  The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 (matrix inversion failed)**

The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.

- **76 (SIG CRC failed)**
  The cyclic redundancy check of a SIGNAL field failed.
- **77 (parity check failed)**
  The parity check of a SIGNAL field failed.
- **78 (bursts not identical)**
  In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 (wrong modulation)**
  The modulation type of the measured signal does not match the configured modulation type.
- **93 (OCXO oven temperature too low)**:
  The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **101 (firmware error)**:
  Indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.

  If the error occurs again, consider the following hints:

  – Firmware errors can often be repaired by restoring the factory default settings. To restore these settings, restart your instrument and press the "Factory Default" softkey during startup.
  – If a software package (update) has not been properly installed, this failure is often indicated in the "Setup" dialog, section "SW/HW-Equipment > Installed Software".
  – Check for software updates correcting the error. Updates are provided in the CMW customer web on GLORIS (registration required): https://gloris.rohde-schwarz.com.

  If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.
- **102 (unidentified error)**:
  Indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 (firmware error)".
- **103 (parameter error)**:
  Indicates that the measurement could not be performed due to internal conflicting parameter settings.
  A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.
  If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see http://www.service.rohde-schwarz.com).
- **104 (not functional)**:
  The application could not be started with the configured parameter set.

8.2 General

The following commands select a radio access network (RAN) and the displayed measurement tab.

CONFigure:DATA:MEAS<i>:RAN 280
CONFigure:DATA:MEAS<i>:RAN:CATaloge? 280
CONFigure:DATA:MEAS<i>:SELect:APP 280

CONFigure:DATA:MEAS<i>:RAN <RAN>

Selects an installed signaling application instance.

You can query a complete list of all supported strings via the command `CONFigure:DATA:MEAS<i>:RAN:CATaloge?`.

Parameters:

<RAN> String parameter, selecting a signaling application instance

Example: See Configuring Global Settings

Firmware/Software: V1.0.15.20

Manual operation: See "Select RAN" on page 111

CONFigure:DATA:MEAS<i>:RAN:CATaloge?

Lists all available signaling applications. You can use the returned strings in other commands to select a RAN.

Return values:

<RAN> Comma-separated list of all supported values. Each value is represented as a string.

Example: See Configuring Global Settings

Usage: Query only

Firmware/Software: V1.0.15.20

Manual operation: See "Select RAN" on page 111

CONFigure:DATA:MEAS<i>:SELect:APP <ApplicationType>

Selects the measurement tab to be displayed.

Parameters:

<ApplicationType> OVERview | PING | IPERf | THRoughput | DNSReq | IPLogging | IPANalysis | IPReplay | AUDiodelay

Firmware/Software: V3.5.30

Manual operation: See "Select Applic (hotkey)" on page 163

8.3 Ping Measurement

The commands in this section are related to the ping measurement. They configure the measurement, control it and return the current measurement state and the measurement results.

CONFigure:DATA:MEAS<i>:PING:DIPaddress....281
CONFigure:DATA:MEAS<i>:PING:INTerval....281
CONFigure:DATA:MEAS<i>:PING:TIMeout....281
CONFigure:DATA:MEAS<i>:PING:PSIZe....282
CONFigure:DATA:MEAS<i>:PING:PCOunt....282
INITiate:DATA:MEAS<i>:PING....282
STOP:DATA:MEAS<i>:PING....282
ABORt:DATA:MEAS<i>:PING....282
FETCh:DATA:MEAS<i>:PING:STATe?....283
FETCh:DATA:MEAS<i>:PING:STATe:ALL?....283
FETCh:DATA:MEAS<i>:PING?....284
READ:DATA:MEAS<i>:PING?....284
FETCh:DATA:MEAS<i>:PING:OVERall?....284
FETCh:DATA:MEAS<i>:PING:NRCount?....285

CONFigure:DATA:MEAS<i>:PING:DIPaddress <IPAddress>

Specifies the destination IP address for the ping command.

Parameters:

<IPAddress> IPv4 or IPv6 address as string
*RST: '172.22.1.100'

Example: See Performing a Ping Measurement

Firmware/Software: V1.0.15.20

Manual operation: See "Config" on page 114

CONFigure:DATA:MEAS<i>:PING:INTerval <Interval>

Specifies the interval between two ping requests.

Parameters:

<Interval> Range: 0.2 s to 10 s
*RST: 1 s
Default unit: s

Example: See Performing a Ping Measurement

Firmware/Software: V1.0.15.20

Manual operation: See "Config" on page 114

CONFigure:DATA:MEAS<i>:PING:TIMeout <Timeout>

Specifies a timeout for ping requests.

Parameters:

<Timeout>
Range: 1 s to 9 s
*RST: 2 s
Default unit: s

Example: See Performing a Ping Measurement

Firmware/Software: V3.2.20

Manual operation: See "Config" on page 114

CONFigure:DATA:MEAS<i>:PING:PSIZe <PacketSize>

Specifies the payload size of echo request packets.

Parameters:

<PacketSize>
Range: 0 bytes to 65507 bytes
*RST: 100 bytes
Default unit: bytes

Example: See Performing a Ping Measurement

Firmware/Software: V1.0.15.20
V3.0.10: range and *RST value changed

Manual operation: See "Config" on page 114

CONFigure:DATA:MEAS<i>:PING:PCOunt <PingCount>

Specifies the number of echo request packets to be sent.

Parameters:

<PingCount>
Range: 1 to 1000
*RST: 100

Example: See Performing a Ping Measurement

Firmware/Software: V1.0.15.20

Manual operation: See "Config" on page 114

INITiate:DATA:MEAS<i>:PING
STOP:DATA:MEAS<i>:PING
ABORt:DATA:MEAS<i>:PING

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.

- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing a Ping Measurement

Usage: Event

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 114

FETCh:DATA:MEAS<i>:PING:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Example: See Performing a Ping Measurement

Usage: Query only

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 114

FETCh:DATA:MEAS<i>:PING:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 114

FETCh:DATA:MEAS<i>:PING?
READ:DATA:MEAS<i>:PING?

Queries the measured ping characteristics.

After the reliability indicator, three results are returned for each ping request:

<Reliability>, {<ReqNo>, <Timestamp>, <Latency>}$_{request\ 1}$, {...}$_{request\ 2}$, ...

The number of ping requests is specified by `CONFigure:DATA:MEAS<i>:PING:PCOunt`.

Return values:

<Reliability> See Reliability Indicator

<ReqNo> Request label, 0 = last request, -1 = previous request, and so on

Range: -1000 to 0

<Timestamp> Timestamp as string in the format 'hh:mm:ss'

<Latency> Round-trip time for sent packets (0 s = no reply)

Range: 0 s to 10 s
Default unit: s

Example: See Performing a Ping Measurement

Usage: Query only

Firmware/Software: V1.0.15.20
V3.0.10: latency unit changed from ms to s

Manual operation: See "Result diagram" on page 113

FETCh:DATA:MEAS<i>:PING:OVERall?

Query the statistical ping results over all ping requests.

Return values:

<Reliability> See Reliability Indicator

<MinPing> Minimum ping latency
Range: 0 s to 10 s
Default unit: s

<MaxPing> Maximum ping latency
Range: 0 s to 10 s
Default unit: s

<AveragePing> Average ping latency
Range: 0 s to 10 s
Default unit: s

Example: See Performing a Ping Measurement

Usage: Query only

Firmware/Software: V3.2.11

Manual operation: See "Average / Minimum / Maximum" on page 113

FETCh:DATA:MEAS<i>:PING:NRCount?

Queries the number of ping requests that have not been answered (no reply).

Return values:

<ReqNo> Range: 0 to 1000

Example: See Performing a Ping Measurement

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "'No Reply' Count" on page 114

8.4 Iperf Measurement

The commands in this section are related to the IP performance measurement. They configure the measurement, control it and return the current measurement state and the measurement results.

CONFigure:DATA:MEAS<i>:IPERf:TDURation 286
CONFigure:DATA:MEAS<i>:IPERf:TYPE 286
CONFigure:DATA:MEAS<i>:IPERf:PSIZe 286
CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:ENABle 287
CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PROTocol 287
CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PORT 287
CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:WSIZe 288
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:ENABle 288
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PROTocol 288

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PORT 289
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:IPADdress 289
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:WSIZe 290
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PCONnection 290
CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:BITRate 290
INITiate:DATA:MEAS<i>:IPERf 291
STOP:DATA:MEAS<i>:IPERf 291
ABORt:DATA:MEAS<i>:IPERf 291
FETCh:DATA:MEAS<i>:IPERf:STATe? 291
FETCh:DATA:MEAS<i>:IPERf:STATe:ALL? 292
FETCh:DATA:MEAS<i>:IPERf:SERVer? 292
READ:DATA:MEAS<i>:IPERf:SERVer? 292
FETCh:DATA:MEAS<i>:IPERf:CLIent? 293
READ:DATA:MEAS<i>:IPERf:CLIent? 293
FETCh:DATA:MEAS<i>:IPERf:ALL? 293

CONFigure:DATA:MEAS<i>:IPERf:TDURation <TestDuration>

Defines the duration of the test.

Parameters:

<TestDuration>
Range: 1 s to 1E+6 s
*RST: 1000 s
Default unit: s

Example: See Performing an Iperf Measurement

Firmware/Software: V1.0.15.20

Manual operation: See "Test Duration" on page 116

CONFigure:DATA:MEAS<i>:IPERf:TYPE <IPERFTYPE>

Selects the iperf version to be used.

Parameters:

<IPERFTYPE>
IPERf | IP3
Iperf or iperf3
*RST: IPER

Example: See Performing an Iperf Measurement

Firmware/Software: V3.5.30

Manual operation: See "IPerf Type" on page 117

CONFigure:DATA:MEAS<i>:IPERf:PSIZe <PacketSize>

Defines the packet size for iperf tests.

Parameters:

<PacketSize> Range: 40 bytes to 65507 bytes
*RST: 1470 bytes
Default unit: bytes

Example: See Performing an Iperf Measurement

Firmware/Software: V3.2.20

Manual operation: See "Packet Size" on page 117

CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:ENABle <Enable>

Activates or deactivates an iperf server instance.

Suffix:

<Index> 1..8
Selects the server instance

Parameters:

<Enable> OFF | ON
*RST: ON for instance 1, else OFF

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Server settings" on page 117

CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PROTocol <Protocol>

Selects the protocol type to be used for an iperf server instance.

Suffix:

<Index> 1..8
Selects the server instance

Parameters:

<Protocol> UDP | TCP
UDP: use the user datagram protocol for data transfer
TCP: use the transport control protocol for data transfer
*RST: TCP

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Server settings" on page 117

CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:PORT <Port>

Defines the LAN DAU port number for an iperf server instance.

Suffix:
<Index> 1..8
Selects the server instance

Parameters:
<Port> Range: 0 to 65535
*RST: 5001 to 5008 for instance 1 to 8

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Server settings" on page 117

CONFigure:DATA:MEAS<i>:IPERf:SERVer<Index>:WSIZe <WindowSize>

Specifies the size of the NACK window for an iperf server instance. Only applicable for protocol type TCP.

Suffix:
<Index> 1..8
Selects the server instance

Parameters:
<WindowSize> Range: 0 kByte to 10.24E+3 kByte
*RST: 0 kByte
Default unit: kByte

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Server settings" on page 117

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:ENABle <Enable>

Activates or deactivates an iperf client instance.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<Enable> OFF | ON
*RST: ON for instance 1, else OFF

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PROTocol <Protocol>

Selects the protocol type to be used for an iperf client instance.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<Protocol> UDP | TCP
UDP: use the user datagram protocol for data transfer
TCP: use the transport control protocol for data transfer
*RST: TCP

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PORT <Port>

Defines the LAN DAU port number for an iperf client instance.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<Port> Range: 0 to 65535
*RST: 5001 to 5008 for instance 1 to 8

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:IPADdress <IPAddress>

Specifies the IP address of the mobile for an iperf client instance.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<IPAddress> String containing the IP address
*RST: '172.22.1.100'

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:WSIZe <WindowSize>

Specifies the size of the NACK window for an iperf client instance. Only applicable for protocol type TCP.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<WindowSize> Range: 0 kByte to 10.24E+3 kByte
*RST: 0 kByte
Default unit: kByte

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:PCONnection <ParConn>

Specifies the number of parallel connections for an iperf client instance. Only applicable for protocol type TCP.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<ParConn> Range: 1 to 4
*RST: 1

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

CONFigure:DATA:MEAS<i>:IPERf:CLIent<Index>:BITRate <BitRate>

Defines the maximum bit rate for an iperf client instance. Only applicable for protocol type UDP.

Suffix:
<Index> 1..8
Selects the client instance

Parameters:
<BitRate> Maximum bit rate to be transferred
Range: 0 bit/s to 1E+9 bit/s
*RST: 1 bit/s
Default unit: bit/s

Example: See Performing an Iperf Measurement

Firmware/Software: V2.0.20

Manual operation: See "Client settings" on page 118

INITiate:DATA:MEAS<i>:IPERf
STOP:DATA:MEAS<i>:IPERf
ABORt:DATA:MEAS<i>:IPERf

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing an Iperf Measurement

Usage: Event

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 118

FETCh:DATA:MEAS<i>:IPERf:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Usage: Query only

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 118

FETCh:DATA:MEAS<i>:IPERf:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate...`, `STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V1.0.15.20

Manual operation: See "Measurement control" on page 118

FETCh:DATA:MEAS<i>:IPERf:SERVer?
READ:DATA:MEAS<i>:IPERf:SERVer?

Queries the uplink throughput for all server instances.

Return values:

<Reliability> See Reliability Indicator

<UpBandwidth> Comma-separated list of eight results (server instance 1 to 8)

Default unit: bit/s

Example: See Performing an Iperf Measurement

Usage: Query only

Firmware/Software: V2.0.20

Manual operation: See "Results" on page 116

FETCh:DATA:MEAS<i>:IPERf:CLIent?
READ:DATA:MEAS<i>:IPERf:CLIent?

Queries the downlink throughput for all client instances.

Return values:

<Reliability> See Reliability Indicator

<DownBandwidth> Comma-separated list of eight results (client instance 1 to 8)
Default unit: bit/s

Example: See Performing an Iperf Measurement

Usage: Query only

Firmware/Software: V2.0.20

Manual operation: See "Results" on page 116

FETCh:DATA:MEAS<i>:IPERf:ALL?

Queries all client and server results of the iperf measurement.

For each server/client instance five results are returned, from instance 1 to instance 8:

<Reliability>, {<ServerCounter>, <ClientCounter>, <UpBandwidth>, <PackErrRate>, <DownBandwidth>}$_{\text{instance 1}}$, {...}$_{\text{instance 2}}$, ..., {...}$_{\text{instance 8}}$

Iperf results are often queried within a loop, to monitor the results over some time. Iperf delivers new results once per second. If your loop is faster, several consecutive queries deliver the same results. Use the <ServerCounter> and <ClientCounter> to identify redundant results and discard them.

Return values:

<Reliability> See Reliability Indicator

<ServerResultCounter> Counter for server results
The first result delivered by iperf is labeled 0, the second result 1, and so on.

<ClientResultCounter> Counter for client results
The first result delivered by iperf is labeled 0, the second result 1, and so on.

<UpBandwidth> Uplink throughput, determined for active server instances
Default unit: bit/s

<PackErrRate> Percentage of lost packets, determined for active server instances with protocol type UDP
Range: 0 % to 100 %
Default unit: %

<DownBandwidth> Downlink throughput, determined for active client instances
Default unit: bit/s

Example: See Performing an Iperf Measurement

Usage: Query only

Firmware/Software: V3.2.20

Manual operation: See "Results" on page 116

8.5 Throughput Measurement

The commands in this section are related to the throughput measurement. They configure the measurement, control it and return the current measurement state and the measurement results.

SENSe:DATA:MEAS<i>:THRoughput:INTerval? 295
CONFigure:DATA:MEAS<i>:THRoughput:MCOunt 295
CONFigure:DATA:MEAS<i>:THRoughput:RAN:MCOunt 295
CONFigure:DATA:MEAS<i>:THRoughput:RAN:CATaloge? 295
CONFigure:DATA:MEAS<i>:THRoughput:RAN<Index> 296
CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:ULINk<Index> 296
CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:DLINk<Index> 296
CONFigure:DATA:MEAS<i>:SELect:THRoughput 297
INITiate:DATA:MEAS<i>:THRoughput 297
STOP:DATA:MEAS<i>:THRoughput 297
ABORt:DATA:MEAS<i>:THRoughput 297
FETCh:DATA:MEAS<i>:THRoughput:STATe? 297
FETCh:DATA:MEAS<i>:THRoughput:STATe:ALL? 298
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk[:CURRent]? 298
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk[:CURRent]? 298
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk[:CURRent]? 299
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk[:CURRent]? 299
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk:EXTended? 299
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk:EXTended? 299
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk:EXTended? 299
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk:EXTended? 299
FETCh:DATA:MEAS<i>:THRoughput:OVERall:ULINk? 300
READ:DATA:MEAS<i>:THRoughput:OVERall:ULINk? 300
FETCh:DATA:MEAS<i>:THRoughput:OVERall:DLINk? 300
READ:DATA:MEAS<i>:THRoughput:OVERall:DLINk? 300
FETCh:DATA:MEAS<i>:THRoughput:TRACe:RAN:ULINk<Index>:CURRent? 301
FETCh:DATA:MEAS<i>:THRoughput:TRACe:RAN:DLINk<Index>:CURRent? 301
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:ULINk<Index>:CURRent? 301
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:DLINk<Index>:CURRent? 301
FETCh:DATA:MEAS<i>:THRoughput:RAN:DLINk<Index>? 301
READ:DATA:MEAS<i>:THRoughput:RAN:DLINk<Index>? 301
FETCh:DATA:MEAS<i>:THRoughput:RAN:ULINk<Index>? 302
READ:DATA:MEAS<i>:THRoughput:RAN:ULINk<Index>? 302
FETCh:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:ULINk? 302
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:ULINk? 302
FETCh:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:DLINk? 303
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:DLINk? 303

SENSe:DATA:MEAS<i>:THRoughput:INTerval?

Queries the time interval between two throughput measurement results.

Return values:

<Interval> In the current software version, the value is fixed.
*RST: 1 s
Default unit: s

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Settings" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:MCOunt <MaxCount>

Specifies the total number of overall throughput results to be measured.

Parameters:

<MaxCount> Range: 5 to 3600
*RST: 5

Example: See Performing a Throughput Measurement

Firmware/Software: V2.0.10
V3.0.10: Range and *RST value changed

Manual operation: See "Settings" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:RAN:MCOunt <MaxCount>

Specifies the total number of RAN throughput results to be measured.

Parameters:

<MaxCount> Range: 5 to 3600
*RST: 5

Example: See Performing a Throughput Measurement

Firmware/Software: V3.5.40

Manual operation: See "Settings" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:RAN:CATaloge?

Lists all available signaling applications. You can use the returned strings in other commands to select a RAN.

Return values:

<RAN> Comma-separated list of all supported values. Each value is represented as a string.

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.40

Manual operation: See "RAN trace selection" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:RAN<Index> <RAN>

Assigns a RAN to the RAN slot number <Index>.

You can query a complete list of all supported strings via the command `CONFigure:DATA:MEAS<i>:THRoughput:RAN:CATaloge?`.

Suffix:
<Index> 1..4

Parameters:
<RAN> String parameter, selecting a signaling application instance

Example: See Performing a Throughput Measurement

Firmware/Software: V3.5.30

Manual operation: See "RAN trace selection" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:ULINk<Index>
<UplinkSelection>

Enables or disables the uplink trace for RAN slot number <Index>.

Suffix:
<Index> 1..4

Parameters:
<UplinkSelection> OFF | ON
*RST: OFF

Example: See Performing a Throughput Measurement

Firmware/Software: V3.5.30

Manual operation: See "RAN trace selection" on page 121

CONFigure:DATA:MEAS<i>:THRoughput:RAN:TRACe:DLINk<Index>
<DlinkSelection>

Enables or disables the downlink trace for RAN slot number <Index>.

Suffix:
<Index> 1..4

Parameters:
<DlinkSelection> OFF | ON
*RST: OFF

Example: See Performing a Throughput Measurement

Firmware/Software: V3.5.30

Manual operation: See "RAN trace selection" on page 121

CONFigure:DATA:MEAS<i>:SELect:THRoughput <ThroughputType>

Selects the overall throughput tab or the RAN throughput tab for display at the GUI.

This command is useful for taking screenshots via remote commands.

Parameters:
<ThroughputType> OVERall | RAN
*RST: OVER

Firmware/Software: V3.5.30

INITiate:DATA:MEAS<i>:THRoughput
STOP:DATA:MEAS<i>:THRoughput
ABORt:DATA:MEAS<i>:THRoughput

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing a Throughput Measurement

Usage: Event

Firmware/Software: V2.0.10

Manual operation: See "Measurement control" on page 121

FETCh:DATA:MEAS<i>:THRoughput:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V2.0.10

Manual operation: See "Measurement control" on page 121

FETCh:DATA:MEAS<i>:THRoughput:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V2.0.10

Manual operation: See "Measurement control" on page 121

FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk[:CURRent]?
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk[:CURRent]?

READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk[:CURRent]?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk[:CURRent]?

Query the values of the overall throughput trace in uplink (ULINk) or downlink (DLINk) direction.

The trace values are returned from left to right (first to last measurement), one result per interval, see `CONFigure:DATA:MEAS<i>:THRoughput:MCOunt` on page 295.

Return values:

<Reliability> See Reliability Indicator

<Results> Comma-separated list of throughput values, one result per interval
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V2.0.10

Manual operation: See "Overall Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk:EXTended?
FETCh:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk:EXTended?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:ULINk:EXTended?
READ:DATA:MEAS<i>:THRoughput:TRACe:OVERall:DLINk:EXTended?

Query the values of the overall throughput trace in uplink (ULINk) or downlink (DLINk) direction.

The trace values are returned from left to right (first to last measurement). There are two values per interval:

<Reliability>, {<Counter>, <Result>}$_{interval\ 1}$, {...}$_{interval\ 2}$, ..., {...}$_{interval\ n}$

The counter is useful if you want to perform repeated queries and combine the returned result traces.

To configure the number of intervals, see `CONFigure:DATA:MEAS<i>:THRoughput:MCOunt` on page 295.

Return values:

<Reliability> See Reliability Indicator

<Counter> Counter for identification of the result value
The counter starts with 1 and is incremented for each result as long as the measurement is running.

<Result> Default unit: bit/s

Usage: Query only

Firmware/Software: V3.5.21

Manual operation: See "Overall Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:OVERall:ULINk?
READ:DATA:MEAS<i>:THRoughput:OVERall:ULINk?

Query the statistical results of the throughput measurement in uplink direction.

Return values:

<Reliability> See Reliability Indicator

<ThCurrUpLink> Current throughput
Default unit: bit/s

<ThMinUpLink> Minimum throughput
Default unit: bit/s

<ThMaxUpLink> Maximum throughput
Default unit: bit/s

<ThAvgUpLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Overall Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:OVERall:DLINk?
READ:DATA:MEAS<i>:THRoughput:OVERall:DLINk?

Query the statistical results of the throughput measurement in downlink direction.

Return values:

<Reliability> See Reliability Indicator

<ThCurrDLink> Current throughput
Default unit: bit/s

<ThMinDLink> Minimum throughput
Default unit: bit/s

<ThMaxDLink> Maximum throughput
Default unit: bit/s

<ThAvgDLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Overall Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:TRACe:RAN:ULINk<Index>:CURRent?
FETCh:DATA:MEAS<i>:THRoughput:TRACe:RAN:DLINk<Index>:CURRent?
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:ULINk<Index>:CURRent?
READ:DATA:MEAS<i>:THRoughput:TRACe:RAN:DLINk<Index>:CURRent?

Query the values of the throughput trace for RAN slot number <Index> in uplink (ULINk) or downlink (DLINk) direction.

The trace values are returned from left to right (first to last measurement), one result per interval, see `CONFigure:DATA:MEAS<i>:THRoughput:MCOunt` on page 295.

Suffix:
<Index> 1..4

Return values:
<Reliability> See Reliability Indicator

<Results> Comma-separated list of throughput values, one result per interval
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.30

Manual operation: See "RAN Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:RAN:DLINk<Index>?
READ:DATA:MEAS<i>:THRoughput:RAN:DLINk<Index>?

Query the statistical results of the throughput measurement for RAN slot number <Index> in downlink direction.

Suffix:
<Index> 1..4

Return values:
<Reliability> See Reliability Indicator

<ThCurrDLink> Current throughput
Default unit: bit/s

<ThMinDLink> Minimum throughput
Default unit: bit/s

<ThMaxDLink> Maximum throughput
Default unit: bit/s

<ThAvgDLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.30

Manual operation: See "RAN Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:RAN:ULINk<Index>?
READ:DATA:MEAS<i>:THRoughput:RAN:ULINk<Index>?

Query the statistical results of the throughput measurement for RAN slot number <Index> in uplink direction.

Suffix:
<Index> 1..4

Return values:
<Reliability> See Reliability Indicator

<ThCurrUpLink> Current throughput
Default unit: bit/s

<ThMinUpLink> Minimum throughput
Default unit: bit/s

<ThMaxUpLink> Maximum throughput
Default unit: bit/s

<ThAvgUpLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.30

Manual operation: See "RAN Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:ULINk?
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:ULINk?

Query the statistical results of the throughput measurement for the sum of all RAN slots in uplink direction.

Return values:
<Reliability> See Reliability Indicator

<SUMThCurrUpLink> Current throughput
Default unit: bit/s

<SUMThMinUpLink> Minimum throughput
Default unit: bit/s

<SUMThMaxUpLink> Maximum throughput
Default unit: bit/s

<SUMThAvgUpLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.30

Manual operation: See "RAN Throughput results" on page 120

FETCh:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:DLINk?
READ:DATA:MEAS<i>:THRoughput:RAN:TOTal:SUM:DLINk?

Query the statistical results of the throughput measurement for the sum of all RAN slots in downlink direction.

Return values:

<Reliability> See Reliability Indicator

<SUMThCurrDLink> Current throughput
Default unit: bit/s

<SUMThMinDLink> Minimum throughput
Default unit: bit/s

<SUMThMaxDLink> Maximum throughput
Default unit: bit/s

<SUMThAvgDLink> Average throughput
Default unit: bit/s

Example: See Performing a Throughput Measurement

Usage: Query only

Firmware/Software: V3.5.30

Manual operation: See "RAN Throughput results" on page 120

8.6 DNS Requests Measurement

The commands in this section are related to the DNS requests measurement. They control the measurement and return the current measurement state and the measurement results.

CONFigure:DATA:MEAS<i>:DNSRequests:MICount 304
INITiate:DATA:MEAS<i>:DNSRequests 304
STOP:DATA:MEAS<i>:DNSRequests 304
ABORt:DATA:MEAS<i>:DNSRequests 304
FETCh:DATA:MEAS<i>:DNSRequests:STATe? 304

FETCh:DATA:MEAS<i>:DNSRequests:STATe:ALL?....305
SENSe:DATA:MEAS<i>:DNSRequests:RCOunt?....305
SENSe:DATA:MEAS<i>:DNSRequests?....306

CONFigure:DATA:MEAS<i>:DNSRequests:MICount <MaxIndexCount>

Specifies the maximum length of the result list for DNS requests measurements. The result list is stored in a ring buffer. When it is full, the first result line is deleted whenever a new result line is added to the end.

Parameters:

<MaxIndexCount> Maximum number of DNS requests in the result list

Range: 1 to 1000
*RST: 1000

Example: See Monitoring DNS Requests

Firmware/Software: V3.0.10

Manual operation: See "Settings" on page 123

INITiate:DATA:MEAS<i>:DNSRequests
STOP:DATA:MEAS<i>:DNSRequests
ABORt:DATA:MEAS<i>:DNSRequests

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Monitoring DNS Requests

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Measurement control" on page 123

FETCh:DATA:MEAS<i>:DNSRequests:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Measurement control" on page 123

FETCh:DATA:MEAS<i>:DNSRequests:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Measurement control" on page 123

SENSe:DATA:MEAS<i>:DNSRequests:RCOunt?

Queries the number of already monitored DNS requests.

Return values:

<ReqCount> Number of requests

Example: See Monitoring DNS Requests

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Results" on page 122

SENSe:DATA:MEAS<i>:DNSRequests?

Queries information about the monitored DNS requests.

After the reliability indicator, four results are returned for each DNS request:

<Reliability>, {<ClientIP>, <URL>, <IP>, <Timestamp>}$_{\text{request 1}}$, {...}$_{\text{request 2}}$, ...

To query the number of monitored requests, see `SENSe:DATA:MEAS<i>:DNSRequests:RCOunt?`.

Return values:

<Reliability> See Reliability Indicator

<ClientIP> String indicating the IP address of the client (mobile) that has sent the DNS request

<URL> String indicating the domain or application to be resolved

<IP> String indicating the IP address or domain returned as answer to the DNS request

<Timestamp> Timestamp as string in the format `'hh:mm:ss'`

Example: See Monitoring DNS Requests

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Results" on page 122

8.7 IP Logging

The commands in this section are related to the IP logging application. They configure and control IP logging and return the current state of the application.

CONFigure:DATA:MEAS<i>:IPLogging:TYPE 307
SENSe:DATA:MEAS<i>:IPLogging:FNAMe? 307
CONFigure:DATA:MEAS<i>:IPLogging:FSIZe 307
CONFigure:DATA:MEAS<i>:IPLogging:PCOunter 308
CONFigure:DATA:MEAS<i>:IPLogging:PSLength 308
INITiate:DATA:MEAS<i>:IPLogging 308
STOP:DATA:MEAS<i>:IPLogging 308

ABORt:DATA:MEAS<i>:IPLogging....308
FETCh:DATA:MEAS<i>:IPLogging:STATe?....309
FETCh:DATA:MEAS<i>:IPLogging:STATe:ALL?....309

CONFigure:DATA:MEAS<i>:IPLogging:TYPE <LoggingType>

Selects the interface to be monitored.

Parameters:

<LoggingType> UPIP | UPPP | LANDau | UPMulti | UIPClient

UPIP: IP unicast traffic from/to the mobile
UPPP: PPP encapsulated IP traffic from/to the mobile
LANDau: IP traffic at the LAN DAU connector
UPMulti: IP multicast traffic to the mobile
UIPClient: IP traffic from/to the mobile with the DAU as client

*RST: UPIP

Example: See Performing IP Logging

Firmware/Software: V3.0.10
V3.2.40 added UPMulti, V3.5.50 added UIPClient

Options: R&S CMW-KA150 for multicast traffic

Manual operation: See "Logging Interface" on page 125

SENSe:DATA:MEAS<i>:IPLogging:FNAMe?

Queries the current or next log file name.

If IP logging is on, the name of the currently used log file is returned.

If IP logging is off, the name of the file to be created in the next logging session is returned.

Return values:

<FileName> File name as string

Example: See Performing IP Logging

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Log File Name" on page 126

CONFigure:DATA:MEAS<i>:IPLogging:FSIZe <FileSize>

Configures the maximum log file size. When this file size is reached, logging stops.

The default value 0 bytes means that no limit is defined.

Parameters:

<FileSize> Range: 0 bytes to 1E+9 bytes
*RST: 0 bytes
Default unit: bytes

Example: See Performing IP Logging

Firmware/Software: V3.0.10

Manual operation: See "Advanced Settings" on page 126

CONFigure:DATA:MEAS<i>:IPLogging:PCOunter <PacketCounter>

Configures the maximum number of IP packets to be logged. When this number of packets is reached, logging stops.

The default value 0 means that no limit is defined.

Parameters:

<PacketCounter> Range: 0 to 1E+6
*RST: 0

Example: See Performing IP Logging

Firmware/Software: V3.0.10

Manual operation: See "Advanced Settings" on page 126

CONFigure:DATA:MEAS<i>:IPLogging:PSLength <PktSnapLength>

Configures the maximum number of bytes to be logged for each IP packet. If the packet is longer, only the specified number of bytes is logged. The remaining bytes of the packet are ignored.

The default value 0 means that no limit is defined.

Parameters:

<PktSnapLength> Range: 0 bytes to 65565 bytes
*RST: 0 bytes
Default unit: bytes

Example: See Performing IP Logging

Firmware/Software: V3.5.20

Manual operation: See "Advanced Settings" on page 126

INITiate:DATA:MEAS<i>:IPLogging
STOP:DATA:MEAS<i>:IPLogging
ABORt:DATA:MEAS<i>:IPLogging

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.

- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing IP Logging

Usage: Event

Firmware/Software: V3.0.10

Manual operation: See "Logging control" on page 126

FETCh:DATA:MEAS<i>:IPLogging:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Example: See Performing IP Logging

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Logging control" on page 126

FETCh:DATA:MEAS<i>:IPLogging:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V3.0.10

Manual operation: See "Logging control" on page 126

8.8 IP Analysis and Security

The commands in this section are related to the IP analysis application.

- Application Control and States, Database Export....310
- Enabling Results and Views....313
- TCP Analysis Settings and Results....314
- IP Connectivity Settings and Results....318
- Data Pie Charts Settings and Results....322
- Flow Throughput and Trigger Results....325
- Voice over IMS Results....327
- IP Connection Security, General Results....332
- SSL/TLS Handshake Information....333
- Keyword Search Settings and Results....338

8.8.1 Application Control and States, Database Export

The following commands control the IP analysis application and return the current application state.

INITiate:DATA:MEAS<i>:IPANalysis 311
STOP:DATA:MEAS<i>:IPANalysis 311
ABORt:DATA:MEAS<i>:IPANalysis 311
FETCh:DATA:MEAS<i>:IPANalysis:STATe? 311
FETCh:DATA:MEAS<i>:IPANalysis:STATe:ALL? 312
CONFigure:DATA:MEAS<i>:IPANalysis:EXPortdb 312

INITiate:DATA:MEAS<i>:IPANalysis
STOP:DATA:MEAS<i>:IPANalysis
ABORt:DATA:MEAS<i>:IPANalysis

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing IP Analysis

Usage: Event

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Measurement control" on page 128

FETCh:DATA:MEAS<i>:IPANalysis:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Measurement control" on page 128

FETCh:DATA:MEAS<i>:IPANalysis:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate...`, `STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Measurement control" on page 128

CONFigure:DATA:MEAS<i>:IPANalysis:EXPortdb

Stores the IP analysis result database to a CSV file on the DAU system drive.

Example: See Performing IP Analysis

Usage: Event

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Export DB (hotkey)" on page 128

8.8.2 Enabling Results and Views

The following commands select the evaluated results and the displayed views.

CONFigure:DATA:MEAS<i>:IPANalysis:RESult[:ALL] 313
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:TCPanalysis 314
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:IPConnect 314
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:DPCP 314
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:VOIMs 314
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:FTTRigger 314

CONFigure:DATA:MEAS<i>:IPANalysis:RESult[:ALL] <TCPAnalysis>, <IPConnect>, <DPCP>, <FTTRigger>, <VOIMs>[, <IPCSecurity>]

Enables or disables the display of the individual detailed views and the evaluation of the related results. This command combines all other `CONFigure:DATA:MEAS<i>:IPANalysis:RESult...` commands.

Parameters:

<TCPAnalysis> OFF | ON
"TCP Analysis" view
OFF: Do not evaluate results, hide the view
ON: Evaluate results and show the view
*RST: ON

<IPConnect> OFF | ON
"IP Connectivity" view
*RST: ON

<DPCP> OFF | ON
"Data Pie Charts" view
*RST: ON

<FTTRigger> OFF | ON
"Flow Throughput and Event Trigger" view
*RST: ON

<VOIMs> OFF | ON
"Voice over IMS" view
*RST: ON

<IPCSecurity> OFF | ON
"IP Connection Security" view
*RST: ON

Example: See Performing IP Analysis

Firmware/Software: V3.2.11
V3.2.40: added <VOIMs>
V3.5.52: added <IPCSecurity>

Options: R&S CMW-KM051, for <IPCSecurity> also R&S CMW-KM052

Manual operation: See "Disabling views / Assign View hotkey" on page 128

CONFigure:DATA:MEAS<i>:IPANalysis:RESult:TCPanalysis <Enable>
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:IPConnect <Enable>
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:DPCP <Enable>
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:VOIMs <Enable>
CONFigure:DATA:MEAS<i>:IPANalysis:RESult:FTTRigger <Enable>

Enables or disables the display of the individual detailed views and the evaluation of the related results.

The mnemonic after "RESult" denotes the view: "TCP Analysis", "IP Connectivity", "Data Pie Charts", "Voice over IMS" and "Flow Throughput and Event Trigger".

Parameters:

<Enable> OFF | ON

OFF: Do not evaluate results, hide the view
ON: Evaluate results and show the view

*RST: ON

Firmware/Software: V3.2.11
V3.2.40: added `...:RESult:VOIMs`

Options: R&S CMW-KM051

Manual operation: See "Disabling views / Assign View hotkey" on page 128

8.8.3 TCP Analysis Settings and Results

The following commands configure the "TCP Analysis" view and return the related results.

CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TWSThreshold....314
CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TRTHreshold....315
CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:RTTThreshold....315
SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:FLOWid?....315
FETCh:DATA:MEAS<i>:IPANalysis:TCPanalysis:ALL?....316
SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:DETails?....317

CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TWSThreshold <Threshold>

Defines a threshold (upper limit) for the current TCP window size as percentage of the negotiated maximum window size.

Parameters:

<Threshold> Range: 0 % to 100 %
*RST: 90 %
Default unit: %

Example: See Performing IP Analysis

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Threshold configuration" on page 129

CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:TRTHreshold <Threshold>

Defines a threshold (upper limit) for TCP retransmissions as percentage of all transmissions.

Parameters:

<Threshold>
Range: 0 % to 100 %
*RST: 10 %
Default unit: %

Example: See Performing IP Analysis

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Threshold configuration" on page 129

CONFigure:DATA:MEAS<i>:IPANalysis:TCPanalysis:RTTThreshold <Threshold>

Defines a threshold (upper limit) for the round-trip time (RTT).

Parameters:

<Threshold>
Range: 0 ms to 200 ms
*RST: 35 ms
Default unit: ms

Example: See Performing IP Analysis

Firmware/Software: V3.5.50

Options: R&S CMW-KM051

Manual operation: See "Threshold configuration" on page 129

SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:FLOWid? <FlowID>

Queries the threshold check and throughput results for a specific connection, selected via its flow ID.

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<ThDown> Throughput in downlink direction
Default unit: bit/s

<TCPWSDown> OK | FULL
Threshold check result for downlink TCP window size

<RetrDown> OK | NOK
Threshold check result for downlink retransmissions

<OverhDown> OK | NOK
Only for backward compatibility - no longer used

<ThUp> Throughput in uplink direction
Default unit: bit/s

<TCPWSUp> OK | FULL
Threshold check result for uplink TCP window size

<RetrUp> OK | NOK
Threshold check result for uplink retransmissions

<OverhUp> OK | NOK
Only for backward compatibility - no longer used

<Destination> Destination address as string

<RTT> OK | NOK
Threshold check result for round-trip time

<PKTsizeUP> Layer 3 uplink packet size
Default unit: byte

<PktSizeDL> Layer 3 downlink packet size
Default unit: byte

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11, V3.5.50 appended <Destination> ...

Options: R&S CMW-KM051

Manual operation: See "Result overview (upper part)" on page 130

FETCh:DATA:MEAS<i>:IPANalysis:TCPanalysis:ALL?

Queries the threshold check and throughput results for all connections.

After the reliability indicator, 13 results are returned for each connection (flow):

<Reliability>, {<FlowID>, <ThDown>, <TCPWSDown>, <RetrDown>, <OverhDown>, <ThUp>, <TCPWSUp>, <RetrUp>, <OverhUp>, <Destination>, <RTTStatus>, <PKTSizeUL>, <PKTSizeDL>}$_{\text{connection 1}}$, {...}$_{\text{connection 2}}$, ...

Return values:

<Reliability> See Reliability Indicator

<FlowID> Flow ID of the connection

<ThDown> Throughput in downlink direction
Default unit: bit/s

<TCPWSDown> OK | FULL
Threshold check result for downlink TCP window size

<RetrDown> OK | NOK
Threshold check result for downlink retransmissions

<OverhDown> OK | NOK
Only for backward compatibility - no longer used

<ThUp> Throughput in uplink direction
Default unit: bit/s

<TCPWSUp> OK | FULL
Threshold check result for uplink TCP window size

<RetrUp> OK | NOK
Threshold check result for uplink retransmissions

<OverhUp> OK | NOK
Only for backward compatibility - no longer used

<Destination> Destination address as string

<RTTStatus> OK | NOK
Threshold check result for round-trip time

<PKTSizeUL> Layer 3 uplink packet size
Default unit: byte

<PKTSizeDL> Layer 3 downlink packet size
Default unit: byte

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11, V3.5.50 appended <Destination> ...

Options: R&S CMW-KM051

Manual operation: See "Result overview (upper part)" on page 130

SENSe:DATA:MEAS<i>:IPANalysis:TCPanalysis:DETails? <FlowID>

Queries the "TCP Analysis" details for a specific connection, selected via its flow ID.

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<IPAddrSource> IP address of the connection source as string

<IPPortSource> Port number of the connection source
Range: 0 to 65654

<IPAddrDest> IP address of the connection destination as string

<IPPortDest> Port number of the connection destination
Range: 0 to 65654

<CurrTCPWSDown> Measured downlink TCP window size
Default unit: byte

<MaxTCPWSDown> Negotiated maximum downlink TCP window size
Default unit: byte

<RetrDown> Downlink retransmission rate
Range: 0 % to 100 %
Default unit: %

<OverhDown> Only for backward compatibility - no longer used
Range: 0 % to 100 %
Default unit: %

<CurrTCPWSUp> Measured uplink TCP window size
Default unit: byte

<MaxTCPWSUp> Negotiated maximum uplink TCP window size
Default unit: byte

<RetrUp> Uplink retransmission rate
Range: 0 % to 100 %
Default unit: %

<OverhUp> Only for backward compatibility - no longer used
Range: 0 % to 100 %
Default unit: %

<RTT> Round-trip time

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11, V3.5.50 added <RTT>

Options: R&S CMW-KM051

Manual operation: See "Result details (lower part)" on page 130

8.8.4 IP Connectivity Settings and Results

The following commands configure the "IP Connectivity" view and return the related results.

CONFigure:DATA:MEAS<i>:IPANalysis:IPConnect:FILTer....319
SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:STATistics?....319
SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:FLOWid?....320
FETCh:DATA:MEAS<i>:IPANalysis:IPConnect:ALL?....320

CONFigure:DATA:MEAS<i>:IPANalysis:IPConnect:FILTer <FilterType>, <FilterString>

Configures a filter for the IP connectivity results.

The other commands return only results for which the attribute selected via `<FilterType>` matches one of the values specified via `<FilterString>`.

Parameters:

<FilterType> FLOWid | IPADd | L4PR | L7PRotocol

FLOWid: flow IDs
IPADd: IP addresses
L4PR: L4 protocol
L7PRotocol: L7 protocol

*RST: FLOW

<FilterString> Enter a single string, containing a single value or a comma-separated list of values.

Example: See Performing IP Analysis

Firmware/Software: V3.5.51

Options: R&S CMW-KM051

Manual operation: See "Filter hotkey" on page 130

SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:STATistics?

Queries the statistical information provided by the "IP Connectivity" view.

Return values:

<NoOfConn> Total number of connections

<NoOfOpenC> Number of open connections

<NoOfClosedC> Number of closed connections

<OpenTCPC> Number of open TCP connections

<ClosedTCPC> Number of closed TCP connections

<OpenUDPC> Number of open UDP connections

<ClosedUDPC> Number of closed UDP connections

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Results" on page 132

SENSe:DATA:MEAS<i>:IPANalysis:IPConnect:FLOWid? <FlowID>

Queries the "IP Connectivity" results for a specific connection, selected via its flow ID.

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<ConnStatus> OPEN | CLOSed
Connection status

<Protocol> Layer 4 protocol as string ("TCP", "UDP", ...)

<DPIProtocol> Layer 7 protocol as string ("HTTP", "FTP", ...)

<IPAddrSource> IP address of the connection source as string

<IPPortSource> Port number of the connection source
Range: 0 to 65654

<IPAddrDest> IP address of the connection destination as string

<IPPortDest> Port number of the connection destination
Range: 0 to 65654

<OverhDown> Downlink overhead as percentage of the packet
Range: 0 % to 100 %
Default unit: %

<OverhUp> Uplink overhead as percentage of the packet
Range: 0 % to 100 %
Default unit: %

<AvgPSDown> Average downlink packet size
Range: 0 bytes to 65535 bytes
Default unit: bytes

<AvgPSUp> Average uplink packet size
Range: 0 bytes to 65535 bytes
Default unit: bytes

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Results" on page 132

FETCh:DATA:MEAS<i>:IPANalysis:IPConnect:ALL?

Queries the "IP Connectivity" results for all connections.

After the reliability indicator, results are returned per connection (flow):

<Reliability>, {<FlowID>, <ConnStatus>, <LST>, <SysClock>, <Protocol>, <DPIProtocol>, <IPAddrSource>, <IPPortSource>, <IPAddrDest>, <IPPortDest>, <OverhDown>, <OverhUp>, <AvgPSDown>, <AvgPSUp>}$_{\text{connection 1}}$, {…}$_{\text{connection 2}}$, …

Return values:

<Reliability> See Reliability Indicator

<FlowID> Flow ID of the connection

<ConnStatus> OPEN | CLOSed
Connection status

<LST> Local system time, incremented every ms
The clock starts when the instrument is switched on.

<SysClock> System clock in units of 10 ns
When 1 ms is reached (100*10 ns), the clock is reset to 0 and the local system time is incremented.
Range: 0 to 99

<Protocol> Layer 4 protocol as string ("TCP", "UDP", ...)

<DPIProtocol> Layer 7 protocol as string ("HTTP", "FTP", ...)

<IPAddrSource> IP address of the connection source as string

<IPPortSource> Port number of the connection source
Range: 0 to 65654

<IPAddrDest> IP address of the connection destination as string

<IPPortDest> Port number of the connection destination
Range: 0 to 65654

<OverhDown> Downlink overhead as percentage of the packet
Range: 0 % to 100 %
Default unit: %

<OverhUp> Uplink overhead as percentage of the packet
Range: 0 % to 100 %
Default unit: %

<AvgPSDown> Average downlink packet size
Range: 0 bytes to 65535 bytes
Default unit: bytes

<AvgPSUp> Average uplink packet size
Range: 0 bytes to 65535 bytes
Default unit: bytes

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11
V3.2.40: added <LST> and <SysClock>

Options: R&S CMW-KM051

Manual operation: See "Results" on page 132

8.8.5 Data Pie Charts Settings and Results

The following commands configure the "Data Pie Charts" view and return the related results.

CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic:APP 322
CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer:LAYer 322
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPConnection? 323
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPPRotocol? 323
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer? 324
FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic? 324

CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic:APP <AppSelected>

Selects a layer of the "Data per Application" pie chart view.

You can navigate from the current layer to the next lower or higher layer. The initial current layer is the application layer. The lower layers are layer 7, layer 4 and layer 3.

To query the entries (strings) of the current layer, see `FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic?` on page 324.

Parameters:

<AppSelected> String with an entry of the current layer: Navigates to the next lower layer for this entry
"Back" or string unknown at the current layer: Navigates back to the next higher layer

Example: See Performing IP Analysis

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Data Per Application" on page 135

CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer:LAYer <Layer>

Selects an analysis layer for the "Data per Layer" pie chart view.

Parameters:

<Layer> FEATure | APP | L7 | L4 | L3
*RST: APP

Firmware/Software: V3.2.50

Options: R&S CMW-KM051

Manual operation: See "Data Per Layer" on page 134

FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPConnection?

Queries the "Data per Connection" results.

After the reliability indicator, three results are returned for each connection:

<Reliability>, {<RemDestination>, <ConnData>, <ConnPercent>}$_{conn\ 1}$, {...}$_{conn\ 2}$, ...

Return values:

<Reliability> See Reliability Indicator

<RemDestination> IP address of the remote destination as string

<ConnData> Data transported via the connection, as absolute number
Default unit: byte

<ConnPercent> Data transported via the connection, as percentage of total transported data
Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Data Per Connection" on page 133

FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPPRotocol?

Queries the "Data per Protocol" results.

After the reliability indicator, three results are returned for each protocol:

<Reliability>, {<Protocol>, <ProtData>, <ProtPercent>}$_{protocol\ 1}$, {...}$_{protocol\ 2}$, ...

Return values:

<Reliability> See Reliability Indicator

<Protocol> Used protocol as string

<ProtData> Data transported via the protocol, as absolute number
Default unit: byte

<ProtPercent> Data transported via the protocol, as percentage of total transported data
Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Data Per Protocol" on page 133

FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPLayer? <LayerDepth>

Queries the "Data per Layer" results.

After the reliability indicator, three values are returned for each result table row:

<Reliability>, {<Layer>, <LayerData>, <LayerPercent>}$_{row\ 1}$, {...}$_{row\ 2}$, ...

Query parameters:

<LayerDepth> FEATure | APP | L7 | L4 | L3
Selects the highest layer at which the packets are analyzed

Return values:

<Reliability> See Reliability Indicator

<Layer> String with the contents of column "Layer" (feature, application or protocol)

<LayerData> Amount of transported data, as absolute value
Default unit: byte

<LayerPercent> Amount of transported data, as percentage of total transported data
Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Data Per Layer" on page 134

FETCh:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic?

Queries the "Data per Application" results of the current layer. To navigate between the layers, see `CONFigure:DATA:MEAS<i>:IPANalysis:DPCP:DPAPplic:APP` on page 322.

After the reliability indicator, three values are returned for each result table row:

<Reliability>, {<App>, <AppData>, <AppPercent>}$_{row\ 1}$, {...}$_{row\ 2}$, ...

Return values:

<Reliability> See Reliability Indicator

<App> String with the contents of column "Application" (application or protocol)

<AppData> Amount of transported data, as absolute value
Default unit: byte

<AppPercent> Amount of transported data, as percentage of total transported data

Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Data Per Application" on page 135

8.8.6 Flow Throughput and Trigger Results

The following commands configure the "Flow Throughput and Event Trigger" view and return the related results.

CONFigure:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACe<TraceIndex>:TFLowid 325
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACes<TraceIndex>:FTHRoughput? 325
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:STARt? 326
FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:END? 326

CONFigure:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACe<TraceIndex>:TFLowid <FlowID>

Assigns a connection (flow ID) to a trace index.

Suffix:
<TraceIndex> 1..10
Selects the trace to which the connection (flow ID) is assigned

Parameters:
<FlowID> Flow ID of the connection to be assigned to the trace index

*RST: <FlowID> = <TraceIndex>

Example: See Performing IP Analysis

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Results" on page 137

FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACes<TraceIndex>:FTHRoughput?

Queries a selected throughput trace.

The trace is selected via its trace index. To assign a specific connection to a trace index, see `CONFigure:DATA:MEAS<i>:IPANalysis:FTTRigger:TRACe<TraceIndex>:TFLowid`.

Suffix:

<TraceIndex> 1..10
Selects the trace to be returned.

Return values:

<Reliability> See Reliability Indicator

<FlowID> Flow ID of the connection assigned to the selected trace index

<Throughput> Comma-separated list of throughput values
Default unit: bit/s

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Results" on page 137

FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:STARt?

Queries the event trigger trace for "open" events.

After the reliability indicator, two values are returned per "open" event:

<Reliability>, {<FlowID>, <TimeElapsed>}$_{event\ 1}$, {<FlowID>, <TimeElapsed>}$_{event\ 2}$, ...

Return values:

<Reliability> See Reliability Indicator

<FlowID> Flow ID of the opened connection as string

<TimeElapsed> X-axis value of the "open" event

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.11

Options: R&S CMW-KM051

Manual operation: See "Results" on page 137

FETCh:DATA:MEAS<i>:IPANalysis:FTTRigger:TRIGger:END?

Queries the event trigger trace for "close" events.

After the reliability indicator, two values are returned per "close" event:

<Reliability>, {<FlowID>, <TimeElapsed>}$_{event\ 1}$, {<FlowID>, <TimeElapsed>}$_{event\ 2}$, ...

Return values:

<Reliability> See Reliability Indicator

<FlowID> Flow ID of the closed connection as string

<TimeElapsed>	X-axis value of the "close" event
Example:	See Performing IP Analysis
Usage:	Query only
Firmware/Software:	V3.2.11
Options:	R&S CMW-KM051
Manual operation:	See "Results" on page 137

8.8.7 Voice over IMS Results

The following commands return the results of the "Voice over IMS" view.

FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL? 327
SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:FLOWs? 328
SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:PERDtx? 329
SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:JITTer? 330
SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:BITRate:CMR? 330

FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?

Queries the call table in the upper part of the "Voice Over IMS" view.

The results are returned row by row (call by call):

<Reliability>, {<ConID>, ..., <UserTo>}$_{call\ 1}$, {...}$_{call\ 2}$, ..., {...}$_{call\ n}$

Return values:

<Reliability>	See Reliability Indicator
<ConID>	Call ID
<FlowsIDs>	String containing a comma-separated list of the flow IDs related to the call (example "1,2" or "128")
<Type>	AUDio | VIDeo | EMER | UNK Call type audio, video, emergency or unknown
<Origin>	MT | MO | UNK **MT**: mobile-terminating call **MO**: mobile-originating call **UNK**: unknown
<State>	RING | EST | REL | HOLD | UNK **RING**: mobile ringing **EST**: call established **REL**: call released **HOLD**: call on hold **UNK**: unknown
<StartTime>	String indicating the time when the call setup was initiated

<SetupTime>	Duration of the call setup procedure Default unit: s
<Duration>	Duration of the call Default unit: s
<UserFrom>	String with the user ID or phone number of the calling party
<UserTo>	String with the user ID or phone number of the called party
Example:	See Performing IP Analysis
Usage:	Query only
Firmware/Software:	V3.2.40
Options:	R&S CMW-KM051
Manual operation:	See "Table (upper part)" on page 138

SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:FLOWs? <SessionID>, <FlowID>, <Direction>, <Type>

Queries flow information related to a voice over IMS call.

A query returns all parameters except the <SessionID>:

<FlowID>, <Direction>, <Type>, <Codec>, <SeqNumber>, ..., <Bitrate>

Parameters:

<FlowID>	Flow ID, as returned by `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?`
<Direction>	UL | DL | UNK Flow direction uplink, downlink or unknown
<Type>	AUDio | VIDeo | UNKNow Flow type audio, video or unknown

Query parameters:

<SessionID>	Call ID, as returned by `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?`

Return values:

<Codec>	String indicating the used codec
<SeqNumber>	Sequence number of the currently processed packet
<NumPack>	Number of already processed packets
<Throughput>	Current audio or video data throughput at the RTP level Default unit: bit/s
<Destport>	Port used at the flow destination
<EVSMode>	String indicating the EVS mode (primary or AMR-WB-IO)
<EVSFormat>	String indicating the EVS format (header-full or compact)

<NumEVSCOMP> Number of EVS packets with compact format

<NumEVSHP> Number of EVS packets with header-full format

<VideoResolution> String indicating the video resolution

<VideoFrate> Video frame rate in frames per second

<VideoOreintation> String indicating the counter-clockwise video rotation

<VideoProfile> String indicating the H.264 profile

<VideoLevel> String indicating the H.264 level

<VideoConstraint> String indicating the H.264 constraint set

<Bitrate> Audio codec rate

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.40, V3.5.50: added <Destport> ... <VideoConstraint>
V3.5.51: added <Bitrate>

Options: R&S CMW-KM051

Manual operation: See "Flows Info tab (lower part)" on page 138

SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:PERDtx? <ConID>

Queries the packets measurement results for a selected voice over IMS call.

To get a list of all calls and their IDs, use `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?` on page 327.

Query parameters:

<ConID> Selects the call for which the results are queried

Return values:

<PacketLossUp> Packet loss rate in the uplink

Range: 0 % to 100 %
Default unit: %

<PacketLossDown> Packet loss rate in the downlink

Range: 0 % to 100 %
Default unit: %

<DTXUp> Discontinuous transmission rate in the uplink

Range: 0 % to 100 %
Default unit: %

<DTXDown> Discontinuous transmission rate in the downlink

Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Packets Meas tab (lower part)" on page 139

SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:JITTer? <ConID>

Queries the jitter results for a selected voice over IMS call.

To get a list of all calls and their IDs, use `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?` on page 327.

Query parameters:

<ConID> Selects the call for which the results are queried

Return values:

<JitterMaxUp> Maximum jitter in the uplink
Default unit: s

<JitterAvgUp> Average jitter in the uplink
Default unit: s

<JitterMinUp> Minimum jitter in the uplink
Default unit: s

<JitterMaxDown> Maximum jitter in the downlink
Default unit: s

<JitterAvgDown> Average jitter in the downlink
Default unit: s

<JitterMinDown> Minimum jitter in the downlink
Default unit: s

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.2.40

Options: R&S CMW-KM051

Manual operation: See "Jitter tab (lower part)" on page 140

SENSe:DATA:MEAS<i>:IPANalysis:VOIMs:BITRate:CMR? <SessionID>, <FlowID>, <Direction>

Queries bitrates and CMR information related to a voice over IMS call.

A query returns all parameters except the <SessionID>:

<FlowID>, <Direction>, <CurrBitrate>, <AvgBitrate>, ..., <MaxCMRBW>

Parameters:

<FlowID> Flow ID, as returned by `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?`

<Direction> UL | DL | UNK
Flow direction uplink, downlink or unknown

Query parameters:

<SessionID> Call ID, as returned by `FETCh:DATA:MEAS<i>:IPANalysis:VOIMs:ALL?`

Return values:

<CurrBitrate> Current measured bitrate
Default unit: bit/s

<AvgBitrate> Average measured bitrate
Default unit: bit/s

<MinBitrate> Minimum measured bitrate
Default unit: bit/s

<MaxBitrate> Maximum measured bitrate
Default unit: bit/s

<CurrCMRBitrate> Bitrate currently requested for the other direction
Default unit: bit/s

<MinCMRBitrate> Minimum of the bitrates requested for the other direction
Default unit: bit/s

<MaxCMRBitrate> Maximum of the bitrates requested for the other direction
Default unit: bit/s

<CurrCMRBW> String indicating the bandwidth currently requested for the other direction

<MinCMRBW> String indicating the minimum of the bandwidths requested for the other direction

<MaxCMRBW> String indicating the maximum of the bandwidths requested for the other direction

Example: See Performing IP Analysis

Usage: Query only

Firmware/Software: V3.5.50

Options: R&S CMW-KM051

Manual operation: See "Bitrate\CMR" on page 140

8.8.8 IP Connection Security, General Results

The following commands query mainly the results presented on the "Applications" tab of the "IP Connection Security" view.

SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:APPLications?.....332
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication?.....332

SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:APPLications?

Queries the list of applications resulting from the IP packet analysis on the application layer.

Return values:

<Applications> Comma-separated list of strings, one string per application

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Application area" on page 143

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication?
<ApplicationName>

Queries information about all connections of a selected application.

The results after the reliability indicator are returned per connection:

<Reliability>, {<Application>, <Flowid>, ..., <HandSkAvailable>, <CertifAvailable>}$_1$, {...}$_2$, ...

Query parameters:

<ApplicationName> Application name as string

Return values:

<Reliability> See Reliability Indicator

<Application> Application name as string

<Flowid> ID of the flow used by the connection

<SourceIP> IP address of the mobile as string

<LocalPort> Port number used at the mobile side

<Destination IP> Destination IP address as string

<Destination Port> Port number of the destination

<FQDN> Fully qualified domain name of the destination as string

<RAN> Used radio access network as string

<APN> Access point name as string

<Protocol>	Used protocol, for example SSL or HTTP, as string
<CountryCode>	Country of the destination as string (two-letter country code)
<Location>	City of the destination, as string
<Latitude>	Latitude of the destination, as string
<Longitude>	Longitude of the destination, as string
<ULData>	Layer 3 UL data exchanged via the connection Default unit: bytes
<ULPKT>	Number of UL packets exchanged via the connection
<DLData>	Layer 3 DL data exchanged via the connection Default unit: bytes
<DLPKT>	Number of DL packets exchanged via the connection
<HandSkAvailable>	OFF | ON Handshake information available for the connection or not
<CertifAvailable>	OFF | ON Certificate information available for the connection or not
Example:	See Performing IP Security Tests
Usage:	Query only
Firmware/Software:	V3.7.10
Options:	R&S CMW-KM052
Manual operation:	See "Connection Group area" on page 143

8.8.9 SSL/TLS Handshake Information

The following commands query information about the handshake for a connection, selected via its flow ID. The information is provided by the "IP Connection Security" view.

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:VERSion? 334
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:CIPSuite? 334
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:COMPression? 335
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SHALgorithm? 335
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECURve? 336

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECPFormat? 336
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SRINdication? 337
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated? 337

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:VERSion? <FlowID>

Queries the protocol version offered during the handshake for a specific connection.

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<Version> Protocol version as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Version" on page 146

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:CIPSuite? <FlowID>

Queries information about the cipher suites offered during the handshake for a specific connection.

After the reliability indicator, two results are returned for each cipher suite:

<Reliability>, $\{<CipherSuitId>, <CiphersuiteName>\}_{Suite\ 1}$, $\{...\}_{Suite\ 2}$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<CipherSuitId> Cipher suite ID as hexadecimal value

<CiphersuiteName> Cipher suite name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Cipher Suites" on page 146

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:COMPression? <FlowID>

Queries information about the compression methods offered during the handshake for a specific connection.

After the reliability indicator, two results are returned for each method:

<Reliability>, {<CompressionId>, <CompressionName>}$_{Method\ 1}$, {...}$_{Method\ 2}$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<CompressionId> Compression method ID as hexadecimal value

<CompressionName> Compression method name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Compression" on page 147

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SHALgorithm? <FlowID>

Queries information about the hash algorithms and signature algorithms offered during the handshake for a specific connection.

After the reliability indicator, four results are returned for each pair of algorithms:

<Reliability>, {<AlgorithmHashID>, <AlgorithmSignID>, <AlgoHashName>, <AlgoSignName>}$_1$, {...}$_2$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<AlgorithmHashID> Hash algorithm ID as hexadecimal value

<AlgorithmSignID> Signature algorithm ID as hexadecimal value

<AlgoHashName> Hash algorithm name as string

<AlgoSignName> Signature algorithm name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Signature Hash Algorithm" on page 147

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECURve? <FlowID>

Queries information about the elliptic curves offered during the handshake for a specific connection.

After the reliability indicator, two results are returned for each elliptic curve:

<Reliability>, {<EllipticCurveID>, <EllipticCurveName>}$_{Curve\ 1}$, {...}$_{Curve\ 2}$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<EllipticCurveID> Elliptic curve ID as hexadecimal value

<EllipticCurveName> Elliptic curve name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Elliptic Curves, Elliptic Curve Point Formats, Elliptic Curve Point Type" on page 147

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:ECPFormat? <FlowID>

Queries information about the elliptic curve point formats offered during the handshake for a specific connection.

After the reliability indicator, two results are returned for each format:

<Reliability>, {<ECPFormatID>, <ECPFormatName>}$_{Format\ 1}$, {...}$_{Format\ 2}$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<ECPFormatID> Elliptic curve point format ID as hexadecimal value

<ECPFormatName> Elliptic curve point format name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Elliptic Curves, Elliptic Curve Point Formats, Elliptic Curve Point Type" on page 147

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:OFFered:SRINdication? <FlowID>

Queries information about the server that the client wants to contact, as sent by the client during the handshake for a specific connection.

After the reliability indicator, three results are returned for each entry:

<Reliability>, {<ServerID>, <ServerName>, <ServerType>}$_1$, {...}$_2$, ...

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<Reliability> See Reliability Indicator

<ServerID> Server ID as hexadecimal value

<ServerName> Server name indication (SNI) as string

<ServerType> Type of the server name as string

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Server Name Indication, Server Type Indication" on page 147

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:CAPPlication:HANDshake:NEGotiated? <FlowID>

Queries the negotiated handshake results for a specific connection.

Query parameters:

<FlowID> Selects the connection for which information is queried

Return values:

<VersionID> Protocol version ID as hexadecimal value

<VersionName> Protocol version as string

<CipherSuiteID> Cipher suite ID as hexadecimal value

<CipherSuiteName> Cipher suite name as string

<CompressionID> Compression method ID as hexadecimal value

<CompressionName> Compression method name as string

<RegisterType> IANA | OID

Type of the register used for the signature hash algorithm pair
If the value is IANA, the fields SignatureAlgoHashID and SignaHashAlgoSignID are filled with the ID values.
If the value is OID, the field OID is filled with a single combined ID value for the signature hash algorithm pair.
In both cases, the fields SignatureAlgoHashName and SignaHashAlgoSignName are filled with the names as strings.

<OID> Signature hash algorithm ID as string

<SignAlgHashID> Hash algorithm ID as hexadecimal value

<SignAlgHashName> Hash algorithm name as string

<SignAlgSignID> Signature algorithm ID as hexadecimal value

<SignAlgSignName> Signature algorithm name as string

<ECurveID> Elliptic curve ID as hexadecimal value

<ECName> Elliptic curve name as string

<ECTypeID> Elliptic curve point type ID as hexadecimal value

<ECTypeName> Elliptic curve point type name as string

<ECPFormatID> Elliptic curve point format ID as hexadecimal value

<ECPFormatName> Elliptic curve point format name as string

<SignLength> Length of the server signature in bits

<PublicLength> Length of the public key of the server in bits

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Version" on page 146

8.8.10 Keyword Search Settings and Results

The following commands configure the keyword search of the "IP Connection Security" view and return the related results.

CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:ADD 339
CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:DELete 339
SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:LIST? 339
FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch? 339

CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:ADD
<Keywd>

Adds an entry to the keyword list.

Setting parameters:

<Keywd> New keyword as string

Example: See Performing IP Security Tests

Usage: Event

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Keyword configuration settings" on page 149

CONFigure:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:DELete

Deletes an entry from the keyword list.

Example: See Performing IP Security Tests

Usage: Event

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Keyword configuration settings" on page 149

SENSe:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch:LIST?

Queries the keyword list.

Return values:

<List> Comma-separated list of strings, one string per keyword

Example: See Performing IP Security Tests

Usage: Query only

Firmware/Software: V3.7.10

Options: R&S CMW-KM052

Manual operation: See "Keyword configuration settings" on page 149

FETCh:DATA:MEAS<i>:IPANalysis:IPCSecurity:KYWord:SEARch?

Queries the keyword search results.

After the reliability indicator, five results are returned for each found keyword:

<Reliability>, {<Keyword>, <Count>, <DstIP>, <FQDN>, <Application>}$_1$, {...}$_2$, ...

Return values:

<Reliability>	See Reliability Indicator
<Keyword>	Found keyword as string
<Count>	How often the keyword was found
<DstIP>	IP address of the destination as string
<FQDN>	FQDN of the destination as string
<Application>	Application using the connection, as string
Example:	See Performing IP Security Tests
Usage:	Query only
Firmware/Software:	V3.7.10
Options:	R&S CMW-KM052
Manual operation:	See "Keyword search results" on page 150

8.9 IP Replay

The commands in this section are related to the IP replay application.

- Playlist Configuration and Control....340
- Application Control and States....343
- File Information....345

8.9.1 Playlist Configuration and Control

The following commands configure the playlist, start/stop replaying the list and return progress information.

CONFigure:DATA:MEAS<i>:IPReplay:CREatelist....340
CONFigure:DATA:MEAS<i>:IPReplay:REMovelist....341
CONFigure:DATA:MEAS<i>:IPReplay:ITERation....341
CONFigure:DATA:MEAS<i>:IPReplay:INTerface....342
CONFigure:DATA:MEAS<i>:IPReplay:PLAYall....342
CONFigure:DATA:MEAS<i>:IPReplay:STOPall....342
SENSe:DATA:MEAS<i>:IPReplay:PROGress?....342

CONFigure:DATA:MEAS<i>:IPReplay:CREatelist <FileName>[, <Iteration>, <NetworkInterface>]

Adds a single file to the playlist (measurement must be OFF).

A query returns all files in the playlist as follows:

{<FileName>, <Iteration>, <NetworkInterface>}$_{file\ 1}$, {...}$_{file\ 2}$, ..., {...}$_{file\ n}$

To query a list of all files in the `ip_replay` directory, see `FETCh:DATA:MEAS<i>:IPReplay:FILelist?` on page 345.

Parameters:

<FileName> File name as a string. Specify the file name with extension but without path, for example "myfile.pcap".

<Iteration> Specifies how often the file is replayed

Range: 0 to 10E+3
*RST: 1

<NetworkInterface> LANDau | IP | MULTicast

LANDau: IP traffic to the LAN DAU connector
IP: IP unicast traffic to the mobile
MULTicast: IP multicast traffic to the mobile

*RST: MULT

Example: See Performing IP Replay

Firmware/Software: V3.2.50

Manual operation: See "Edit Playlist (hotkey)" on page 154

CONFigure:DATA:MEAS<i>:IPReplay:REMovelist

Removes all files from the playlist (measurement must be OFF).

Example: See Performing IP Replay

Usage: Event

Firmware/Software: V3.2.50

Manual operation: See "Clear Playlist (hotkey)" on page 153

CONFigure:DATA:MEAS<i>:IPReplay:ITERation <FileName>, <Iteration>

Specifies how often a selected file in the playlist is replayed.

A query returns all files in the playlist as follows:

$\{<FileName>, <Iteration>\}_{file\ 1}, \{...\}_{file\ 2}, ..., \{...\}_{file\ n}$

Parameters:

<FileName> File name as a string. Specify the file name with extension but without path, for example "myfile.pcap".

<Iteration> Specifies how often the file is replayed

Range: 0 to 10E+3
*RST: 1

Example: See Performing IP Replay

Firmware/Software: V3.2.50

Manual operation: See "Edit Playlist (hotkey)" on page 154

CONFigure:DATA:MEAS<i>:IPReplay:INTerface <FileName>, <NetworkInterface>

Specifies the network interface for a selected file in the playlist.

A query returns all files in the playlist as follows:

{<FileName>, <NetworkInterface>}$_{file\ 1}$, {...}$_{file\ 2}$, ..., {...}$_{file\ n}$

Setting parameters:

<FileName> File name as a string. Specify the file name with extension but without path, for example "myfile.pcap".

<NetworkInterface> LANDau | IP | MULTicast

LANDau: IP traffic to the LAN DAU connector
IP: IP unicast traffic to the mobile
MULTicast: IP multicast traffic to the mobile

*RST: MULT

Example: See Performing IP Replay

Firmware/Software: V3.2.50

Manual operation: See "Edit Playlist (hotkey)" on page 154

CONFigure:DATA:MEAS<i>:IPReplay:PLAYall

Starts replaying the playlist.

Example: See Performing IP Replay

Usage: Event

Firmware/Software: V3.2.50

Manual operation: See "Play/Stop (hotkeys)" on page 153

CONFigure:DATA:MEAS<i>:IPReplay:STOPall

Stops replaying the playlist.

Example: See Performing IP Replay

Usage: Event

Firmware/Software: V3.2.50

Manual operation: See "Play/Stop (hotkeys)" on page 153

SENSe:DATA:MEAS<i>:IPReplay:PROGress?

Queries the replay progress for all files in the playlist.

The results are returned as follows:

{<FileName>, <Progress>}$_{file\ 1}$, {...}$_{file\ 2}$, ..., {...}$_{file\ n}$

Return values:

<FileName> File name as a string

<Progress> Range: 0 % to 100 %
Default unit: %

Example: See Performing IP Replay

Usage: Query only

Firmware/Software: V3.2.50

Manual operation: See "Playlist (upper part of tab)" on page 153

8.9.2 Application Control and States

The following commands control the IP replay application and return the current application state.

INITiate:DATA:MEAS<i>:IPReplay....343
STOP:DATA:MEAS<i>:IPReplay....343
ABORt:DATA:MEAS<i>:IPReplay....343
FETCh:DATA:MEAS<i>:IPReplay:STATe?....344
FETCh:DATA:MEAS<i>:IPReplay:STATe:ALL?....344

INITiate:DATA:MEAS<i>:IPReplay
STOP:DATA:MEAS<i>:IPReplay
ABORt:DATA:MEAS<i>:IPReplay

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing IP Replay

Usage: Event

Firmware/Software: V3.2.50

Manual operation: See "Application control" on page 154

FETCh:DATA:MEAS<i>:IPReplay:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Example: See Performing IP Replay

Usage: Query only

Firmware/Software: V3.2.50

Manual operation: See "Application control" on page 154

FETCh:DATA:MEAS<i>:IPReplay:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V3.2.50

Manual operation: See "Application control" on page 154

8.9.3 File Information

The following commands query information about the files in the `ip_replay` directory and in the playlist.

FETCh:DATA:MEAS<i>:IPReplay:FILelist?.....345
SENSe:DATA:MEAS<i>:IPReplay:INFofile?.....345
SENSe:DATA:MEAS<i>:IPReplay:TRAFficfile?.....346

FETCh:DATA:MEAS<i>:IPReplay:FILelist?

Queries a list of all files in the directory `ip_replay` of the samba share.

Return values:

<Files> Comma-separated list of strings, one string per filename

Example: See Performing IP Replay

Usage: Query only

Firmware/Software: V3.5.10

Manual operation: See "Edit Playlist (hotkey)" on page 154

SENSe:DATA:MEAS<i>:IPReplay:INFofile? <FileName>

Queries information about a selected file in the playlist.

If the file has not yet been analyzed and the measurement state is RUN, the command triggers file analysis. The analysis takes some time. Repeat the command until the analysis results are available.

Query parameters:

<FileName> File name as a string. Specify the file name with extension but without path, for example "myfile.pcap".

Return values:

<NumberOfPackets> Number of IP packets in the file

<FileSize> Default unit: byte

<Bitrate> Default unit: bit/s

<Duration> Default unit: s

<Type> String indicating the file type and information about the capturing application

<Encapsulation> **"Raw IP"**: file contains raw IP traffic
"Ethernet": file contains IP traffic plus Ethernet headers

Example: See Performing IP Replay

Usage: Query only

Firmware/Software: V3.2.50

Manual operation: See "Playlist (upper part of tab)" on page 153

SENSe:DATA:MEAS<i>:IPReplay:TRAFficfile? <FileName>

Queries information about all IP connections contained in a selected file of the playlist.

If the file has not yet been analyzed and the measurement state is RUN, the command triggers file analysis. The analysis takes some time. Repeat the command until the analysis results are available.

The results are returned as follows:

{<L4Protocol>, <NoOfPackets>, ..., <IPDstPort>}$_{conn\ 1}$, {...}$_{conn\ 2}$, ...

Query parameters:

<FileName> File name as a string. Specify the file name with extension but without path, for example "myfile.pcap".

Return values:

<L4Protocol> Layer 4 protocol as string ("TCP", "UDP", ...)

<NoOfPackets> Number of IP packets for the connection

<IPSrcAddress> IP address of the connection source as string

<IPSrcPort> Port number of the connection source

<IPDstAddress> IP address of the connection destination as string

<IPDstPort> Port number of the connection destination

Example: See Performing IP Replay

Usage: Query only

Firmware/Software: V3.2.50

Manual operation: See "Connection information (lower part of tab)" on page 153

8.10 Audio Delay Measurement

The commands in this section are related to the audio delay measurement. They configure the measurement, control it and return the current measurement state and the measurement results.

CONFigure:DATA:MEAS<i>:ADELay:SAMPles? 347
CONFigure:DATA:MEAS<i>:ADELay:SPINterval? 347
CONFigure:DATA:MEAS<i>:ADELay:MSAMples 348
INITiate:DATA:MEAS<i>:ADELay 348
STOP:DATA:MEAS<i>:ADELay 348
ABORt:DATA:MEAS<i>:ADELay 348
FETCh:DATA:MEAS<i>:ADELay:STATe? 348
FETCh:DATA:MEAS<i>:ADELay:STATe:ALL? 349

FETCh:DATA:MEAS<i>:ADELay:TRACe:ULINk[:CURRent]? 349
FETCh:DATA:MEAS<i>:ADELay:TRACe:DLINk[:CURRent]? 349
FETCh:DATA:MEAS<i>:ADELay:TRACe:LOOPback[:CURRent]? 349
READ:DATA:MEAS<i>:ADELay:TRACe:ULINk[:CURRent]? 350
READ:DATA:MEAS<i>:ADELay:TRACe:DLINk[:CURRent]? 350
READ:DATA:MEAS<i>:ADELay:TRACe:LOOPback[:CURRent]? 350
FETCh:DATA:MEAS<i>:ADELay:TRACe:TAULink[:CURRent]? 350
FETCh:DATA:MEAS<i>:ADELay:TRACe:TALoopback[:CURRent]? 350
READ:DATA:MEAS<i>:ADELay:TRACe:TAULink[:CURRent]? 350
READ:DATA:MEAS<i>:ADELay:TRACe:TALoopback[:CURRent]? 350
FETCh:DATA:MEAS<i>:ADELay:ULINk? 350
FETCh:DATA:MEAS<i>:ADELay:DLINk? 350
FETCh:DATA:MEAS<i>:ADELay:LOOPback? 350
READ:DATA:MEAS<i>:ADELay:ULINk? 350
READ:DATA:MEAS<i>:ADELay:DLINk? 351
READ:DATA:MEAS<i>:ADELay:LOOPback? 351
FETCh:DATA:MEAS<i>:ADELay:TAULink? 351
FETCh:DATA:MEAS<i>:ADELay:TALoopback? 351
READ:DATA:MEAS<i>:ADELay:TAULink? 351
READ:DATA:MEAS<i>:ADELay:TALoopback? 351

CONFigure:DATA:MEAS<i>:ADELay:SAMPles?

Queries the fixed duration of a measurement interval.

Return values:

<Interval> Range: 1 s
Default unit: s

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Settings" on page 157

CONFigure:DATA:MEAS<i>:ADELay:SPINterval?

Queries the fixed number of measurement samples per interval.

Return values:

<SmplPerInterval> Range: 50

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Settings" on page 157

CONFigure:DATA:MEAS<i>:ADELay:MSAMples <MaxSamples>

Configures the maximum number of samples that can be displayed in the result diagrams. The traces cover the sample range *-<MaxSamples> + 1 to 0*.

Parameters:

<MaxSamples>	Range:	1500 to 6000
	*RST:	3000

Example: See Performing an Audio Delay Measurement

Firmware/Software: V3.5.20

Manual operation: See "Settings" on page 157

INITiate:DATA:MEAS<i>:ADELay
STOP:DATA:MEAS<i>:ADELay
ABORt:DATA:MEAS<i>:ADELay

Starts, stops, or aborts the measurement:

- `INITiate...` starts or restarts the measurement. The measurement enters the "RUN" state.
- `STOP...` halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- `ABORt...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCh...STATe?` to query the current measurement state.

See also: "Measurement Control" in the R&S CMW base unit manual, chapter "Remote Control"

Example: See Performing an Audio Delay Measurement

Usage: Event

Firmware/Software: V3.5.20

Manual operation: See "Measurement control" on page 158

FETCh:DATA:MEAS<i>:ADELay:STATe?

Queries the main measurement state. Use `FETCh:...:STATe:ALL?` to query the measurement state including the substates. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MeasState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Measurement control" on page 158

FETCh:DATA:MEAS<i>:ADELay:STATe:ALL?

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCh:...:STATe?` to query the main measurement state only. Use `INITiate..., STOP..., ABORt...` to change the measurement state.

Return values:

<MainState> OFF | RUN | RDY

OFF: measurement off, no resources allocated, no results
RUN: measurement running, synchronization pending or adjusted, resources active or queued
RDY: measurement terminated, valid results can be available

*RST: OFF

<SyncState> PEND | ADJ | INV

PEND: waiting for resource allocation, adjustment, hardware switching ("pending")
ADJ: adjustments finished, measurement running ("adjusted")
INV: not applicable, <MainState> OFF or RDY ("invalid")

<ResourceState> QUE | ACT | INV

QUE: measurement without resources, no results available ("queued")
ACT: resources allocated, acquisition of results in progress but not complete ("active")
INV: not applicable, <MainState> OFF or RDY ("invalid")

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Measurement control" on page 158

FETCh:DATA:MEAS<i>:ADELay:TRACe:ULINk[:CURRent]?
FETCh:DATA:MEAS<i>:ADELay:TRACe:DLINk[:CURRent]?
FETCh:DATA:MEAS<i>:ADELay:TRACe:LOOPback[:CURRent]?

READ:DATA:MEAS<i>:ADELay:TRACe:ULINk[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:DLINk[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:LOOPback[:CURRent]?

Query the values of the audio delay traces "Uplink", "Downlink" and "Loopback".

The trace values are returned from left to right, from sample number -N to sample number 0. N equals the configured maximum number of samples minus one, see `CONFigure:DATA:MEAS<i>:ADELay:MSAMples` on page 348.

Return values:

<Reliability> See Reliability Indicator

<Results> Comma-separated list of delay values, one result per sample
Default unit: s

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Delay result diagram" on page 156

FETCh:DATA:MEAS<i>:ADELay:TRACe:TAULink[:CURRent]?
FETCh:DATA:MEAS<i>:ADELay:TRACe:TALoopback[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:TAULink[:CURRent]?
READ:DATA:MEAS<i>:ADELay:TRACe:TALoopback[:CURRent]?

Query the values of the time of arrival traces "Uplink" and "Loopback".

The trace values are returned from left to right, from sample number -N to sample number 0. N equals the configured maximum number of samples minus one, see `CONFigure:DATA:MEAS<i>:ADELay:MSAMples` on page 348.

Return values:

<Reliability> See Reliability Indicator

<Results> Comma-separated list of time error values, one result per sample
Default unit: s

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Time of Arrival result diagram" on page 157

FETCh:DATA:MEAS<i>:ADELay:ULINk?
FETCh:DATA:MEAS<i>:ADELay:DLINk?
FETCh:DATA:MEAS<i>:ADELay:LOOPback?
READ:DATA:MEAS<i>:ADELay:ULINk?

READ:DATA:MEAS<i>:ADELay:DLINk?
READ:DATA:MEAS<i>:ADELay:LOOPback?

Query the statistical audio delay results for "Uplink", "Downlink" and "Loopback".

Return values:

<Reliability> See Reliability Indicator

<Results> Comma-separated list of four results: <Current>, <Average>, <Minimum>, <Maximum> value
Default unit: s

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Result table" on page 157

FETCh:DATA:MEAS<i>:ADELay:TAULink?
FETCh:DATA:MEAS<i>:ADELay:TALoopback?
READ:DATA:MEAS<i>:ADELay:TAULink?
READ:DATA:MEAS<i>:ADELay:TALoopback?

Query the statistical time of arrival results for "Uplink" and "Loopback".

Return values:

<Reliability> See Reliability Indicator

<Results> Comma-separated list of four results: <Current>, <Average>, <Minimum>, <Maximum> value
Default unit: s

Example: See Performing an Audio Delay Measurement

Usage: Query only

Firmware/Software: V3.5.20

Manual operation: See "Result table" on page 157

8.11 Quality-of-Service Settings

The commands in this section are related to the handling of outgoing IP packets.

SOURce:DATA:MEAS<i>:QOS:STATe ... 352
CONFigure:DATA:MEAS<i>:QOS:FILTer:ADD ... 352
SOURce:DATA:MEAS<i>:QOS:FILTer:CATalog? ... 352
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:REMove ... 353
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:ENABle ... 353
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:IPADdress ... 353
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PROTocol ... 354
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:SRCPrange ... 354

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PRANge....354
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:TCPackprio:ENABle....355
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:HOPLmt....355
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PLRate....355
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DELay....356
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JITTer....356
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JDIStribut....356
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:RRATe....357
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DRATe....357
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:CRATe....358
CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:BITRate....358

SOURce:DATA:MEAS<i>:QOS:STATe <Control>

Switches the QoS feature on or off. To enable QoS profiles, see `CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:ENABle` on page 353.

Setting parameters:

<Control> ON | OFF

Switch the QoS feature on or off

Return values:

<NIState> OFF | PENDing | ON

OFF: QoS feature switched off
PEND: switching on/off is ongoing
ON: QoS feature switched on

*RST: OFF

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "QoS activation" on page 162

CONFigure:DATA:MEAS<i>:QOS:FILTer:ADD

Creates a QoS profile.

Example: See Configuring QoS Profiles

Usage: Event

Firmware/Software: V3.5.30

Manual operation: See "Add or delete a tab" on page 160

SOURce:DATA:MEAS<i>:QOS:FILTer:CATalog?

Queries a list of all existing QoS profiles.

Return values:

<Catalog> String with comma-separated list of profile names, for example:
`"Filter 1,Filter 2,Filter 3"`

Usage: Query only

Firmware/Software: V3.5.40

Manual operation: See "Add or delete a tab" on page 160

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:REMove

Deletes the QoS profile number <Index>.

Suffix:
<Index> 2..15

Usage: Event

Firmware/Software: V3.5.30

Manual operation: See "Add or delete a tab" on page 160

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:ENABle <Enable>

Enables or disables a QoS profile. To use QoS profiles, you must also activate the QoS feature, see `SOURce:DATA:MEAS<i>:QOS:STATe` on page 352.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<Enable> OFF | ON
*RST: OFF

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Used" on page 160

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:IPADdress <IPAddress>

Specifies the destination address as filter criterion for IP packets.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<IPAddress> String indicating a full IPv4 address or a full IPv6 address or an IPv6 prefix
*RST: '172.22.1.100'

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "IP Address" on page 160

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PROTocol <Protocol>

Specifies the protocol as filter criterion for IP packets.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<Protocol> ALL | TCP | UDP
No filtering, TCP only, UDP only
*RST: ALL

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Protocol" on page 160

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:SRCPrange <StartPort>, <EndPort>

Specifies a source port range as filter criterion for IP packets.

To disable source port filtering, set both values to zero.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<StartPort> Range: 0 to 65535
*RST: 0

<EndPort> Range: 0 to 65535
*RST: 0

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Src Port Range / Dst Port Range" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PRANge <StartPort>, <EndPort>

Specifies a destination port range as filter criterion for IP packets.

To disable destination port filtering, set both values to zero.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<StartPort> Range: 0 to 65535
*RST: 0

<EndPort> Range: 0 to 65535
*RST: 0

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Src Port Range / Dst Port Range" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:TCPackprio:ENABle <Enable>

Enables the prioritization of TCP acknowledgments over other packets.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<Enable> OFF | ON
*RST: OFF

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "TCP ACK Prioritization" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:HOPLmt <HOPLmt>

Sets the hop limit of packets matching the filter criteria. The setting 0 means that the hop limit of the packets is left unchanged.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<HOPLmt> Range: 0 to 255
*RST: 0

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Hop Limit" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:PLRate <PacketLossRate>

Specifies the packet loss rate for a QoS profile.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:

<PacketLossRate> Range: 0 % to 100 %
*RST: 0 %
Default unit: %

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Packetloss Rate" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DELay <Delay>

Specifies the delay for a QoS profile.

Suffix:

<Index> 1..15
Selects the QoS profile

Parameters:

<Delay> Range: 0 s to 10 s
*RST: 0 s
Default unit: s

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Delay" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JITTer <Jitter>

Specifies the jitter for a QoS profile. The jitter must be smaller than or equal to the configured delay.

Suffix:

<Index> 1..15
Selects the QoS profile

Parameters:

<Jitter> Range: 0 s to 10 s
*RST: 0 s
Default unit: s

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Jitter, Jitter Distribution" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:JDIStribut <JitterDistr>

Specifies the jitter distribution for a QoS profile.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<JitterDistr> UNIForm | NORMal | PAReto | PNORmal
Uniform, normal, pareto, pareto normal
*RST: NORM

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Jitter, Jitter Distribution" on page 161

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:RRATe <ReorderRate>

Specifies the reordering rate for a QoS profile.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<ReorderRate> Range: 0 % to 100 %
*RST: 0 %
Default unit: %

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Reordering Rate" on page 162

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:DRATe <DuplicateRate>

Specifies the percentage of packets to be duplicated.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<DuplicateRate> Range: 0 % to 100 %
*RST: 0 %
Default unit: %

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Duplicates Rate" on page 162

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:CRATe <CorruptRate>

Specifies the percentage of packets to be corrupted.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<CorruptRate> Range: 0 % to 100 %
*RST: 0 %
Default unit: %

Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Corruption Rate" on page 162

CONFigure:DATA:MEAS<i>:QOS:FILTer<Index>:BITRate <QosBitrate>

Specifies the maximum bit rate for a QoS profile.

Suffix:
<Index> 1..15
Selects the QoS profile

Parameters:
<QosBitrate> Range: 0 bit/s to 100E+9 bit/s
*RST: 0 bit/s
Default unit: bit/s
Additional values: OFF | ON (disables | enables the bit-rate limitation)

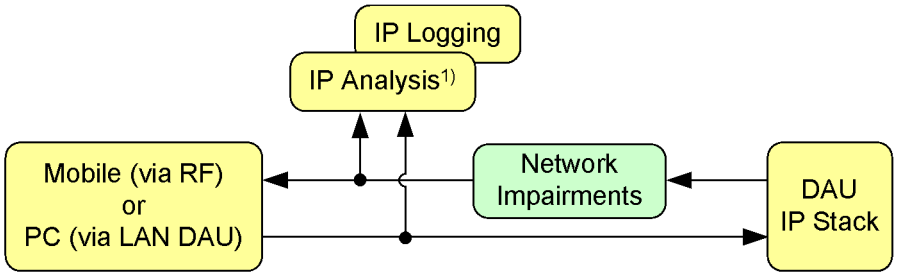
Example: See Configuring QoS Profiles

Firmware/Software: V3.5.30

Manual operation: See "Maximum Bit Rate" on page 162

9 Annex: Network Impairments

The DAU allows you to simulate network impairments for any outgoing IP traffic. You can, for example, impair traffic from the DAU to a mobile or traffic from the DAU to a host reachable via the LAN DAU connector.



1) IP Analysis is only supported for traffic to/from mobile

The DAU applications "IP Analysis/Security" and "IP Logging" monitor the outgoing traffic after the network impairments have been applied. They have no information about the unimpaired traffic.

Analysis of packet retransmission with network impairments

To recognize a retransmitted packet, the IP analysis application must receive both the initially transmitted packet and the retransmitted packet.

If the initially transmitted packet is lost **after** passing the application, the retransmitted packet is counted correctly by the IP analysis application. Example: packet loss on the RF connection to a mobile.

If the initially transmitted packet is lost **before** passing the application, the retransmitted packet is counted as initial transmission, not as retransmission. Example: packet loss rate configured as network impairment.

In the latter case, you can use the IP logging application to analyze the retransmissions, for example by checking for retransmission requests.

The network impairments functionality of the DAU is based on the netem Linux kernel module (see http://www.linuxfoundation.org/collaborate/workgroups/networking/netem).

For configuration and control of network impairments, see Chapter 5.11, "Quality-of-Service Settings", on page 158.

The following sections illustrate the effect of the individual network impairment settings.

9.1 Delay and Jitter

To show the effect of configured delays and jitter distribution types, the round-trip time (RTT) resulting from the settings has been analyzed with the ping application of the DAU.

The results are shown in the following sections. Dashed gray lines in the figures indicate the configured delay value ± the configured jitter value.

9.1.1 RTT without Impairments

Without any configured delay or jitter, the following RTT have been measured by sending 300 ping requests. For the used setup, the RTT is less than 1 ms and fairly constant.

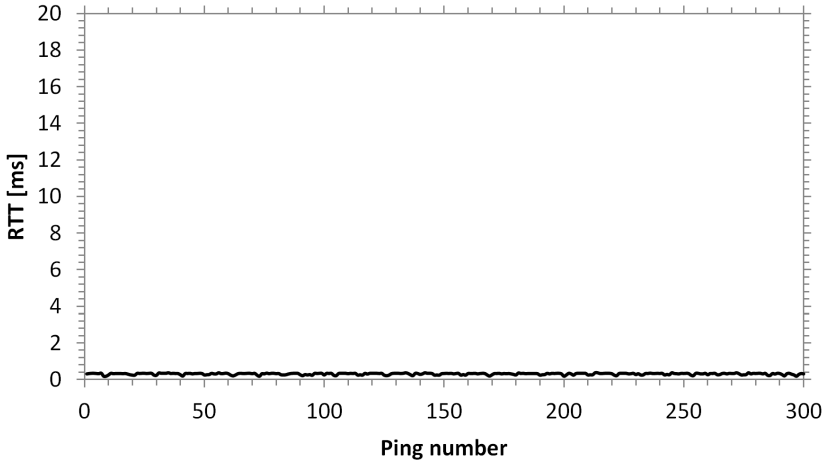


Figure 9-1: RTT vs. ping request, no impairments

Minimum RTT	Maximum RTT	Average RTT
0.16 ms	0.55 ms	0.31 ms

9.1.2 RTT with Delay (No Jitter)

With a configured delay of 70 ms, but without jitter, the measured RTT is increased by 70 ms.

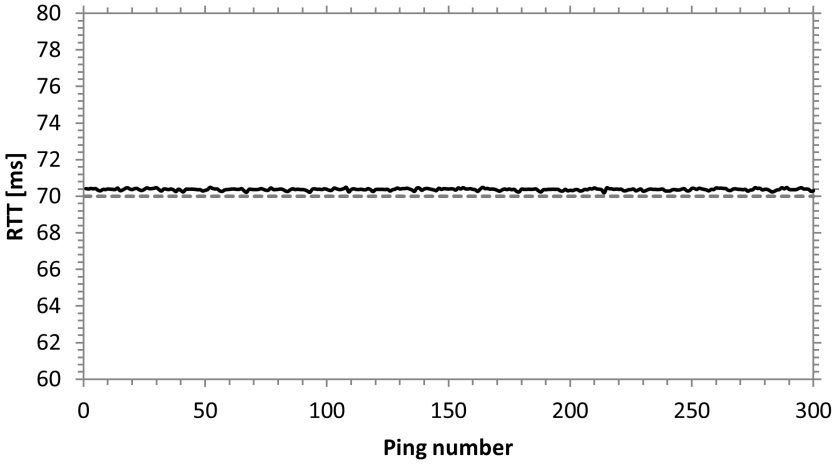


Figure 9-2: RTT vs. ping request, delay without jitter

Minimum RTT	Maximum RTT	Average RTT
70.19 ms	70.80 ms	70.38 ms

9.1.3 RTT with Uniform Jitter Distribution

A delay of 70 ms is configured and also a jitter of 10 ms with a uniform distribution.

In comparison to the previous setup (delay without jitter), the maximum RTT is increased by the jitter value and the minimum RTT is decreased by the jitter value.

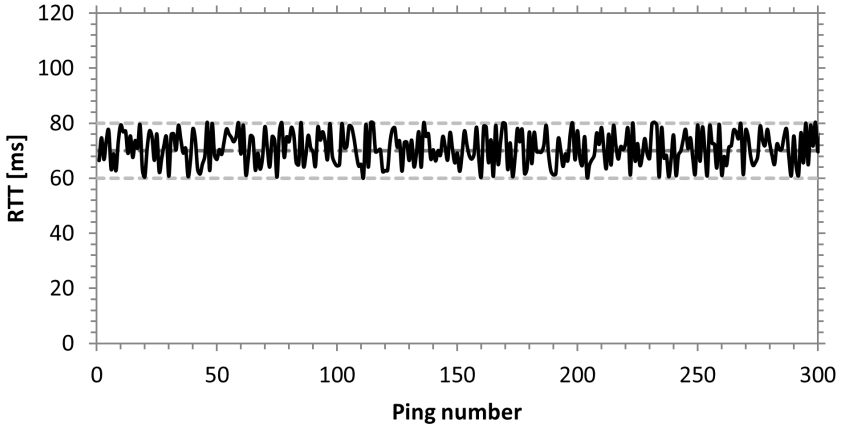


Figure 9-3: RTT vs. ping request, uniform jitter distribution

Within the interval [delay - jitter, delay + jitter], the measured RTT values are distributed uniformly. The following figure shows the distribution of 1000 measured RTT values (previous figure shows the first 300 of these 1000 values).

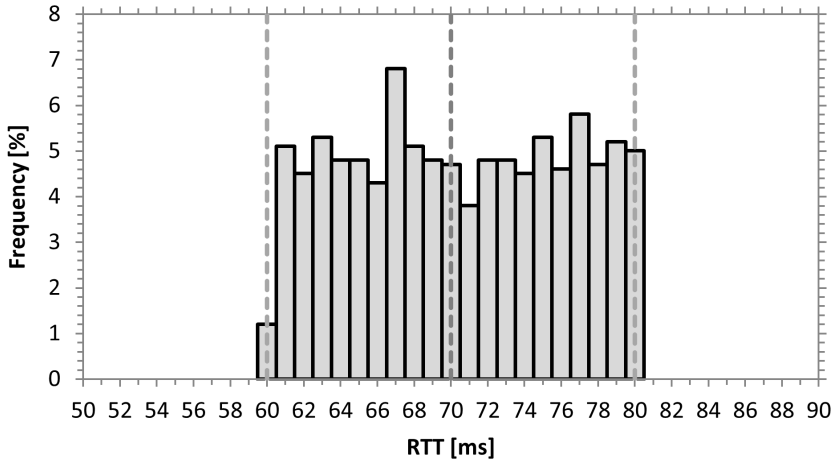


Figure 9-4: Histogram of uniform jitter distribution

Minimum RTT	Maximum RTT	Average RTT
60.32 ms	80.38 ms	70.39 ms

9.1.4 RTT with Normal Jitter Distribution

A delay of 70 ms is configured and also a jitter of 10 ms with a normal (Gaussian) distribution.

With this configuration, the following RTT values have been measured.

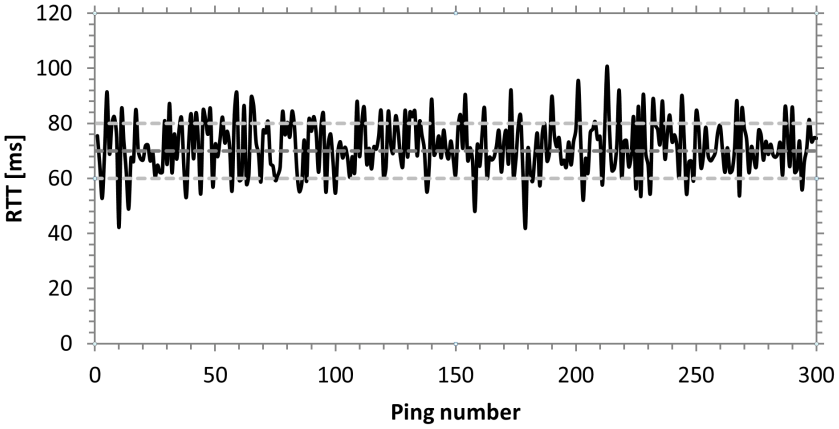


Figure 9-5: RTT vs. ping request, normal jitter distribution

For an ideal Gaussian distribution of the RTT values, the mean value μ of the distribution equals the delay and the standard deviation σ of the distribution equals the jitter. The interval [μ - σ, μ + σ] contains 68.27 % of all RTT values. The interval [μ - 2σ, μ + 2σ] contains 95.45 %.

The following figure shows the distribution of 1000 measured RTT values (previous figure shows the first 300 of these 1000 values).

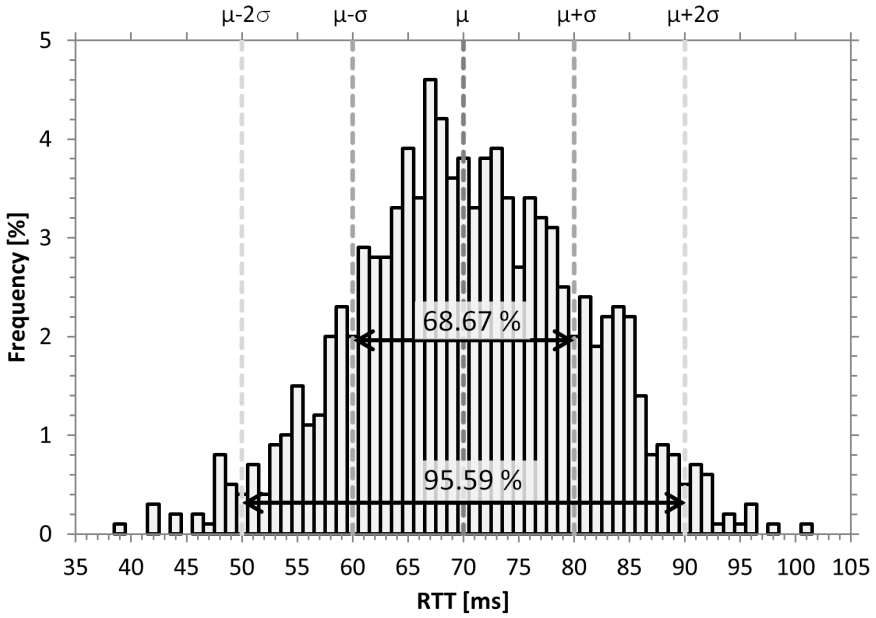


Figure 9-6: Histogram of normal jitter distribution

Minimum RTT	Maximum RTT	Average RTT
39.45 ms	100.79 ms	70.51 ms

9.1.5 RTT with Pareto Jitter Distribution

Modeling of the delay jitter in a real network is still a matter of research. However, some papers claim that the pareto distribution is suitable to model the delay jitter of the Internet.

A delay of 70 ms is configured and also a jitter of 10 ms with a pareto distribution.

With this configuration, the following RTT values have been measured.

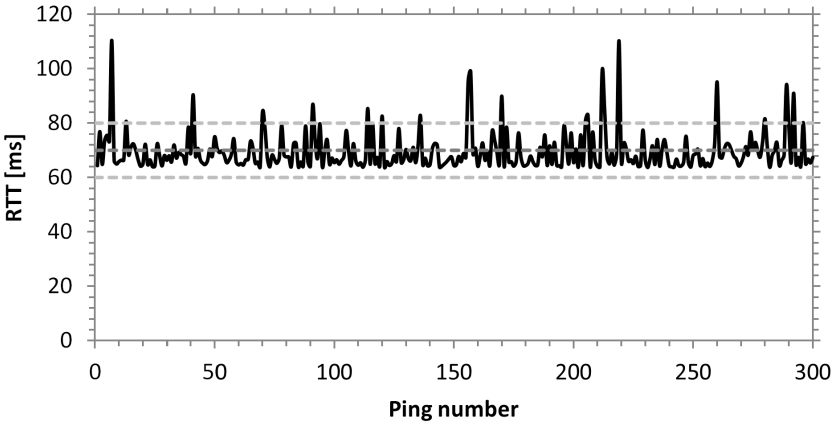


Figure 9-7: RTT vs. ping request, pareto jitter distribution

The following figure shows the distribution of 1000 measured RTT values (previous figure shows the first 300 of these 1000 values).

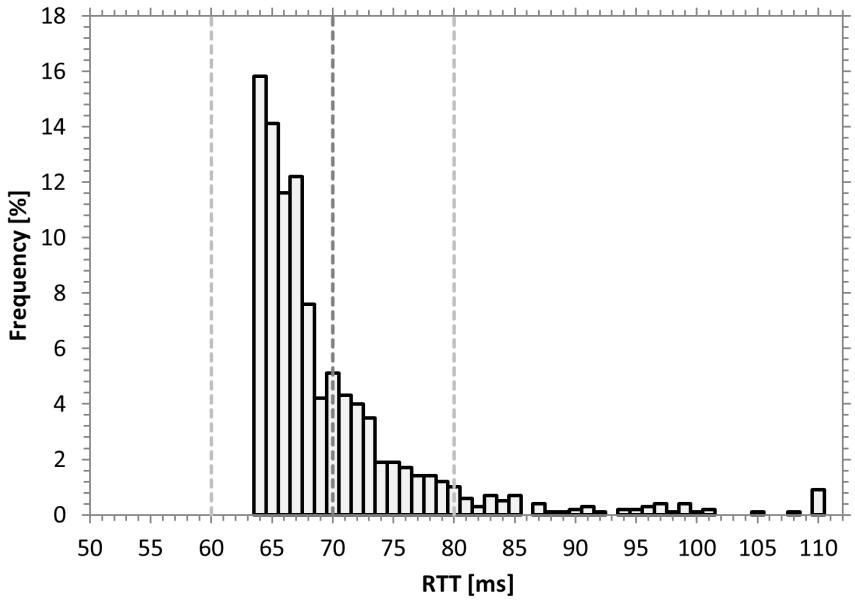


Figure 9-8: Histogram of pareto jitter distribution

Minimum RTT	Maximum RTT	Average RTT
63.67 ms	110.50 ms	69.81 ms

9.1.6 RTT with Pareto Normal Jitter Distribution

A delay of 70 ms is configured and also a jitter of 10 ms with a pareto normal distribution (a combination of a normal distribution and a pareto distribution).

With this configuration, the following RTT values have been measured.

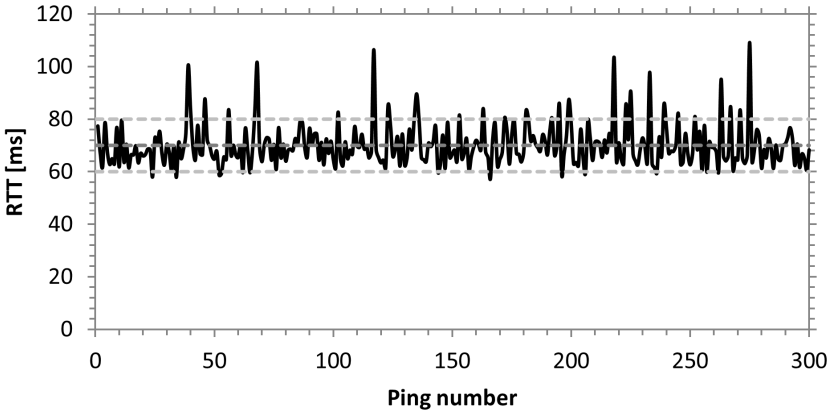


Figure 9-9: RTT vs. ping request, pareto normal jitter distribution

The following figure shows the distribution of 1000 measured RTT values (previous figure shows the first 300 of these 1000 values).

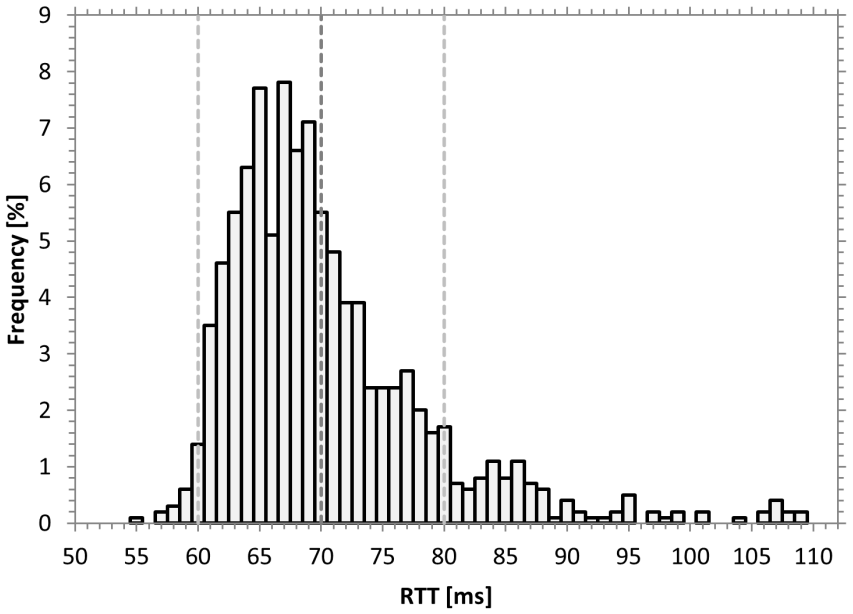


Figure 9-10: Histogram of pareto normal jitter distribution

Minimum RTT	Maximum RTT	Average RTT
55.30 ms	109.19 ms	70.41 ms

9.2 Reordering Rate

Reordering of packets requires buffering of packets. So a configured delay is a prerequisite for packet reordering. For the examples in this section, a delay of 50 ms without jitter is configured. UDP is used with an average packet rate of one packet per ms.

Each reordering rate result value is derived from a measurement interval of 100 packets. The following figures show the measured reordering rates for the first 300 measurement intervals (30000 packets). The configured reordering rates are 10 % and 25 %, as indicated by the gray dashed lines.

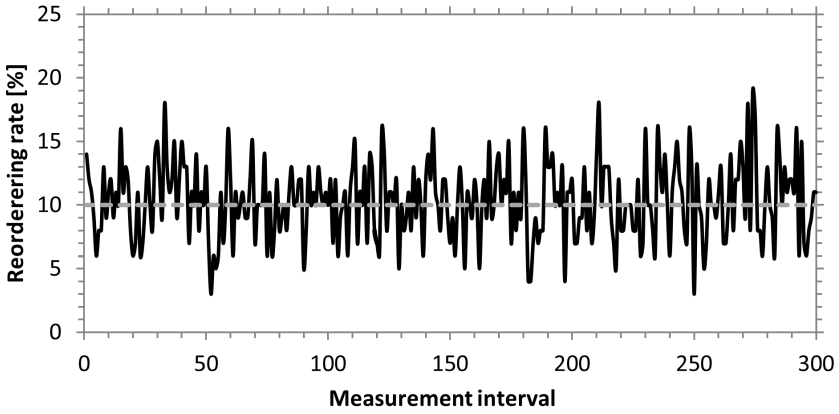


Figure 9-11: Reordering rate vs. measurement interval (10 % reordering configured)

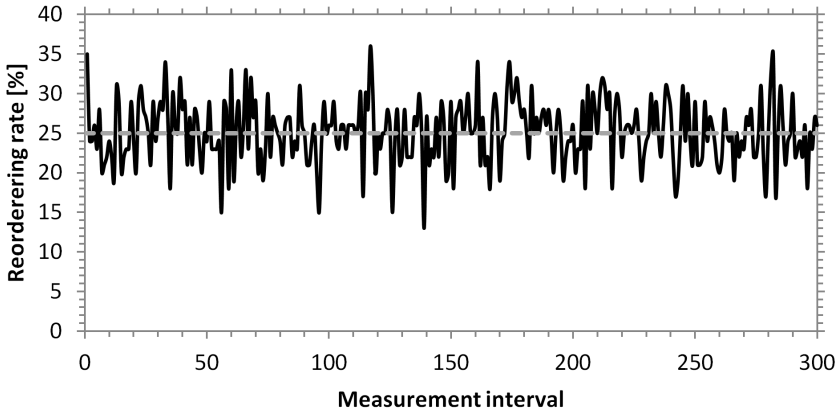


Figure 9-12: Reordering rate vs. measurement interval (25 % reordering configured)

The following figures show the distribution of the reordering rates over all measured intervals.

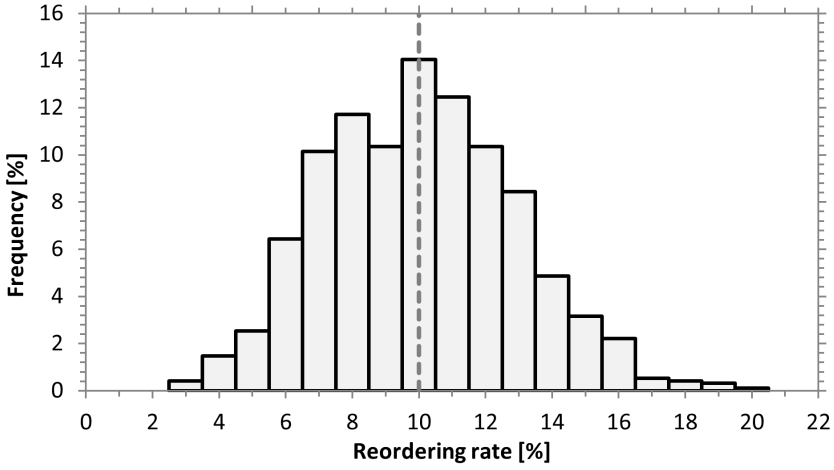


Figure 9-13: Reordering rate distribution (10 % reordering configured)

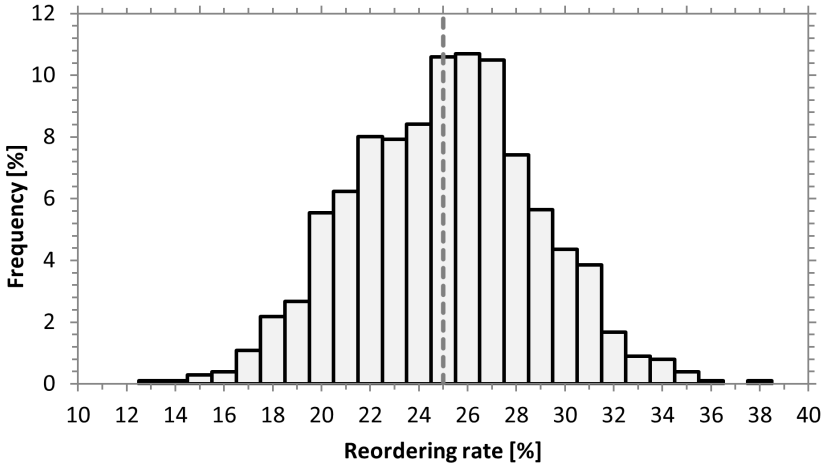


Figure 9-14: Reordering rate distribution (25 % reordering configured)

9.3 Duplicates Rate

The effect of configured packet duplication rates is tested without configured delay, with UDP and with an average packet rate of one packet per ms.

Each duplicates rate result value is derived from a measurement interval of 100 packets. The following figures show the measured duplicates rates for the first 300 measurement intervals (30000 packets). The configured duplicates rates are 10 % and 25 %, as indicated by the gray dashed lines.

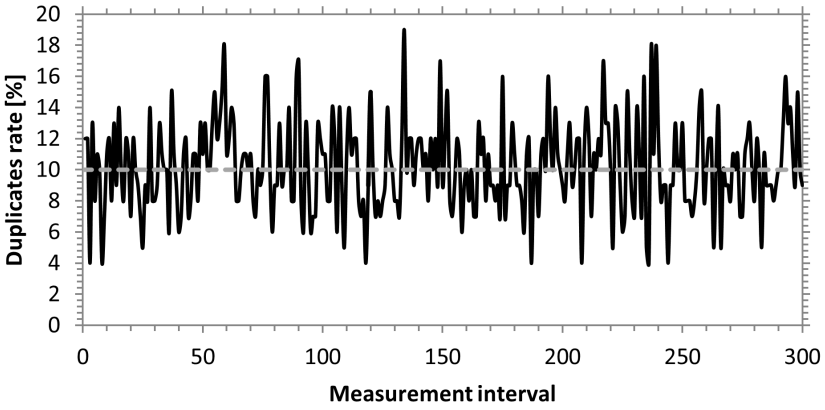


Figure 9-15: Duplicates rate vs. measurement interval (10 % duplicates configured)

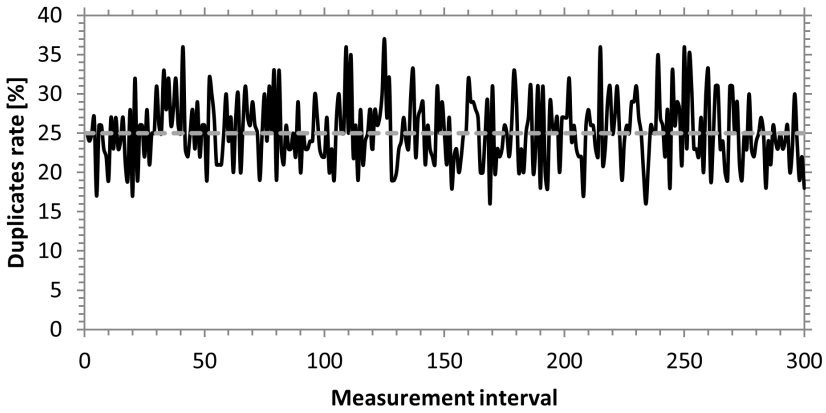


Figure 9-16: Duplicates rate vs. measurement interval (25 % duplicates configured)

The following figures show the distribution of the duplicates rates over all measured intervals.

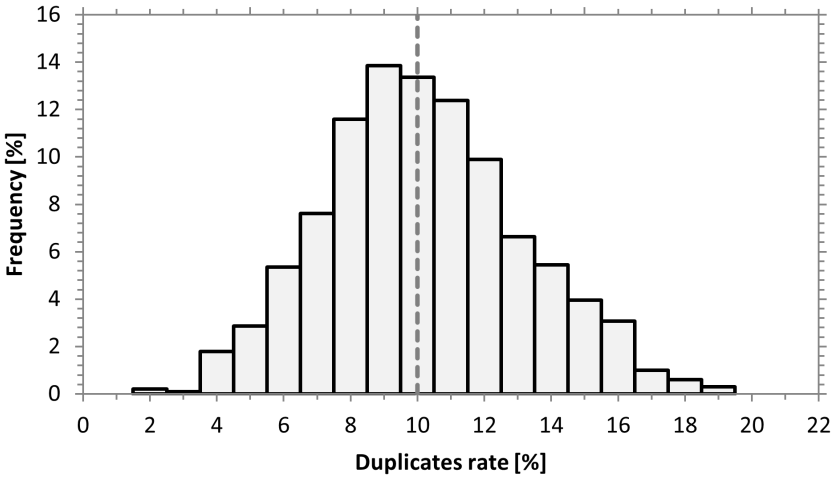


Figure 9-17: Duplicates rate distribution (10 % duplicates configured)

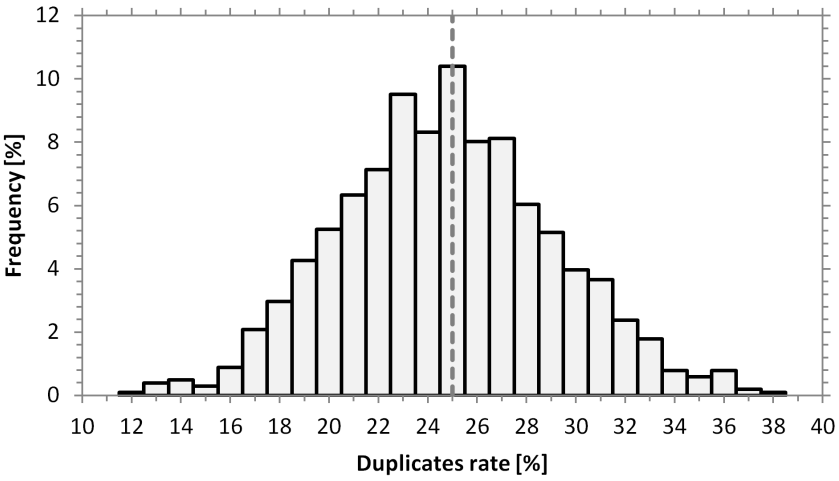


Figure 9-18: Duplicates rate distribution (25 % duplicates configured)

9.4 Packet Loss Rate

The effect of configured packet loss rates is tested without configured delay, with UDP and with an average packet rate of one packet per ms.

Each packet loss rate result value is derived from a measurement interval of 100 packets. The following figures show the measured packet loss rates for the first 300 measurement intervals (30000 packets). The configured packet loss rates are 10 % and 25 %, as indicated by the gray dashed lines.

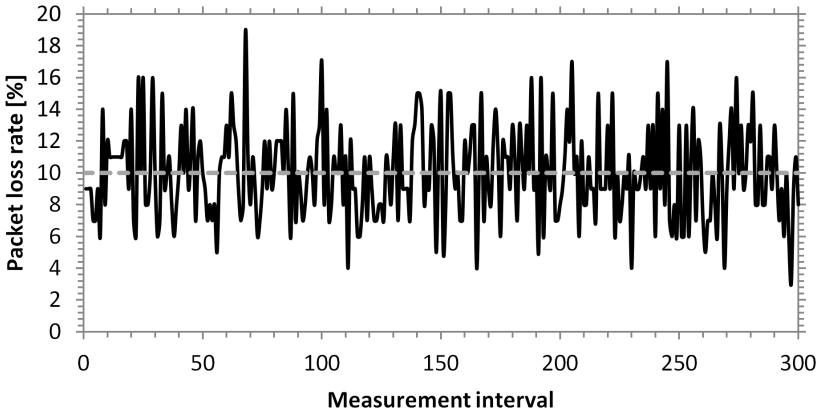


Figure 9-19: Packet loss rate vs. measurement interval (10 % loss configured)

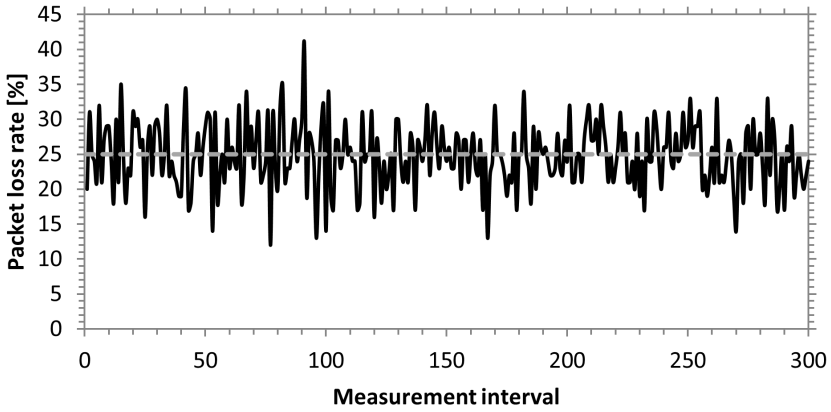


Figure 9-20: Packet loss rate vs. measurement interval (25 % loss configured)

The following figures show the distribution of the packet loss rates over all measured intervals.

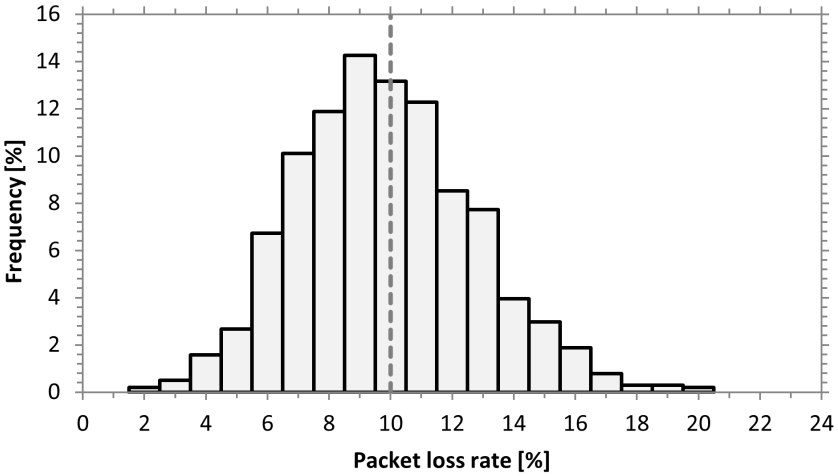


Figure 9-21: Packet loss rate distribution (10 % packet loss configured)

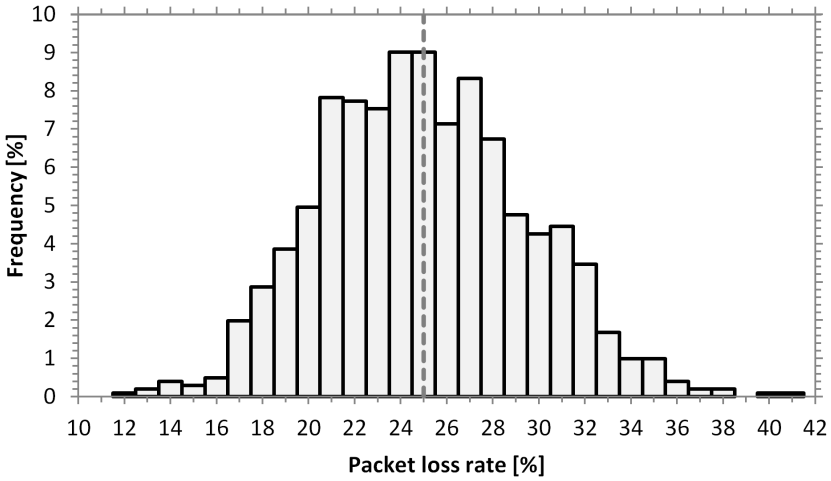


Figure 9-22: Packet loss rate distribution (25 % packet loss configured)

9.5 Corruption Rate

The effect of configured corruption rates is tested without configured delay, with UDP and with an average packet rate of one packet per ms.

Each corruption rate result value is derived from a measurement interval of 100 packets. The value indicates the percentage of packets having a bit error.

The following figures show the measured corruption rates for the first 300 measurement intervals (30000 packets). The configured corruption rates are 10 % and 25 %, as indicated by the gray dashed lines.

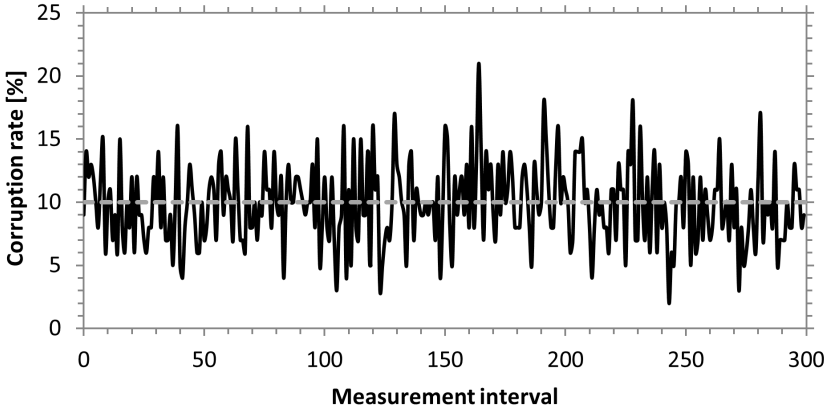


Figure 9-23: Corruption rate vs. measurement interval (10 % corruption configured)

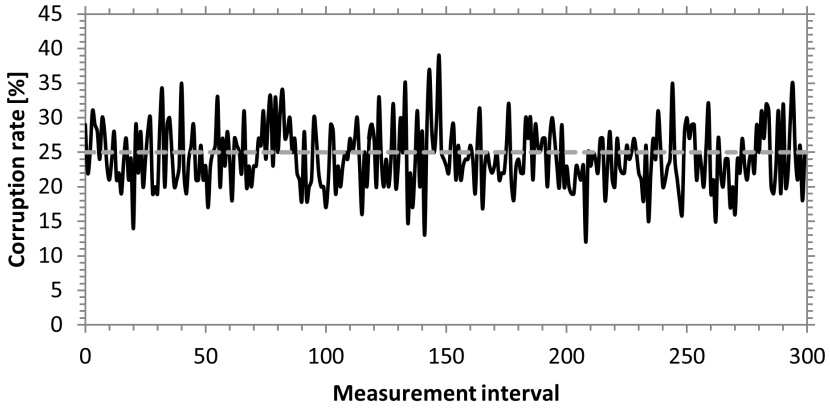


Figure 9-24: Corruption rate vs. measurement interval (25 % corruption configured)

The following figures show the distribution of the corruption rates over all measured intervals.

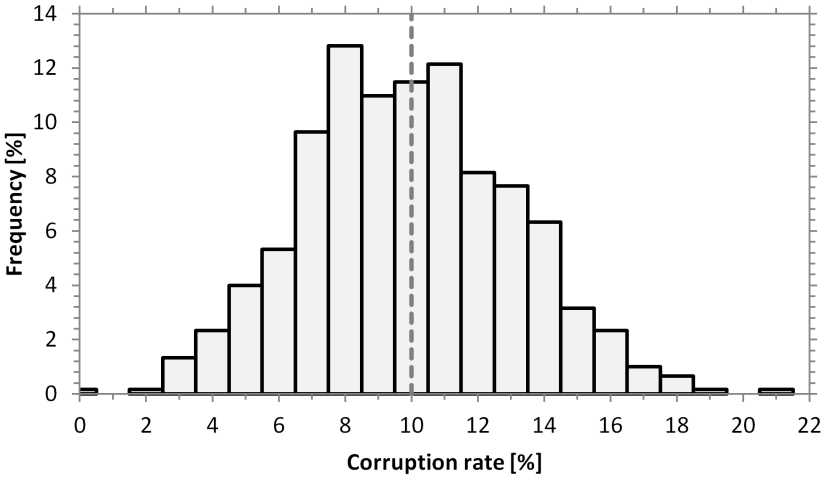


Figure 9-25: Corruption rate distribution (10 % corruption configured)

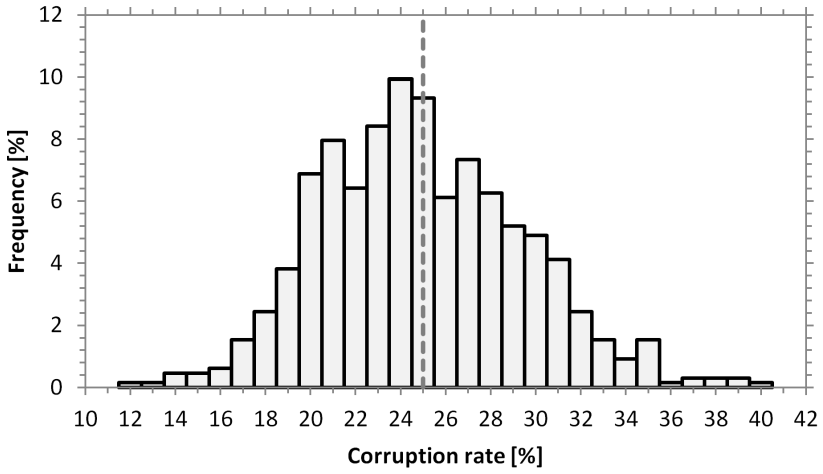


Figure 9-26: Corruption rate distribution (25 % corruption configured)

List of Commands

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ABORt:DATA:MEAS<i>:DNSRequests....304
ABORt:DATA:MEAS<i>:IPANalysis....311
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